

CONTRACTOR SUBMITTAL TRANSMITTAL FORM REV. A

Owner: County of Hawaii
Contractor: Nan, Inc. **Project No.:** WW-4705R
Project Name: Hilo WWTP Phase 1 **Submittal Number:**
Submittal Title: FOR INFORMATION ONLY
TO:
From: Nan Inc.

Specification No. and Subject of Submittal / Equipment Supplier	
Spec:	Paragraph:
Authored By:	Date Submitted:

Submittal Certification	
Check Either (A) or (B):	
☐ (A)	We have verified that the equipment or material contained in this submittal meets all the requirements specified in the project manual or shown on the contract drawings with no exceptions.
☐ (B)	We have verified that the equipment or material contained in this submittal meets all the requirements specified in the project manual or shown on the contract drawings except for the deviations listed.
Certification Statement: By this submittal, I hereby represent that I have determined and verified all field measurements, field construction criteria, materials, dimensions, catalog numbers and similar data, and I have checked and coordinated each item with other applicable approved shop drawings and all Contract requirements.	
General Contractor's Reviewer's Signature: 	
Printed Name and Title:	
In the event, Contractor believes the Submittal response does or will cause a change to the requirements of the Contract, Contractor shall immediately give written notice stating that Contractor considers the response to be a Change Order.	
Firm:	Signature: Date Returned:

PM/CM Office Use
Date Received GC to PM/CM:
Date Received PM/CM to Reviewer:
Date Received Reviewer to PM/CM:
Date Sent PM/CM to GC:

Nan, Inc

 PROJECT: HILO WWTP REHABILITATION
 AND REPLACEMENT PROJECT - PHASE 1

JOB NO. WW-4705R

 THIS SUBMITTAL HAS BEEN CHECKED BY
 THIS CONTRACTOR. IT IS CERTIFIED
 CORRECT, COMPLETE, AND IN
 COMPLIANCE WITH CONTRACT
 DRAWINGS AND SPECIFICATIONS. ALL
 AFFECTED CONTRACTORS AND
 SUPPLIERS ARE AWARE OF, AND WILL
 INTEGRATE THIS SUBMITTAL (UPON
 APPROVAL) INTO THEIR OWN WORK.

 DATE RECEIVED _____
 SPECIFICATION SECTION # _____
 SPECIFICATION _____
 PARAGRAPH _____
 DRAWING _____
 SUBCONTRACTOR _____
 SUPPLIER _____
 MANUFACTURER _____
CERTIFIED BY CQCM or Designee : 

Submittal



SUB Reference No : MECI-SUB-0009

Status: GC AND CLIENT REVIEW

For Action: Darrin Lee, MECI
David Wieseler, MECI

Project: HILO WWTP REHABILITATION AND REPLACEMENT PROJECT

Subject: **16305 - ELECTRICAL SYSTEMS STUDIES - INITIAL SHORT-CIRCUIT FAULT ANALYSIS**

Submittal on initial short-circuit fault analysis.
Information provided is derived from Specification 16305 Section 1.04.B.

Discipline: ELEC **Area:** 00

Submitted For: Approval

Specification Reference : 16305 - ELECTRICAL SYSTEM STUDIES

Paragraph No.: 1.04.B

Description : Submittal on initial short-circuit fault analysis.
Information provided is derived from Specification 16305 Section 1.04.B.

Discipline: ELEC **Area:** 00

Attachments: 16305 ELECTRICAL SYSTEM STUDIES 1.0.pdf

SUB by:	Hannah Anderson, MECI	On: 24 September 2025
Created By:	Hannah Anderson, MECI	On: 24 September 2025, 10:44:43 AM -10:00
Last Edited By:	Hannah Anderson, MECI	On: 24 September 2025, 11:47:18 AM -10:00

01 - QC/Project Engineer Review

Approved without comment.
Requested pages removed. Approved without comment.

QC/Project Engineer Review by:	David Wieseler, MECI	On: 24 September 2025
Created By:	David Wieseler, MECI	On: 24 September 2025, 10:51:25 AM -10:00
Last Edited By:	David Wieseler, MECI	On: 24 September 2025, 11:49:38 AM -10:00

02 - Project Manager Review

approved without comment

Project Manager Darrin Lee, MECI
Review by:
Created By: Darrin Lee, MECI
Last Edited By: Darrin Lee, MECI

On: 24 September 2025
On: 24 September 2025, 11:27:15 AM -10:00
On: 25 September 2025, 10:43:04 PM -10:00

MASS ELECTRIC CONSTRUCTION COMPANY PRODUCT DATA SUBMITTAL

REVISION	DATE	PREPARED BY	APPROVED BY
1	09/24/2025	Hannah Anderson	David Wieseler

HILO WASTEWATER TREATMENT PLANT REHABILITATION AND REPLACEMENT PROJECT



SPECIFICATION - 16305 ELECTRICAL SYSTEMS STUDIES

DOCUMENT REVISION LOG						
Revision Number	Revision Date	Description	Approvals			
			PE INITIAL		PM INITIAL	
0	09/24/2025	Issued for Review	DCW	09/24/2025		
1	09/24/2025	Issued for Review – Requested pages removed	DCW	09/24/2025		

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SECTION 16305
ELECTRICAL SYSTEM STUDIES

PART 1 GENERAL

1.01 SUMMARY

- A. Section includes:
1. Short-circuit fault analysis study.
 2. Protective device coordination study.
 3. Arc-flash hazard study.

1.02 REFERENCES

- A. Definitions:
1. Modified: Equipment with load additions or with loads being removed that affect fault current, include new overcurrent protective devices that require settings and device coordination, or require additional/removal/replacement of arc flash labels.
- B. Standards:
1. American National Standards Institute (ANSI).
 2. Institute of Electrical and Electronics Engineers (IEEE):
 - a. 1584 - IEEE Guide for Specification of Scope and Deliverable Requirements for an Arc-Flash Hazard Calculations Study in Accordance with IEEE Std 1584(TM).
 3. National Fire Protection Association (NFPA):
 - a. 70E - Standard for Electrical Safety in the Workplace.

1.03 DELEGATED DESIGN

- A. As specified in Section 01357 - Delegated Design Procedures.
- B. Signed and Sealed electrical system study reports.

1.04 SUBMITTALS

- A. Furnish submittals as specified in Section 01330 - Submittal Procedures.

- ✓ B. Initial studies and reports:
- ✓ 1. Include the following in the initial short-circuit current report:
 - ✓ a. List of all devices included in the studies.
 - ✓ b. A description of all operating scenarios.
 - c. Form and format of arc-flash labels

ARC-FLASH LABELS WILL
BE SUBMITTED WITH A
FUTURE SUBMITTAL

- C. Delegated Design final studies and reports:
1. Format and quantity:
 - a. Provide 3 bound copies of all final reports.

- b. Provide 3 complete sets of electronic files on CD or DVD media, including the electrical system model(s), configuration files, custom libraries, and any other files used to perform the studies and produce the reports. Also provide an electronic version of the bound reports in PDF format.
- 2. Include the sections below in the final report:
 - a. Copies of correspondence and data obtained from the electric utility company.
 - b. Letter certifying the inspection and verification of existing equipment and incorporation of applicable RFI's and change orders.
 - c. One-line diagrams:
 - 1) The following information shall be included at a minimum:
 - a) Motor horsepower.
 - b) Transformer data:
 - (1) kVA.
 - (2) Configuration.
 - c) Cable data:
 - (1) Insulation.
 - (2) Size.
 - (3) Length.
 - 2) One-line diagrams shall be fully legible at 11-inch by 17-inch size.
 - d. Include in the short-circuit fault analysis study:
 - 1) Descriptions, purpose, basis, assumptions, recommendations, and scope of the study.
 - 2) Normal system connections and those that result in maximum fault conditions.
 - 3) Tabulation of circuit breaker, fuse, and other protective device ratings compared to maximum calculated short-circuit duties.
 - 4) Fault current calculations for the cases run including a definition of terms and guide for interpretation of computer software printouts.
 - e. Protective device coordination study shall include:
 - 1) Descriptions, purpose, basis, assumptions, recommendations, and scope of the study.
 - 2) List all requirements used in the selection and setting criteria for any protective devices.
 - 3) TCCs graphically indicating the coordination proposed for the system on log-log graphs:
 - a) All TCCs shall be in color.
 - 4) Tabulation of relay, fuse, circuit breaker, and other protective devices in graphical form with a one-line diagram to display area coordination.
 - 5) Where coordination could not be achieved, an explanation shall be included in the report to support the statement along with recommendations to improve coordination. Recommended equipment modifications or settings shall be in a tabulated form.
 - 6) All protective device settings in a tabulated format.
 - f. Include in the arc-flash hazard study:
 - 1) Descriptions, purpose, basis, assumptions, recommendations, and scope of the study.
 - 2) Normal system connections and those that result in maximum arc-flash conditions.
 - 3) Arc-flash raw data, calculations, and assumptions.

- 4) Arc-flash label data:
 - a) Identifying the content of each label.
 - b) Identifying the location of each label.
- D. Submit the credentials of the individual(s) performing the study and the individual in responsible charge of the study.
- E. Submit course outline for Owner's training.

1.05 QUALITY ASSURANCE

- A. Qualifications of the entity responsible for electrical system studies:
 - 1. A minimum of 5 years of experience in power system analysis is required for the individual in responsible charge of the studies.
 - 2. The short-circuit fault analysis, protective device coordination, and arc-flash hazard studies shall be performed with the aid of a digital computer program:
 - a. Point-to-point calculations are not acceptable.
- B. The study shall be performed by an independent firm or the equipment manufacturer of the Switchboard addition.

1.06 ADMINISTRATIVE REQUIREMENTS

- A. Coordination:
 - 1. The individual performing the studies shall visit the site and collect all necessary field data in order to perform and complete comprehensive electrical system studies.
 - 2. Obtain, for all equipment, the required data for preparation of the study including, but not limited to:
 - a. Transformer kilovolt-ampere (kVA), nameplate data, and impedances.
 - b. Generator impedances, kilovolt-ampere (kVA), and voltage.
 - c. Generator decrement curves.
 - d. Bus withstand ratings.
 - e. Cable and bus data.
 - f. Protective device taps, time dials, instantaneous pickups, and time-delay settings.
 - g. As-built lengths for all new wire and cable installations covered by the studies.
 - h. Grounding schemes (solidly grounded or resistive grounded).
 - 3. Obtain the Electric Utility information on the minimum and maximum available fault current, minimum and maximum utility impedances, utility protective device settings including manufacturer and model number, interrupting ratings, X/R ratios, and model information one level above the point of connection:
 - a. Utility tolerances and voltage variations.
 - 4. Obtain equipment layouts and configurations from the manufacturer's final submittal requirements and project layout drawings as required.
 - 5. Bus and conductor data:
 - a. Use impedances of the actual installed or specified conductors, unless otherwise indicated.
 - b. Use cable and bus impedances calculated at 25 degrees Celsius, unless otherwise indicated.

E. Sequencing:

1. Below is an outline of the typical work sequence. Proposed changes to the work sequence may be reviewed and approved by the Engineer.
 - a. Site visit to gather data on the existing facility systems for all studies:
 - 1) Make multiple trips as required to obtain all data for the short-circuit fault analysis, protection device coordination, and arc flash hazard studies.
 - b. Initial electrical system study meeting.
 - c. Submit the initial short-circuit fault analysis study before submittal of any electrical equipment.
 - 1) Only the initial short-circuit results will be reviewed.
 - d. Submit the preliminary short-circuit fault analysis, protective device coordination, and arc-flash hazard studies after the approval of all electrical equipment.
 - e. Second electrical system study meeting for preliminary results.
 - f. Update the model with all changes to the electrical system made during start-up and commissioning.
 - g. Final arc-flash meeting and final short-circuit fault analysis, protective device coordination, and arc-flash hazard studies.
 - h. Submit the final electrical system studies.
 - i. Label equipment with approved arc-flash labels.
 - j. Owner's training.

PART 2 PRODUCTS

2.01 DESIGN AND PERFORMANCE CRITERIA

A. General study requirements:

1. Scope:
 - a. The short-circuit fault analysis, protective device coordination, and arc-flash hazard studies shall include all equipment in the power distribution system including, but not limited to:
 - 1) Service entrance equipment (service entrance transformer and primary protective device).
 - 2) Available utility fault contribution current.
 - 3) All electrical equipment including:
 - a) Dry-type transformers.
 - b) 240- and 208-volt panelboards.
 - 4) Generators.
 - 5) Motors.
 - 6) Vendor control panels.
 - 7) HVAC equipment.
 - b. Study scenarios:
 - 1) The studies shall include all possible electrical system configurations, for example:
 - a) Operation on normal (utility) source.
 - b) Operation on generator source.
 - c) Operating on single utility source – One source out of service and both ties closed.

2. Motors:
 - a. Each motor shall be individually modeled:
 - 1) Grouping of motors for fault contribution current is not acceptable.
 - b. Motors with variable frequency drives (VFDs) may be assumed to have no contribution to fault current, unless VFDs are equipped with Bypass Contactors.
 - 1) If VFDs are equipped with bypass contactors, perform the fault current and arc flash study assuming all of bypass contactors are in the closed position.
 3. Use the equipment, bus, and device designations as indicated on the Drawings for all studies.
- B. Short-circuit fault analysis study additional requirements:
1. The short-circuit fault analysis shall be performed and submitted in 2 phases:
 - ✓ a. Initial short-circuit fault analysis:
 - ✓ 1) Based on the Contract Documents and Electric Utility information.
 - ✓ 2) The initial short-circuit fault analysis study shall indicate the estimated available short-circuit current at the line side terminals of each piece of equipment covered by the scope of the study.
 - ✓ a) Measure conductor lengths from the Drawings. Use of arbitrary short conductor lengths is not allowed.
 - ✓ 3) Provide a list of assumptions used in the initial study.
 - b. Final short-circuit fault analysis:
 - 1) The final short-circuit fault analysis shall modify the initial analysis as follows:
 - a) Utilize the actual equipment provided on the project.
 - b) Utilize conductor lengths based on installation.
 2. Calculate 3-phase bolted fault, line-to-line fault, line-to-ground fault, double line-to-ground fault, short-circuit 1/2 cycle momentary symmetrical and asymmetrical RMS, 1-1/2 to 4 cycle interrupting symmetrical RMS, and 30-cycle steady-state short-circuit current values at each piece of equipment in the distribution system.
 3. Evaluate bus bracing, short-circuit ratings, fuse interrupting capacity and circuit-breaker-adjusted interrupting capacities against the fault currents, and calculate X/R values:
 - a. Identify and document all devices and equipment as either inadequate or acceptable.
 4. Calculate line-to-ground and double line-to-ground momentary short-circuit values at all buses having ground-fault devices.
 5. Provide calculation methods, assumptions, one-line diagrams, and source impedance data, including utility X/R ratios, typical values, recommendations, and areas of concern.
- C. Protective device coordination study additional requirements:
1. Furnish protective device settings for all functions indicated on the Drawings including, but not limited to:
 - a. Current.
 - b. Voltage:
 - 1) Provide settings for all voltage relays based upon actual utility and generator tolerances and specifications.

SEE ONE-LINE
DIAGRAMS

SEE SHORT-CIRCUIT
INPUT REPORT

- c. Frequency:
 - 1) Provide settings for all frequency relays based upon actual utility and generator tolerances and specifications.
- d. Negative sequence.
- e. Reverse power.
- f. Machine protection functions:
 - 1) Provide settings for all motor and generator protective relays based on the manufacturer's recommended protection requirements.
- 2. Provide log-log form time-current curves (TCCs) graphically indicating the coordination proposed for the system:
 - a. Include with each TCC a complete title and one-line diagram with legend identifying the specific portion of the system covered by the particular TCC:
 - 1) Typical TCCs for identical portions of the system, such as motor circuits, are acceptable as allowed by the Engineer.
 - b. Include a detailed description of each protective device identifying its type, function, manufacturer, and time-current characteristics:
 - 1) These details can be included on the TCC.
 - c. Include a detailed description of each protective device tap, time dial, pickup, instantaneous, and time delay settings:
 - 1) These details can be included on the TCC.
- 3. TCCs shall include all equipment in the power distribution system where required to demonstrate coordination. Include utility relay and fuse characteristics, medium-voltage equipment protective relay and fuse characteristics, low-voltage equipment circuit breaker trip device characteristics, transformer characteristics, motor and generator characteristics, and characteristics of other system load protective devices:
 - a. Include all devices down to the largest branch circuit and largest feeder circuit breaker in each motor control center, main breaker in branch panelboards, and fused disconnect switches.
 - b. Provide ground fault TCCs with all adjustable settings for ground fault protective devices.
 - c. Include manufacturing tolerances and damage bands in plotted fuse and circuit breaker characteristics.
 - d. On the TCCs, show transformer full load currents, transformer magnetizing inrush, ANSI transformer withstand parameters, and transformer damage curves.
 - e. Cable damage curves.
 - f. Terminate device characteristic curves at a point reflecting the maximum symmetrical or asymmetrical fault current to which the device is exposed based on the short-circuit fault analysis study.
 - g. Coordinate time interval medium-voltage relay characteristics with upstream and downstream devices to avoid nuisance tripping.
- 4. Site generation: When site generation (including cogeneration, standby, and emergency generators) is part of the electrical system, include phase and ground coordination of the generator protective devices:
 - a. Show the generator decrement curve and damage curve along with the operating characteristic of the protective devices.
- 5. Suggest modifications or additions to equipment rating or settings in a tabulated form.

- D. Arc-flash hazard study additional requirements:
1. Include the calculated arc-flash boundary and incident energy (calories/square centimeter) at each piece of equipment in the distribution system:
 - a. Perform study with 15 percent arcing fault variation in accordance with IEEE 1584.
 - b. Perform arc-flash calculations at minimum and maximum utility and generator fault contributions.
 - c. Perform arc-flash calculations for both the line side and load side of the switchgear, switchboard, motor control center, and panelboard main breakers.
 - d. Perform arc-flash calculations for all short-circuit scenarios with all motors on for 3 to 5 cycles and with all motors off.
 2. Provide executive summary of the study results:
 - a. Provide summary based upon worst case results.
 3. Provide a detailed written discussion and explanation of the tabulated outputs:
 - a. Include all scenarios.
 4. Provide alternative device settings to allow the Owner to select the desired functionality of the system:
 - a. Minimize the arc-flash energy by selective trip and time settings for equipment maintenance purposes.
 - b. Identify the arc-flash energy based upon the criteria of maintaining coordination and selectivity of the protective devices.

2.02 MANUFACTURERS

- A. Electrical system study software: One of the following or equal:
1. Operation Technology, Inc., ETAP.
 2. SKM Systems Analysis, Powertools.

2.03 COMPONENTS

- A. Arc-flash hazard labels:
1. Dimensions:
 - a. Minimum 5 inches by 3.5 inches.
 2. Materials:
 - a. Polyester with polyvinyl polymer over-laminate.
 - b. Self-adhesive.
 - c. Resistant to:
 - 1) UV.
 - 2) Chemicals and common cleaning solvents.
 - 3) Scuffing.
 - 4) Wide temperature changes.
 3. Contents:
 - a. Short-circuit bus identification.
 - b. Calculated incident energy (calories/square centimeter) range:
 - 1) Based on worst-case study results.
 - c. Arc-flash protection boundary.
 - d. Shock hazard boundary:
 - 1) The Contractor may provide separate labels for indication of the shock hazard boundary.

- e. Fed from:
 - 1) Identify the tag number of the upstream equipment providing power.
- 4. Color scheme:
 - a. For locations above 40 calories/square centimeter:
 - 1) White label with red "DANGER" strip across the top.
 - 2) Black lettering.
 - b. For locations below 40 calories/square centimeter:
 - 1) White label with orange "WARNING" strip across the top.
 - 2) Black lettering.

PART 3 EXECUTION

3.01 INSTALLATION

- A. After review and acceptance of the arc-flash hazard study by the Engineer, install all arc-flash hazard labels:
 - 1. Install labels at all locations required by NFPA, ANSI, or IEEE standards.
 - 2. At a minimum, install labels in the following locations:
 - a. The front of each main or incoming service compartment.
 - b. The front of each low-voltage switchgear section.
 - c. The front of each medium-voltage circuit breaker door.
 - d. The front of each accessible auxiliary or conductor compartment.
 - e. Each accessible rear or side vertical section.
 - f. Each motor control center vertical section.
 - g. Each panelboard covered by the study.
 - h. Each control panel, individual starter or VFD, or other equipment covered by the scope of the study.

3.02 COMMISSIONING

- A. As specified in Section 01756 - Commissioning.
- B. Owner Training:
 - 1. Include, but is not limited to:
 - a. Introduction and basics of NFPA 70E: 2 hours 1 sessions.
 - b. Detailed review of the electrical system study 2 hours 1 session.

3.03 FIELD QUALITY CONTROL

- A. The individual performing the arc-flash hazard study shall direct the installation of the arc-flash hazard labels:
 - 1. Remove and replace any improperly applied labels.
 - 2. Repair the equipment finish damaged by removal of any label.
 - 3. Install labels level or plumb across the entire dimension of the label.

3.04 ADJUSTING

- A. After review and acceptance of the draft arc-flash hazard study and protective device coordination study by the Engineer, adjust protective device settings per final study prior to equipment energization.
 - 1. Devices that require power for configuration may be set during energization, but before any subfed loads are energized.
Ensure that settings for upstream equipment are set prior to energizing downstream devices. Provide documentation that protective devices are set per the study recommendation.

END OF SECTION

1.0 UTILITY CORRESPONDENCE

SPECIFICATION – 16305 ELECTRICAL SYSTEMS STUDIES



REQUEST FOR INFORMATION (RFI) RESPONSE

Owner: County of Hawai'i **Project No.:** 203795
Contractor: Nan, Inc. **RFI No.:** 0027
Project Name: Hilo WWTP Rehabilitation and Replacement
Project Phase 1 **Spec/Dwg
Reference:** 16305
RFI Title: HECO_EQUT_IMPEDANCE (2)

Information Requested	
Spec section 16305 1.06 A indicates that contractor shall obtain information from electrical utility (HECO) to conduct the initial short-circuit fault current analysis. Through initial coordination with the HECO contacts provided in spec section 16210 1.01 B, HECO cannot confirm the exact equipment and components that will be used at this time. The short-circuit fault current analysis is a critical first step to submittal approval and procurement of long-lead electrical equipment.	
Authored By: <u>Nan, Inc.</u>	Date Submitted: <u>06/03/2025</u>

Response to Request:	
HECO has provided a minimum impedance for their 2000kVA transformers at 5.32%. Please proceed with this value to determine a worst case fault contribution.	
In the event, Contractor believes the RFI response does or will cause a change to the requirements of the Contract, Contractor shall immediately give written notice stating that Contractor considers the response to be a Change Order.	
Firm Name: <u>Carollo Engineers</u>	Date Returned: <u>6/4/2025</u>
Signature: <u>Kellen Gross</u>	

END OF DOCUMENT

RE: [External]RE: Hilo Waste Water treatment Plant

From Gonzalez, Aaron <aaron.gonzalez@hawaiianelectric.com>

Date Tue 6/3/2025 1:45 PM

To Darrin.Lee <Darrin.Lee@MassElec.com>; Mark McCarthy <mmccarthy@nanhawaii.com>

Cc Matt Chun <mchun@nanhawaii.com>; Makoa Ng <mng@nanhawaii.com>; Kellen Gross <kgross@carollo.com>

 2 attachments (18 KB)

Fault Summary WWTP 1101_25-05.txt; Fault Summary WWTP 13_25-05.txt;

Hi Darrin,

+Kellen Gross

There are two circuits going to the site. The two circuits are Kanoelehua 1101 and Kanoelehua 13. The two file are for each circuit coming to the site.

The minimum impedance for the transformer is - 13800 delta-480Y/277V → %Zmin = 5.32

Please let me know if you have any questions.

Thanks,

AARON GONZALEZ

Senior Designer, Transmission & Distribution Engineering

O: 808.969.0350

aaron.gonzalez@hawaiianelectric.com

Hawaiian Electric

PO Box 1027, Hilo, HI 96721



From: Darrin.Lee <darrin.lee@masselec.com>

Sent: Tuesday, June 3, 2025 1:34 PM

To: Gonzalez, Aaron <aaron.gonzalez@hawaiianelectric.com>; Mark McCarthy <mmccarthy@nanhawaii.com>

Cc: Matt Chun <mchun@nanhawaii.com>; Makoa Ng <mng@nanhawaii.com>

Subject: RE: [External]RE: Hilo Waste Water treatment Plant

Hi Aaron,

[External]RE: Hilo WWTP Utility Transformer Data

From Gonzalez, Aaron <aaron.gonzalez@hawaiianelectric.com>

Date Mon 8/18/2025 9:14 AM

To Darrin.Lee <Darrin.Lee@MassElec.com>

Cc Mark McCarthy <mmccarthy@nanhawaii.com>

Hi Darren,

The transformer information is as follows:

2000KVA-13800delta-480Y/277V with a minimum %Z of 5.32%.

Please let me know if you have any questions.

Thanks,

AARON GONZALEZ

Senior Designer, Transmission & Distribution Engineering

O: 808.969.0350

aaron.gonzalez@hawaiianelectric.com

Hawaiian Electric

PO Box 1027, Hilo, HI 96721



From: Darrin.Lee <Darrin.Lee@MassElec.com>

Sent: Wednesday, August 13, 2025 9:40 AM

To: Gonzalez, Aaron <aaron.gonzalez@hawaiianelectric.com>

Cc: Mark McCarthy <mmccarthy@nanhawaii.com>

Subject: Hilo WWTP Utility Transformer Data

[This email is coming from an EXTERNAL source. Please use caution when opening attachments or links in suspicious email.]

Hey Aaron,

Looking for below info for our systems studies group. Is this something you could shoot over fairly quickly?

1. Utility Data

- a. ~~Provide calculated available fault current at the HECO points of connection to Primary Switch 1~~
- b. T-1 and T-2 Service transformer kVA rating, Primary and Secondary voltages, % impedance

Looks like the information sent over a while back was just for the primary side.

Thanks,
Darrin

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2.0 EATON INITIAL SHORT-CIRCUIT FAULT ANALYSIS

SPECIFICATION – 16305 ELECTRICAL SYSTEMS STUDIES



Submittal Package

GEAR

Hilo WWTP Phase 1 - Hawaii

INITIAL SHORT-CIRCUIT FAULT ANALYSIS ANALYSES

Sep 16, 2025

Prepared For

MASS. ELECTRIC CONSTRUCTION CO.

Darren Lee

Darrin.Lee@MassElec.com



Prepared By

Vicki Newell

vnewell@wesco.com



**Electrical Engineering Services & Systems
4863 Shawline Street #E
San Diego, CA 92111
(619) 692-6558**

GENERAL ORDER NUMBER: MHL0013035
REPORT NUMBER: PRELIMINARY
SUBMITTED BY: STEVEN MCGOVERN, P.E.

**INITIAL SHORT-CIRCUIT FAULT ANALYSIS ANALYSES
FOR
HILO WWTP REHABILITATION AND REPLACEMENT PROJECT
PHASE 1
HILO, HI
REVISION 1
SEPTEMBER 2025**

Eaton.com/PSE

REVISION HISTORY

Rev #	Report Number	Issued	Revision / Modification Description
-	PRELIMINARY	08/2025	Initial Study Report Issue
1	PRELIMINARY	09/2025	Calculations updated with new cable data provided by the electrical contractor.

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1.0 EXECUTIVE SUMMARY

16305-2.01.B.1.a.2

1.1 General

This summary report contains the initial short-circuit fault analysis calculations performed on the electrical distribution system for the HILO WWTP Replacement Phase 1 project in Hilo, HI. The purpose of this analysis and report is to provide the fault current calculations using the power system configuration per design one-line drawings and using the provided utility fault current data and cable data provided by the electrical contractor. This report is to fulfill the requested requirement for an initial short-circuit fault current analysis per part c of sequencing.

This report included a table with the calculated available short-circuit current at the line side terminals at each bus modeled. It also included the model input report and short-circuit out report files from the SKM PowerTools software. The utility data provided is also included as well as the current model one-line drawings.

Cable data as provided by the electrical contractor is used in the analysis. The cable lengths from the disconnects to the motors was not provided and is assumed at nine feet. Cables are assumed to have THWN insulation applied at 75C per typical ratings of terminals.

VFDs are assumed to have a bypass configuration.

MCC tie breaker normal state is not noted on the project one-line drawings and are assumed to be normally open.

16305-2.01.B.1.a.3

2.0 SHORT-CIRCUIT CURRENT CALCULATIONS TABLE

**Low Voltage Short-Circuit Calculations
Maximum Utility Sources**

Equipment I.D.	Equipment Voltage (V)	Calculated Isc (kA)
05-VFD-1510	480	10.49
05-VFD-1520	480	9.09
05-VFD-1630	480	10.49
06-VFD-1610	480	10.49
06-VFD-1615	480	9.09
06-VFD-1625	480	9.09
06-VFD-1630	480	10.49
06-VFD-1635	480	9.09
07-VFD-1110	480	17.13
07-VFD-1120	480	16.42
09-VFD-1100	480	9.17
09-VFD-1200	480	4.01
11-VFD-2312	480	9.67
11-VFD-2321	480	9.35
11-VFD-2410	480	9.67
11-VFD-2420	480	9.35
12-VCP1-4110	480	17.23
12-VCP1-4210	480	16.10
12-VFD-3320	480	21.25
12-VFD-3420	480	19.74
12-VFD-3520	480	21.25
21-GEN-1100 [Ex]	480	18.71
21-GEN-1200	480	15.87
21-PNL-1200	480	2.37
21-VFD-1100	480	13.49
21-VFD-1200	480	12.88
21-VFD-1300	480	13.49
22-VFD-2022	480	9.09
22-VFD-2201	480	10.49
22-VFD-2203	480	10.49
23-VCP-2810	480	2.50
ADMIN	480	8.21
AHF-10A	480	8.14
AHF-10B	480	5.71
AHF-8A	480	17.25
AHF-8B	480	15.62
CL BLDG PORT GEN	480	7.62
DCB 1 PORT GEN	480	8.60
DP-10	480	7.92
DP-13	480	13.62
DP-5A	480	15.06

**Low Voltage Short-Circuit Calculations
Maximum Utility Sources**

Equipment I.D.	Equipment Voltage (V)	Calculated Isc (kA)
DP-5B	480	13.96
DP-6	480	8.64
DP-8A-1	480	19.81
DP-8A-2	480	18.45
DP-8B-1	480	18.27
DP-8B-2	480	17.10
DP-9	480	16.62
DSC 02-EF-1320	480	0.96
DSC 02-EF-1330	480	1.15
DSC 02-EF-1340	480	1.15
DSC 02-EF-1370	480	1.42
DSC 02-EF-1380	480	1.14
DSC 02-EF-1390	480	1.78
DSC 02-EF-1396	480	1.49
DSC 02-SF-1310	480	1.14
DSC 02-SF-1350	480	1.16
DSC 02-SF-1360	480	1.16
DSC 02-SF-1570	480	1.41
DSC 02-SF-1575	480	1.28
DSC 02-SF-1810	480	1.41
DSC 03-EF-1420	480	1.85
DSC 03-HSS-1140	480	0.83
DSC 03-HSS-1240	480	0.73
DSC 03-HSS-1340	480	0.69
DSC 03-PMP-1510	480	1.45
DSC 03-PMP-1520	480	1.13
DSC 03-PMP-1530	480	0.98
DSC 03-PMP-1610	480	0.75
DSC 03-PMP-1620	480	0.85
DSC 03-SCC-1150	480	1.16
DSC 03-SCC-1250	480	1.11
DSC 03-SCC-1350	480	0.83
DSC 03-SF-1410	480	0.73
DSC 03-SF-1575	480	0.72
DSC 03-SLC-1120	480	1.44
DSC 03-SLC-1220	480	1.11
DSC 03-SLC-1320	480	0.83
DSC 04-PMP-1140	480	2.34
DSC 06-CLA-1510	480	2.04
DSC 06-CLA-1520	480	1.23
DSC 06-CLA-1530	480	0.89

**Low Voltage Short-Circuit Calculations
Maximum Utility Sources**

Equipment I.D.	Equipment Voltage (V)	Calculated Isc (kA)
DSC 06-PMP-1515	480	2.80
DSC 06-PMP-1525	480	1.74
DSC 06-PMP-1535	480	1.28
DSC 08-EF-1230	480	3.51
DSC 09-MIX-1110	480	0.80
DSC 09-PMP-903	480	0.86
DSC 11-EF-1530	480	1.76
DSC 11-EF-1540	480	2.13
DSC 11-EF-1560	480	1.75
DSC 11-PMP-2311	480	1.37
DSC 11-PMP-2314	480	1.76
DSC 11-PMP-2322	480	1.37
DSC 11-PMP-2324	480	2.13
DSC 11-PMP-2610	480	1.36
DSC 11-PMP-2620	480	1.36
DSC 11-PMP-2711	480	1.36
DSC 11-PMP-2721	480	1.66
DSC 11-SF-1510	480	1.76
DSC 11-SF-1520	480	2.13
DSC 11-SF-1550	480	1.76
DSC 11-SF-1575	480	1.76
DSC 12-AUG-4615	480	2.64
DSC 12-AUG-4617	480	3.21
DSC 12-CON-4610	480	1.46
DSC 12-CON-4620	480	1.80
DSC 12-EF-2530	480	4.15
DSC 12-EF-2540	480	4.12
DSC 12-HPU-4750	480	3.23
DSC 12-SF-2510	480	4.15
DSC 12-SF-2520	480	4.12
DSC 12-SF-2550	480	2.35
DSC 12-SF-2560	480	1.16
DSC 12-SF-2570	480	1.44
DSC 12-SF-2580	480	1.16
DSC 12-SRW-4612	480	1.44
DSC 12-SRW-4613	480	1.44
DSC 19-PMP-1110	480	1.92
DSC 19-PMP-1120	480	2.21
DSC 21-EF-1250	480	2.35
DSC 21-EF-1310	480	1.16
DSC 2P-07	480	0.95

**Low Voltage Short-Circuit Calculations
Maximum Utility Sources**

Equipment I.D.	Equipment Voltage (V)	Calculated Isc (kA)
DSC 3G-01	480	0.63
DSC 3G-02	480	0.73
DSC 3G-03	480	0.77
DSC 3P-05	480	0.97
DSC 3P-07	480	1.07
DSC 4P-01	480	2.19
DSC 4P-02	480	2.22
DSC 4P-03	480	2.28
DSC SHB CRANE	480	3.14
GEN SWBD [Ex]	480	15.71
LP-10	208	1.72
LP-11	208	1.53
LP-13A	208	4.25
LP-13B	208	4.03
LP-4PA	240	0.28
LP-5	208	4.31
LP-6	208	1.73
LP-8	208	4.33
LP-9	208	3.28
MCC-10A	480	10.50
MCC-10B	480	9.10
MCC-11A	480	9.18
MCC-11B	480	4.01
MCC-5A	480	15.59
MCC-5B	480	14.72
MCC-6A	480	9.68
MCC-6B	480	9.36
MCC-8A	480	21.30
MCC-8B	480	19.78
MCC-9A	480	23.40
MCC-9B	480	21.86
MTS-10	480	8.19
MTS-11	480	3.72
MTS-13	480	14.00
MTS-5	480	12.77
MTS-6	480	8.84
MTS-8	480	17.40
MTS-9	480	17.49
SHB CRANE	480	2.90
SWBD-MAIN-A	480	30.12
SWBD-MAIN-B	480	27.96

3.0 SHORT-CIRCUIT INPUT REPORT

Input Report Interpretation

Input Data Tables are provided on the following pages. The following is a guide for interpreting the input data.

- Feeder Data

Feeder data includes the following cable and bus data: length, impedance in ohms per 1,000 feet, and per-unit impedance on a 100 MVA base. Impedance values for conductors were obtained from Tables 4A-7 and 4A-8 of IEEE Std 3002.3.

- Transformer Data

Transformer data includes the transformer kVA rating and per-unit impedance on a 100 MVA base. Unless otherwise provided, transformer X/R ratios are obtained from IEEE Std C37.010. Typical design impedances published by Eaton were used to model the Eaton dry-type transformers.

- Generation Contribution Data

- Utility contribution data includes the available fault current in MVA and amps, per unit impedance on a 100 MVA base, X/R, and the line-to-line bus voltage.
- Generator data includes the generator kW rating, X''d, X/R, line-to-line voltage and per unit impedance on a 100 MVA base.

- Motor Contribution Data

Motor Contribution Data includes the horsepower rating (base kVA rating), speed, subtransient reactance adjusted per the *First Cycle Duty* multipliers described in IEEE Std 3002.3, per-unit impedance on a 100 MVA base, and the bus voltage. X/R ratios for induction motors are obtained from IEEE Std C37.010.

ALL PU VALUES ARE EXPRESSED ON A 100 MVA BASE.

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FEEDER INPUT DATA

CABLE NAME	FEEDER FROM NAME	FEEDER TO NAME	QTY /PH	VOLTS L-L	LENGTH	FEEDER SIZE	FEEDER TYPE
CBL-02-CON-1400	MCC-5A	BUS-02-CON-1400	1	480	150.0 FEET	10	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.18 + J 0.0854	Ohms/1000 ft	76.82 + J 5.56	PU			
Z0 Impedance:	3.72 + J 0.2103	Ohms/1000 ft	242.11 + J 13.69	PU			
CBL-02-EF-1320	DSC 02-EF-1320	BUS-02-EF-1320	1	480	9.0 FEET	12	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.87 + J 0.0910	Ohms/1000 ft	7.30 + J 0.3555	PU			
Z0 Impedance:	5.89 + J 0.2241	Ohms/1000 ft	23.02 + J 0.8754	PU			
CBL-02-EF-1330	DSC 02-EF-1330	BUS-02-EF-1330	1	480	9.0 FEET	12	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.87 + J 0.0910	Ohms/1000 ft	7.30 + J 0.3555	PU			
Z0 Impedance:	5.89 + J 0.2241	Ohms/1000 ft	23.02 + J 0.8754	PU			
CBL-02-EF-1340	DSC 02-EF-1340	BUS-02-EF-1340	1	480	9.0 FEET	12	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.87 + J 0.0910	Ohms/1000 ft	7.30 + J 0.3555	PU			
Z0 Impedance:	5.89 + J 0.2241	Ohms/1000 ft	23.02 + J 0.8754	PU			
CBL-02-EF-1370	DSC 02-EF-1370	BUS-02-EF-1370	1	480	9.0 FEET	12	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.87 + J 0.0910	Ohms/1000 ft	7.30 + J 0.3555	PU			
Z0 Impedance:	5.89 + J 0.2241	Ohms/1000 ft	23.02 + J 0.8754	PU			
CBL-02-EF-1380	DSC 02-EF-1380	BUS-02-EF-1380	1	480	9.0 FEET	12	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.87 + J 0.0910	Ohms/1000 ft	7.30 + J 0.3555	PU			
Z0 Impedance:	5.89 + J 0.2241	Ohms/1000 ft	23.02 + J 0.8754	PU			
CBL-02-EF-1390	DSC 02-EF-1390	BUS-02-EF-1390	1	480	9.0 FEET	10	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.18 + J 0.0854	Ohms/1000 ft	4.61 + J 0.3336	PU			
Z0 Impedance:	3.72 + J 0.2103	Ohms/1000 ft	14.53 + J 0.8215	PU			
CBL-02-EF-1396	DSC 02-EF-1396	BUS-02-EF-1396	1	480	9.0 FEET	10	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.18 + J 0.0854	Ohms/1000 ft	4.61 + J 0.3336	PU			
Z0 Impedance:	3.72 + J 0.2103	Ohms/1000 ft	14.53 + J 0.8215	PU			
CBL-02-MIX-1210	MCC-5A	BUS-02-MIX-1210	1	480	100.0 FEET	12	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.87 + J 0.0910	Ohms/1000 ft	81.16 + J 3.95	PU			
Z0 Impedance:	5.89 + J 0.2241	Ohms/1000 ft	255.79 + J 9.73	PU			

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FEEDER INPUT DATA

CABLE NAME	FEEDER FROM NAME	FEEDER TO NAME	QTY /PH	VOLTS L-L	LENGTH	FEEDER SIZE	FEEDER TYPE
CBL-02-MIX-1220	MCC-5B	BUS-02-MIX-1220	1	480	100.0 FEET	12	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.87 + J 0.0910	Ohms/1000 ft	81.16 + J 3.95	PU			
Z0 Impedance:	5.89 + J 0.2241	Ohms/1000 ft	255.79 + J 9.73	PU			
CBL-02-PMP-1211	MCC-5A	BUS-02-PMP-1211	1	480	100.0 FEET	8	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	0.8110 + J 0.0754	Ohms/1000 ft	35.20 + J 3.27	PU			
Z0 Impedance:	2.56 + J 0.1856	Ohms/1000 ft	110.93 + J 8.06	PU			
CBL-02-PMP-1221	MCC-5B	BUS-02-PMP-1221	1	480	100.0 FEET	8	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	0.8110 + J 0.0754	Ohms/1000 ft	35.20 + J 3.27	PU			
Z0 Impedance:	2.56 + J 0.1856	Ohms/1000 ft	110.93 + J 8.06	PU			
CBL-02-SF-1310	DSC 02-SF-1310	BUS-02-SF-1310	1	480	9.0 FEET	12	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.87 + J 0.0910	Ohms/1000 ft	7.30 + J 0.3555	PU			
Z0 Impedance:	5.89 + J 0.2241	Ohms/1000 ft	23.02 + J 0.8754	PU			
CBL-02-SF-1350	DSC 02-SF-1350	BUS-02-SF-1350	1	480	9.0 FEET	12	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.87 + J 0.0910	Ohms/1000 ft	7.30 + J 0.3555	PU			
Z0 Impedance:	5.89 + J 0.2241	Ohms/1000 ft	23.02 + J 0.8754	PU			
CBL-02-SF-1360	DSC 02-SF-1360	BUS-02-SF-1360	1	480	9.0 FEET	12	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.87 + J 0.0910	Ohms/1000 ft	7.30 + J 0.3555	PU			
Z0 Impedance:	5.89 + J 0.2241	Ohms/1000 ft	23.02 + J 0.8754	PU			
CBL-02-SF-1570	DSC 02-SF-1570	BUS-02-SF-1570	1	480	9.0 FEET	12	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.87 + J 0.0910	Ohms/1000 ft	7.30 + J 0.3555	PU			
Z0 Impedance:	5.89 + J 0.2241	Ohms/1000 ft	23.02 + J 0.8754	PU			
CBL-02-SF-1575	DSC 02-SF-1575	BUS-02-SF-1575	1	480	9.0 FEET	10	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.18 + J 0.0854	Ohms/1000 ft	4.61 + J 0.3336	PU			
Z0 Impedance:	3.72 + J 0.2103	Ohms/1000 ft	14.53 + J 0.8215	PU			
CBL-02-SF-1810	DSC 02-SF-1810	BUS-02-SF-1810	1	480	9.0 FEET	12	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.87 + J 0.0910	Ohms/1000 ft	7.30 + J 0.3555	PU			
Z0 Impedance:	5.89 + J 0.2241	Ohms/1000 ft	23.02 + J 0.8754	PU			

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FEEDER INPUT DATA

CABLE NAME	FEEDER FROM NAME	FEEDER TO NAME	QTY /PH	VOLTS L-L	LENGTH	FEEDER SIZE	FEEDER TYPE
CBL-03-EF-1420	DSC 03-EF-1420	BUS-03-EF-1420	1	480	9.0 FEET	12	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.87 + J 0.0910	Ohms/1000 ft	7.30 + J 0.3555	PU			
Z0 Impedance:	5.89 + J 0.2241	Ohms/1000 ft	23.02 + J 0.8754	PU			
CBL-03-HSS-1140	DSC 03-HSS-1140	BUS-03-HSS-1140	1	480	9.0 FEET	12	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.87 + J 0.0910	Ohms/1000 ft	7.30 + J 0.3555	PU			
Z0 Impedance:	5.89 + J 0.2241	Ohms/1000 ft	23.02 + J 0.8754	PU			
CBL-03-HSS-1240	DSC 03-HSS-1240	BUS-03-HSS-1240	1	480	9.0 FEET	12	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.87 + J 0.0910	Ohms/1000 ft	7.30 + J 0.3555	PU			
Z0 Impedance:	5.89 + J 0.2241	Ohms/1000 ft	23.02 + J 0.8754	PU			
CBL-03-HSS-1340	DSC 03-HSS-1340	BUS-03-HSS-1340	1	480	9.0 FEET	12	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.87 + J 0.0910	Ohms/1000 ft	7.30 + J 0.3555	PU			
Z0 Impedance:	5.89 + J 0.2241	Ohms/1000 ft	23.02 + J 0.8754	PU			
CBL-03-PMP-1510	DSC 03-PMP-1510	BUS-03-PMP-1510	1	480	9.0 FEET	12	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.87 + J 0.0910	Ohms/1000 ft	7.30 + J 0.3555	PU			
Z0 Impedance:	5.89 + J 0.2241	Ohms/1000 ft	23.02 + J 0.8754	PU			
CBL-03-PMP-1520	DSC 03-PMP-1520	BUS-03-PMP-1520	1	480	9.0 FEET	12	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.87 + J 0.0910	Ohms/1000 ft	7.30 + J 0.3555	PU			
Z0 Impedance:	5.89 + J 0.2241	Ohms/1000 ft	23.02 + J 0.8754	PU			
CBL-03-PMP-1530	DSC 03-PMP-1530	BUS-03-PMP-1530	1	480	9.0 FEET	12	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.87 + J 0.0910	Ohms/1000 ft	7.30 + J 0.3555	PU			
Z0 Impedance:	5.89 + J 0.2241	Ohms/1000 ft	23.02 + J 0.8754	PU			
CBL-03-PMP-1610	DSC 03-PMP-1610	BUS-03-PMP-1610	1	480	9.0 FEET	12	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.87 + J 0.0910	Ohms/1000 ft	7.30 + J 0.3555	PU			
Z0 Impedance:	5.89 + J 0.2241	Ohms/1000 ft	23.02 + J 0.8754	PU			
CBL-03-PMP-1620	DSC 03-PMP-1620	BUS-03-PMP-1620	1	480	9.0 FEET	12	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.87 + J 0.0910	Ohms/1000 ft	7.30 + J 0.3555	PU			
Z0 Impedance:	5.89 + J 0.2241	Ohms/1000 ft	23.02 + J 0.8754	PU			

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FEEDER INPUT DATA

CABLE NAME	FEEDER FROM NAME	FEEDER TO NAME	QTY /PH	VOLTS L-L	LENGTH	FEEDER SIZE	FEEDER TYPE
CBL-03-SCC-1150	DSC 03-SCC-1150	BUS-03-SCC-1150	1	480	9.0 FEET	12	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.87 + J 0.0910	Ohms/1000 ft	7.30 + J 0.3555	PU			
Z0 Impedance:	5.89 + J 0.2241	Ohms/1000 ft	23.02 + J 0.8754	PU			
CBL-03-SCC-1250	DSC 03-SCC-1250	BUS-03-SCC-1250	1	480	9.0 FEET	12	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.87 + J 0.0910	Ohms/1000 ft	7.30 + J 0.3555	PU			
Z0 Impedance:	5.89 + J 0.2241	Ohms/1000 ft	23.02 + J 0.8754	PU			
CBL-03-SCC-1350	DSC 03-SCC-1350	BUS-03-SCC-1350	1	480	9.0 FEET	12	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.87 + J 0.0910	Ohms/1000 ft	7.30 + J 0.3555	PU			
Z0 Impedance:	5.89 + J 0.2241	Ohms/1000 ft	23.02 + J 0.8754	PU			
CBL-03-SF-1410	DSC 03-SF-1410	BUS-03-SF-1410	1	480	9.0 FEET	12	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.87 + J 0.0910	Ohms/1000 ft	7.30 + J 0.3555	PU			
Z0 Impedance:	5.89 + J 0.2241	Ohms/1000 ft	23.02 + J 0.8754	PU			
CBL-03-SF-1575	DSC 03-SF-1575	BUS-03-SF-1575	1	480	9.0 FEET	12	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.87 + J 0.0910	Ohms/1000 ft	7.30 + J 0.3555	PU			
Z0 Impedance:	5.89 + J 0.2241	Ohms/1000 ft	23.02 + J 0.8754	PU			
CBL-03-SLC-1120	DSC 03-SLC-1120	BUS-03-SLC-1120	1	480	9.0 FEET	12	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.87 + J 0.0910	Ohms/1000 ft	7.30 + J 0.3555	PU			
Z0 Impedance:	5.89 + J 0.2241	Ohms/1000 ft	23.02 + J 0.8754	PU			
CBL-03-SLC-1220	DSC 03-SLC-1220	BUS-03-SLC-1220	1	480	9.0 FEET	12	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.87 + J 0.0910	Ohms/1000 ft	7.30 + J 0.3555	PU			
Z0 Impedance:	5.89 + J 0.2241	Ohms/1000 ft	23.02 + J 0.8754	PU			
CBL-03-SLC-1320	DSC 03-SLC-1320	BUS-03-SLC-1320	1	480	9.0 FEET	12	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.87 + J 0.0910	Ohms/1000 ft	7.30 + J 0.3555	PU			
Z0 Impedance:	5.89 + J 0.2241	Ohms/1000 ft	23.02 + J 0.8754	PU			
CBL-04-PMP-1140	DSC 04-PMP-1140	BUS-04-PMP-1140	1	480	9.0 FEET	6	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	0.5100 + J 0.0685	Ohms/1000 ft	1.99 + J 0.2676	PU			
Z0 Impedance:	1.61 + J 0.1687	Ohms/1000 ft	6.28 + J 0.6590	PU			

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CABLE NAME	FEEDER FROM NAME	FEEDER TO NAME	QTY /PH	VOLTS L-L	LENGTH	FEEDER SIZE	FEEDER TYPE
CBL-05-PMP-1510	05-VFD-1510	BUS-05-PMP-1510	1	480	100.0 FEET	12	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.87 + J 0.0910	Ohms/1000 ft	81.16 + J 3.95	PU			
Z0 Impedance:	5.89 + J 0.2241	Ohms/1000 ft	255.79 + J 9.73	PU			
CBL-05-PMP-1520	05-VFD-1520	BUS-05-PMP-1520	1	480	100.0 FEET	12	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.87 + J 0.0910	Ohms/1000 ft	81.16 + J 3.95	PU			
Z0 Impedance:	5.89 + J 0.2241	Ohms/1000 ft	255.79 + J 9.73	PU			
CBL-05-PMP-1630	05-VFD-1630	BUS-05-PMP-1630	1	480	250.0 FEET	10	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.18 + J 0.0854	Ohms/1000 ft	128.04 + J 9.27	PU			
Z0 Impedance:	3.72 + J 0.2103	Ohms/1000 ft	403.52 + J 22.82	PU			
CBL-06-CLA-1510	DSC 06-CLA-1510	BUS-06-CLA-1510	1	480	9.0 FEET	10	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.18 + J 0.0854	Ohms/1000 ft	4.61 + J 0.3336	PU			
Z0 Impedance:	3.72 + J 0.2103	Ohms/1000 ft	14.53 + J 0.8215	PU			
CBL-06-CLA-1520	DSC 06-CLA-1520	BUS-06-CLA-1520	1	480	9.0 FEET	10	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.18 + J 0.0854	Ohms/1000 ft	4.61 + J 0.3336	PU			
Z0 Impedance:	3.72 + J 0.2103	Ohms/1000 ft	14.53 + J 0.8215	PU			
CBL-06-CLA-1530	DSC 06-CLA-1530	BUS-06-CLA-1530	1	480	9.0 FEET	10	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.18 + J 0.0854	Ohms/1000 ft	4.61 + J 0.3336	PU			
Z0 Impedance:	3.72 + J 0.2103	Ohms/1000 ft	14.53 + J 0.8215	PU			
CBL-06-PMP-1515	DSC 06-PMP-1515	BUS-06-PMP-1515	1	480	9.0 FEET	8	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	0.8110 + J 0.0754	Ohms/1000 ft	3.17 + J 0.2945	PU			
Z0 Impedance:	2.56 + J 0.1856	Ohms/1000 ft	9.98 + J 0.7250	PU			
CBL-06-PMP-1525	DSC 06-PMP-1525	BUS-06-PMP-1525	1	480	9.0 FEET	8	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	0.8110 + J 0.0754	Ohms/1000 ft	3.17 + J 0.2945	PU			
Z0 Impedance:	2.56 + J 0.1856	Ohms/1000 ft	9.98 + J 0.7250	PU			
CBL-06-PMP-1535	DSC 06-PMP-1535	BUS-06-PMP-1535	1	480	9.0 FEET	8	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	0.8110 + J 0.0754	Ohms/1000 ft	3.17 + J 0.2945	PU			
Z0 Impedance:	2.56 + J 0.1856	Ohms/1000 ft	9.98 + J 0.7250	PU			

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CABLE NAME	FEEDER FROM NAME	FEEDER TO NAME	QTY /PH	VOLTS L-L	LENGTH	FEEDER SIZE	FEEDER TYPE
CBL-06-PMP-1610	06-VFD-1610	BUS-06-PMP-1610	1	480	100.0 FEET	10	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.18 + J 0.0854	Ohms/1000 ft	51.22 + J 3.71	PU			
Z0 Impedance:	3.72 + J 0.2103	Ohms/1000 ft	161.41 + J 9.13	PU			
CBL-06-PMP-1615	06-VFD-1615	BUS-06-PMP-1615	1	480	100.0 FEET	10	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.18 + J 0.0854	Ohms/1000 ft	51.22 + J 3.71	PU			
Z0 Impedance:	3.72 + J 0.2103	Ohms/1000 ft	161.41 + J 9.13	PU			
CBL-06-PMP-1625	06-VFD-1625	BUS-06-PMP-1625	1	480	175.0 FEET	10	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.18 + J 0.0854	Ohms/1000 ft	89.63 + J 6.49	PU			
Z0 Impedance:	3.72 + J 0.2103	Ohms/1000 ft	282.46 + J 15.97	PU			
CBL-06-PMP-1630	06-VFD-1630	BUS-06-PMP-1630	1	480	250.0 FEET	10	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.18 + J 0.0854	Ohms/1000 ft	128.04 + J 9.27	PU			
Z0 Impedance:	3.72 + J 0.2103	Ohms/1000 ft	403.52 + J 22.82	PU			
CBL-06-PMP-1635	06-VFD-1635	BUS-06-PMP-1635	1	480	200.0 FEET	10	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.18 + J 0.0854	Ohms/1000 ft	102.43 + J 7.41	PU			
Z0 Impedance:	3.72 + J 0.2103	Ohms/1000 ft	322.81 + J 18.26	PU			
CBL-07-FAN-1110	07-VFD-1110	BUS-07-FAN-1110	1	480	200.0 FEET	2	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	0.2020 + J 0.0585	Ohms/1000 ft	17.53 + J 5.08	PU			
Z0 Impedance:	0.6366 + J 0.1440	Ohms/1000 ft	55.26 + J 12.50	PU			
CBL-07-FAN-1120	07-VFD-1120	BUS-07-FAN-1120	1	480	200.0 FEET	2	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	0.2020 + J 0.0585	Ohms/1000 ft	17.53 + J 5.08	PU			
Z0 Impedance:	0.6366 + J 0.1440	Ohms/1000 ft	55.26 + J 12.50	PU			
CBL-07-VFD-1110	SWBD-MAIN-A	07-VFD-1110	1	480	50.0 FEET	1	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	0.1600 + J 0.0570	Ohms/1000 ft	3.47 + J 1.24	PU			
Z0 Impedance:	0.5042 + J 0.1403	Ohms/1000 ft	10.94 + J 3.04	PU			
CBL-07-VFD-1120	SWBD-MAIN-B	07-VFD-1120	1	480	50.0 FEET	1	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	0.1600 + J 0.0570	Ohms/1000 ft	3.47 + J 1.24	PU			
Z0 Impedance:	0.5042 + J 0.1403	Ohms/1000 ft	10.94 + J 3.04	PU			

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CABLE NAME	FEEDER FROM NAME	FEEDER TO NAME	QTY /PH	VOLTS L-L	LENGTH	FEEDER SIZE	FEEDER TYPE
CBL-08-EF-1230	DSC 08-EF-1230	BUS-08-EF-1230	1	480	9.0 FEET	12	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.87 + J 0.0910	Ohms/1000 ft	7.30 + J 0.3555	PU			
Z0 Impedance:	5.89 + J 0.2241	Ohms/1000 ft	23.02 + J 0.8754	PU			
CBL-09-MIX-1110	DSC 09-MIX-1110	BUS-09-MIX-1110	1	480	9.0 FEET	12	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.87 + J 0.0910	Ohms/1000 ft	7.30 + J 0.3555	PU			
Z0 Impedance:	5.89 + J 0.2241	Ohms/1000 ft	23.02 + J 0.8754	PU			
CBL-09-PMP-1100	09-VFD-1100	BUS-09-PMP-1100	1	480	175.0 FEET	1	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	0.1600 + J 0.0570	Ohms/1000 ft	12.15 + J 4.33	PU			
Z0 Impedance:	0.5042 + J 0.1403	Ohms/1000 ft	38.30 + J 10.66	PU			
CBL-09-PMP-1200	09-VFD-1200	BUS-09-PMP-1200	1	480	175.0 FEET	1	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	0.1600 + J 0.0570	Ohms/1000 ft	12.15 + J 4.33	PU			
Z0 Impedance:	0.5042 + J 0.1403	Ohms/1000 ft	38.30 + J 10.66	PU			
CBL-09-PMP-903	DSC 09-PMP-903	BUS-09-PMP-903	1	480	9.0 FEET	12	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.87 + J 0.0910	Ohms/1000 ft	7.30 + J 0.3555	PU			
Z0 Impedance:	5.89 + J 0.2241	Ohms/1000 ft	23.02 + J 0.8754	PU			
CBL-11-EF-1530	DSC 11-EF-1530	BUS-11-EF-1530	1	480	9.0 FEET	12	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.87 + J 0.0910	Ohms/1000 ft	7.30 + J 0.3555	PU			
Z0 Impedance:	5.89 + J 0.2241	Ohms/1000 ft	23.02 + J 0.8754	PU			
CBL-11-EF-1540	DSC 11-EF-1540	BUS-11-EF-1540	1	480	9.0 FEET	12	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.87 + J 0.0910	Ohms/1000 ft	7.30 + J 0.3555	PU			
Z0 Impedance:	5.89 + J 0.2241	Ohms/1000 ft	23.02 + J 0.8754	PU			
CBL-11-EF-1560	DSC 11-EF-1560	BUS-11-EF-1560	1	480	9.0 FEET	12	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.87 + J 0.0910	Ohms/1000 ft	7.30 + J 0.3555	PU			
Z0 Impedance:	5.89 + J 0.2241	Ohms/1000 ft	23.02 + J 0.8754	PU			
CBL-11-PMP-2311	DSC 11-PMP-2311	BUS-11-PMP-2311	1	480	9.0 FEET	12	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.87 + J 0.0910	Ohms/1000 ft	7.30 + J 0.3555	PU			
Z0 Impedance:	5.89 + J 0.2241	Ohms/1000 ft	23.02 + J 0.8754	PU			

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CABLE NAME	FEEDER FROM NAME	FEEDER TO NAME	QTY /PH	VOLTS L-L	LENGTH	FEEDER SIZE	FEEDER TYPE
CBL-11-PMP-2312	11-VFD-2312	BUS-11-PMP-2312	1	480	100.0 FEET	4	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	0.3210 + J 0.0632	Ohms/1000 ft	13.93 + J 2.74	PU			
Z0 Impedance:	1.01 + J 0.1556	Ohms/1000 ft	43.91 + J 6.75	PU			
CBL-11-PMP-2314	DSC 11-PMP-2314	BUS-11-PMP-2314	1	480	9.0 FEET	12	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.87 + J 0.0910	Ohms/1000 ft	7.30 + J 0.3555	PU			
Z0 Impedance:	5.89 + J 0.2241	Ohms/1000 ft	23.02 + J 0.8754	PU			
CBL-11-PMP-2321	11-VFD-2321	BUS-11-PMP-2321	1	480	80.0 FEET	4	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	0.3210 + J 0.0632	Ohms/1000 ft	11.15 + J 2.19	PU			
Z0 Impedance:	1.01 + J 0.1556	Ohms/1000 ft	35.13 + J 5.40	PU			
CBL-11-PMP-2322	DSC 11-PMP-2322	BUS-11-PMP-2322	1	480	9.0 FEET	12	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.87 + J 0.0910	Ohms/1000 ft	7.30 + J 0.3555	PU			
Z0 Impedance:	5.89 + J 0.2241	Ohms/1000 ft	23.02 + J 0.8754	PU			
CBL-11-PMP-2324	DSC 11-PMP-2324	BUS-11-PMP-2324	1	480	9.0 FEET	12	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.87 + J 0.0910	Ohms/1000 ft	7.30 + J 0.3555	PU			
Z0 Impedance:	5.89 + J 0.2241	Ohms/1000 ft	23.02 + J 0.8754	PU			
CBL-11-PMP-2410	11-VFD-2410	BUS-11-PMP-2410	1	480	75.0 FEET	8	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	0.8110 + J 0.0754	Ohms/1000 ft	26.40 + J 2.45	PU			
Z0 Impedance:	2.56 + J 0.1856	Ohms/1000 ft	83.20 + J 6.04	PU			
CBL-11-PMP-2420	11-VFD-2420	BUS-11-PMP-2420	1	480	65.0 FEET	8	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	0.8110 + J 0.0754	Ohms/1000 ft	22.88 + J 2.13	PU			
Z0 Impedance:	2.56 + J 0.1856	Ohms/1000 ft	72.11 + J 5.24	PU			
CBL-11-PMP-2610	DSC 11-PMP-2610	BUS-11-PMP-2610	1	480	9.0 FEET	12	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.87 + J 0.0910	Ohms/1000 ft	7.30 + J 0.3555	PU			
Z0 Impedance:	5.89 + J 0.2241	Ohms/1000 ft	23.02 + J 0.8754	PU			
CBL-11-PMP-2620	DSC 11-PMP-2620	BUS-11-PMP-2620	1	480	9.0 FEET	12	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.87 + J 0.0910	Ohms/1000 ft	7.30 + J 0.3555	PU			
Z0 Impedance:	5.89 + J 0.2241	Ohms/1000 ft	23.02 + J 0.8754	PU			

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CABLE NAME	FEEDER FROM NAME	FEEDER TO NAME	QTY /PH	VOLTS L-L	LENGTH	FEEDER SIZE	FEEDER TYPE
CBL-11-PMP-2711	DSC 11-PMP-2711	BUS-11-PMP-2711	1	480	9.0 FEET	12	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.87 + J 0.0910	Ohms/1000 ft	7.30 + J 0.3555	PU			
Z0 Impedance:	5.89 + J 0.2241	Ohms/1000 ft	23.02 + J 0.8754	PU			
CBL-11-PMP-2721	DSC 11-PMP-2721	BUS-11-PMP-2721	1	480	9.0 FEET	12	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.87 + J 0.0910	Ohms/1000 ft	7.30 + J 0.3555	PU			
Z0 Impedance:	5.89 + J 0.2241	Ohms/1000 ft	23.02 + J 0.8754	PU			
CBL-11-SF-1510	DSC 11-SF-1510	BUS-11-SF-1510	1	480	9.0 FEET	12	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.87 + J 0.0910	Ohms/1000 ft	7.30 + J 0.3555	PU			
Z0 Impedance:	5.89 + J 0.2241	Ohms/1000 ft	23.02 + J 0.8754	PU			
CBL-11-SF-1520	DSC 11-SF-1520	BUS-11-SF-1520	1	480	9.0 FEET	12	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.87 + J 0.0910	Ohms/1000 ft	7.30 + J 0.3555	PU			
Z0 Impedance:	5.89 + J 0.2241	Ohms/1000 ft	23.02 + J 0.8754	PU			
CBL-11-SF-1550	DSC 11-SF-1550	BUS-11-SF-1550	1	480	9.0 FEET	12	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.87 + J 0.0910	Ohms/1000 ft	7.30 + J 0.3555	PU			
Z0 Impedance:	5.89 + J 0.2241	Ohms/1000 ft	23.02 + J 0.8754	PU			
CBL-11-SF-1575	DSC 11-SF-1575	BUS-11-SF-1575	1	480	9.0 FEET	12	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.87 + J 0.0910	Ohms/1000 ft	7.30 + J 0.3555	PU			
Z0 Impedance:	5.89 + J 0.2241	Ohms/1000 ft	23.02 + J 0.8754	PU			
CBL-12-AUG-4615	DSC 12-AUG-4615	BUS-12-AUG-4615	1	480	9.0 FEET	8	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	0.8110 + J 0.0754	Ohms/1000 ft	3.17 + J 0.2945	PU			
Z0 Impedance:	2.56 + J 0.1856	Ohms/1000 ft	9.98 + J 0.7250	PU			
CBL-12-AUG-4617	DSC 12-AUG-4617	BUS-12-AUG-4617	1	480	9.0 FEET	8	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	0.8110 + J 0.0754	Ohms/1000 ft	3.17 + J 0.2945	PU			
Z0 Impedance:	2.56 + J 0.1856	Ohms/1000 ft	9.98 + J 0.7250	PU			
CBL-12-CON-4610	DSC 12-CON-4610	BUS-12-CON-4610	1	480	9.0 FEET	12	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.87 + J 0.0910	Ohms/1000 ft	7.30 + J 0.3555	PU			
Z0 Impedance:	5.89 + J 0.2241	Ohms/1000 ft	23.02 + J 0.8754	PU			

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CABLE NAME	FEEDER FROM NAME	FEEDER TO NAME	QTY /PH	VOLTS L-L	LENGTH	FEEDER SIZE	FEEDER TYPE
CBL-12-CON-4620	DSC 12-CON-4620	BUS-12-CON-4620	1	480	9.0 FEET	12	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.87 + J 0.0910	Ohms/1000 ft	7.30 + J 0.3555	PU			
Z0 Impedance:	5.89 + J 0.2241	Ohms/1000 ft	23.02 + J 0.8754	PU			
CBL-12-EF-2530	DSC 12-EF-2530	BUS-12-EF-2530	1	480	9.0 FEET	8	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	0.8110 + J 0.0754	Ohms/1000 ft	3.17 + J 0.2945	PU			
Z0 Impedance:	2.56 + J 0.1856	Ohms/1000 ft	9.98 + J 0.7250	PU			
CBL-12-EF-2540	DSC 12-EF-2540	BUS-12-EF-2540	1	480	9.0 FEET	8	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	0.8110 + J 0.0754	Ohms/1000 ft	3.17 + J 0.2945	PU			
Z0 Impedance:	2.56 + J 0.1856	Ohms/1000 ft	9.98 + J 0.7250	PU			
CBL-12-FG-4115	12-VCP1-4110	BUS-12-FG-4115	1	480	80.0 FEET	12	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.87 + J 0.0910	Ohms/1000 ft	64.93 + J 3.16	PU			
Z0 Impedance:	5.89 + J 0.2241	Ohms/1000 ft	204.63 + J 7.78	PU			
CBL-12-FG-4215	12-VCP1-4210	BUS-12-FG-4215	1	480	80.0 FEET	12	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.87 + J 0.0910	Ohms/1000 ft	64.93 + J 3.16	PU			
Z0 Impedance:	5.89 + J 0.2241	Ohms/1000 ft	204.63 + J 7.78	PU			
CBL-12-FV-4117	12-VCP1-4110	BUS-12-FV-4117	1	480	80.0 FEET	12	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.87 + J 0.0910	Ohms/1000 ft	64.93 + J 3.16	PU			
Z0 Impedance:	5.89 + J 0.2241	Ohms/1000 ft	204.63 + J 7.78	PU			
CBL-12-FV-4217	12-VCP1-4210	BUS-12-FV-4217	1	480	80.0 FEET	12	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.87 + J 0.0910	Ohms/1000 ft	64.93 + J 3.16	PU			
Z0 Impedance:	5.89 + J 0.2241	Ohms/1000 ft	204.63 + J 7.78	PU			
CBL-12-HPU-4750	DSC 12-HPU-4750	BUS-12-HPU-4750	1	480	9.0 FEET	8	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	0.8110 + J 0.0754	Ohms/1000 ft	3.17 + J 0.2945	PU			
Z0 Impedance:	2.56 + J 0.1856	Ohms/1000 ft	9.98 + J 0.7250	PU			
CBL-12-MTR1-4110	12-VCP1-4110	BUS-12-MTR1-4110	1	480	100.0 FEET	2/0	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	0.1020 + J 0.0533	Ohms/1000 ft	4.43 + J 2.31	PU			
Z0 Impedance:	0.3214 + J 0.1312	Ohms/1000 ft	13.95 + J 5.69	PU			

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CABLE NAME	FEEDER FROM NAME	FEEDER TO NAME	QTY /PH	VOLTS L-L	LENGTH	FEEDER SIZE	FEEDER TYPE
CBL-12-MTR1-4210	12-VCP1-4210	BUS-12-MTR1-4210	1	480	100.0 FEET	2/0	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	0.1020 + J 0.0533	Ohms/1000 ft	4.43 + J 2.31	PU			
Z0 Impedance:	0.3214 + J 0.1312	Ohms/1000 ft	13.95 + J 5.69	PU			
CBL-12-MTR2-4110	12-VCP1-4110	BUS-12-MTR2-4110	1	480	100.0 FEET	8	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	0.8110 + J 0.0754	Ohms/1000 ft	35.20 + J 3.27	PU			
Z0 Impedance:	2.56 + J 0.1856	Ohms/1000 ft	110.93 + J 8.06	PU			
CBL-12-MTR2-4210	12-VCP1-4210	BUS-12-MTR2-4210	1	480	100.0 FEET	8	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	0.8110 + J 0.0754	Ohms/1000 ft	35.20 + J 3.27	PU			
Z0 Impedance:	2.56 + J 0.1856	Ohms/1000 ft	110.93 + J 8.06	PU			
CBL-12-PMP-3320	12-VFD-3320	BUS-12-PMP-3320	1	480	85.0 FEET	8	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	0.8110 + J 0.0754	Ohms/1000 ft	29.92 + J 2.78	PU			
Z0 Impedance:	2.56 + J 0.1856	Ohms/1000 ft	94.29 + J 6.85	PU			
CBL-12-PMP-3420	12-VFD-3420	BUS-12-PMP-3420	1	480	80.0 FEET	8	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	0.8110 + J 0.0754	Ohms/1000 ft	28.16 + J 2.62	PU			
Z0 Impedance:	2.56 + J 0.1856	Ohms/1000 ft	88.75 + J 6.44	PU			
CBL-12-PMP-3520	12-VFD-3520	BUS-12-PMP-3520	1	480	100.0 FEET	8	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	0.8110 + J 0.0754	Ohms/1000 ft	35.20 + J 3.27	PU			
Z0 Impedance:	2.56 + J 0.1856	Ohms/1000 ft	110.93 + J 8.06	PU			
CBL-12-SF-2510	DSC 12-SF-2510	BUS-12-SF-2510	1	480	9.0 FEET	8	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	0.8110 + J 0.0754	Ohms/1000 ft	3.17 + J 0.2945	PU			
Z0 Impedance:	2.56 + J 0.1856	Ohms/1000 ft	9.98 + J 0.7250	PU			
CBL-12-SF-2520	DSC 12-SF-2520	BUS-12-SF-2520	1	480	9.0 FEET	8	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	0.8110 + J 0.0754	Ohms/1000 ft	3.17 + J 0.2945	PU			
Z0 Impedance:	2.56 + J 0.1856	Ohms/1000 ft	9.98 + J 0.7250	PU			
CBL-12-SF-2550	DSC 12-SF-2550	BUS-12-SF-2550	1	480	9.0 FEET	12	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.87 + J 0.0910	Ohms/1000 ft	7.30 + J 0.3555	PU			
Z0 Impedance:	5.89 + J 0.2241	Ohms/1000 ft	23.02 + J 0.8754	PU			

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CABLE NAME	FEEDER FROM NAME	FEEDER TO NAME	QTY /PH	VOLTS L-L	LENGTH	FEEDER SIZE	FEEDER TYPE
CBL-12-SF-2560	DSC 12-SF-2560	BUS-12-SF-2560	1	480	9.0 FEET	12	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.87 + J 0.0910	Ohms/1000 ft	7.30 + J 0.3555	PU			
Z0 Impedance:	5.89 + J 0.2241	Ohms/1000 ft	23.02 + J 0.8754	PU			
CBL-12-SF-2570	DSC 12-SF-2570	BUS-12-SF-2570	1	480	9.0 FEET	12	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.87 + J 0.0910	Ohms/1000 ft	7.30 + J 0.3555	PU			
Z0 Impedance:	5.89 + J 0.2241	Ohms/1000 ft	23.02 + J 0.8754	PU			
CBL-12-SF-2580	DSC 12-SF-2580	BUS-12-SF-2580	1	480	9.0 FEET	12	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.87 + J 0.0910	Ohms/1000 ft	7.30 + J 0.3555	PU			
Z0 Impedance:	5.89 + J 0.2241	Ohms/1000 ft	23.02 + J 0.8754	PU			
CBL-12-SRW-4612	DSC 12-SRW-4612	BUS-12-SRW-4612	1	480	9.0 FEET	12	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.87 + J 0.0910	Ohms/1000 ft	7.30 + J 0.3555	PU			
Z0 Impedance:	5.89 + J 0.2241	Ohms/1000 ft	23.02 + J 0.8754	PU			
CBL-12-SRW-4613	DSC 12-SRW-4613	BUS-12-SRW-4613	1	480	9.0 FEET	12	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.87 + J 0.0910	Ohms/1000 ft	7.30 + J 0.3555	PU			
Z0 Impedance:	5.89 + J 0.2241	Ohms/1000 ft	23.02 + J 0.8754	PU			
CBL-12-VCP1-4110	MCC-8A	12-VCP1-4110	1	480	50.0 FEET	250	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	0.0552 + J 0.0495	Ohms/1000 ft	1.20 + J 1.07	PU			
Z0 Impedance:	0.1739 + J 0.1219	Ohms/1000 ft	3.77 + J 2.65	PU			
CBL-12-VCP1-4210	MCC-8B	12-VCP1-4210	1	480	50.0 FEET	250	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	0.0552 + J 0.0495	Ohms/1000 ft	1.20 + J 1.07	PU			
Z0 Impedance:	0.1739 + J 0.1219	Ohms/1000 ft	3.77 + J 2.65	PU			
CBL-19-PMP-1110	DSC 19-PMP-1110	BUS-19-PMP-1110	1	480	9.0 FEET	8	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	0.8110 + J 0.0754	Ohms/1000 ft	3.17 + J 0.2945	PU			
Z0 Impedance:	2.56 + J 0.1856	Ohms/1000 ft	9.98 + J 0.7250	PU			
CBL-19-PMP-1120	DSC 19-PMP-1120	BUS-19-PMP-1120	1	480	9.0 FEET	8	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	0.8110 + J 0.0754	Ohms/1000 ft	3.17 + J 0.2945	PU			
Z0 Impedance:	2.56 + J 0.1856	Ohms/1000 ft	9.98 + J 0.7250	PU			

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CABLE NAME	FEEDER FROM NAME	FEEDER TO NAME	QTY /PH	VOLTS L-L	LENGTH	FEEDER SIZE	FEEDER TYPE
CBL-21-EF-1250	DSC 21-EF-1250	BUS-21-EF-1250	1	480	9.0 FEET	12	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.87 + J 0.0910	Ohms/1000 ft	7.30 + J 0.3555	PU			
Z0 Impedance:	5.89 + J 0.2241	Ohms/1000 ft	23.02 + J 0.8754	PU			
CBL-21-EF-1310	DSC 21-EF-1310	BUS-21-EF-1310	1	480	9.0 FEET	12	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.87 + J 0.0910	Ohms/1000 ft	7.30 + J 0.3555	PU			
Z0 Impedance:	5.89 + J 0.2241	Ohms/1000 ft	23.02 + J 0.8754	PU			
CBL-21-PMP-1100	21-VFD-1100	BUS-21-PMP-1100	1	480	100.0 FEET	2	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	0.2020 + J 0.0585	Ohms/1000 ft	8.77 + J 2.54	PU			
Z0 Impedance:	0.6366 + J 0.1440	Ohms/1000 ft	27.63 + J 6.25	PU			
CBL-21-PMP-1200	21-VFD-1200	BUS-21-PMP-1200	1	480	120.0 FEET	2	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	0.2020 + J 0.0585	Ohms/1000 ft	10.52 + J 3.05	PU			
Z0 Impedance:	0.6366 + J 0.1440	Ohms/1000 ft	33.16 + J 7.50	PU			
CBL-21-PMP-1300	21-VFD-1300	BUS-21-PMP-1300	1	480	100.0 FEET	2	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	0.2020 + J 0.0585	Ohms/1000 ft	8.77 + J 2.54	PU			
Z0 Impedance:	0.6366 + J 0.1440	Ohms/1000 ft	27.63 + J 6.25	PU			
CBL-21-PNL-1200	MCC-9B	21-PNL-1200	1	480	135.0 FEET	8	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	0.8110 + J 0.0754	Ohms/1000 ft	47.52 + J 4.42	PU			
Z0 Impedance:	2.56 + J 0.1856	Ohms/1000 ft	149.76 + J 10.88	PU			
CBL-21-VFD-1100	MCC-9A	21-VFD-1100	1	480	60.0 FEET	1	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	0.1600 + J 0.0570	Ohms/1000 ft	4.17 + J 1.48	PU			
Z0 Impedance:	0.5042 + J 0.1403	Ohms/1000 ft	13.13 + J 3.65	PU			
CBL-21-VFD-1200	MCC-9B	21-VFD-1200	1	480	60.0 FEET	1	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	0.1600 + J 0.0570	Ohms/1000 ft	4.17 + J 1.48	PU			
Z0 Impedance:	0.5042 + J 0.1403	Ohms/1000 ft	13.13 + J 3.65	PU			
CBL-21-VFD-1300	MCC-9A	21-VFD-1300	1	480	60.0 FEET	1	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	0.1600 + J 0.0570	Ohms/1000 ft	4.17 + J 1.48	PU			
Z0 Impedance:	0.5042 + J 0.1403	Ohms/1000 ft	13.13 + J 3.65	PU			

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CABLE NAME	FEEDER FROM NAME	FEEDER TO NAME	QTY /PH	VOLTS L-L	LENGTH	FEEDER SIZE	FEEDER TYPE
CBL-22-BLO-2022	22-VFD-2022	BUS-22-BLO-2022	1	480	75.0 FEET	8	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	0.8110 + J 0.0754	Ohms/1000 ft	26.40 + J 2.45	PU			
Z0 Impedance:	2.56 + J 0.1856	Ohms/1000 ft	83.20 + J 6.04	PU			
CBL-22-BLO-2201	22-VFD-2201	BUS-22-BLO-2201	1	480	60.0 FEET	8	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	0.8110 + J 0.0754	Ohms/1000 ft	21.12 + J 1.96	PU			
Z0 Impedance:	2.56 + J 0.1856	Ohms/1000 ft	66.56 + J 4.83	PU			
CBL-22-BLO-2203	22-VFD-2203	BUS-22-BLO-2203	1	480	80.0 FEET	8	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	0.8110 + J 0.0754	Ohms/1000 ft	28.16 + J 2.62	PU			
Z0 Impedance:	2.56 + J 0.1856	Ohms/1000 ft	88.75 + J 6.44	PU			
CBL-23-VCP-2810	MCC-11A	23-VCP-2810	1	480	75.0 FEET	10	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.18 + J 0.0854	Ohms/1000 ft	38.41 + J 2.78	PU			
Z0 Impedance:	3.72 + J 0.2103	Ohms/1000 ft	121.05 + J 6.85	PU			
CBL-2P-07	DSC 2P-07	BUS-2P-07	1	480	9.0 FEET	12	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.87 + J 0.0910	Ohms/1000 ft	7.30 + J 0.3555	PU			
Z0 Impedance:	5.89 + J 0.2241	Ohms/1000 ft	23.02 + J 0.8754	PU			
CBL-3G-01	DSC 3G-01	BUS-3G-01	1	480	9.0 FEET	12	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.87 + J 0.0910	Ohms/1000 ft	7.30 + J 0.3555	PU			
Z0 Impedance:	5.89 + J 0.2241	Ohms/1000 ft	23.02 + J 0.8754	PU			
CBL-3G-02	DSC 3G-02	BUS-3G-02	1	480	9.0 FEET	12	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.87 + J 0.0910	Ohms/1000 ft	7.30 + J 0.3555	PU			
Z0 Impedance:	5.89 + J 0.2241	Ohms/1000 ft	23.02 + J 0.8754	PU			
CBL-3G-03	DSC 3G-03	BUS-3G-03	1	480	9.0 FEET	12	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.87 + J 0.0910	Ohms/1000 ft	7.30 + J 0.3555	PU			
Z0 Impedance:	5.89 + J 0.2241	Ohms/1000 ft	23.02 + J 0.8754	PU			
CBL-3P-05	DSC 3P-05	BUS-3P-05	1	480	9.0 FEET	12	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.87 + J 0.0910	Ohms/1000 ft	7.30 + J 0.3555	PU			
Z0 Impedance:	5.89 + J 0.2241	Ohms/1000 ft	23.02 + J 0.8754	PU			

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CABLE NAME	FEEDER FROM NAME	FEEDER TO NAME	QTY /PH	VOLTS L-L	LENGTH	FEEDER SIZE	FEEDER TYPE
CBL-3P-07	DSC 3P-07	BUS-3P-07	1	480	9.0 FEET	12	Copper
	Duct Material: Magnetic	Insulation Type: PVC	Insulation Class: THWN				
	+/- Impedance: 1.87 + J 0.0910	Ohms/1000 ft 7.30 + J 0.3555	PU				
	Z0 Impedance: 5.89 + J 0.2241	Ohms/1000 ft 23.02 + J 0.8754	PU				
CBL-4P-01	DSC 4P-01	BUS-4P-01	1	480	9.0 FEET	6	Copper
	Duct Material: Magnetic	Insulation Type: PVC	Insulation Class: THWN				
	+/- Impedance: 0.5100 + J 0.0685	Ohms/1000 ft 1.99 + J 0.2676	PU				
	Z0 Impedance: 1.61 + J 0.1687	Ohms/1000 ft 6.28 + J 0.6590	PU				
CBL-4P-02	DSC 4P-02	BUS-4P-02	1	480	9.0 FEET	6	Copper
	Duct Material: Magnetic	Insulation Type: PVC	Insulation Class: THWN				
	+/- Impedance: 0.5100 + J 0.0685	Ohms/1000 ft 1.99 + J 0.2676	PU				
	Z0 Impedance: 1.61 + J 0.1687	Ohms/1000 ft 6.28 + J 0.6590	PU				
CBL-4P-03	DSC 4P-03	BUS-4P-03	1	480	9.0 FEET	6	Copper
	Duct Material: Magnetic	Insulation Type: PVC	Insulation Class: THWN				
	+/- Impedance: 0.5100 + J 0.0685	Ohms/1000 ft 1.99 + J 0.2676	PU				
	Z0 Impedance: 1.61 + J 0.1687	Ohms/1000 ft 6.28 + J 0.6590	PU				
CBL-ADMIN	SWBD-MAIN-A	ADMIN	1	480	335.0 FEET	250	Copper
	Duct Material: Magnetic	Insulation Type: PVC	Insulation Class: THWN				
	+/- Impedance: 0.0552 + J 0.0495	Ohms/1000 ft 8.03 + J 7.20	PU				
	Z0 Impedance: 0.1739 + J 0.1219	Ohms/1000 ft 25.28 + J 17.72	PU				
CBL-AHF-10A	MCC-10A	AHF-10A	1	480	30.0 FEET	4	Copper
	Duct Material: Magnetic	Insulation Type: PVC	Insulation Class: THWN				
	+/- Impedance: 0.3210 + J 0.0632	Ohms/1000 ft 4.18 + J 0.8229	PU				
	Z0 Impedance: 1.01 + J 0.1556	Ohms/1000 ft 13.17 + J 2.03	PU				
CBL-AHF-10B	MCC-10B	AHF-10B	1	480	30.0 FEET	8	Copper
	Duct Material: Magnetic	Insulation Type: PVC	Insulation Class: THWN				
	+/- Impedance: 0.8110 + J 0.0754	Ohms/1000 ft 10.56 + J 0.9818	PU				
	Z0 Impedance: 2.56 + J 0.1856	Ohms/1000 ft 33.28 + J 2.42	PU				
CBL-AHF-8A	MCC-8A	AHF-8A	1	480	25.0 FEET	1	Copper
	Duct Material: Magnetic	Insulation Type: PVC	Insulation Class: THWN				
	+/- Impedance: 0.1600 + J 0.0570	Ohms/1000 ft 1.74 + J 0.6185	PU				
	Z0 Impedance: 0.5042 + J 0.1403	Ohms/1000 ft 5.47 + J 1.52	PU				
CBL-AHF-8B	MCC-8B	AHF-8B	1	480	30.0 FEET	1	Copper
	Duct Material: Magnetic	Insulation Type: PVC	Insulation Class: THWN				
	+/- Impedance: 0.1600 + J 0.0570	Ohms/1000 ft 2.08 + J 0.7422	PU				
	Z0 Impedance: 0.5042 + J 0.1403	Ohms/1000 ft 6.57 + J 1.83	PU				

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CABLE NAME	FEEDER FROM NAME	FEEDER TO NAME	QTY /PH	VOLTS L-L	LENGTH	FEEDER SIZE	FEEDER TYPE
CBL-DP-10	MTS-10	DP-10	1	480	10.0 FEET	1/0	Copper
	Duct Material: Magnetic	Insulation Type: PVC	Insulation Class: THWN				
	+/- Impedance: 0.1280 + J 0.0540 Ohms/1000 ft	0.5556 + J 0.2344 PU					
	Z0 Impedance: 0.4034 + J 0.1329 Ohms/1000 ft	1.75 + J 0.5768 PU					
CBL-DP-13	MTS-13	DP-13	1	480	10.0 FEET	500	Copper
	Duct Material: Magnetic	Insulation Type: PVC	Insulation Class: THWN				
	+/- Impedance: 0.0294 + J 0.0466 Ohms/1000 ft	0.1276 + J 0.2023 PU					
	Z0 Impedance: 0.0926 + J 0.1147 Ohms/1000 ft	0.4019 + J 0.4978 PU					
CBL-DP-5A	MCC-5A	DP-5A	2	480	15.0 FEET	3/0	Copper
	Duct Material: Magnetic	Insulation Type: PVC	Insulation Class: THWN				
	+/- Impedance: 0.0805 + J 0.0519 Ohms/1000 ft	0.2620 + J 0.1689 PU					
	Z0 Impedance: 0.2537 + J 0.1278 Ohms/1000 ft	0.8258 + J 0.4160 PU					
CBL-DP-5B	MCC-5B	DP-5B	2	480	25.0 FEET	3/0	Copper
	Duct Material: Magnetic	Insulation Type: PVC	Insulation Class: THWN				
	+/- Impedance: 0.0805 + J 0.0519 Ohms/1000 ft	0.4367 + J 0.2816 PU					
	Z0 Impedance: 0.2537 + J 0.1278 Ohms/1000 ft	1.38 + J 0.6934 PU					
CBL-DP-6	MTS-6	DP-6	1	480	10.0 FEET	250	Copper
	Duct Material: Magnetic	Insulation Type: PVC	Insulation Class: THWN				
	+/- Impedance: 0.0552 + J 0.0495 Ohms/1000 ft	0.2396 + J 0.2148 PU					
	Z0 Impedance: 0.1739 + J 0.1219 Ohms/1000 ft	0.7548 + J 0.5291 PU					
CBL-DP-8A-1	MCC-8A	DP-8A-1	2	480	25.0 FEET	3/0	Copper
	Duct Material: Magnetic	Insulation Type: PVC	Insulation Class: THWN				
	+/- Impedance: 0.0805 + J 0.0519 Ohms/1000 ft	0.4367 + J 0.2816 PU					
	Z0 Impedance: 0.2537 + J 0.1278 Ohms/1000 ft	1.38 + J 0.6934 PU					
CBL-DP-8A-2	DP-8A-1	DP-8A-2	1	480	10.0 FEET	1/0	Copper
	Duct Material: Magnetic	Insulation Type: PVC	Insulation Class: THWN				
	+/- Impedance: 0.1280 + J 0.0540 Ohms/1000 ft	0.5556 + J 0.2344 PU					
	Z0 Impedance: 0.4034 + J 0.1329 Ohms/1000 ft	1.75 + J 0.5768 PU					
CBL-DP-8B-1	MCC-8B	DP-8B-1	2	480	30.0 FEET	3/0	Copper
	Duct Material: Magnetic	Insulation Type: PVC	Insulation Class: THWN				
	+/- Impedance: 0.0805 + J 0.0519 Ohms/1000 ft	0.5241 + J 0.3379 PU					
	Z0 Impedance: 0.2537 + J 0.1278 Ohms/1000 ft	1.65 + J 0.8320 PU					
CBL-DP-8B-2	DP-8B-1	DP-8B-2	1	480	10.0 FEET	1/0	Copper
	Duct Material: Magnetic	Insulation Type: PVC	Insulation Class: THWN				
	+/- Impedance: 0.1280 + J 0.0540 Ohms/1000 ft	0.5556 + J 0.2344 PU					
	Z0 Impedance: 0.4034 + J 0.1329 Ohms/1000 ft	1.75 + J 0.5768 PU					

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CABLE NAME	FEEDER FROM NAME	FEEDER TO NAME	QTY /PH	VOLTS L-L	LENGTH	FEEDER SIZE	FEEDER TYPE
CBL-DP-9	MTS-9	DP-9	1	480	10.0 FEET	4/0	Copper
	Duct Material: Magnetic	Insulation Type: PVC	Insulation Class: THWN				
	+/- Impedance: 0.0640 + J 0.0497 Ohms/1000 ft	0.2778 + J 0.2157 PU					
	Z0 Impedance: 0.2017 + J 0.1224 Ohms/1000 ft	0.8754 + J 0.5313 PU					
CBL-DSC 02-EF-1320	MCC-5B	DSC 02-EF-1320	1	480	150.0 FEET	12	Copper
	Duct Material: Magnetic	Insulation Type: PVC	Insulation Class: THWN				
	+/- Impedance: 1.87 + J 0.0910 Ohms/1000 ft	121.74 + J 5.92 PU					
	Z0 Impedance: 5.89 + J 0.2241 Ohms/1000 ft	383.68 + J 14.59 PU					
CBL-DSC 02-EF-1330	MCC-5A	DSC 02-EF-1330	1	480	125.0 FEET	12	Copper
	Duct Material: Magnetic	Insulation Type: PVC	Insulation Class: THWN				
	+/- Impedance: 1.87 + J 0.0910 Ohms/1000 ft	101.45 + J 4.94 PU					
	Z0 Impedance: 5.89 + J 0.2241 Ohms/1000 ft	319.74 + J 12.16 PU					
CBL-DSC 02-EF-1340	MCC-5B	DSC 02-EF-1340	1	480	125.0 FEET	12	Copper
	Duct Material: Magnetic	Insulation Type: PVC	Insulation Class: THWN				
	+/- Impedance: 1.87 + J 0.0910 Ohms/1000 ft	101.45 + J 4.94 PU					
	Z0 Impedance: 5.89 + J 0.2241 Ohms/1000 ft	319.74 + J 12.16 PU					
CBL-DSC 02-EF-1370	MCC-5A	DSC 02-EF-1370	1	480	100.0 FEET	12	Copper
	Duct Material: Magnetic	Insulation Type: PVC	Insulation Class: THWN				
	+/- Impedance: 1.87 + J 0.0910 Ohms/1000 ft	81.16 + J 3.95 PU					
	Z0 Impedance: 5.89 + J 0.2241 Ohms/1000 ft	255.79 + J 9.73 PU					
CBL-DSC 02-EF-1380	MCC-5B	DSC 02-EF-1380	1	480	125.0 FEET	12	Copper
	Duct Material: Magnetic	Insulation Type: PVC	Insulation Class: THWN				
	+/- Impedance: 1.87 + J 0.0910 Ohms/1000 ft	101.45 + J 4.94 PU					
	Z0 Impedance: 5.89 + J 0.2241 Ohms/1000 ft	319.74 + J 12.16 PU					
CBL-DSC 02-EF-1390	MCC-5A	DSC 02-EF-1390	1	480	125.0 FEET	10	Copper
	Duct Material: Magnetic	Insulation Type: PVC	Insulation Class: THWN				
	+/- Impedance: 1.18 + J 0.0854 Ohms/1000 ft	64.02 + J 4.63 PU					
	Z0 Impedance: 3.72 + J 0.2103 Ohms/1000 ft	201.76 + J 11.41 PU					
CBL-DSC 02-EF-1396	MCC-5B	DSC 02-EF-1396	1	480	150.0 FEET	10	Copper
	Duct Material: Magnetic	Insulation Type: PVC	Insulation Class: THWN				
	+/- Impedance: 1.18 + J 0.0854 Ohms/1000 ft	76.82 + J 5.56 PU					
	Z0 Impedance: 3.72 + J 0.2103 Ohms/1000 ft	242.11 + J 13.69 PU					
CBL-DSC 02-SF-1310	MCC-5B	DSC 02-SF-1310	1	480	125.0 FEET	12	Copper
	Duct Material: Magnetic	Insulation Type: PVC	Insulation Class: THWN				
	+/- Impedance: 1.87 + J 0.0910 Ohms/1000 ft	101.45 + J 4.94 PU					
	Z0 Impedance: 5.89 + J 0.2241 Ohms/1000 ft	319.74 + J 12.16 PU					

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CABLE NAME	FEEDER FROM NAME	FEEDER TO NAME	QTY /PH	VOLTS L-L	LENGTH	FEEDER SIZE	FEEDER TYPE
CBL-DSC 02-SF-1350	MCC-5A	DSC 02-SF-1350	1	480	125.0 FEET	12	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.87 + J 0.0910	Ohms/1000 ft	101.45 + J 4.94	PU			
Z0 Impedance:	5.89 + J 0.2241	Ohms/1000 ft	319.74 + J 12.16	PU			
CBL-DSC 02-SF-1360	MCC-5B	DSC 02-SF-1360	1	480	125.0 FEET	12	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.87 + J 0.0910	Ohms/1000 ft	101.45 + J 4.94	PU			
Z0 Impedance:	5.89 + J 0.2241	Ohms/1000 ft	319.74 + J 12.16	PU			
CBL-DSC 02-SF-1570	MCC-5A	DSC 02-SF-1570	1	480	100.0 FEET	12	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.87 + J 0.0910	Ohms/1000 ft	81.16 + J 3.95	PU			
Z0 Impedance:	5.89 + J 0.2241	Ohms/1000 ft	255.79 + J 9.73	PU			
CBL-DSC 02-SF-1575	MCC-5B	DSC 02-SF-1575	1	480	175.0 FEET	10	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.18 + J 0.0854	Ohms/1000 ft	89.63 + J 6.49	PU			
Z0 Impedance:	3.72 + J 0.2103	Ohms/1000 ft	282.46 + J 15.97	PU			
CBL-DSC 02-SF-1810	MCC-5B	DSC 02-SF-1810	1	480	100.0 FEET	12	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.87 + J 0.0910	Ohms/1000 ft	81.16 + J 3.95	PU			
Z0 Impedance:	5.89 + J 0.2241	Ohms/1000 ft	255.79 + J 9.73	PU			
CBL-DSC 03-EF-1420	MCC-5B	DSC 03-EF-1420	1	480	75.0 FEET	12	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.87 + J 0.0910	Ohms/1000 ft	60.87 + J 2.96	PU			
Z0 Impedance:	5.89 + J 0.2241	Ohms/1000 ft	191.84 + J 7.29	PU			
CBL-DSC 03-HSS-1140	MCC-9A	DSC 03-HSS-1140	1	480	175.0 FEET	12	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.87 + J 0.0910	Ohms/1000 ft	142.04 + J 6.91	PU			
Z0 Impedance:	5.89 + J 0.2241	Ohms/1000 ft	447.63 + J 17.02	PU			
CBL-DSC 03-HSS-1240	MCC-9A	DSC 03-HSS-1240	1	480	200.0 FEET	12	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.87 + J 0.0910	Ohms/1000 ft	162.33 + J 7.90	PU			
Z0 Impedance:	5.89 + J 0.2241	Ohms/1000 ft	511.58 + J 19.45	PU			
CBL-DSC 03-HSS-1340	MCC-9B	DSC 03-HSS-1340	1	480	213.0 FEET	12	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.87 + J 0.0910	Ohms/1000 ft	172.88 + J 8.41	PU			
Z0 Impedance:	5.89 + J 0.2241	Ohms/1000 ft	544.83 + J 20.72	PU			

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CABLE NAME	FEEDER FROM NAME	FEEDER TO NAME	QTY /PH	VOLTS L-L	LENGTH	FEEDER SIZE	FEEDER TYPE
CBL-DSC 03-PMP-1510	MCC-9A	DSC 03-PMP-1510	1	480	100.0 FEET	12	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.87 + J 0.0910	Ohms/1000 ft	81.16 + J 3.95	PU			
Z0 Impedance:	5.89 + J 0.2241	Ohms/1000 ft	255.79 + J 9.73	PU			
CBL-DSC 03-PMP-1520	MCC-9B	DSC 03-PMP-1520	1	480	130.0 FEET	12	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.87 + J 0.0910	Ohms/1000 ft	105.51 + J 5.13	PU			
Z0 Impedance:	5.89 + J 0.2241	Ohms/1000 ft	332.53 + J 12.64	PU			
CBL-DSC 03-PMP-1530	MCC-9A	DSC 03-PMP-1530	1	480	150.0 FEET	12	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.87 + J 0.0910	Ohms/1000 ft	121.74 + J 5.92	PU			
Z0 Impedance:	5.89 + J 0.2241	Ohms/1000 ft	383.68 + J 14.59	PU			
CBL-DSC 03-PMP-1610	MCC-9A	DSC 03-PMP-1610	1	480	200.0 FEET	12	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.87 + J 0.0910	Ohms/1000 ft	162.33 + J 7.90	PU			
Z0 Impedance:	5.89 + J 0.2241	Ohms/1000 ft	511.58 + J 19.45	PU			
CBL-DSC 03-PMP-1620	MCC-9B	DSC 03-PMP-1620	1	480	175.0 FEET	12	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.87 + J 0.0910	Ohms/1000 ft	142.04 + J 6.91	PU			
Z0 Impedance:	5.89 + J 0.2241	Ohms/1000 ft	447.63 + J 17.02	PU			
CBL-DSC 03-SCC-1150	MCC-9A	DSC 03-SCC-1150	1	480	125.0 FEET	12	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.87 + J 0.0910	Ohms/1000 ft	101.45 + J 4.94	PU			
Z0 Impedance:	5.89 + J 0.2241	Ohms/1000 ft	319.74 + J 12.16	PU			
CBL-DSC 03-SCC-1250	MCC-9B	DSC 03-SCC-1250	1	480	130.0 FEET	12	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.87 + J 0.0910	Ohms/1000 ft	105.51 + J 5.13	PU			
Z0 Impedance:	5.89 + J 0.2241	Ohms/1000 ft	332.53 + J 12.64	PU			
CBL-DSC 03-SCC-1350	MCC-9A	DSC 03-SCC-1350	1	480	175.0 FEET	12	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.87 + J 0.0910	Ohms/1000 ft	142.04 + J 6.91	PU			
Z0 Impedance:	5.89 + J 0.2241	Ohms/1000 ft	447.63 + J 17.02	PU			
CBL-DSC 03-SF-1410	MCC-5B	DSC 03-SF-1410	1	480	200.0 FEET	12	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.87 + J 0.0910	Ohms/1000 ft	162.33 + J 7.90	PU			
Z0 Impedance:	5.89 + J 0.2241	Ohms/1000 ft	511.58 + J 19.45	PU			

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CABLE NAME	FEEDER FROM NAME	FEEDER TO NAME	QTY /PH	VOLTS L-L	LENGTH	FEEDER SIZE	FEEDER TYPE
CBL-DSC 03-SF-1575	MCC-5B	DSC 03-SF-1575	1	480	200.0 FEET	12	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.87 + J 0.0910	Ohms/1000 ft	162.33 + J 7.90	PU			
Z0 Impedance:	5.89 + J 0.2241	Ohms/1000 ft	511.58 + J 19.45	PU			
CBL-DSC 03-SLC-1120	MCC-9A	DSC 03-SLC-1120	1	480	100.0 FEET	12	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.87 + J 0.0910	Ohms/1000 ft	81.16 + J 3.95	PU			
Z0 Impedance:	5.89 + J 0.2241	Ohms/1000 ft	255.79 + J 9.73	PU			
CBL-DSC 03-SLC-1220	MCC-9B	DSC 03-SLC-1220	1	480	130.0 FEET	12	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.87 + J 0.0910	Ohms/1000 ft	105.51 + J 5.13	PU			
Z0 Impedance:	5.89 + J 0.2241	Ohms/1000 ft	332.53 + J 12.64	PU			
CBL-DSC 03-SLC-1320	MCC-9A	DSC 03-SLC-1320	1	480	175.0 FEET	12	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.87 + J 0.0910	Ohms/1000 ft	142.04 + J 6.91	PU			
Z0 Impedance:	5.89 + J 0.2241	Ohms/1000 ft	447.63 + J 17.02	PU			
CBL-DSC 04-PMP-1140	MCC-10B	DSC 04-PMP-1140	1	480	205.0 FEET	6	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	0.5100 + J 0.0685	Ohms/1000 ft	45.38 + J 6.09	PU			
Z0 Impedance:	1.61 + J 0.1687	Ohms/1000 ft	143.00 + J 15.01	PU			
CBL-DSC 06-CLA-1510	MCC-10A	DSC 06-CLA-1510	1	480	100.0 FEET	10	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.18 + J 0.0854	Ohms/1000 ft	51.22 + J 3.71	PU			
Z0 Impedance:	3.72 + J 0.2103	Ohms/1000 ft	161.41 + J 9.13	PU			
CBL-DSC 06-CLA-1520	MCC-10B	DSC 06-CLA-1520	1	480	175.0 FEET	10	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.18 + J 0.0854	Ohms/1000 ft	89.63 + J 6.49	PU			
Z0 Impedance:	3.72 + J 0.2103	Ohms/1000 ft	282.46 + J 15.97	PU			
CBL-DSC 06-CLA-1530	MCC-10A	DSC 06-CLA-1530	1	480	250.0 FEET	10	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.18 + J 0.0854	Ohms/1000 ft	128.04 + J 9.27	PU			
Z0 Impedance:	3.72 + J 0.2103	Ohms/1000 ft	403.52 + J 22.82	PU			
CBL-DSC 06-PMP-1515	MCC-10A	DSC 06-PMP-1515	1	480	100.0 FEET	8	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	0.8110 + J 0.0754	Ohms/1000 ft	35.20 + J 3.27	PU			
Z0 Impedance:	2.56 + J 0.1856	Ohms/1000 ft	110.93 + J 8.06	PU			

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CABLE NAME	FEEDER FROM NAME	FEEDER TO NAME	QTY /PH	VOLTS L-L	LENGTH	FEEDER SIZE	FEEDER TYPE
CBL-DSC 06-PMP-1525	MCC-10B	DSC 06-PMP-1525	1	480	175.0 FEET	8	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	0.8110 + J 0.0754	Ohms/1000 ft	61.60 + J 5.73	PU			
Z0 Impedance:	2.56 + J 0.1856	Ohms/1000 ft	194.13 + J 14.10	PU			
CBL-DSC 06-PMP-1535	MCC-10A	DSC 06-PMP-1535	1	480	250.0 FEET	8	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	0.8110 + J 0.0754	Ohms/1000 ft	88.00 + J 8.18	PU			
Z0 Impedance:	2.56 + J 0.1856	Ohms/1000 ft	277.33 + J 20.14	PU			
CBL-DSC 08-EF-1230	MCC-11A	DSC 08-EF-1230	1	480	30.0 FEET	12	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.87 + J 0.0910	Ohms/1000 ft	24.35 + J 1.18	PU			
Z0 Impedance:	5.89 + J 0.2241	Ohms/1000 ft	76.74 + J 2.92	PU			
CBL-DSC 09-MIX-1110	MCC-11A	DSC 09-MIX-1110	1	480	175.0 FEET	12	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.87 + J 0.0910	Ohms/1000 ft	142.04 + J 6.91	PU			
Z0 Impedance:	5.89 + J 0.2241	Ohms/1000 ft	447.63 + J 17.02	PU			
CBL-DSC 09-PMP-903	MCC-11B	DSC 09-PMP-903	1	480	145.0 FEET	12	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.87 + J 0.0910	Ohms/1000 ft	117.69 + J 5.73	PU			
Z0 Impedance:	5.89 + J 0.2241	Ohms/1000 ft	370.90 + J 14.10	PU			
CBL-DSC 11-EF-1530	MCC-6A	DSC 11-EF-1530	1	480	75.0 FEET	12	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.87 + J 0.0910	Ohms/1000 ft	60.87 + J 2.96	PU			
Z0 Impedance:	5.89 + J 0.2241	Ohms/1000 ft	191.84 + J 7.29	PU			
CBL-DSC 11-EF-1540	MCC-6B	DSC 11-EF-1540	1	480	60.0 FEET	12	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.87 + J 0.0910	Ohms/1000 ft	48.70 + J 2.37	PU			
Z0 Impedance:	5.89 + J 0.2241	Ohms/1000 ft	153.47 + J 5.84	PU			
CBL-DSC 11-EF-1560	MCC-6B	DSC 11-EF-1560	1	480	75.0 FEET	12	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.87 + J 0.0910	Ohms/1000 ft	60.87 + J 2.96	PU			
Z0 Impedance:	5.89 + J 0.2241	Ohms/1000 ft	191.84 + J 7.29	PU			
CBL-DSC 11-PMP-2311	MCC-6A	DSC 11-PMP-2311	1	480	100.0 FEET	12	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.87 + J 0.0910	Ohms/1000 ft	81.16 + J 3.95	PU			
Z0 Impedance:	5.89 + J 0.2241	Ohms/1000 ft	255.79 + J 9.73	PU			

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CABLE NAME	FEEDER FROM NAME	FEEDER TO NAME	QTY /PH	VOLTS L-L	LENGTH	FEEDER SIZE	FEEDER TYPE
CBL-DSC 11-PMP-2314	MCC-6A	DSC 11-PMP-2314	1	480	75.0 FEET	12	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.87 + J 0.0910	Ohms/1000 ft	60.87 + J 2.96	PU			
Z0 Impedance:	5.89 + J 0.2241	Ohms/1000 ft	191.84 + J 7.29	PU			
CBL-DSC 11-PMP-2322	MCC-6B	DSC 11-PMP-2322	1	480	100.0 FEET	12	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.87 + J 0.0910	Ohms/1000 ft	81.16 + J 3.95	PU			
Z0 Impedance:	5.89 + J 0.2241	Ohms/1000 ft	255.79 + J 9.73	PU			
CBL-DSC 11-PMP-2324	MCC-6B	DSC 11-PMP-2324	1	480	60.0 FEET	12	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.87 + J 0.0910	Ohms/1000 ft	48.70 + J 2.37	PU			
Z0 Impedance:	5.89 + J 0.2241	Ohms/1000 ft	153.47 + J 5.84	PU			
CBL-DSC 11-PMP-2610	MCC-6A	DSC 11-PMP-2610	1	480	100.0 FEET	12	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.87 + J 0.0910	Ohms/1000 ft	81.16 + J 3.95	PU			
Z0 Impedance:	5.89 + J 0.2241	Ohms/1000 ft	255.79 + J 9.73	PU			
CBL-DSC 11-PMP-2620	MCC-6B	DSC 11-PMP-2620	1	480	100.0 FEET	12	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.87 + J 0.0910	Ohms/1000 ft	81.16 + J 3.95	PU			
Z0 Impedance:	5.89 + J 0.2241	Ohms/1000 ft	255.79 + J 9.73	PU			
CBL-DSC 11-PMP-2711	MCC-6A	DSC 11-PMP-2711	1	480	100.0 FEET	12	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.87 + J 0.0910	Ohms/1000 ft	81.16 + J 3.95	PU			
Z0 Impedance:	5.89 + J 0.2241	Ohms/1000 ft	255.79 + J 9.73	PU			
CBL-DSC 11-PMP-2721	MCC-6B	DSC 11-PMP-2721	1	480	80.0 FEET	12	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.87 + J 0.0910	Ohms/1000 ft	64.93 + J 3.16	PU			
Z0 Impedance:	5.89 + J 0.2241	Ohms/1000 ft	204.63 + J 7.78	PU			
CBL-DSC 11-SF-1510	MCC-6A	DSC 11-SF-1510	1	480	75.0 FEET	12	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.87 + J 0.0910	Ohms/1000 ft	60.87 + J 2.96	PU			
Z0 Impedance:	5.89 + J 0.2241	Ohms/1000 ft	191.84 + J 7.29	PU			
CBL-DSC 11-SF-1520	MCC-6B	DSC 11-SF-1520	1	480	60.0 FEET	12	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.87 + J 0.0910	Ohms/1000 ft	48.70 + J 2.37	PU			
Z0 Impedance:	5.89 + J 0.2241	Ohms/1000 ft	153.47 + J 5.84	PU			

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CBL-DSC 11-SF-1550	MCC-6B	DSC 11-SF-1550	1	480	75.0 FEET	12	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.87 + J 0.0910	Ohms/1000 ft	60.87 + J 2.96	PU			
Z0 Impedance:	5.89 + J 0.2241	Ohms/1000 ft	191.84 + J 7.29	PU			
CBL-DSC 11-SF-1575	MCC-6A	DSC 11-SF-1575	1	480	75.0 FEET	12	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.87 + J 0.0910	Ohms/1000 ft	60.87 + J 2.96	PU			
Z0 Impedance:	5.89 + J 0.2241	Ohms/1000 ft	191.84 + J 7.29	PU			
CBL-DSC 12-AUG-4615	MCC-8A	DSC 12-AUG-4615	1	480	125.0 FEET	8	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	0.8110 + J 0.0754	Ohms/1000 ft	44.00 + J 4.09	PU			
Z0 Impedance:	2.56 + J 0.1856	Ohms/1000 ft	138.67 + J 10.07	PU			
CBL-DSC 12-AUG-4617	MCC-8B	DSC 12-AUG-4617	1	480	100.0 FEET	8	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	0.8110 + J 0.0754	Ohms/1000 ft	35.20 + J 3.27	PU			
Z0 Impedance:	2.56 + J 0.1856	Ohms/1000 ft	110.93 + J 8.06	PU			
CBL-DSC 12-CON-4610	MCC-8A	DSC 12-CON-4610	1	480	100.0 FEET	12	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.87 + J 0.0910	Ohms/1000 ft	81.16 + J 3.95	PU			
Z0 Impedance:	5.89 + J 0.2241	Ohms/1000 ft	255.79 + J 9.73	PU			
CBL-DSC 12-CON-4620	MCC-8B	DSC 12-CON-4620	1	480	80.0 FEET	12	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.87 + J 0.0910	Ohms/1000 ft	64.93 + J 3.16	PU			
Z0 Impedance:	5.89 + J 0.2241	Ohms/1000 ft	204.63 + J 7.78	PU			
CBL-DSC 12-EF-2530	MCC-8A	DSC 12-EF-2530	1	480	75.0 FEET	8	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	0.8110 + J 0.0754	Ohms/1000 ft	26.40 + J 2.45	PU			
Z0 Impedance:	2.56 + J 0.1856	Ohms/1000 ft	83.20 + J 6.04	PU			
CBL-DSC 12-EF-2540	MCC-8B	DSC 12-EF-2540	1	480	75.0 FEET	8	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	0.8110 + J 0.0754	Ohms/1000 ft	26.40 + J 2.45	PU			
Z0 Impedance:	2.56 + J 0.1856	Ohms/1000 ft	83.20 + J 6.04	PU			
CBL-DSC 12-HPU-4750	MCC-8A	DSC 12-HPU-4750	1	480	100.0 FEET	8	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	0.8110 + J 0.0754	Ohms/1000 ft	35.20 + J 3.27	PU			
Z0 Impedance:	2.56 + J 0.1856	Ohms/1000 ft	110.93 + J 8.06	PU			

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CBL-DSC 12-SF-2510	MCC-8A	DSC 12-SF-2510	1	480	75.0 FEET	8	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	0.8110 + J 0.0754	Ohms/1000 ft	26.40 + J 2.45	PU			
Z0 Impedance:	2.56 + J 0.1856	Ohms/1000 ft	83.20 + J 6.04	PU			
CBL-DSC 12-SF-2520	MCC-8B	DSC 12-SF-2520	1	480	75.0 FEET	8	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	0.8110 + J 0.0754	Ohms/1000 ft	26.40 + J 2.45	PU			
Z0 Impedance:	2.56 + J 0.1856	Ohms/1000 ft	83.20 + J 6.04	PU			
CBL-DSC 12-SF-2550	MCC-8A	DSC 12-SF-2550	1	480	60.0 FEET	12	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.87 + J 0.0910	Ohms/1000 ft	48.70 + J 2.37	PU			
Z0 Impedance:	5.89 + J 0.2241	Ohms/1000 ft	153.47 + J 5.84	PU			
CBL-DSC 12-SF-2560	MCC-8B	DSC 12-SF-2560	1	480	125.0 FEET	12	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.87 + J 0.0910	Ohms/1000 ft	101.45 + J 4.94	PU			
Z0 Impedance:	5.89 + J 0.2241	Ohms/1000 ft	319.74 + J 12.16	PU			
CBL-DSC 12-SF-2570	MCC-8A	DSC 12-SF-2570	1	480	100.0 FEET	12	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.87 + J 0.0910	Ohms/1000 ft	81.16 + J 3.95	PU			
Z0 Impedance:	5.89 + J 0.2241	Ohms/1000 ft	255.79 + J 9.73	PU			
CBL-DSC 12-SF-2580	MCC-8B	DSC 12-SF-2580	1	480	125.0 FEET	12	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.87 + J 0.0910	Ohms/1000 ft	101.45 + J 4.94	PU			
Z0 Impedance:	5.89 + J 0.2241	Ohms/1000 ft	319.74 + J 12.16	PU			
CBL-DSC 12-SRW-4612	MCC-8A	DSC 12-SRW-4612	1	480	100.0 FEET	12	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.87 + J 0.0910	Ohms/1000 ft	81.16 + J 3.95	PU			
Z0 Impedance:	5.89 + J 0.2241	Ohms/1000 ft	255.79 + J 9.73	PU			
CBL-DSC 12-SRW-4613	MCC-8B	DSC 12-SRW-4613	1	480	100.0 FEET	12	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	1.87 + J 0.0910	Ohms/1000 ft	81.16 + J 3.95	PU			
Z0 Impedance:	5.89 + J 0.2241	Ohms/1000 ft	255.79 + J 9.73	PU			
CBL-DSC 19-PMP-1110	MCC-8A	DSC 19-PMP-1110	1	480	175.0 FEET	8	Copper
Duct Material:	Magnetic	Insulation Type:	PVC	Insulation Class:	THWN		
+/- Impedance:	0.8110 + J 0.0754	Ohms/1000 ft	61.60 + J 5.73	PU			
Z0 Impedance:	2.56 + J 0.1856	Ohms/1000 ft	194.13 + J 14.10	PU			

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CBL-DSC 19-PMP-1120	MCC-8B	DSC 19-PMP-1120	1	480	150.0 FEET	8	Copper
	Duct Material: Magnetic	Insulation Type: PVC	Insulation Class: THWN				
	+/- Impedance: 0.8110 + J 0.0754	Ohms/1000 ft 52.80 + J 4.91	PU				
	Z0 Impedance: 2.56 + J 0.1856	Ohms/1000 ft 166.40 + J 12.08	PU				
CBL-DSC 21-EF-1250	MCC-9A	DSC 21-EF-1250	1	480	60.0 FEET	12	Copper
	Duct Material: Magnetic	Insulation Type: PVC	Insulation Class: THWN				
	+/- Impedance: 1.87 + J 0.0910	Ohms/1000 ft 48.70 + J 2.37	PU				
	Z0 Impedance: 5.89 + J 0.2241	Ohms/1000 ft 153.47 + J 5.84	PU				
CBL-DSC 21-EF-1310	MCC-9A	DSC 21-EF-1310	1	480	125.0 FEET	12	Copper
	Duct Material: Magnetic	Insulation Type: PVC	Insulation Class: THWN				
	+/- Impedance: 1.87 + J 0.0910	Ohms/1000 ft 101.45 + J 4.94	PU				
	Z0 Impedance: 5.89 + J 0.2241	Ohms/1000 ft 319.74 + J 12.16	PU				
CBL-DSC 2P-07	MCC-5B	DSC 2P-07	1	480	155.0 FEET	12	Copper
	Duct Material: Magnetic	Insulation Type: PVC	Insulation Class: THWN				
	+/- Impedance: 1.87 + J 0.0910	Ohms/1000 ft 125.80 + J 6.12	PU				
	Z0 Impedance: 5.89 + J 0.2241	Ohms/1000 ft 396.47 + J 15.08	PU				
CBL-DSC 3G-01	MCC-9A	DSC 3G-01	1	480	235.0 FEET	12	Copper
	Duct Material: Magnetic	Insulation Type: PVC	Insulation Class: THWN				
	+/- Impedance: 1.87 + J 0.0910	Ohms/1000 ft 190.73 + J 9.28	PU				
	Z0 Impedance: 5.89 + J 0.2241	Ohms/1000 ft 601.11 + J 22.86	PU				
CBL-DSC 3G-02	MCC-9B	DSC 3G-02	1	480	200.0 FEET	12	Copper
	Duct Material: Magnetic	Insulation Type: PVC	Insulation Class: THWN				
	+/- Impedance: 1.87 + J 0.0910	Ohms/1000 ft 162.33 + J 7.90	PU				
	Z0 Impedance: 5.89 + J 0.2241	Ohms/1000 ft 511.58 + J 19.45	PU				
CBL-DSC 3G-03	MCC-9A	DSC 3G-03	1	480	191.0 FEET	12	Copper
	Duct Material: Magnetic	Insulation Type: PVC	Insulation Class: THWN				
	+/- Impedance: 1.87 + J 0.0910	Ohms/1000 ft 155.02 + J 7.54	PU				
	Z0 Impedance: 5.89 + J 0.2241	Ohms/1000 ft 488.56 + J 18.58	PU				
CBL-DSC 3P-05	MCC-9A	DSC 3P-05	1	480	150.0 FEET	12	Copper
	Duct Material: Magnetic	Insulation Type: PVC	Insulation Class: THWN				
	+/- Impedance: 1.87 + J 0.0910	Ohms/1000 ft 121.74 + J 5.92	PU				
	Z0 Impedance: 5.89 + J 0.2241	Ohms/1000 ft 383.68 + J 14.59	PU				
CBL-DSC 3P-07	MCC-9B	DSC 3P-07	1	480	135.0 FEET	12	Copper
	Duct Material: Magnetic	Insulation Type: PVC	Insulation Class: THWN				
	+/- Impedance: 1.87 + J 0.0910	Ohms/1000 ft 109.57 + J 5.33	PU				
	Z0 Impedance: 5.89 + J 0.2241	Ohms/1000 ft 345.32 + J 13.13	PU				

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CBL-DSC 4P-01	MCC-10A	DSC 4P-01	1	480	225.0 FEET	6	Copper
	Duct Material: Magnetic	Insulation Type: PVC	Insulation Class: THWN				
	+/- Impedance: 0.5100 + J 0.0685	Ohms/1000 ft 49.80 + J 6.69	PU				
	Z0 Impedance: 1.61 + J 0.1687	Ohms/1000 ft 156.95 + J 16.47	PU				
CBL-DSC 4P-02	MCC-10B	DSC 4P-02	1	480	220.0 FEET	6	Copper
	Duct Material: Magnetic	Insulation Type: PVC	Insulation Class: THWN				
	+/- Impedance: 0.5100 + J 0.0685	Ohms/1000 ft 48.70 + J 6.54	PU				
	Z0 Impedance: 1.61 + J 0.1687	Ohms/1000 ft 153.47 + J 16.11	PU				
CBL-DSC 4P-03	MCC-10A	DSC 4P-03	1	480	215.0 FEET	6	Copper
	Duct Material: Magnetic	Insulation Type: PVC	Insulation Class: THWN				
	+/- Impedance: 0.5100 + J 0.0685	Ohms/1000 ft 47.59 + J 6.39	PU				
	Z0 Impedance: 1.61 + J 0.1687	Ohms/1000 ft 149.98 + J 15.74	PU				
CBL-DSC SHB CRANE	MCC-8B	DSC SHB CRANE	1	480	100.0 FEET	8	Copper
	Duct Material: Magnetic	Insulation Type: PVC	Insulation Class: THWN				
	+/- Impedance: 0.8110 + J 0.0754	Ohms/1000 ft 35.20 + J 3.27	PU				
	Z0 Impedance: 2.56 + J 0.1856	Ohms/1000 ft 110.93 + J 8.06	PU				
CBL-GEN SWBD-1 [Ex]	21-GEN-1100 [Ex]	GEN SWBD [Ex]	6	480	55.0 FEET	500	Copper
	Duct Material: Magnetic	Insulation Type: PVC	Insulation Class: THWN				
	+/- Impedance: 0.0294 + J 0.0466	Ohms/1000 ft 0.1170 + J 0.1854	PU				
	Z0 Impedance: 0.0926 + J 0.1147	Ohms/1000 ft 0.3684 + J 0.4563	PU				
CBL-GEN SWBD-2	21-GEN-1200	GEN SWBD [Ex]	6	480	100.0 FEET	500	Copper
	Duct Material: Magnetic	Insulation Type: PVC	Insulation Class: THWN				
	+/- Impedance: 0.0294 + J 0.0466	Ohms/1000 ft 0.2127 + J 0.3371	PU				
	Z0 Impedance: 0.0926 + J 0.1147	Ohms/1000 ft 0.6698 + J 0.8297	PU				
CBL-LP-10	BUS-Sec-XFMR-10	LP-10	1	208	10.0 FEET	2	Copper
	Duct Material: Magnetic	Insulation Type: PVC	Insulation Class: THWN				
	+/- Impedance: 0.2020 + J 0.0585	Ohms/1000 ft 4.67 + J 1.35	PU				
	Z0 Impedance: 0.6366 + J 0.1440	Ohms/1000 ft 14.71 + J 3.33	PU				
CBL-LP-11	BUS-Sec-XFMR-11	LP-11	1	208	20.0 FEET	2	Copper
	Duct Material: Magnetic	Insulation Type: PVC	Insulation Class: THWN				
	+/- Impedance: 0.2020 + J 0.0585	Ohms/1000 ft 9.34 + J 2.70	PU				
	Z0 Impedance: 0.6366 + J 0.1440	Ohms/1000 ft 29.43 + J 6.66	PU				
CBL-LP-13A	BUS-Sec-XFMR-13	LP-13A	1	208	10.0 FEET	250	Copper
	Duct Material: Magnetic	Insulation Type: PVC	Insulation Class: THWN				
	+/- Impedance: 0.0552 + J 0.0495	Ohms/1000 ft 1.28 + J 1.14	PU				
	Z0 Impedance: 0.1739 + J 0.1219	Ohms/1000 ft 4.02 + J 2.82	PU				

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CBL-LP-13B	LP-13A	LP-13B	1	208	10.0 FEET	2	Copper
	Duct Material: Magnetic	Insulation Type: PVC	Insulation Class: THWN				
	+/- Impedance: 0.2020 + J 0.0585 Ohms/1000 ft	4.67 + J 1.35 PU					
	Z0 Impedance: 0.6366 + J 0.1440 Ohms/1000 ft	14.71 + J 3.33 PU					
CBL-LP-4PA	BUS-Sec-XFMR-4PA	LP-4PA	1	240	180.0 FEET	8	Copper
	Duct Material: Non-Magnetic	Insulation Type: PVC	Insulation Class: THWN				
	+/- Impedance: 0.8110 + J 0.0603 Ohms/1000 ft	253.44 + J 18.84 PU					
	Z0 Impedance: 1.29 + J 0.1534 Ohms/1000 ft	402.91 + J 47.94 PU					
CBL-LP-5	BUS-Sec-XFMR-5	LP-5	1	208	5.0 FEET	250	Copper
	Duct Material: Magnetic	Insulation Type: PVC	Insulation Class: THWN				
	+/- Impedance: 0.0552 + J 0.0495 Ohms/1000 ft	0.6379 + J 0.5721 PU					
	Z0 Impedance: 0.1739 + J 0.1219 Ohms/1000 ft	2.01 + J 1.41 PU					
CBL-LP-6	BUS-Sec-XFMR-6	LP-6	1	208	10.0 FEET	2	Copper
	Duct Material: Magnetic	Insulation Type: PVC	Insulation Class: THWN				
	+/- Impedance: 0.2020 + J 0.0585 Ohms/1000 ft	4.67 + J 1.35 PU					
	Z0 Impedance: 0.6366 + J 0.1440 Ohms/1000 ft	14.71 + J 3.33 PU					
CBL-LP-8	BUS-Sec-XFMR-8	LP-8	1	208	10.0 FEET	250	Copper
	Duct Material: Magnetic	Insulation Type: PVC	Insulation Class: THWN				
	+/- Impedance: 0.0552 + J 0.0495 Ohms/1000 ft	1.28 + J 1.14 PU					
	Z0 Impedance: 0.1739 + J 0.1219 Ohms/1000 ft	4.02 + J 2.82 PU					
CBL-LP-9	BUS-Sec-XFMR-9	LP-9	1	208	10.0 FEET	1/0	Copper
	Duct Material: Magnetic	Insulation Type: PVC	Insulation Class: THWN				
	+/- Impedance: 0.1280 + J 0.0540 Ohms/1000 ft	2.96 + J 1.25 PU					
	Z0 Impedance: 0.4034 + J 0.1329 Ohms/1000 ft	9.32 + J 3.07 PU					
CBL-MCC-10A	SWBD-MAIN-A	MCC-10A	2	480	760.0 FEET	350	Copper
	Duct Material: Magnetic	Insulation Type: PVC	Insulation Class: THWN				
	+/- Impedance: 0.0378 + J 0.0491 Ohms/1000 ft	6.23 + J 8.10 PU					
	Z0 Impedance: 0.1191 + J 0.1209 Ohms/1000 ft	19.64 + J 19.94 PU					
CBL-MCC-10B	SWBD-MAIN-B	MCC-10B	2	480	760.0 FEET	350	Copper
	Duct Material: Magnetic	Insulation Type: PVC	Insulation Class: THWN				
	+/- Impedance: 0.0378 + J 0.0491 Ohms/1000 ft	6.23 + J 8.10 PU					
	Z0 Impedance: 0.1191 + J 0.1209 Ohms/1000 ft	19.64 + J 19.94 PU					
CBL-MCC-11A	SWBD-MAIN-A	MCC-11A	3	480	920.0 FEET	250	Copper
	Duct Material: Magnetic	Insulation Type: PVC	Insulation Class: THWN				
	+/- Impedance: 0.0552 + J 0.0495 Ohms/1000 ft	7.35 + J 6.59 PU					
	Z0 Impedance: 0.1739 + J 0.1219 Ohms/1000 ft	23.15 + J 16.23 PU					

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CBL-MCC-11A-PG	CL BLDG PORT GEN	MCC-11A	1	480	50.0 FEET	1/0	Copper
	Duct Material: Magnetic	Insulation Type: PVC	Insulation Class: THWN				
	+/- Impedance: 0.1280 + J 0.0540	Ohms/1000 ft 2.78 + J 1.17	PU				
	Z0 Impedance: 0.4034 + J 0.1329	Ohms/1000 ft 8.75 + J 2.88	PU				
CBL-MCC-11B	SWBD-MAIN-B	MCC-11B	1	480	920.0 FEET	250	Copper
	Duct Material: Magnetic	Insulation Type: PVC	Insulation Class: THWN				
	+/- Impedance: 0.0552 + J 0.0495	Ohms/1000 ft 22.04 + J 19.77	PU				
	Z0 Impedance: 0.1739 + J 0.1219	Ohms/1000 ft 69.44 + J 48.68	PU				
CBL-MCC-5A	SWBD-MAIN-A	MCC-5A	2	480	270.0 FEET	350	Copper
	Duct Material: Magnetic	Insulation Type: PVC	Insulation Class: THWN				
	+/- Impedance: 0.0378 + J 0.0491	Ohms/1000 ft 2.21 + J 2.88	PU				
	Z0 Impedance: 0.1191 + J 0.1209	Ohms/1000 ft 6.98 + J 7.08	PU				
CBL-MCC-5B	SWBD-MAIN-B	MCC-5B	2	480	270.0 FEET	350	Copper
	Duct Material: Magnetic	Insulation Type: PVC	Insulation Class: THWN				
	+/- Impedance: 0.0378 + J 0.0491	Ohms/1000 ft 2.21 + J 2.88	PU				
	Z0 Impedance: 0.1191 + J 0.1209	Ohms/1000 ft 6.98 + J 7.08	PU				
CBL-MCC-6A	SWBD-MAIN-A	MCC-6A	2	480	665.0 FEET	350	Copper
	Duct Material: Magnetic	Insulation Type: PVC	Insulation Class: THWN				
	+/- Impedance: 0.0378 + J 0.0491	Ohms/1000 ft 5.46 + J 7.09	PU				
	Z0 Impedance: 0.1191 + J 0.1209	Ohms/1000 ft 17.19 + J 17.45	PU				
CBL-MCC-6B	SWBD-MAIN-B	MCC-6B	2	480	665.0 FEET	350	Copper
	Duct Material: Magnetic	Insulation Type: PVC	Insulation Class: THWN				
	+/- Impedance: 0.0378 + J 0.0491	Ohms/1000 ft 5.46 + J 7.09	PU				
	Z0 Impedance: 0.1191 + J 0.1209	Ohms/1000 ft 17.19 + J 17.45	PU				
CBL-MCC-6B-PG	DCB 1 PORT GEN	MCC-6B	2	480	75.0 FEET	250	Copper
	Duct Material: Magnetic	Insulation Type: PVC	Insulation Class: THWN				
	+/- Impedance: 0.0552 + J 0.0495	Ohms/1000 ft 0.8984 + J 0.8057	PU				
	Z0 Impedance: 0.1739 + J 0.1219	Ohms/1000 ft 2.83 + J 1.98	PU				
CBL-MCC-8A	SWBD-MAIN-A	MCC-8A	5	480	340.0 FEET	500	Copper
	Duct Material: Magnetic	Insulation Type: PVC	Insulation Class: THWN				
	+/- Impedance: 0.0294 + J 0.0466	Ohms/1000 ft 0.8677 + J 1.38	PU				
	Z0 Impedance: 0.0926 + J 0.1147	Ohms/1000 ft 2.73 + J 3.39	PU				
CBL-MCC-8B	SWBD-MAIN-B	MCC-8B	5	480	340.0 FEET	500	Copper
	Duct Material: Magnetic	Insulation Type: PVC	Insulation Class: THWN				
	+/- Impedance: 0.0294 + J 0.0466	Ohms/1000 ft 0.8677 + J 1.38	PU				
	Z0 Impedance: 0.0926 + J 0.1147	Ohms/1000 ft 2.73 + J 3.39	PU				

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FEEDER INPUT DATA

CABLE NAME	FEEDER FROM NAME	FEEDER TO NAME	QTY /PH	VOLTS L-L	LENGTH	FEEDER SIZE	FEEDER TYPE
CBL-MCC-9A	SWBD-MAIN-A	MCC-9A	2	480	70.0 FEET	250	Copper
	Duct Material: Magnetic	Insulation Type: PVC	Insulation Class: THWN				
	+/- Impedance: 0.0552 + J 0.0495 Ohms/1000 ft	0.8385 + J 0.7520 PU					
	Z0 Impedance: 0.1739 + J 0.1219 Ohms/1000 ft	2.64 + J 1.85 PU					
CBL-MCC-9B	SWBD-MAIN-B	MCC-9B	2	480	70.0 FEET	250	Copper
	Duct Material: Magnetic	Insulation Type: PVC	Insulation Class: THWN				
	+/- Impedance: 0.0552 + J 0.0495 Ohms/1000 ft	0.8385 + J 0.7520 PU					
	Z0 Impedance: 0.1739 + J 0.1219 Ohms/1000 ft	2.64 + J 1.85 PU					
CBL-MTS-10 E	MCC-10A	BUS-MTS-10 E	1	480	40.0 FEET	1/0	Copper
	Duct Material: Magnetic	Insulation Type: PVC	Insulation Class: THWN				
	+/- Impedance: 0.1280 + J 0.0540 Ohms/1000 ft	2.22 + J 0.9375 PU					
	Z0 Impedance: 0.4034 + J 0.1329 Ohms/1000 ft	7.00 + J 2.31 PU					
CBL-MTS-10 N	MCC-10B	BUS-MTS-10 N	1	480	30.0 FEET	1/0	Copper
	Duct Material: Magnetic	Insulation Type: PVC	Insulation Class: THWN				
	+/- Impedance: 0.1280 + J 0.0540 Ohms/1000 ft	1.67 + J 0.7031 PU					
	Z0 Impedance: 0.4034 + J 0.1329 Ohms/1000 ft	5.25 + J 1.73 PU					
CBL-MTS-11 E	MCC-11A	BUS-MTS-11 E	1	480	30.0 FEET	2	Copper
	Duct Material: Magnetic	Insulation Type: PVC	Insulation Class: THWN				
	+/- Impedance: 0.2020 + J 0.0585 Ohms/1000 ft	2.63 + J 0.7617 PU					
	Z0 Impedance: 0.6366 + J 0.1440 Ohms/1000 ft	8.29 + J 1.88 PU					
CBL-MTS-11 N	MCC-11B	BUS-MTS-11 N	1	480	30.0 FEET	2	Copper
	Duct Material: Magnetic	Insulation Type: PVC	Insulation Class: THWN				
	+/- Impedance: 0.2020 + J 0.0585 Ohms/1000 ft	2.63 + J 0.7617 PU					
	Z0 Impedance: 0.6366 + J 0.1440 Ohms/1000 ft	8.29 + J 1.88 PU					
CBL-MTS-13 E	MCC-8B	BUS-MTS-13 E	1	480	125.0 FEET	500	Copper
	Duct Material: Magnetic	Insulation Type: PVC	Insulation Class: THWN				
	+/- Impedance: 0.0294 + J 0.0466 Ohms/1000 ft	1.60 + J 2.53 PU					
	Z0 Impedance: 0.0926 + J 0.1147 Ohms/1000 ft	5.02 + J 6.22 PU					
CBL-MTS-13 N	MCC-8A	BUS-MTS-13 N	1	480	125.0 FEET	500	Copper
	Duct Material: Magnetic	Insulation Type: PVC	Insulation Class: THWN				
	+/- Impedance: 0.0294 + J 0.0466 Ohms/1000 ft	1.60 + J 2.53 PU					
	Z0 Impedance: 0.0926 + J 0.1147 Ohms/1000 ft	5.02 + J 6.22 PU					
CBL-MTS-5 E	DP-5A	BUS-MTS-5 E	1	480	15.0 FEET	4/0	Copper
	Duct Material: Magnetic	Insulation Type: PVC	Insulation Class: THWN				
	+/- Impedance: 0.0640 + J 0.0497 Ohms/1000 ft	0.4167 + J 0.3236 PU					
	Z0 Impedance: 0.2017 + J 0.1224 Ohms/1000 ft	1.31 + J 0.7969 PU					

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FEEDER INPUT DATA

CABLE NAME	FEEDER FROM NAME	FEEDER TO NAME	QTY /PH	VOLTS L-L	LENGTH	FEEDER SIZE	FEEDER TYPE
CBL-MTS-5 N	DP-5B	BUS-MTS-5 N	1	480	25.0 FEET	4/0	Copper
	Duct Material: Magnetic	Insulation Type: PVC	Insulation Class: THWN				
	+/- Impedance: 0.0640 + J 0.0497	Ohms/1000 ft 0.6944 + J 0.5393	PU				
	Z0 Impedance: 0.2017 + J 0.1224	Ohms/1000 ft 2.19 + J 1.33	PU				
CBL-MTS-6 E	MCC-6A	BUS-MTS-6 E	1	480	50.0 FEET	250	Copper
	Duct Material: Magnetic	Insulation Type: PVC	Insulation Class: THWN				
	+/- Impedance: 0.0552 + J 0.0495	Ohms/1000 ft 1.20 + J 1.07	PU				
	Z0 Impedance: 0.1739 + J 0.1219	Ohms/1000 ft 3.77 + J 2.65	PU				
CBL-MTS-6 N	MCC-6B	BUS-MTS-6 N	1	480	25.0 FEET	250	Copper
	Duct Material: Magnetic	Insulation Type: PVC	Insulation Class: THWN				
	+/- Impedance: 0.0552 + J 0.0495	Ohms/1000 ft 0.5990 + J 0.5371	PU				
	Z0 Impedance: 0.1739 + J 0.1219	Ohms/1000 ft 1.89 + J 1.32	PU				
CBL-MTS-8 E	DP-8A-1	BUS-MTS-8 E	1	480	10.0 FEET	4/0	Copper
	Duct Material: Magnetic	Insulation Type: PVC	Insulation Class: THWN				
	+/- Impedance: 0.0640 + J 0.0497	Ohms/1000 ft 0.2778 + J 0.2157	PU				
	Z0 Impedance: 0.2017 + J 0.1224	Ohms/1000 ft 0.8754 + J 0.5313	PU				
CBL-MTS-8 N	DP-8B-1	BUS-MTS-8 N	1	480	10.0 FEET	4/0	Copper
	Duct Material: Magnetic	Insulation Type: PVC	Insulation Class: THWN				
	+/- Impedance: 0.0640 + J 0.0497	Ohms/1000 ft 0.2778 + J 0.2157	PU				
	Z0 Impedance: 0.2017 + J 0.1224	Ohms/1000 ft 0.8754 + J 0.5313	PU				
CBL-MTS-9 E	MCC-9A	BUS-MTS-9 E	1	480	40.0 FEET	4/0	Copper
	Duct Material: Magnetic	Insulation Type: PVC	Insulation Class: THWN				
	+/- Impedance: 0.0640 + J 0.0497	Ohms/1000 ft 1.11 + J 0.8628	PU				
	Z0 Impedance: 0.2017 + J 0.1224	Ohms/1000 ft 3.50 + J 2.13	PU				
CBL-MTS-9 N	MCC-9B	BUS-MTS-9 N	1	480	40.0 FEET	4/0	Copper
	Duct Material: Magnetic	Insulation Type: PVC	Insulation Class: THWN				
	+/- Impedance: 0.0640 + J 0.0497	Ohms/1000 ft 1.11 + J 0.8628	PU				
	Z0 Impedance: 0.2017 + J 0.1224	Ohms/1000 ft 3.50 + J 2.13	PU				
CBL-SHB CRANE	DSC SHB CRANE	SHB CRANE	1	480	9.0 FEET	8	Copper
	Duct Material: Magnetic	Insulation Type: PVC	Insulation Class: THWN				
	+/- Impedance: 0.8110 + J 0.0754	Ohms/1000 ft 3.17 + J 0.2945	PU				
	Z0 Impedance: 2.56 + J 0.1856	Ohms/1000 ft 9.98 + J 0.7250	PU				
CBL-SWBD-MAIN-A	BUS-Sec-HECO PRIMARY TX	SWBD-MAIN-A	8	480	95.0 FEET	500	Copper
	Duct Material: Magnetic	Insulation Type: PVC	Insulation Class: THWN				
	+/- Impedance: 0.0294 + J 0.0466	Ohms/1000 ft 0.1515 + J 0.2402	PU				
	Z0 Impedance: 0.0926 + J 0.1147	Ohms/1000 ft 0.4773 + J 0.5912	PU				

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FEEDER INPUT DATA

CABLE NAME	FEEDER FROM NAME	FEEDER TO NAME	QTY /PH	VOLTS L-L	LENGTH	FEEDER SIZE	FEEDER TYPE
CBL-SWBD-MAIN-B	BUS-Sec-HILO SECONDARY TX	SWBD-MAIN-B	8	480	85.0 FEET	500	Copper
	Duct Material: Magnetic Insulation Type: PVC Insulation Class: THWN						
	+/- Impedance: 0.0294 + J 0.0466 Ohms/1000 ft 0.1356 + J 0.2149 PU						
	Z0 Impedance: 0.0926 + J 0.1147 Ohms/1000 ft 0.4270 + J 0.5289 PU						
CBL-XFMR-10	DP-10	BUS-Pri-XFMR-10	1	480	10.0 FEET	2	Copper
	Duct Material: Magnetic Insulation Type: PVC Insulation Class: THWN						
	+/- Impedance: 0.2020 + J 0.0585 Ohms/1000 ft 0.8767 + J 0.2539 PU						
	Z0 Impedance: 0.6366 + J 0.1440 Ohms/1000 ft 2.76 + J 0.6250 PU						
CBL-XFMR-11	MTS-11	BUS-Pri-XFMR-11	1	480	20.0 FEET	2	Copper
	Duct Material: Magnetic Insulation Type: PVC Insulation Class: THWN						
	+/- Impedance: 0.2020 + J 0.0585 Ohms/1000 ft 1.75 + J 0.5078 PU						
	Z0 Impedance: 0.6366 + J 0.1440 Ohms/1000 ft 5.53 + J 1.25 PU						
CBL-XFMR-13	DP-13	BUS-Pri-XFMR-13	1	480	10.0 FEET	4/0	Copper
	Duct Material: Magnetic Insulation Type: PVC Insulation Class: THWN						
	+/- Impedance: 0.0640 + J 0.0497 Ohms/1000 ft 0.2778 + J 0.2157 PU						
	Z0 Impedance: 0.2017 + J 0.1224 Ohms/1000 ft 0.8754 + J 0.5313 PU						
CBL-XFMR-4PA	DP-10	BUS-Pri-XFMR-4PA	1	480	35.0 FEET	10	Copper
	Duct Material: Magnetic Insulation Type: PVC Insulation Class: THWN						
	+/- Impedance: 1.18 + J 0.0854 Ohms/1000 ft 17.93 + J 1.30 PU						
	Z0 Impedance: 3.72 + J 0.2103 Ohms/1000 ft 56.49 + J 3.19 PU						
CBL-XFMR-5	MTS-5	BUS-Pri-XFMR-5	1	480	10.0 FEET	4/0	Copper
	Duct Material: Magnetic Insulation Type: PVC Insulation Class: THWN						
	+/- Impedance: 0.0640 + J 0.0497 Ohms/1000 ft 0.2778 + J 0.2157 PU						
	Z0 Impedance: 0.2017 + J 0.1224 Ohms/1000 ft 0.8754 + J 0.5313 PU						
CBL-XFMR-6	DP-6	BUS-Pri-XFMR-6	1	480	10.0 FEET	2	Copper
	Duct Material: Magnetic Insulation Type: PVC Insulation Class: THWN						
	+/- Impedance: 0.2020 + J 0.0585 Ohms/1000 ft 0.8767 + J 0.2539 PU						
	Z0 Impedance: 0.6366 + J 0.1440 Ohms/1000 ft 2.76 + J 0.6250 PU						
CBL-XFMR-8	MTS-8	BUS-Pri-XFMR-8	1	480	10.0 FEET	4/0	Copper
	Duct Material: Magnetic Insulation Type: PVC Insulation Class: THWN						
	+/- Impedance: 0.0640 + J 0.0497 Ohms/1000 ft 0.2778 + J 0.2157 PU						
	Z0 Impedance: 0.2017 + J 0.1224 Ohms/1000 ft 0.8754 + J 0.5313 PU						
CBL-XFMR-9	DP-9	BUS-Pri-XFMR-9	1	480	10.0 FEET	1	Copper
	Duct Material: Magnetic Insulation Type: PVC Insulation Class: THWN						
	+/- Impedance: 0.1600 + J 0.0570 Ohms/1000 ft 0.6944 + J 0.2474 PU						
	Z0 Impedance: 0.5042 + J 0.1403 Ohms/1000 ft 2.19 + J 0.6089 PU						

Pos. Seq. Z: 0.0100+J 0.0100 PU ; Zero Seq. Z: 0.0100+J 0.0100 PU
From Shunt Y: 0.00000+J 0.00000 PU ; To Shunt Y: 0.00000+J 0.00000 PU

PI-0028

MCC-11B 09-VFD-1200 480.00
Pos. Seq. Z: 0.0100+J 0.0100 PU ; Zero Seq. Z: 0.0100+J 0.0100 PU
From Shunt Y: 0.00000+J 0.00000 PU ; To Shunt Y: 0.00000+J 0.00000 PU

EQUIVALENT PI DATA

PI NAME	FROM NAME	TO NAME	VOLTS
PI-0004	MCC-6A	11-VFD-2312	480.00
	Pos. Seq. Z: 0.0100+J 0.0100 PU ;	Zero Seq. Z: 0.0100+J 0.0100 PU	
	From Shunt Y: 0.00000+J 0.00000 PU ;	To Shunt Y: 0.00000+J 0.00000 PU	
PI-0005	MCC-6A	11-VFD-2410	480.00
	Pos. Seq. Z: 0.0100+J 0.0100 PU ;	Zero Seq. Z: 0.0100+J 0.0100 PU	
	From Shunt Y: 0.00000+J 0.00000 PU ;	To Shunt Y: 0.00000+J 0.00000 PU	
PI-0007	MCC-6B	11-VFD-2420	480.00
	Pos. Seq. Z: 0.0100+J 0.0100 PU ;	Zero Seq. Z: 0.0100+J 0.0100 PU	
	From Shunt Y: 0.00000+J 0.00000 PU ;	To Shunt Y: 0.00000+J 0.00000 PU	
PI-0008	MCC-6B	11-VFD-2321	480.00
	Pos. Seq. Z: 0.0100+J 0.0100 PU ;	Zero Seq. Z: 0.0100+J 0.0100 PU	
	From Shunt Y: 0.00000+J 0.00000 PU ;	To Shunt Y: 0.00000+J 0.00000 PU	
PI-0009	MCC-8A	12-VFD-3320	480.00
	Pos. Seq. Z: 0.0100+J 0.0100 PU ;	Zero Seq. Z: 0.0100+J 0.0100 PU	
	From Shunt Y: 0.00000+J 0.00000 PU ;	To Shunt Y: 0.00000+J 0.00000 PU	
PI-0010	MCC-8A	12-VFD-3520	480.00
	Pos. Seq. Z: 0.0100+J 0.0100 PU ;	Zero Seq. Z: 0.0100+J 0.0100 PU	
	From Shunt Y: 0.00000+J 0.00000 PU ;	To Shunt Y: 0.00000+J 0.00000 PU	
PI-0012	MCC-8B	12-VFD-3420	480.00
	Pos. Seq. Z: 0.0100+J 0.0100 PU ;	Zero Seq. Z: 0.0100+J 0.0100 PU	
	From Shunt Y: 0.00000+J 0.00000 PU ;	To Shunt Y: 0.00000+J 0.00000 PU	
PI-0014	MCC-10A	06-VFD-1610	480.00
	Pos. Seq. Z: 0.0100+J 0.0100 PU ;	Zero Seq. Z: 0.0100+J 0.0100 PU	
	From Shunt Y: 0.00000+J 0.00000 PU ;	To Shunt Y: 0.00000+J 0.00000 PU	
PI-0015	MCC-10A	05-VFD-1630	480.00
	Pos. Seq. Z: 0.0100+J 0.0100 PU ;	Zero Seq. Z: 0.0100+J 0.0100 PU	
	From Shunt Y: 0.00000+J 0.00000 PU ;	To Shunt Y: 0.00000+J 0.00000 PU	
PI-0016	MCC-10A	06-VFD-1630	480.00
	Pos. Seq. Z: 0.0100+J 0.0100 PU ;	Zero Seq. Z: 0.0100+J 0.0100 PU	
	From Shunt Y: 0.00000+J 0.00000 PU ;	To Shunt Y: 0.00000+J 0.00000 PU	
PI-0017	MCC-10A	05-VFD-1510	480.00
	Pos. Seq. Z: 0.0100+J 0.0100 PU ;	Zero Seq. Z: 0.0100+J 0.0100 PU	
	From Shunt Y: 0.00000+J 0.00000 PU ;	To Shunt Y: 0.00000+J 0.00000 PU	
PI-0018	MCC-10A	22-VFD-2201	480.00
	Pos. Seq. Z: 0.0100+J 0.0100 PU ;	Zero Seq. Z: 0.0100+J 0.0100 PU	
	From Shunt Y: 0.00000+J 0.00000 PU ;	To Shunt Y: 0.00000+J 0.00000 PU	
PI-0019	MCC-10A	22-VFD-2203	480.00
	Pos. Seq. Z: 0.0100+J 0.0100 PU ;	Zero Seq. Z: 0.0100+J 0.0100 PU	
	From Shunt Y: 0.00000+J 0.00000 PU ;	To Shunt Y: 0.00000+J 0.00000 PU	
PI-0021	MCC-10B	06-VFD-1615	480.00
	Pos. Seq. Z: 0.0100+J 0.0100 PU ;	Zero Seq. Z: 0.0100+J 0.0100 PU	
	From Shunt Y: 0.00000+J 0.00000 PU ;	To Shunt Y: 0.00000+J 0.00000 PU	
PI-0022	MCC-10B	06-VFD-1625	480.00
	Pos. Seq. Z: 0.0100+J 0.0100 PU ;	Zero Seq. Z: 0.0100+J 0.0100 PU	
	From Shunt Y: 0.00000+J 0.00000 PU ;	To Shunt Y: 0.00000+J 0.00000 PU	
PI-0023	MCC-10B	06-VFD-1635	480.00
	Pos. Seq. Z: 0.0100+J 0.0100 PU ;	Zero Seq. Z: 0.0100+J 0.0100 PU	
	From Shunt Y: 0.00000+J 0.00000 PU ;	To Shunt Y: 0.00000+J 0.00000 PU	
PI-0024	MCC-10B	05-VFD-1520	480.00
	Pos. Seq. Z: 0.0100+J 0.0100 PU ;	Zero Seq. Z: 0.0100+J 0.0100 PU	
	From Shunt Y: 0.00000+J 0.00000 PU ;	To Shunt Y: 0.00000+J 0.00000 PU	
PI-0025	MCC-10B	22-VFD-2022	480.00
	Pos. Seq. Z: 0.0100+J 0.0100 PU ;	Zero Seq. Z: 0.0100+J 0.0100 PU	
	From Shunt Y: 0.00000+J 0.00000 PU ;	To Shunt Y: 0.00000+J 0.00000 PU	
PI-0026	MCC-11A	08-VFD-1100	480.00

TRANSFORMER INPUT DATA

TRANSFORMER NAME	PRIMARY RECORD NO NAME	VOLTS L-L	* SECONDARY RECORD NO NAME	VOLTS L-L	FULL-LOAD KVA	NOMINAL KVA
HECO PRIMARY TX	BUS-Pri-HECO PRIMARY TX	D 13800.0	BUS-Sec-HECO PRIMARY TX	YG 480.00	2000.00	2000.00
	Pos. Seq. Z%: 0.722 + J 5.27 (Zpu 0.361 + j 2.64)		Shell Type			
	Zero Seq. Z%: 0.722 + J 5.27 (Sec 0.361 + j 2.64 Pri Open)					
	Taps Pri. 0.000 % Sec. 0.000 % Phase Shift (Pri. Leading Sec.): 30.00 Deg.					
HILO SECONDARY TX	BUS-Pri-HILO SECONDARY TX	D 13800.0	BUS-Sec-HILO SECONDARY TX	YG 480.00	2000.00	2000.00
	Pos. Seq. Z%: 0.722 + J 5.27 (Zpu 0.361 + j 2.64)		Shell Type			
	Zero Seq. Z%: 0.722 + J 5.27 (Sec 0.361 + j 2.64 Pri Open)					
	Taps Pri. 0.000 % Sec. 0.000 % Phase Shift (Pri. Leading Sec.): 30.00 Deg.					
XFMR-10	BUS-Pri-XFMR-10	D 480.00	BUS-Sec-XFMR-10	YG 208.00	30.00	30.00
	Pos. Seq. Z%: 3.22 + J 2.93 (Zpu 107.4 + j 97.82)		Shell Type			
	Zero Seq. Z%: 3.22 + J 2.93 (Sec 107.4 + j 97.82 Pri Open)					
	Taps Pri. 0.000 % Sec. 0.000 % Phase Shift (Pri. Leading Sec.): 30.00 Deg.					
XFMR-11	BUS-Pri-XFMR-11	D 480.00	BUS-Sec-XFMR-11	YG 208.00	30.00	30.00
	Pos. Seq. Z%: 3.22 + J 2.93 (Zpu 107.4 + j 97.82)		Shell Type			
	Zero Seq. Z%: 3.22 + J 2.93 (Sec 107.4 + j 97.82 Pri Open)					
	Taps Pri. 0.000 % Sec. 0.000 % Phase Shift (Pri. Leading Sec.): 30.00 Deg.					
XFMR-13	BUS-Pri-XFMR-13	D 480.00	BUS-Sec-XFMR-13	YG 208.00	75.00	75.00
	Pos. Seq. Z%: 2.37 + J 3.53 (Zpu 31.58 + j 47.05)		Shell Type			
	Zero Seq. Z%: 2.37 + J 3.53 (Sec 31.58 + j 47.05 Pri Open)					
	Taps Pri. 0.000 % Sec. 0.000 % Phase Shift (Pri. Leading Sec.): 30.00 Deg.					
XFMR-4PA	BUS-Pri-XFMR-4PA	D 480.00	BUS-Sec-XFMR-4PA	YG 240.00	5.00	5.00
	Pos. Seq. Z%: 1.48 + J 3.17 (Zpu 295.1 + j 634.7)		Shell Type			
	Zero Seq. Z%: 1.48 + J 3.17 (Sec 295.1 + j 634.7 Pri Open)					
	Taps Pri. 0.000 % Sec. 0.000 % Phase Shift (Pri. Leading Sec.): 30.00 Deg.					

TRANSFORMER INPUT DATA

TRANSFORMER NAME	PRIMARY RECORD NO NAME	VOLTS L-L	* SECONDARY RECORD NO NAME	VOLTS L-L	FULL-LOAD KVA	NOMINAL KVA
XFMR-5	BUS-Pri-XFMR-5	D 480.00	BUS-Sec-XFMR-5	YG 208.00	75.00	75.00
	Pos. Seq. Z%: 2.37 + J 3.53 (Zpu 31.58 + j 47.05)		Shell Type			
	Zero Seq. Z%: 2.37 + J 3.53 (Sec 31.58 + j 47.05 Pri Open)					
	Taps Pri. 0.000 % Sec. 0.000 % Phase Shift (Pri. Leading Sec.): 30.00 Deg.					
XFMR-6	BUS-Pri-XFMR-6	D 480.00	BUS-Sec-XFMR-6	YG 208.00	30.00	30.00
	Pos. Seq. Z%: 3.22 + J 2.93 (Zpu 107.4 + j 97.82)		Shell Type			
	Zero Seq. Z%: 3.22 + J 2.93 (Sec 107.4 + j 97.82 Pri Open)					
	Taps Pri. 0.000 % Sec. 0.000 % Phase Shift (Pri. Leading Sec.): 30.00 Deg.					
XFMR-8	BUS-Pri-XFMR-8	D 480.00	BUS-Sec-XFMR-8	YG 208.00	75.00	75.00
	Pos. Seq. Z%: 2.37 + J 3.53 (Zpu 31.58 + j 47.05)		Shell Type			
	Zero Seq. Z%: 2.37 + J 3.53 (Sec 31.58 + j 47.05 Pri Open)					
	Taps Pri. 0.000 % Sec. 0.000 % Phase Shift (Pri. Leading Sec.): 30.00 Deg.					
XFMR-9	BUS-Pri-XFMR-9	D 480.00	BUS-Sec-XFMR-9	YG 208.00	45.00	45.00
	Pos. Seq. Z%: 2.62 + J 2.18 (Zpu 58.31 + j 48.40)		Shell Type			
	Zero Seq. Z%: 2.62 + J 2.18 (Sec 58.31 + j 48.40 Pri Open)					
	Taps Pri. 0.000 % Sec. 0.000 % Phase Shift (Pri. Leading Sec.): 30.00 Deg.					

GENERATION CONTRIBUTION DATA

BUS NAME	CONTRIBUTION NAME	VOLTAGE		X*d	X/R
		L-L	MVA		
21-GEN-1100 [Ex]	21-GEN-1100-[Ex]	480.00	0.750	0.1500	23.15
	Pos Sequence Impedance (100 MVA Base)	0.8639 + J	20.00 PU		
21-GEN-1200	21-GEN_1200	480.00	1.25	0.1500	20.00
	Pos Sequence Impedance (100 MVA Base)	0.6000 + J	12.00 PU		
BUS-Pri-HECO PRIMARY TX	HECO PRIMARY	13800.0	39.20		
	Three Phase Contribution:	1640.20 AMPS	2.63		
	Single Line to Ground Contribution:	286.40 AMPS	4.01		
	Pos Sequence Impedance (100 MVA Base)	0.9061 + J	2.38 PU		
	Zero Sequence Impedance (100 MVA Base)	8.80 + J	37.75 PU		
BUS-Pri-HILO SECONDARY TX	HILO SECONDARY	13800.0	32.38		
	Three Phase Contribution:	1354.70 AMPS	2.40		
	Single Line to Ground Contribution:	237.20 AMPS	3.59		
	Pos Sequence Impedance (100 MVA Base)	1.19 + J	2.85 PU		
	Zero Sequence Impedance (100 MVA Base)	11.84 + J	45.27 PU		

MOTOR CONTRIBUTION DATA

BUS NAME	CONTRIBUTION NAME			VOLTAGE L-L	BASE kVA	X*d	X/R	Motor Number
BUS-02-CON-1400	02-CON-1400			480	7.50	0.1670	4.90	1.00
	Pos Sequence Impedance (100 MVA Base)	454.51 + j	2226.67 PU					
BUS-02-EF-1320	02-EF-1320			480	2.00	0.1670	4.90	1.00
	Pos Sequence Impedance (100 MVA Base)	1704.43 + j	8350.00 PU					
BUS-02-EF-1330	02-EF-1330			480	5.00	0.1670	4.90	1.00
	Pos Sequence Impedance (100 MVA Base)	681.77 + j	3340.00 PU					
BUS-02-EF-1340	02-EF-1340			480	5.00	0.1670	4.90	1.00
	Pos Sequence Impedance (100 MVA Base)	681.77 + j	3340.00 PU					
BUS-02-EF-1370	02-EF-1370			480	2.00	0.1670	4.90	1.00
	Pos Sequence Impedance (100 MVA Base)	1704.43 + j	8350.00 PU					
BUS-02-EF-1380	02-EF-1380			480	2.00	0.1670	4.90	1.00
	Pos Sequence Impedance (100 MVA Base)	1704.43 + j	8350.00 PU					
BUS-02-EF-1390	02-EF-1390			480	7.50	0.1670	4.90	1.00
	Pos Sequence Impedance (100 MVA Base)	454.51 + j	2226.67 PU					
BUS-02-EF-1396	02-EF-1396			480	4.50	0.1670	4.90	1.00
	Pos Sequence Impedance (100 MVA Base)	757.52 + j	3711.11 PU					
BUS-02-MIX-1210	02-MIX-1210			480	2.00	0.1670	4.90	1.00
	Pos Sequence Impedance (100 MVA Base)	1704.43 + j	8350.00 PU					
BUS-02-MIX-1220	02-MIX-1220			480	2.00	0.1670	4.90	1.00
	Pos Sequence Impedance (100 MVA Base)	1704.43 + j	8350.00 PU					

MOTOR CONTRIBUTION DATA

BUS NAME	CONTRIBUTION NAME			VOLTAGE L-L	BASE kVA	X*d	X/R	Motor Number
BUS-02-PMP-1211	02-PMP-1211			480	15.00	0.1670	4.90	1.00
	Pos	Sequence Impedance (100 MVA Base)	227.26 + j 1113.33 PU					
BUS-02-PMP-1221	02-PMP-1221			480	15.00	0.1670	4.90	1.00
	Pos	Sequence Impedance (100 MVA Base)	227.26 + j 1113.33 PU					
BUS-02-SF-1310	02-SF-1310			480	2.00	0.1670	4.90	1.00
	Pos	Sequence Impedance (100 MVA Base)	1704.43 + j 8350.00 PU					
BUS-02-SF-1350	02-SF-1350			480	7.50	0.1670	4.90	1.00
	Pos	Sequence Impedance (100 MVA Base)	454.51 + j 2226.67 PU					
BUS-02-SF-1360	02-SF-1360			480	7.50	0.1670	4.90	1.00
	Pos	Sequence Impedance (100 MVA Base)	454.51 + j 2226.67 PU					
BUS-02-SF-1570	02-SF-1570			480	0.5000	0.1670	4.90	1.00
	Pos	Sequence Impedance (100 MVA Base)	6817.72 + j 33400.0 PU					
BUS-02-SF-1575	02-SF-1575			480	0.5000	0.1670	4.90	1.00
	Pos	Sequence Impedance (100 MVA Base)	6817.72 + j 33400.0 PU					
BUS-02-SF-1810	02-SF-1810			480	0.5000	0.1670	4.90	1.00
	Pos	Sequence Impedance (100 MVA Base)	6817.72 + j 33400.0 PU					
BUS-03-EF-1420	03-EF-1420			480	2.00	0.1670	4.90	1.00
	Pos	Sequence Impedance (100 MVA Base)	1704.43 + j 8350.00 PU					
BUS-03-HSS-1140	03-HSS-1140			480	0.5000	0.1670	4.90	1.00
	Pos	Sequence Impedance (100 MVA Base)	6817.72 + j 33400.0 PU					

MOTOR CONTRIBUTION DATA

BUS NAME	CONTRIBUTION NAME			VOLTAGE L-L	BASE kVA	X*d	X/R	Motor Number
BUS-03-HSS-1240	Pos	Sequence Impedance (100 MVA Base)	03-HSS-1240	6817.72 + j	480 0.5000 33400.0 PU	0.1670	4.90	1.00
BUS-03-HSS-1340	Pos	Sequence Impedance (100 MVA Base)	03-HSS-1340	6817.72 + j	480 0.5000 33400.0 PU	0.1670	4.90	1.00
BUS-03-PMP-1510	Pos	Sequence Impedance (100 MVA Base)	03-PMP-1510	454.51 + j	480 7.50 2226.67 PU	0.1670	4.90	1.00
BUS-03-PMP-1520	Pos	Sequence Impedance (100 MVA Base)	03-PMP-1520	340.89 + j	480 10.00 1670.00 PU	0.1670	4.90	1.00
BUS-03-PMP-1530	Pos	Sequence Impedance (100 MVA Base)	03-PMP-1530	454.51 + j	480 7.50 2226.67 PU	0.1670	4.90	1.00
BUS-03-PMP-1610	Pos	Sequence Impedance (100 MVA Base)	03-PMP-1610	454.51 + j	480 7.50 2226.67 PU	0.1670	4.90	1.00
BUS-03-PMP-1620	Pos	Sequence Impedance (100 MVA Base)	03-PMP-1620	454.51 + j	480 7.50 2226.67 PU	0.1670	4.90	1.00
BUS-03-SCC-1150	Pos	Sequence Impedance (100 MVA Base)	03-SCC-1150	4545.15 + j	480 0.7500 22266.6 PU	0.1670	4.90	1.00
BUS-03-SCC-1250	Pos	Sequence Impedance (100 MVA Base)	03-SCC-1250	4545.15 + j	480 0.7500 22266.6 PU	0.1670	4.90	1.00
BUS-03-SCC-1350	Pos	Sequence Impedance (100 MVA Base)	03-SCC-1350	6817.72 + j	480 0.5000 33400.0 PU	0.1670	4.90	1.00

MOTOR CONTRIBUTION DATA

BUS NAME	CONTRIBUTION NAME			VOLTAGE L-L	BASE kVA	X*d	X/R	Motor Number
BUS-03-SF-1410	Pos	Sequence	Impedance (100 MVA Base)	2272.57 + j 11133.3 PU	480 1.50	0.1670	4.90	1.00
BUS-03-SF-1575	Pos	Sequence	Impedance (100 MVA Base)	6817.72 + j 33400.0 PU	480 0.5000	0.1670	4.90	1.00
BUS-03-SLC-1120	Pos	Sequence	Impedance (100 MVA Base)	4545.15 + j 22266.6 PU	480 0.7500	0.1670	4.90	1.00
BUS-03-SLC-1220	Pos	Sequence	Impedance (100 MVA Base)	4545.15 + j 22266.6 PU	480 0.7500	0.1670	4.90	1.00
BUS-03-SLC-1320	Pos	Sequence	Impedance (100 MVA Base)	4545.15 + j 22266.6 PU	480 0.7500	0.1670	4.90	1.00
BUS-04-PMP-1140	Pos	Sequence	Impedance (100 MVA Base)	85.22 + j 417.50 PU	480 40.00	0.1670	4.90	1.00
BUS-05-PMP-1510	Pos	Sequence	Impedance (100 MVA Base)	4.84 + j 23.69 PU	480 705.00	0.1670	4.90	1.00
BUS-05-PMP-1520	Pos	Sequence	Impedance (100 MVA Base)	454.51 + j 2226.67 PU	480 7.50	0.1670	4.90	1.00
BUS-05-PMP-1630	Pos	Sequence	Impedance (100 MVA Base)	227.26 + j 1113.33 PU	480 15.00	0.1670	4.90	1.00
BUS-06-CLA-1510	Pos	Sequence	Impedance (100 MVA Base)	3408.86 + j 16700.0 PU	480 1.0000	0.1670	4.90	1.00

MOTOR CONTRIBUTION DATA

BUS NAME	CONTRIBUTION NAME			VOLTAGE L-L	BASE kVA	X*d	X/R	Motor Number
BUS-06-CLA-1520	Pos	Sequence	Impedance (100 MVA Base)	480 1.0000	3408.86 + j 16700.0 PU	0.1670	4.90	1.00
BUS-06-CLA-1530	Pos	Sequence	Impedance (100 MVA Base)	480 1.0000	3408.86 + j 16700.0 PU	0.1670	4.90	1.00
BUS-06-PMP-1515	Pos	Sequence	Impedance (100 MVA Base)	480 10.00	340.89 + j 1670.00 PU	0.1670	4.90	1.00
BUS-06-PMP-1525	Pos	Sequence	Impedance (100 MVA Base)	480 10.00	340.89 + j 1670.00 PU	0.1670	4.90	1.00
BUS-06-PMP-1535	Pos	Sequence	Impedance (100 MVA Base)	480 10.00	340.89 + j 1670.00 PU	0.1670	4.90	1.00
BUS-06-PMP-1610	Pos	Sequence	Impedance (100 MVA Base)	480 15.00	227.26 + j 1113.33 PU	0.1670	4.90	1.00
BUS-06-PMP-1615	Pos	Sequence	Impedance (100 MVA Base)	480 15.00	227.26 + j 1113.33 PU	0.1670	4.90	1.00
BUS-06-PMP-1625	Pos	Sequence	Impedance (100 MVA Base)	480 15.00	227.26 + j 1113.33 PU	0.1670	4.90	1.00
BUS-06-PMP-1630	Pos	Sequence	Impedance (100 MVA Base)	480 15.00	227.26 + j 1113.33 PU	0.1670	4.90	1.00
BUS-06-PMP-1635	Pos	Sequence	Impedance (100 MVA Base)	480 15.00	227.26 + j 1113.33 PU	0.1670	4.90	1.00

MOTOR CONTRIBUTION DATA

BUS NAME	CONTRIBUTION NAME		VOLTAGE L-L	BASE kVA	X*d	X/R	Motor Number
BUS-07-FAN-1110	07-FAN-1110		480	60.00	0.1670	4.90	1.00
	Pos Sequence Impedance (100 MVA Base)	56.81 + j	278.33	PU			
BUS-07-FAN-1120	07-FAN-1120		480	60.00	0.1670	4.90	1.00
	Pos Sequence Impedance (100 MVA Base)	56.81 + j	278.33	PU			
BUS-08-EF-1230	08-EF-1230		480	0.2500	0.1670	4.90	1.00
	Pos Sequence Impedance (100 MVA Base)	13635.4 + j	66800.0	PU			
BUS-09-MIX-1110	09-MIX-1110		480	1.0000	0.1670	4.90	1.00
	Pos Sequence Impedance (100 MVA Base)	3408.86 + j	16700.0	PU			
BUS-09-PMP-1100	09-PMP-1100		480	75.00	0.1670	4.90	1.00
	Pos Sequence Impedance (100 MVA Base)	45.45 + j	222.67	PU			
BUS-09-PMP-1200	09-PMP-1200		480	75.00	0.1670	4.90	1.00
	Pos Sequence Impedance (100 MVA Base)	45.45 + j	222.67	PU			
BUS-09-PMP-903	09-PMP-903		480	1.0000	0.1670	4.90	1.00
	Pos Sequence Impedance (100 MVA Base)	3408.86 + j	16700.0	PU			
BUS-11-EF-1530	11-EF-1530		480	1.50	0.1670	4.90	1.00
	Pos Sequence Impedance (100 MVA Base)	2272.57 + j	11133.3	PU			
BUS-11-EF-1540	11-EF-1540		480	1.50	0.1670	4.90	1.00
	Pos Sequence Impedance (100 MVA Base)	2272.57 + j	11133.3	PU			
BUS-11-EF-1560	11-EF-1560		480	0.5000	0.1670	4.90	1.00
	Pos Sequence Impedance (100 MVA Base)	6817.72 + j	33400.0	PU			

MOTOR CONTRIBUTION DATA

BUS NAME	CONTRIBUTION NAME			VOLTAGE L-L	BASE kVA	X*d	X/R	Motor Number
BUS-11-PMP-2311	11-PMP-2311			480	5.00	0.1670	4.90	1.00
	Pos	Sequence Impedance (100 MVA Base)	681.77 + j	3340.00	PU			
BUS-11-PMP-2312	11-PMP-2312			480	50.00	0.1670	4.90	1.00
	Pos	Sequence Impedance (100 MVA Base)	68.18 + j	334.00	PU			
BUS-11-PMP-2314	11-PMP-2314			480	1.0000	0.1670	4.90	1.00
	Pos	Sequence Impedance (100 MVA Base)	3408.86 + j	16700.0	PU			
BUS-11-PMP-2321	11-PMP-2321			480	50.00	0.1670	4.90	1.00
	Pos	Sequence Impedance (100 MVA Base)	68.18 + j	334.00	PU			
BUS-11-PMP-2322	11-PMP-2322			480	5.00	0.1670	4.90	1.00
	Pos	Sequence Impedance (100 MVA Base)	681.77 + j	3340.00	PU			
BUS-11-PMP-2324	11-PMP-2324			480	1.0000	0.1670	4.90	1.00
	Pos	Sequence Impedance (100 MVA Base)	3408.86 + j	16700.0	PU			
BUS-11-PMP-2410	11-PMP-2410			480	20.00	0.1670	4.90	1.00
	Pos	Sequence Impedance (100 MVA Base)	170.44 + j	835.00	PU			
BUS-11-PMP-2420	11-PMP-2420			480	20.00	0.1670	4.90	1.00
	Pos	Sequence Impedance (100 MVA Base)	170.44 + j	835.00	PU			
BUS-11-PMP-2610	11-PMP-2610			480	1.0000	0.1670	4.90	1.00
	Pos	Sequence Impedance (100 MVA Base)	3408.86 + j	16700.0	PU			
BUS-11-PMP-2620	11-PMP-2620			480	1.0000	0.1670	4.90	1.00
	Pos	Sequence Impedance (100 MVA Base)	3408.86 + j	16700.0	PU			

MOTOR CONTRIBUTION DATA

BUS NAME	CONTRIBUTION NAME			VOLTAGE L-L	BASE kVA	X*d	X/R	Motor Number
BUS-11-PMP-2711	11-PMP-2711			480	1.50	0.1670	4.90	1.00
	Pos	Sequence Impedance (100 MVA Base)	2272.57 + j 11133.3 PU					
BUS-11-PMP-2721	11-PMP-2721			480	1.50	0.1670	4.90	1.00
	Pos	Sequence Impedance (100 MVA Base)	2272.57 + j 11133.3 PU					
BUS-11-SF-1510	11-SF-1510			480	1.50	0.1670	4.90	1.00
	Pos	Sequence Impedance (100 MVA Base)	2272.57 + j 11133.3 PU					
BUS-11-SF-1520	11-SF-1520			480	1.50	0.1670	4.90	1.00
	Pos	Sequence Impedance (100 MVA Base)	2272.57 + j 11133.3 PU					
BUS-11-SF-1550	11-SF-1550			480	1.0000	0.1670	4.90	1.00
	Pos	Sequence Impedance (100 MVA Base)	3408.86 + j 16700.0 PU					
BUS-11-SF-1575	11-SF-1575			480	0.5000	0.1670	4.90	1.00
	Pos	Sequence Impedance (100 MVA Base)	6817.72 + j 33400.0 PU					
BUS-12-AUG-4615	12-AUG-4615			480	25.00	0.1670	4.90	1.00
	Pos	Sequence Impedance (100 MVA Base)	136.35 + j 668.00 PU					
BUS-12-AUG-4617	12-AUG-4617			480	25.00	0.1670	4.90	1.00
	Pos	Sequence Impedance (100 MVA Base)	136.35 + j 668.00 PU					
BUS-12-CON-4610	12-CON-4610			480	7.50	0.1670	4.90	1.00
	Pos	Sequence Impedance (100 MVA Base)	454.51 + j 2226.67 PU					
BUS-12-CON-4620	12-CON-4620			480	7.50	0.1670	4.90	1.00
	Pos	Sequence Impedance (100 MVA Base)	454.51 + j 2226.67 PU					

MOTOR CONTRIBUTION DATA

BUS NAME	CONTRIBUTION NAME			VOLTAGE L-L	BASE kVA	X*d	X/R	Motor Number
BUS-12-EF-2530	Pos	Sequence	Impedance (100 MVA Base)	170.44 + j	835.00 PU	0.1670	4.90	1.00
BUS-12-EF-2530			12-EF-2530	480	20.00			
BUS-12-EF-2540	Pos	Sequence	Impedance (100 MVA Base)	170.44 + j	835.00 PU	0.1670	4.90	1.00
BUS-12-EF-2540			12-EF-2540	480	20.00			
BUS-12-FG-4115	Pos	Sequence	Impedance (100 MVA Base)	3399.72 + j	16655.2 PU	0.1670	4.90	1.00
BUS-12-FG-4115			12-FG-4115	480	1.00			
BUS-12-FG-4215	Pos	Sequence	Impedance (100 MVA Base)	3399.72 + j	16655.2 PU	0.1670	4.90	1.00
BUS-12-FG-4215			12-FG-4215	480	1.00			
BUS-12-FV-4117	Pos	Sequence	Impedance (100 MVA Base)	3399.72 + j	16655.2 PU	0.1670	4.90	1.00
BUS-12-FV-4117			12-FV-4117	480	1.00			
BUS-12-FV-4217	Pos	Sequence	Impedance (100 MVA Base)	3399.72 + j	16655.2 PU	0.1670	4.90	1.00
BUS-12-FV-4217			12-FV-4217	480	1.00			
BUS-12-HPU-4750	Pos	Sequence	Impedance (100 MVA Base)	136.35 + j	668.00 PU	0.1670	4.90	1.00
BUS-12-HPU-4750			12-HPU-4750	480	25.00			
BUS-12-MTR1-4110	Pos	Sequence	Impedance (100 MVA Base)	34.00 + j	166.55 PU	0.1670	4.90	1.00
BUS-12-MTR1-4110			12-MTR1-4110	480	100.27			
BUS-12-MTR1-4210	Pos	Sequence	Impedance (100 MVA Base)	34.00 + j	166.55 PU	0.1670	4.90	1.00
BUS-12-MTR1-4210			12-MTR1-4210	480	100.27			
BUS-12-MTR2-4110	Pos	Sequence	Impedance (100 MVA Base)	113.32 + j	555.17 PU	0.1670	4.90	1.00
BUS-12-MTR2-4110			12-MTR2-4110	480	30.08			

MOTOR CONTRIBUTION DATA

BUS NAME	CONTRIBUTION NAME			VOLTAGE L-L	BASE kVA	X*d	X/R	Motor Number
BUS-12-MTR2-4210	Pos	Sequence	Impedance (100 MVA Base)	113.32 + j	480 30.08 555.17 PU	0.1670	4.90	1.00
BUS-12-PMP-3320	Pos	Sequence	Impedance (100 MVA Base)	170.44 + j	480 20.00 835.00 PU	0.1670	4.90	1.00
BUS-12-PMP-3420	Pos	Sequence	Impedance (100 MVA Base)	170.44 + j	480 20.00 835.00 PU	0.1670	4.90	1.00
BUS-12-PMP-3520	Pos	Sequence	Impedance (100 MVA Base)	170.44 + j	480 20.00 835.00 PU	0.1670	4.90	1.00
BUS-12-SF-2510	Pos	Sequence	Impedance (100 MVA Base)	170.44 + j	480 20.00 835.00 PU	0.1670	4.90	1.00
BUS-12-SF-2520	Pos	Sequence	Impedance (100 MVA Base)	170.44 + j	480 20.00 835.00 PU	0.1670	4.90	1.00
BUS-12-SF-2550	Pos	Sequence	Impedance (100 MVA Base)	13635.4 + j	480 0.2500 66800.0 PU	0.1670	4.90	1.00
BUS-12-SF-2560	Pos	Sequence	Impedance (100 MVA Base)	13635.4 + j	480 0.2500 66800.0 PU	0.1670	4.90	1.00
BUS-12-SF-2570	Pos	Sequence	Impedance (100 MVA Base)	1136.29 + j	480 3.00 5566.67 PU	0.1670	4.90	1.00
BUS-12-SF-2580	Pos	Sequence	Impedance (100 MVA Base)	1136.29 + j	480 3.00 5566.67 PU	0.1670	4.90	1.00

MOTOR CONTRIBUTION DATA

BUS NAME	CONTRIBUTION NAME			VOLTAGE L-L	BASE kVA	X*d	X/R	Motor Number
BUS-12-SRW-4612	12-SRW-4612			480	3.00	0.1670	4.90	1.00
	Pos Sequence Impedance (100 MVA Base)	1136.29 + j	5566.67 PU					
BUS-12-SRW-4613	12-SRW-4613			480	3.00	0.1670	4.90	1.00
	Pos Sequence Impedance (100 MVA Base)	1136.29 + j	5566.67 PU					
BUS-19-PMP-1110	19-PMP-1110			480	20.00	0.1670	4.90	1.00
	Pos Sequence Impedance (100 MVA Base)	170.44 + j	835.00 PU					
BUS-19-PMP-1120	19-PMP-1120			480	20.00	0.1670	4.90	1.00
	Pos Sequence Impedance (100 MVA Base)	170.44 + j	835.00 PU					
BUS-21-EF-1250	21-EF-1250			480	0.5000	0.1670	4.90	1.00
	Pos Sequence Impedance (100 MVA Base)	6817.72 + j	33400.0 PU					
BUS-21-EF-1310	21-EF-1310			480	0.7500	0.1670	4.90	1.00
	Pos Sequence Impedance (100 MVA Base)	4545.15 + j	22266.6 PU					
BUS-22-BLO-2022	22-BLO-2022			480	20.00	0.1670	4.90	1.00
	Pos Sequence Impedance (100 MVA Base)	170.44 + j	835.00 PU					
BUS-22-BLO-2201	22-BLO-2201			480	20.00	0.1670	4.90	1.00
	Pos Sequence Impedance (100 MVA Base)	170.44 + j	835.00 PU					
BUS-22-BLO-2203	22-BLO-2203			480	20.00	0.1670	4.90	1.00
	Pos Sequence Impedance (100 MVA Base)	170.44 + j	835.00 PU					
BUS-2P-07	2P-07			480	10.00	0.1670	4.90	1.00
	Pos Sequence Impedance (100 MVA Base)	340.89 + j	1670.00 PU					

MOTOR CONTRIBUTION DATA

BUS NAME	CONTRIBUTION NAME			VOLTAGE L-L	BASE kVA	X*d	X/R	Motor Number
BUS-3G-01	3G-01			480	3.00	0.1670	4.90	1.00
	Pos	Sequence Impedance (100 MVA Base)	1136.29 + j	5566.67	PU			
BUS-3G-02	3G-02			480	3.00	0.1670	4.90	1.00
	Pos	Sequence Impedance (100 MVA Base)	1136.29 + j	5566.67	PU			
BUS-3G-03	3G-03			480	3.00	0.1670	4.90	1.00
	Pos	Sequence Impedance (100 MVA Base)	1136.29 + j	5566.67	PU			
BUS-3P-05	3P-05			480	1.50	0.1670	4.90	1.00
	Pos	Sequence Impedance (100 MVA Base)	2272.57 + j	11133.3	PU			
BUS-3P-07	3P-07			480	1.50	0.1670	4.90	1.00
	Pos	Sequence Impedance (100 MVA Base)	2272.57 + j	11133.3	PU			
BUS-4P-01	4P-01			480	40.00	0.1670	4.90	1.00
	Pos	Sequence Impedance (100 MVA Base)	85.22 + j	417.50	PU			
BUS-4P-02	4P-02			480	40.00	0.1670	4.90	1.00
	Pos	Sequence Impedance (100 MVA Base)	85.22 + j	417.50	PU			
BUS-4P-03	4P-03			480	40.00	0.1670	4.90	1.00
	Pos	Sequence Impedance (100 MVA Base)	85.22 + j	417.50	PU			

4.0 SHORT-CIRCUIT OUTPUT RESULTS

THREE PHASE FAULT REPORT
(FOR APPLICATION OF LOW VOLTAGE BREAKERS)
PRE FAULT VOLTAGE: 1.0000
MODEL TRANSFORMER TAPS: NO

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05-VFD-1510      3P Duty: 10.489 KA AT -56.31 DEG ( 8.72 MVA) X/R: 1.46
VOLTAGE: 480. EQUIV. IMPEDANCE= 0.0147 + J 0.0220 OHMS
LOW VOLTAGE POWER CIRCUIT BREAKER 10.489 KA
MOLDED CASE CIRCUIT BREAKER < 20KA 10.489 KA
MOLDED CASE CIRCUIT BREAKER > 20KA 10.489 KA
CBL-05-PMP-1510      BUS-05-PMP-1510      1.332 KA      ANG: -197.82
PI-0017              MCC-10A              9.483 KA      ANG: -61.33

05-VFD-1520      3P Duty: 9.089 KA AT -61.20 DEG ( 7.56 MVA) X/R: 1.90
VOLTAGE: 480. EQUIV. IMPEDANCE= 0.0147 + J 0.0267 OHMS
LOW VOLTAGE POWER CIRCUIT BREAKER 9.089 KA
MOLDED CASE CIRCUIT BREAKER < 10KA 9.309 KA
MOLDED CASE CIRCUIT BREAKER < 20KA 9.089 KA
MOLDED CASE CIRCUIT BREAKER > 20KA 9.089 KA
CBL-05-PMP-1520      BUS-05-PMP-1520      0.052 KA      ANG: 103.50
PI-0024              MCC-10B              9.038 KA      ANG: -61.11

05-VFD-1630      3P Duty: 10.486 KA AT -56.32 DEG ( 8.72 MVA) X/R: 1.46
VOLTAGE: 480. EQUIV. IMPEDANCE= 0.0147 + J 0.0220 OHMS
LOW VOLTAGE POWER CIRCUIT BREAKER 10.486 KA
MOLDED CASE CIRCUIT BREAKER < 20KA 10.486 KA
MOLDED CASE CIRCUIT BREAKER > 20KA 10.486 KA
CBL-05-PMP-1630      BUS-05-PMP-1630      0.102 KA      ANG: 107.56
PI-0015              MCC-10A              10.388 KA     ANG: -56.16

06-VFD-1610      3P Duty: 10.486 KA AT -56.32 DEG ( 8.72 MVA) X/R: 1.46
VOLTAGE: 480. EQUIV. IMPEDANCE= 0.0147 + J 0.0220 OHMS
LOW VOLTAGE POWER CIRCUIT BREAKER 10.486 KA
MOLDED CASE CIRCUIT BREAKER < 20KA 10.486 KA
MOLDED CASE CIRCUIT BREAKER > 20KA 10.486 KA
CBL-06-PMP-1610      BUS-06-PMP-1610      0.104 KA      ANG: 104.00
PI-0014              MCC-10A              10.388 KA     ANG: -56.13

06-VFD-1615      3P Duty: 9.089 KA AT -61.20 DEG ( 7.56 MVA) X/R: 1.90
VOLTAGE: 480. EQUIV. IMPEDANCE= 0.0147 + J 0.0267 OHMS
LOW VOLTAGE POWER CIRCUIT BREAKER 9.089 KA
MOLDED CASE CIRCUIT BREAKER < 10KA 9.309 KA
MOLDED CASE CIRCUIT BREAKER < 20KA 9.089 KA
MOLDED CASE CIRCUIT BREAKER > 20KA 9.089 KA
CBL-06-PMP-1615      BUS-06-PMP-1615      0.104 KA      ANG: 104.00
PI-0021              MCC-10B              8.988 KA      ANG: -61.03

06-VFD-1625      3P Duty: 9.089 KA AT -61.20 DEG ( 7.56 MVA) X/R: 1.90
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THREE PHASE FAULT REPORT
(FOR APPLICATION OF LOW VOLTAGE BREAKERS)
PRE FAULT VOLTAGE: 1.0000
MODEL TRANSFORMER TAPS: NO

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	VOLTAGE: 480.	EQUIV. IMPEDANCE=	0.0147 + J 0.0267	OHMS	
	LOW VOLTAGE POWER CIRCUIT BREAKER	9.089 KA			
	MOLDED CASE CIRCUIT BREAKER < 10KA	9.309 KA			
	MOLDED CASE CIRCUIT BREAKER < 20KA	9.089 KA			
	MOLDED CASE CIRCUIT BREAKER > 20KA	9.089 KA			
	CBL-06-PMP-1625	BUS-06-PMP-1625	0.103 KA	ANG: 105.80	
	PI-0022	MCC-10B	8.988 KA	ANG: -61.05	
06-VFD-1630	3P Duty: 10.486 KA AT -56.32 DEG (8.72 MVA) X/R: 1.46				
	VOLTAGE: 480.	EQUIV. IMPEDANCE=	0.0147 + J 0.0220	OHMS	
	LOW VOLTAGE POWER CIRCUIT BREAKER	10.486 KA			
	MOLDED CASE CIRCUIT BREAKER < 20KA	10.486 KA			
	MOLDED CASE CIRCUIT BREAKER > 20KA	10.486 KA			
	CBL-06-PMP-1630	BUS-06-PMP-1630	0.102 KA	ANG: 107.56	
	PI-0016	MCC-10A	10.388 KA	ANG: -56.16	
06-VFD-1635	3P Duty: 9.089 KA AT -61.20 DEG (7.56 MVA) X/R: 1.90				
	VOLTAGE: 480.	EQUIV. IMPEDANCE=	0.0147 + J 0.0267	OHMS	
	LOW VOLTAGE POWER CIRCUIT BREAKER	9.089 KA			
	MOLDED CASE CIRCUIT BREAKER < 10KA	9.309 KA			
	MOLDED CASE CIRCUIT BREAKER < 20KA	9.089 KA			
	MOLDED CASE CIRCUIT BREAKER > 20KA	9.089 KA			
	CBL-06-PMP-1635	BUS-06-PMP-1635	0.103 KA	ANG: 106.39	
	PI-0023	MCC-10B	8.989 KA	ANG: -61.06	
07-VFD-1110	3P Duty: 17.030 KA AT -49.46 DEG (14.16 MVA) X/R: 1.16				
	VOLTAGE: 480.	EQUIV. IMPEDANCE=	0.0106 + J 0.0124	OHMS	
	LOW VOLTAGE POWER CIRCUIT BREAKER	17.030 KA			
	MOLDED CASE CIRCUIT BREAKER < 20KA	17.030 KA			
	MOLDED CASE CIRCUIT BREAKER > 20KA	17.030 KA			
	CBL-07-FAN-1110	BUS-07-FAN-1110	0.411 KA	ANG: 104.70	
	CBL-07-VFD-1110	SWBD-MAIN-A	16.662 KA	ANG: -48.85	
07-VFD-1120	3P Duty: 16.058 KA AT -51.42 DEG (13.35 MVA) X/R: 1.28				
	VOLTAGE: 480.	EQUIV. IMPEDANCE=	0.0108 + J 0.0135	OHMS	
	LOW VOLTAGE POWER CIRCUIT BREAKER	16.058 KA			
	MOLDED CASE CIRCUIT BREAKER < 20KA	16.058 KA			
	MOLDED CASE CIRCUIT BREAKER > 20KA	16.058 KA			
	CBL-07-FAN-1120	BUS-07-FAN-1120	0.411 KA	ANG: 104.70	
	CBL-07-VFD-1120	SWBD-MAIN-B	15.684 KA	ANG: -50.81	
09-VFD-1100	3P Duty: 9.169 KA AT -52.72 DEG (7.62 MVA) X/R: 1.36				
	VOLTAGE: 480.	EQUIV. IMPEDANCE=	0.0183 + J 0.0240	OHMS	

THREE PHASE FAULT REPORT
(FOR APPLICATION OF LOW VOLTAGE BREAKERS)
PRE FAULT VOLTAGE: 1.0000
MODEL TRANSFORMER TAPS: NO

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	LOW VOLTAGE POWER CIRCUIT BREAKER	9.169 KA			
	MOLDED CASE CIRCUIT BREAKER < 10KA	9.169 KA			
	MOLDED CASE CIRCUIT BREAKER < 20KA	9.169 KA			
	MOLDED CASE CIRCUIT BREAKER > 20KA	9.169 KA			
	CBL-09-PMP-1100	BUS-09-PMP-1100	0.514 KA	ANG:	104.24
	PI-0026	MCC-11A	8.699 KA	ANG:	-51.40
09-VFD-1200	3P Duty: 4.010 KA AT -49.95 DEG (3.33 MVA) X/R: 1.33				
	VOLTAGE: 480. EQUIV. IMPEDANCE=	0.0445 + J 0.0529 OHMS			
	LOW VOLTAGE POWER CIRCUIT BREAKER	4.010 KA			
	MOLDED CASE CIRCUIT BREAKER < 10KA	4.010 KA			
	MOLDED CASE CIRCUIT BREAKER < 20KA	4.010 KA			
	MOLDED CASE CIRCUIT BREAKER > 20KA	4.010 KA			
	CBL-09-PMP-1200	BUS-09-PMP-1200	0.514 KA	ANG:	104.24
	PI-0028	MCC-11B	3.555 KA	ANG:	-46.34
11-VFD-2312	3P Duty: 9.674 KA AT -60.19 DEG (8.04 MVA) X/R: 1.79				
	VOLTAGE: 480. EQUIV. IMPEDANCE=	0.0142 + J 0.0249 OHMS			
	LOW VOLTAGE POWER CIRCUIT BREAKER	9.674 KA			
	MOLDED CASE CIRCUIT BREAKER < 10KA	9.761 KA			
	MOLDED CASE CIRCUIT BREAKER < 20KA	9.674 KA			
	MOLDED CASE CIRCUIT BREAKER > 20KA	9.674 KA			
	CBL-11-PMP-2312	BUS-11-PMP-2312	0.347 KA	ANG:	103.70
	PI-0004	MCC-6A	9.341 KA	ANG:	-59.60
11-VFD-2321	3P Duty: 9.354 KA AT -60.97 DEG (7.78 MVA) X/R: 1.87				
	VOLTAGE: 480. EQUIV. IMPEDANCE=	0.0144 + J 0.0259 OHMS			
	LOW VOLTAGE POWER CIRCUIT BREAKER	9.354 KA			
	MOLDED CASE CIRCUIT BREAKER < 10KA	9.543 KA			
	MOLDED CASE CIRCUIT BREAKER < 20KA	9.354 KA			
	MOLDED CASE CIRCUIT BREAKER > 20KA	9.354 KA			
	CBL-11-PMP-2321	BUS-11-PMP-2321	0.348 KA	ANG:	103.28
	PI-0008	MCC-6B	9.019 KA	ANG:	-60.37
11-VFD-2410	3P Duty: 9.674 KA AT -60.19 DEG (8.04 MVA) X/R: 1.79				
	VOLTAGE: 480. EQUIV. IMPEDANCE=	0.0142 + J 0.0249 OHMS			
	LOW VOLTAGE POWER CIRCUIT BREAKER	9.674 KA			
	MOLDED CASE CIRCUIT BREAKER < 10KA	9.761 KA			
	MOLDED CASE CIRCUIT BREAKER < 20KA	9.674 KA			
	MOLDED CASE CIRCUIT BREAKER > 20KA	9.674 KA			
	CBL-11-PMP-2410	BUS-11-PMP-2410	0.140 KA	ANG:	103.23
	PI-0005	MCC-6A	9.540 KA	ANG:	-59.95
11-VFD-2420	3P Duty: 9.353 KA AT -60.97 DEG (7.78 MVA) X/R: 1.87				

THREE PHASE FAULT REPORT
(FOR APPLICATION OF LOW VOLTAGE BREAKERS)
PRE FAULT VOLTAGE: 1.0000
MODEL TRANSFORMER TAPS: NO

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	VOLTAGE: 480.	EQUIV. IMPEDANCE=	0.0144 + J 0.0259	OHMS	
	LOW VOLTAGE POWER CIRCUIT BREAKER		9.353 KA		
	MOLDED CASE CIRCUIT BREAKER < 10KA		9.542 KA		
	MOLDED CASE CIRCUIT BREAKER < 20KA		9.353 KA		
	MOLDED CASE CIRCUIT BREAKER > 20KA		9.353 KA		
	CBL-11-PMP-2420	BUS-11-PMP-2420	0.140 KA	ANG:	103.00
	PI-0007	MCC-6B	9.219 KA	ANG:	-60.73
12-VCP1-4110	3P Duty: 17.234 KA AT -64.28 DEG (14.33 MVA) X/R: 2.11				
	VOLTAGE: 480.	EQUIV. IMPEDANCE=	0.0070 + J 0.0145	OHMS	
	LOW VOLTAGE POWER CIRCUIT BREAKER		17.234 KA		
	MOLDED CASE CIRCUIT BREAKER < 20KA		17.234 KA		
	MOLDED CASE CIRCUIT BREAKER > 20KA		17.234 KA		
	CBL-12-FV-4117	BUS-12-FV-4117	0.007 KA	ANG:	101.75
	CBL-12-MTR1-4110	BUS-12-MTR1-4110	0.695 KA	ANG:	102.82
	CBL-12-MTR2-4110	BUS-12-MTR2-4110	0.208 KA	ANG:	104.89
	CBL-12-FG-4115	BUS-12-FG-4115	0.007 KA	ANG:	101.75
	CBL-12-VCP1-4110	MCC-8A	16.340 KA	ANG:	-63.59
12-VCP1-4210	3P Duty: 16.102 KA AT -65.27 DEG (13.39 MVA) X/R: 2.23				
	VOLTAGE: 480.	EQUIV. IMPEDANCE=	0.0072 + J 0.0156	OHMS	
	LOW VOLTAGE POWER CIRCUIT BREAKER		16.102 KA		
	MOLDED CASE CIRCUIT BREAKER < 20KA		16.102 KA		
	MOLDED CASE CIRCUIT BREAKER > 20KA		16.102 KA		
	CBL-12-FV-4217	BUS-12-FV-4217	0.007 KA	ANG:	101.75
	CBL-12-MTR1-4210	BUS-12-MTR1-4210	0.695 KA	ANG:	102.82
	CBL-12-MTR2-4210	BUS-12-MTR2-4210	0.208 KA	ANG:	104.89
	CBL-12-FG-4215	BUS-12-FG-4215	0.007 KA	ANG:	101.75
	CBL-12-VCP1-4210	MCC-8B	15.205 KA	ANG:	-64.59
12-VFD-3320	3P Duty: 21.253 KA AT -69.62 DEG (17.67 MVA) X/R: 2.74				
	VOLTAGE: 480.	EQUIV. IMPEDANCE=	0.0045 + J 0.0122	OHMS	
	LOW VOLTAGE POWER CIRCUIT BREAKER		21.253 KA		
	MOLDED CASE CIRCUIT BREAKER > 20KA		21.253 KA		
	CBL-12-PMP-3320	BUS-12-PMP-3320	0.140 KA	ANG:	103.45
	PI-0009	MCC-8A	21.114 KA	ANG:	-69.57
12-VFD-3420	3P Duty: 19.493 KA AT -70.34 DEG (16.21 MVA) X/R: 2.84				
	VOLTAGE: 480.	EQUIV. IMPEDANCE=	0.0048 + J 0.0134	OHMS	
	LOW VOLTAGE POWER CIRCUIT BREAKER		19.493 KA		
	MOLDED CASE CIRCUIT BREAKER < 20KA		19.493 KA		
	MOLDED CASE CIRCUIT BREAKER > 20KA		19.493 KA		
	CBL-12-PMP-3420	BUS-12-PMP-3420	0.140 KA	ANG:	103.34
	PI-0012	MCC-8B	19.354 KA	ANG:	-70.30

THREE PHASE FAULT REPORT
(FOR APPLICATION OF LOW VOLTAGE BREAKERS)
PRE FAULT VOLTAGE: 1.0000
MODEL TRANSFORMER TAPS: NO

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12-VFD-3520          3P Duty: 21.253 KA AT -69.62 DEG ( 17.67 MVA) X/R: 2.74
VOLTAGE: 480. EQUIV. IMPEDANCE= 0.0045 + J 0.0122 OHMS
LOW VOLTAGE POWER CIRCUIT BREAKER 21.253 KA
MOLDED CASE CIRCUIT BREAKER > 20KA 21.253 KA
CBL-12-PMP-3520      BUS-12-PMP-3520          0.139 KA    ANG: 103.78
PI-0010              MCC-8A                    21.114 KA    ANG: -69.57

21-GEN-1100 [Ex]     3P Duty: 15.586 KA AT -86.44 DEG ( 12.96 MVA) X/R: 17.20
VOLTAGE: 480. EQUIV. IMPEDANCE= 0.0011 + J 0.0177 OHMS
LOW VOLTAGE POWER CIRCUIT BREAKER 17.626 KA
MOLDED CASE CIRCUIT BREAKER > 20KA 18.714 KA
CONTRIBUTIONS: 21-GEN-1100-[Ex]          6.008 KA    ANG: -87.53
CBL-GEN SWBD-1 [Ex]    GEN SWBD [Ex]        9.579 KA    ANG: -85.75

21-GEN-1200          3P Duty: 15.862 KA AT -86.97 DEG ( 13.19 MVA) X/R: 18.96
VOLTAGE: 480. EQUIV. IMPEDANCE= 0.0009 + J 0.0174 OHMS
LOW VOLTAGE POWER CIRCUIT BREAKER 18.078 KA
MOLDED CASE CIRCUIT BREAKER > 20KA 19.194 KA
CONTRIBUTIONS: 21-GEN_1200          10.011 KA    ANG: -87.14
CBL-GEN SWBD-2        GEN SWBD [Ex]        5.851 KA    ANG: -86.67

21-PNL-1200          3P Duty: 2.375 KA AT -11.15 DEG ( 1.97 MVA) X/R: 0.20
VOLTAGE: 480. EQUIV. IMPEDANCE= 0.1145 + J 0.0226 OHMS
LOW VOLTAGE POWER CIRCUIT BREAKER 2.375 KA
MOLDED CASE CIRCUIT BREAKER < 10KA 2.375 KA
MOLDED CASE CIRCUIT BREAKER < 20KA 2.375 KA
MOLDED CASE CIRCUIT BREAKER > 20KA 2.375 KA
CBL-21-PNL-1200      MCC-9B                    2.375 KA    ANG: -11.15

21-VFD-1100          3P Duty: 13.493 KA AT -45.53 DEG ( 11.22 MVA) X/R: 0.99
VOLTAGE: 480. EQUIV. IMPEDANCE= 0.0144 + J 0.0147 OHMS
LOW VOLTAGE POWER CIRCUIT BREAKER 13.493 KA
MOLDED CASE CIRCUIT BREAKER < 20KA 13.493 KA
MOLDED CASE CIRCUIT BREAKER > 20KA 13.493 KA
CBL-21-VFD-1100      MCC-9A                    13.493 KA    ANG: -45.53

21-VFD-1200          3P Duty: 12.878 KA AT -47.24 DEG ( 10.71 MVA) X/R: 1.08
VOLTAGE: 480. EQUIV. IMPEDANCE= 0.0146 + J 0.0158 OHMS
LOW VOLTAGE POWER CIRCUIT BREAKER 12.878 KA
MOLDED CASE CIRCUIT BREAKER < 20KA 12.878 KA
MOLDED CASE CIRCUIT BREAKER > 20KA 12.878 KA
CBL-21-VFD-1200      MCC-9B                    12.878 KA    ANG: -47.24
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THREE PHASE FAULT REPORT
(FOR APPLICATION OF LOW VOLTAGE BREAKERS)
PRE FAULT VOLTAGE: 1.0000
MODEL TRANSFORMER TAPS: NO

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21-VFD-1300      3P Duty: 13.493 KA AT -45.53 DEG ( 11.22 MVA) X/R: 0.99
VOLTAGE: 480. EQUIV. IMPEDANCE= 0.0144 + J 0.0147 OHMS
LOW VOLTAGE POWER CIRCUIT BREAKER 13.493 KA
MOLDED CASE CIRCUIT BREAKER < 20KA 13.493 KA
MOLDED CASE CIRCUIT BREAKER > 20KA 13.493 KA
CBL-21-VFD-1300      MCC-9A                      13.493 KA      ANG: -45.53

22-VFD-2022      3P Duty: 9.089 KA AT -61.20 DEG ( 7.56 MVA) X/R: 1.90
VOLTAGE: 480. EQUIV. IMPEDANCE= 0.0147 + J 0.0267 OHMS
LOW VOLTAGE POWER CIRCUIT BREAKER 9.089 KA
MOLDED CASE CIRCUIT BREAKER < 10KA 9.309 KA
MOLDED CASE CIRCUIT BREAKER < 20KA 9.089 KA
MOLDED CASE CIRCUIT BREAKER > 20KA 9.089 KA
CBL-22-BLO-2022      BUS-22-BLO-2022              0.140 KA      ANG: 103.23
PI-0025              MCC-10B                      8.955 KA      ANG: -60.96

22-VFD-2201      3P Duty: 10.486 KA AT -56.32 DEG ( 8.72 MVA) X/R: 1.46
VOLTAGE: 480. EQUIV. IMPEDANCE= 0.0147 + J 0.0220 OHMS
LOW VOLTAGE POWER CIRCUIT BREAKER 10.486 KA
MOLDED CASE CIRCUIT BREAKER < 20KA 10.486 KA
MOLDED CASE CIRCUIT BREAKER > 20KA 10.486 KA
CBL-22-BLO-2201      BUS-22-BLO-2201              0.140 KA      ANG: 102.89
PI-0018              MCC-10A                      10.355 KA     ANG: -56.05

22-VFD-2203      3P Duty: 10.486 KA AT -56.32 DEG ( 8.72 MVA) X/R: 1.46
VOLTAGE: 480. EQUIV. IMPEDANCE= 0.0147 + J 0.0220 OHMS
LOW VOLTAGE POWER CIRCUIT BREAKER 10.486 KA
MOLDED CASE CIRCUIT BREAKER < 20KA 10.486 KA
MOLDED CASE CIRCUIT BREAKER > 20KA 10.486 KA
CBL-22-BLO-2203      BUS-22-BLO-2203              0.140 KA      ANG: 103.34
PI-0019              MCC-10A                      10.355 KA     ANG: -56.05

23-VCP-2810      3P Duty: 2.496 KA AT -15.91 DEG ( 2.07 MVA) X/R: 0.28
VOLTAGE: 480. EQUIV. IMPEDANCE= 0.1068 + J 0.0304 OHMS
LOW VOLTAGE POWER CIRCUIT BREAKER 2.496 KA
MOLDED CASE CIRCUIT BREAKER < 10KA 2.496 KA
MOLDED CASE CIRCUIT BREAKER < 20KA 2.496 KA
MOLDED CASE CIRCUIT BREAKER > 20KA 2.496 KA
CBL-23-VCP-2810      MCC-11A                      2.496 KA      ANG: -15.91

ADMIN            3P Duty: 8.209 KA AT -50.68 DEG ( 6.83 MVA) X/R: 1.21
VOLTAGE: 480. EQUIV. IMPEDANCE= 0.0214 + J 0.0261 OHMS
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THREE PHASE FAULT REPORT
(FOR APPLICATION OF LOW VOLTAGE BREAKERS)
PRE FAULT VOLTAGE: 1.0000
MODEL TRANSFORMER TAPS: NO

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	LOW VOLTAGE POWER CIRCUIT BREAKER	8.209 KA			
	MOLDED CASE CIRCUIT BREAKER < 10KA	8.209 KA			
	MOLDED CASE CIRCUIT BREAKER < 20KA	8.209 KA			
	MOLDED CASE CIRCUIT BREAKER > 20KA	8.209 KA			
	CBL-ADMIN	SWBD-MAIN-A	8.209 KA	ANG:	-50.68
AHF-10A	3P Duty: 8.143 KA AT -44.53 DEG (6.77 MVA) X/R: 0.91				
	VOLTAGE: 480. EQUIV. IMPEDANCE=	0.0243 + J 0.0239 OHMS			
	LOW VOLTAGE POWER CIRCUIT BREAKER	8.143 KA			
	MOLDED CASE CIRCUIT BREAKER < 10KA	8.143 KA			
	MOLDED CASE CIRCUIT BREAKER < 20KA	8.143 KA			
	MOLDED CASE CIRCUIT BREAKER > 20KA	8.143 KA			
	CBL-AHF-10A	MCC-10A	8.143 KA	ANG:	-44.53
AHF-10B	3P Duty: 5.705 KA AT -36.60 DEG (4.74 MVA) X/R: 0.75				
	VOLTAGE: 480. EQUIV. IMPEDANCE=	0.0390 + J 0.0290 OHMS			
	LOW VOLTAGE POWER CIRCUIT BREAKER	5.705 KA			
	MOLDED CASE CIRCUIT BREAKER < 10KA	5.705 KA			
	MOLDED CASE CIRCUIT BREAKER < 20KA	5.705 KA			
	MOLDED CASE CIRCUIT BREAKER > 20KA	5.705 KA			
	CBL-AHF-10B	MCC-10B	5.705 KA	ANG:	-36.60
AHF-8A	3P Duty: 17.246 KA AT -57.98 DEG (14.34 MVA) X/R: 1.58				
	VOLTAGE: 480. EQUIV. IMPEDANCE=	0.0085 + J 0.0136 OHMS			
	LOW VOLTAGE POWER CIRCUIT BREAKER	17.246 KA			
	MOLDED CASE CIRCUIT BREAKER < 20KA	17.246 KA			
	MOLDED CASE CIRCUIT BREAKER > 20KA	17.246 KA			
	CBL-AHF-8A	MCC-8A	17.246 KA	ANG:	-57.98
AHF-8B	3P Duty: 15.524 KA AT -57.62 DEG (12.91 MVA) X/R: 1.59				
	VOLTAGE: 480. EQUIV. IMPEDANCE=	0.0096 + J 0.0151 OHMS			
	LOW VOLTAGE POWER CIRCUIT BREAKER	15.524 KA			
	MOLDED CASE CIRCUIT BREAKER < 20KA	15.524 KA			
	MOLDED CASE CIRCUIT BREAKER > 20KA	15.524 KA			
	CBL-AHF-8B	MCC-8B	15.524 KA	ANG:	-57.62
CL BLDG PORT GEN	3P Duty: 7.617 KA AT -47.27 DEG (6.33 MVA) X/R: 1.10				
	VOLTAGE: 480. EQUIV. IMPEDANCE=	0.0247 + J 0.0267 OHMS			
	LOW VOLTAGE POWER CIRCUIT BREAKER	7.617 KA			
	MOLDED CASE CIRCUIT BREAKER < 10KA	7.617 KA			
	MOLDED CASE CIRCUIT BREAKER < 20KA	7.617 KA			
	MOLDED CASE CIRCUIT BREAKER > 20KA	7.617 KA			
	CBL-MCC-11A-PG	MCC-11A	7.617 KA	ANG:	-47.27

THREE PHASE FAULT REPORT
(FOR APPLICATION OF LOW VOLTAGE BREAKERS)
PRE FAULT VOLTAGE: 1.0000
MODEL TRANSFORMER TAPS: NO

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DCB 1 PORT GEN          3P Duty: 8.596 KA AT -59.37 DEG ( 7.15 MVA) X/R: 1.74
VOLTAGE: 480. EQUIV. IMPEDANCE= 0.0164 + J 0.0277 OHMS
LOW VOLTAGE POWER CIRCUIT BREAKER 8.596 KA
MOLDED CASE CIRCUIT BREAKER < 10KA 8.612 KA
MOLDED CASE CIRCUIT BREAKER < 20KA 8.596 KA
MOLDED CASE CIRCUIT BREAKER > 20KA 8.596 KA
CBL-MCC-6B-PG          MCC-6B          8.596 KA      ANG: -59.37

DP-10                   3P Duty: 7.921 KA AT -55.56 DEG ( 6.59 MVA) X/R: 1.50
VOLTAGE: 480. EQUIV. IMPEDANCE= 0.0198 + J 0.0289 OHMS
LOW VOLTAGE POWER CIRCUIT BREAKER 7.921 KA
MOLDED CASE CIRCUIT BREAKER < 10KA 7.921 KA
MOLDED CASE CIRCUIT BREAKER < 20KA 7.921 KA
MOLDED CASE CIRCUIT BREAKER > 20KA 7.921 KA
CBL-DP-10              MTS-10          7.921 KA      ANG: -55.56

DP-13                   3P Duty: 13.620 KA AT -65.34 DEG ( 11.32 MVA) X/R: 2.18
VOLTAGE: 480. EQUIV. IMPEDANCE= 0.0085 + J 0.0185 OHMS
LOW VOLTAGE POWER CIRCUIT BREAKER 13.620 KA
MOLDED CASE CIRCUIT BREAKER < 20KA 13.620 KA
MOLDED CASE CIRCUIT BREAKER > 20KA 13.620 KA
CBL-DP-13              MTS-13          13.620 KA     ANG: -65.34

DP-5A                   3P Duty: 15.064 KA AT -62.82 DEG ( 12.52 MVA) X/R: 1.95
VOLTAGE: 480. EQUIV. IMPEDANCE= 0.0084 + J 0.0164 OHMS
LOW VOLTAGE POWER CIRCUIT BREAKER 15.064 KA
MOLDED CASE CIRCUIT BREAKER < 20KA 15.064 KA
MOLDED CASE CIRCUIT BREAKER > 20KA 15.064 KA
CBL-DP-5A              MCC-5A          15.064 KA     ANG: -62.82

DP-5B                   3P Duty: 13.962 KA AT -63.13 DEG ( 11.61 MVA) X/R: 1.99
VOLTAGE: 480. EQUIV. IMPEDANCE= 0.0090 + J 0.0177 OHMS
LOW VOLTAGE POWER CIRCUIT BREAKER 13.962 KA
MOLDED CASE CIRCUIT BREAKER < 20KA 13.962 KA
MOLDED CASE CIRCUIT BREAKER > 20KA 13.962 KA
CBL-DP-5B              MCC-5B          13.962 KA     ANG: -63.13

DP-6                    3P Duty: 8.643 KA AT -59.47 DEG ( 7.19 MVA) X/R: 1.75
VOLTAGE: 480. EQUIV. IMPEDANCE= 0.0163 + J 0.0276 OHMS
LOW VOLTAGE POWER CIRCUIT BREAKER 8.643 KA
MOLDED CASE CIRCUIT BREAKER < 10KA 8.669 KA
MOLDED CASE CIRCUIT BREAKER < 20KA 8.643 KA
MOLDED CASE CIRCUIT BREAKER > 20KA 8.643 KA

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THREE PHASE FAULT REPORT
(FOR APPLICATION OF LOW VOLTAGE BREAKERS)
PRE FAULT VOLTAGE: 1.0000
MODEL TRANSFORMER TAPS: NO

CONTRIBUTIONS TO DP-6		(CONTINUED)			
CBL-DP-6	MTS-6	8.643 KA	ANG:	-59.47	
DP-8A-1	3P Duty: 19.814 KA AT -66.73 DEG (16.47 MVA) X/R: 2.34				
	VOLTAGE: 480. EQUIV. IMPEDANCE= 0.0055 + J 0.0128 OHMS				
	LOW VOLTAGE POWER CIRCUIT BREAKER 19.814 KA				
	MOLDED CASE CIRCUIT BREAKER < 20KA 19.814 KA				
	MOLDED CASE CIRCUIT BREAKER > 20KA 19.814 KA				
CBL-DP-8A-1	MCC-8A	19.814 KA	ANG:	-66.73	
DP-8A-2	3P Duty: 18.451 KA AT -63.06 DEG (15.34 MVA) X/R: 1.96				
	VOLTAGE: 480. EQUIV. IMPEDANCE= 0.0068 + J 0.0134 OHMS				
	LOW VOLTAGE POWER CIRCUIT BREAKER 18.451 KA				
	MOLDED CASE CIRCUIT BREAKER < 20KA 18.451 KA				
	MOLDED CASE CIRCUIT BREAKER > 20KA 18.451 KA				
CBL-DP-8A-2	DP-8A-1	18.451 KA	ANG:	-63.06	
DP-8B-1	3P Duty: 18.052 KA AT -67.12 DEG (15.01 MVA) X/R: 2.39				
	VOLTAGE: 480. EQUIV. IMPEDANCE= 0.0060 + J 0.0141 OHMS				
	LOW VOLTAGE POWER CIRCUIT BREAKER 18.052 KA				
	MOLDED CASE CIRCUIT BREAKER < 20KA 18.052 KA				
	MOLDED CASE CIRCUIT BREAKER > 20KA 18.052 KA				
CBL-DP-8B-1	MCC-8B	18.052 KA	ANG:	-67.12	
DP-8B-2	3P Duty: 16.924 KA AT -63.73 DEG (14.07 MVA) X/R: 2.04				
	VOLTAGE: 480. EQUIV. IMPEDANCE= 0.0072 + J 0.0147 OHMS				
	LOW VOLTAGE POWER CIRCUIT BREAKER 16.924 KA				
	MOLDED CASE CIRCUIT BREAKER < 20KA 16.924 KA				
	MOLDED CASE CIRCUIT BREAKER > 20KA 16.924 KA				
CBL-DP-8B-2	DP-8B-1	16.924 KA	ANG:	-63.73	
DP-9	3P Duty: 16.320 KA AT -61.08 DEG (13.57 MVA) X/R: 1.81				
	VOLTAGE: 480. EQUIV. IMPEDANCE= 0.0082 + J 0.0149 OHMS				
	LOW VOLTAGE POWER CIRCUIT BREAKER 16.320 KA				
	MOLDED CASE CIRCUIT BREAKER < 20KA 16.320 KA				
	MOLDED CASE CIRCUIT BREAKER > 20KA 16.320 KA				
CBL-DP-9	MTS-9	16.320 KA	ANG:	-61.08	
DSC 02-EF-1320	3P Duty: 0.960 KA AT -6.88 DEG (0.80 MVA) X/R: 0.11				
	VOLTAGE: 480. EQUIV. IMPEDANCE= 0.2867 + J 0.0346 OHMS				
	LOW VOLTAGE POWER CIRCUIT BREAKER 0.960 KA				
	MOLDED CASE CIRCUIT BREAKER < 10KA 0.960 KA				
	MOLDED CASE CIRCUIT BREAKER < 20KA 0.960 KA				
	MOLDED CASE CIRCUIT BREAKER > 20KA 0.960 KA				

THREE PHASE FAULT REPORT
(FOR APPLICATION OF LOW VOLTAGE BREAKERS)
PRE FAULT VOLTAGE: 1.0000
MODEL TRANSFORMER TAPS: NO

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CONTRIBUTIONS TO DSC 02-EF-1320			(CONTINUED)		
CBL-02-EF-1320	BUS-02-EF-1320	0.014 KA	ANG:	101.58	
CBL-DSC 02-EF-1320	MCC-5B	0.955 KA	ANG:	-6.08	
DSC 02-EF-1330					
3P Duty: 1.151 KA AT -8.13 DEG (0.96 MVA) X/R: 0.13					
VOLTAGE: 480. EQUIV. IMPEDANCE= 0.2383 + J 0.0341 OHMS					
LOW VOLTAGE POWER CIRCUIT BREAKER 1.151 KA					
MOLDED CASE CIRCUIT BREAKER < 10KA 1.151 KA					
MOLDED CASE CIRCUIT BREAKER < 20KA 1.151 KA					
MOLDED CASE CIRCUIT BREAKER > 20KA 1.151 KA					
CBL-02-EF-1330	BUS-02-EF-1330	0.035 KA	ANG:	-78.34	
CBL-DSC 02-EF-1330	MCC-5A	1.140 KA	ANG:	-6.47	
DSC 02-EF-1340					
3P Duty: 1.150 KA AT -8.38 DEG (0.96 MVA) X/R: 0.14					
VOLTAGE: 480. EQUIV. IMPEDANCE= 0.2384 + J 0.0351 OHMS					
LOW VOLTAGE POWER CIRCUIT BREAKER 1.150 KA					
MOLDED CASE CIRCUIT BREAKER < 10KA 1.150 KA					
MOLDED CASE CIRCUIT BREAKER < 20KA 1.150 KA					
MOLDED CASE CIRCUIT BREAKER > 20KA 1.150 KA					
CBL-02-EF-1340	BUS-02-EF-1340	0.035 KA	ANG:	101.66	
CBL-DSC 02-EF-1340	MCC-5B	1.139 KA	ANG:	-6.72	
DSC 02-EF-1370					
3P Duty: 1.416 KA AT -7.88 DEG (1.18 MVA) X/R: 0.13					
VOLTAGE: 480. EQUIV. IMPEDANCE= 0.1939 + J 0.0268 OHMS					
LOW VOLTAGE POWER CIRCUIT BREAKER 1.416 KA					
MOLDED CASE CIRCUIT BREAKER < 10KA 1.416 KA					
MOLDED CASE CIRCUIT BREAKER < 20KA 1.416 KA					
MOLDED CASE CIRCUIT BREAKER > 20KA 1.416 KA					
CBL-02-EF-1370	BUS-02-EF-1370	0.014 KA	ANG:	-78.42	
CBL-DSC 02-EF-1370	MCC-5A	1.411 KA	ANG:	-7.34	
DSC 02-EF-1380					
3P Duty: 1.143 KA AT -7.38 DEG (0.95 MVA) X/R: 0.12					
VOLTAGE: 480. EQUIV. IMPEDANCE= 0.2404 + J 0.0312 OHMS					
LOW VOLTAGE POWER CIRCUIT BREAKER 1.143 KA					
MOLDED CASE CIRCUIT BREAKER < 10KA 1.143 KA					
MOLDED CASE CIRCUIT BREAKER < 20KA 1.143 KA					
MOLDED CASE CIRCUIT BREAKER > 20KA 1.143 KA					
CBL-02-EF-1380	BUS-02-EF-1380	0.014 KA	ANG:	101.58	
CBL-DSC 02-EF-1380	MCC-5B	1.139 KA	ANG:	-6.71	
DSC 02-EF-1390					
3P Duty: 1.778 KA AT -11.34 DEG (1.48 MVA) X/R: 0.19					
VOLTAGE: 480. EQUIV. IMPEDANCE= 0.1528 + J 0.0306 OHMS					

THREE PHASE FAULT REPORT
(FOR APPLICATION OF LOW VOLTAGE BREAKERS)
PRE FAULT VOLTAGE: 1.0000
MODEL TRANSFORMER TAPS: NO

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LOW VOLTAGE POWER CIRCUIT BREAKER	1.778 KA				
MOLDED CASE CIRCUIT BREAKER < 10KA	1.778 KA				
MOLDED CASE CIRCUIT BREAKER < 20KA	1.778 KA				
MOLDED CASE CIRCUIT BREAKER > 20KA	1.778 KA				
CBL-02-EF-1390	BUS-02-EF-1390	0.053 KA	ANG:	-78.35	
CBL-DSC 02-EF-1390	MCC-5A	1.758 KA	ANG:	-9.75	
DSC 02-EF-1396 3P Duty: 1.490 KA AT -10.32 DEG (1.24 MVA) X/R: 0.18					
VOLTAGE: 480. EQUIV. IMPEDANCE= 0.1829 + J 0.0333 OHMS					
LOW VOLTAGE POWER CIRCUIT BREAKER	1.490 KA				
MOLDED CASE CIRCUIT BREAKER < 10KA	1.490 KA				
MOLDED CASE CIRCUIT BREAKER < 20KA	1.490 KA				
MOLDED CASE CIRCUIT BREAKER > 20KA	1.490 KA				
CBL-02-EF-1396	BUS-02-EF-1396	0.032 KA	ANG:	101.60	
CBL-DSC 02-EF-1396	MCC-5B	1.479 KA	ANG:	-9.18	
DSC 02-SF-1310 3P Duty: 1.143 KA AT -7.38 DEG (0.95 MVA) X/R: 0.12					
VOLTAGE: 480. EQUIV. IMPEDANCE= 0.2404 + J 0.0312 OHMS					
LOW VOLTAGE POWER CIRCUIT BREAKER	1.143 KA				
MOLDED CASE CIRCUIT BREAKER < 10KA	1.143 KA				
MOLDED CASE CIRCUIT BREAKER < 20KA	1.143 KA				
MOLDED CASE CIRCUIT BREAKER > 20KA	1.143 KA				
CBL-02-SF-1310	BUS-02-SF-1310	0.014 KA	ANG:	101.58	
CBL-DSC 02-SF-1310	MCC-5B	1.139 KA	ANG:	-6.71	
DSC 02-SF-1350 3P Duty: 1.157 KA AT -8.96 DEG (0.96 MVA) X/R: 0.14					
VOLTAGE: 480. EQUIV. IMPEDANCE= 0.2365 + J 0.0373 OHMS					
LOW VOLTAGE POWER CIRCUIT BREAKER	1.157 KA				
MOLDED CASE CIRCUIT BREAKER < 10KA	1.157 KA				
MOLDED CASE CIRCUIT BREAKER < 20KA	1.157 KA				
MOLDED CASE CIRCUIT BREAKER > 20KA	1.157 KA				
CBL-02-SF-1350	BUS-02-SF-1350	0.053 KA	ANG:	-78.28	
CBL-DSC 02-SF-1350	MCC-5A	1.140 KA	ANG:	-6.47	
DSC 02-SF-1360 3P Duty: 1.156 KA AT -9.21 DEG (0.96 MVA) X/R: 0.14					
VOLTAGE: 480. EQUIV. IMPEDANCE= 0.2366 + J 0.0383 OHMS					
LOW VOLTAGE POWER CIRCUIT BREAKER	1.156 KA				
MOLDED CASE CIRCUIT BREAKER < 10KA	1.156 KA				
MOLDED CASE CIRCUIT BREAKER < 20KA	1.156 KA				
MOLDED CASE CIRCUIT BREAKER > 20KA	1.156 KA				
CBL-02-SF-1360	BUS-02-SF-1360	0.053 KA	ANG:	-78.28	
CBL-DSC 02-SF-1360	MCC-5B	1.138 KA	ANG:	-6.72	
DSC 02-SF-1570 3P Duty: 1.412 KA AT -7.47 DEG (1.17 MVA) X/R: 0.13					

THREE PHASE FAULT REPORT
(FOR APPLICATION OF LOW VOLTAGE BREAKERS)
PRE FAULT VOLTAGE: 1.0000
MODEL TRANSFORMER TAPS: NO

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VOLTAGE:	480.	EQUIV. IMPEDANCE=	0.1946 + J 0.0255	OHMS	
LOW VOLTAGE POWER CIRCUIT BREAKER	1.412 KA				
MOLDED CASE CIRCUIT BREAKER < 10KA	1.412 KA				
MOLDED CASE CIRCUIT BREAKER < 20KA	1.412 KA				
MOLDED CASE CIRCUIT BREAKER > 20KA	1.412 KA				
CBL-02-SF-1570	BUS-02-SF-1570	0.004 KA	ANG:	-78.45	
CBL-DSC 02-SF-1570	MCC-5A	1.411 KA	ANG:	-7.34	
DSC 02-SF-1575 3P Duty: 1.279 KA AT -8.64 DEG (1.06 MVA) X/R: 0.15					
VOLTAGE:	480.	EQUIV. IMPEDANCE=	0.2142 + J 0.0325	OHMS	
LOW VOLTAGE POWER CIRCUIT BREAKER	1.279 KA				
MOLDED CASE CIRCUIT BREAKER < 10KA	1.279 KA				
MOLDED CASE CIRCUIT BREAKER < 20KA	1.279 KA				
MOLDED CASE CIRCUIT BREAKER > 20KA	1.279 KA				
CBL-02-SF-1575	BUS-02-SF-1575	0.004 KA	ANG:	-78.46	
CBL-DSC 02-SF-1575	MCC-5B	1.278 KA	ANG:	-8.49	
DSC 02-SF-1810 3P Duty: 1.410 KA AT -7.78 DEG (1.17 MVA) X/R: 0.14					
VOLTAGE:	480.	EQUIV. IMPEDANCE=	0.1947 + J 0.0266	OHMS	
LOW VOLTAGE POWER CIRCUIT BREAKER	1.410 KA				
MOLDED CASE CIRCUIT BREAKER < 10KA	1.410 KA				
MOLDED CASE CIRCUIT BREAKER < 20KA	1.410 KA				
MOLDED CASE CIRCUIT BREAKER > 20KA	1.410 KA				
CBL-02-SF-1810	BUS-02-SF-1810	0.004 KA	ANG:	-78.45	
CBL-DSC 02-SF-1810	MCC-5B	1.409 KA	ANG:	-7.64	
DSC 03-EF-1420 3P Duty: 1.851 KA AT -9.57 DEG (1.54 MVA) X/R: 0.17					
VOLTAGE:	480.	EQUIV. IMPEDANCE=	0.1476 + J 0.0249	OHMS	
LOW VOLTAGE POWER CIRCUIT BREAKER	1.851 KA				
MOLDED CASE CIRCUIT BREAKER < 10KA	1.851 KA				
MOLDED CASE CIRCUIT BREAKER < 20KA	1.851 KA				
MOLDED CASE CIRCUIT BREAKER > 20KA	1.851 KA				
CBL-03-EF-1420	BUS-03-EF-1420	0.014 KA	ANG:	101.58	
CBL-DSC 03-EF-1420	MCC-5B	1.846 KA	ANG:	-9.16	
DSC 03-HSS-1140 3P Duty: 0.833 KA AT -4.91 DEG (0.69 MVA) X/R: 0.08					
VOLTAGE:	480.	EQUIV. IMPEDANCE=	0.3315 + J 0.0285	OHMS	
LOW VOLTAGE POWER CIRCUIT BREAKER	0.833 KA				
MOLDED CASE CIRCUIT BREAKER < 10KA	0.833 KA				
MOLDED CASE CIRCUIT BREAKER < 20KA	0.833 KA				
MOLDED CASE CIRCUIT BREAKER > 20KA	0.833 KA				
CBL-03-HSS-1140	BUS-03-HSS-1140	0.004 KA	ANG:	101.55	
CBL-DSC 03-HSS-1140	MCC-9A	0.832 KA	ANG:	-4.68	

THREE PHASE FAULT REPORT
(FOR APPLICATION OF LOW VOLTAGE BREAKERS)
PRE FAULT VOLTAGE: 1.0000
MODEL TRANSFORMER TAPS: NO

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DSC 03-HSS-1240      3P Duty: 0.730 KA AT -4.71 DEG ( 0.61 MVA) X/R: 0.08
VOLTAGE: 480. EQUIV. IMPEDANCE= 0.3781 + J 0.0312 OHMS
LOW VOLTAGE POWER CIRCUIT BREAKER 0.730 KA
MOLDED CASE CIRCUIT BREAKER < 10KA 0.730 KA
MOLDED CASE CIRCUIT BREAKER < 20KA 0.730 KA
MOLDED CASE CIRCUIT BREAKER > 20KA 0.730 KA
CBL-03-HSS-1240      BUS-03-HSS-1240      0.004 KA      ANG: 101.55
CBL-DSC 03-HSS-1240      MCC-9A      0.729 KA      ANG: -4.44

DSC 03-HSS-1340      3P Duty: 0.686 KA AT -4.79 DEG ( 0.57 MVA) X/R: 0.08
VOLTAGE: 480. EQUIV. IMPEDANCE= 0.4026 + J 0.0337 OHMS
LOW VOLTAGE POWER CIRCUIT BREAKER 0.686 KA
MOLDED CASE CIRCUIT BREAKER < 10KA 0.686 KA
MOLDED CASE CIRCUIT BREAKER < 20KA 0.686 KA
MOLDED CASE CIRCUIT BREAKER > 20KA 0.686 KA
CBL-03-HSS-1340      BUS-03-HSS-1340      0.004 KA      ANG: 101.55
CBL-DSC 03-HSS-1340      MCC-9B      0.685 KA      ANG: -4.50

DSC 03-PMP-1510      3P Duty: 1.454 KA AT -8.04 DEG ( 1.21 MVA) X/R: 0.12
VOLTAGE: 480. EQUIV. IMPEDANCE= 0.1887 + J 0.0267 OHMS
LOW VOLTAGE POWER CIRCUIT BREAKER 1.454 KA
MOLDED CASE CIRCUIT BREAKER < 10KA 1.454 KA
MOLDED CASE CIRCUIT BREAKER < 20KA 1.454 KA
MOLDED CASE CIRCUIT BREAKER > 20KA 1.454 KA
CBL-03-PMP-1510      BUS-03-PMP-1510      0.053 KA      ANG: 101.72
CBL-DSC 03-PMP-1510      MCC-9A      1.437 KA      ANG: -6.06

DSC 03-PMP-1520      3P Duty: 1.135 KA AT -8.98 DEG ( 0.94 MVA) X/R: 0.13
VOLTAGE: 480. EQUIV. IMPEDANCE= 0.2413 + J 0.0381 OHMS
LOW VOLTAGE POWER CIRCUIT BREAKER 1.135 KA
MOLDED CASE CIRCUIT BREAKER < 10KA 1.135 KA
MOLDED CASE CIRCUIT BREAKER < 20KA 1.135 KA
MOLDED CASE CIRCUIT BREAKER > 20KA 1.135 KA
CBL-03-PMP-1520      BUS-03-PMP-1520      0.070 KA      ANG: 101.77
CBL-DSC 03-PMP-1520      MCC-9B      1.112 KA      ANG: -5.58

DSC 03-PMP-1530      3P Duty: 0.984 KA AT -7.94 DEG ( 0.82 MVA) X/R: 0.11
VOLTAGE: 480. EQUIV. IMPEDANCE= 0.2789 + J 0.0389 OHMS
LOW VOLTAGE POWER CIRCUIT BREAKER 0.984 KA
MOLDED CASE CIRCUIT BREAKER < 10KA 0.984 KA
MOLDED CASE CIRCUIT BREAKER < 20KA 0.984 KA
MOLDED CASE CIRCUIT BREAKER > 20KA 0.984 KA
CBL-03-PMP-1530      BUS-03-PMP-1530      0.053 KA      ANG: 101.72

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THREE PHASE FAULT REPORT
(FOR APPLICATION OF LOW VOLTAGE BREAKERS)
PRE FAULT VOLTAGE: 1.0000
MODEL TRANSFORMER TAPS: NO

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CONTRIBUTIONS TO DSC 03-PMP-1530 (CONTINUED)
CBL-DSC 03-PMP-1530 MCC-9A 0.968 KA ANG: -4.99

DSC 03-PMP-1610 3P Duty: 0.746 KA AT -8.35 DEG ( 0.62 MVA) X/R: 0.10
VOLTAGE: 480. EQUIV. IMPEDANCE= 0.3676 + J 0.0540 OHMS
LOW VOLTAGE POWER CIRCUIT BREAKER 0.746 KA
MOLDED CASE CIRCUIT BREAKER < 10KA 0.746 KA
MOLDED CASE CIRCUIT BREAKER < 20KA 0.746 KA
MOLDED CASE CIRCUIT BREAKER > 20KA 0.746 KA
CBL-03-PMP-1610 BUS-03-PMP-1610 0.053 KA ANG: 101.72
CBL-DSC 03-PMP-1610 MCC-9A 0.729 KA ANG: -4.45

DSC 03-PMP-1620 3P Duty: 0.848 KA AT -8.30 DEG ( 0.70 MVA) X/R: 0.11
VOLTAGE: 480. EQUIV. IMPEDANCE= 0.3235 + J 0.0472 OHMS
LOW VOLTAGE POWER CIRCUIT BREAKER 0.848 KA
MOLDED CASE CIRCUIT BREAKER < 10KA 0.848 KA
MOLDED CASE CIRCUIT BREAKER < 20KA 0.848 KA
MOLDED CASE CIRCUIT BREAKER > 20KA 0.848 KA
CBL-03-PMP-1620 BUS-03-PMP-1620 0.053 KA ANG: 101.72
CBL-DSC 03-PMP-1620 MCC-9B 0.831 KA ANG: -4.87

DSC 03-SCC-1150 3P Duty: 1.158 KA AT -5.67 DEG ( 0.96 MVA) X/R: 0.09
VOLTAGE: 480. EQUIV. IMPEDANCE= 0.2381 + J 0.0236 OHMS
LOW VOLTAGE POWER CIRCUIT BREAKER 1.158 KA
MOLDED CASE CIRCUIT BREAKER < 10KA 1.158 KA
MOLDED CASE CIRCUIT BREAKER < 20KA 1.158 KA
MOLDED CASE CIRCUIT BREAKER > 20KA 1.158 KA
CBL-03-SCC-1150 BUS-03-SCC-1150 0.005 KA ANG: 101.55
CBL-DSC 03-SCC-1150 MCC-9A 1.157 KA ANG: -5.42

DSC 03-SCC-1250 3P Duty: 1.113 KA AT -5.83 DEG ( 0.93 MVA) X/R: 0.10
VOLTAGE: 480. EQUIV. IMPEDANCE= 0.2477 + J 0.0253 OHMS
LOW VOLTAGE POWER CIRCUIT BREAKER 1.113 KA
MOLDED CASE CIRCUIT BREAKER < 10KA 1.113 KA
MOLDED CASE CIRCUIT BREAKER < 20KA 1.113 KA
MOLDED CASE CIRCUIT BREAKER > 20KA 1.113 KA
CBL-03-SCC-1250 BUS-03-SCC-1250 0.005 KA ANG: 101.55
CBL-DSC 03-SCC-1250 MCC-9B 1.112 KA ANG: -5.57

DSC 03-SCC-1350 3P Duty: 0.833 KA AT -4.91 DEG ( 0.69 MVA) X/R: 0.08
VOLTAGE: 480. EQUIV. IMPEDANCE= 0.3315 + J 0.0285 OHMS

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THREE PHASE FAULT REPORT
(FOR APPLICATION OF LOW VOLTAGE BREAKERS)
PRE FAULT VOLTAGE: 1.0000
MODEL TRANSFORMER TAPS: NO

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LOW VOLTAGE POWER CIRCUIT BREAKER	0.833 KA				
MOLDED CASE CIRCUIT BREAKER < 10KA	0.833 KA				
MOLDED CASE CIRCUIT BREAKER < 20KA	0.833 KA				
MOLDED CASE CIRCUIT BREAKER > 20KA	0.833 KA				
CBL-03-SCC-1350	BUS-03-SCC-1350	0.004 KA	ANG:	101.55	
CBL-DSC 03-SCC-1350	MCC-9A	0.832 KA	ANG:	-4.68	
DSC 03-SF-1410 3P Duty: 0.726 KA AT -6.08 DEG (0.60 MVA) X/R: 0.10					
VOLTAGE: 480. EQUIV. IMPEDANCE= 0.3798 + J 0.0404 OHMS					
LOW VOLTAGE POWER CIRCUIT BREAKER	0.726 KA				
MOLDED CASE CIRCUIT BREAKER < 10KA	0.726 KA				
MOLDED CASE CIRCUIT BREAKER < 20KA	0.726 KA				
MOLDED CASE CIRCUIT BREAKER > 20KA	0.726 KA				
CBL-03-SF-1410	BUS-03-SF-1410	0.011 KA	ANG:	101.57	
CBL-DSC 03-SF-1410	MCC-5B	0.722 KA	ANG:	-5.28	
DSC 03-SF-1575 3P Duty: 0.723 KA AT -5.54 DEG (0.60 MVA) X/R: 0.09					
VOLTAGE: 480. EQUIV. IMPEDANCE= 0.3813 + J 0.0370 OHMS					
LOW VOLTAGE POWER CIRCUIT BREAKER	0.723 KA				
MOLDED CASE CIRCUIT BREAKER < 10KA	0.723 KA				
MOLDED CASE CIRCUIT BREAKER < 20KA	0.723 KA				
MOLDED CASE CIRCUIT BREAKER > 20KA	0.723 KA				
CBL-03-SF-1575	BUS-03-SF-1575	0.004 KA	ANG:	101.55	
CBL-DSC 03-SF-1575	MCC-5B	0.722 KA	ANG:	-5.27	
DSC 03-SLC-1120 3P Duty: 1.439 KA AT -6.25 DEG (1.20 MVA) X/R: 0.10					
VOLTAGE: 480. EQUIV. IMPEDANCE= 0.1915 + J 0.0210 OHMS					
LOW VOLTAGE POWER CIRCUIT BREAKER	1.439 KA				
MOLDED CASE CIRCUIT BREAKER < 10KA	1.439 KA				
MOLDED CASE CIRCUIT BREAKER < 20KA	1.439 KA				
MOLDED CASE CIRCUIT BREAKER > 20KA	1.439 KA				
CBL-03-SLC-1120	BUS-03-SLC-1120	0.005 KA	ANG:	101.55	
CBL-DSC 03-SLC-1120	MCC-9A	1.437 KA	ANG:	-6.05	
DSC 03-SLC-1220 3P Duty: 1.113 KA AT -5.83 DEG (0.93 MVA) X/R: 0.10					
VOLTAGE: 480. EQUIV. IMPEDANCE= 0.2477 + J 0.0253 OHMS					
LOW VOLTAGE POWER CIRCUIT BREAKER	1.113 KA				
MOLDED CASE CIRCUIT BREAKER < 10KA	1.113 KA				
MOLDED CASE CIRCUIT BREAKER < 20KA	1.113 KA				
MOLDED CASE CIRCUIT BREAKER > 20KA	1.113 KA				
CBL-03-SLC-1220	BUS-03-SLC-1220	0.005 KA	ANG:	101.55	
CBL-DSC 03-SLC-1220	MCC-9B	1.112 KA	ANG:	-5.57	
DSC 03-SLC-1320 3P Duty: 0.833 KA AT -5.03 DEG (0.69 MVA) X/R: 0.08					

THREE PHASE FAULT REPORT
(FOR APPLICATION OF LOW VOLTAGE BREAKERS)
PRE FAULT VOLTAGE: 1.0000
MODEL TRANSFORMER TAPS: NO

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VOLTAGE:	480.	EQUIV. IMPEDANCE=	0.3313 + J 0.0291	OHMS	
LOW VOLTAGE POWER CIRCUIT BREAKER			0.833 KA		
MOLDED CASE CIRCUIT BREAKER < 10KA			0.833 KA		
MOLDED CASE CIRCUIT BREAKER < 20KA			0.833 KA		
MOLDED CASE CIRCUIT BREAKER > 20KA			0.833 KA		
CBL-03-SLC-1320		BUS-03-SLC-1320	0.005 KA	ANG:	101.55
CBL-DSC 03-SLC-1320		MCC-9A	0.832 KA	ANG:	-4.68
DSC 04-PMP-1140 3P Duty: 2.343 KA AT -25.01 DEG (1.95 MVA) X/R: 0.53					
VOLTAGE:	480.	EQUIV. IMPEDANCE=	0.1072 + J 0.0500	OHMS	
LOW VOLTAGE POWER CIRCUIT BREAKER			2.343 KA		
MOLDED CASE CIRCUIT BREAKER < 10KA			2.343 KA		
MOLDED CASE CIRCUIT BREAKER < 20KA			2.343 KA		
MOLDED CASE CIRCUIT BREAKER > 20KA			2.343 KA		
CBL-04-PMP-1140		BUS-04-PMP-1140	0.282 KA	ANG:	101.79
CBL-DSC 04-PMP-1140		MCC-10B	2.186 KA	ANG:	-19.08
DSC 06-CLA-1510 3P Duty: 2.039 KA AT -13.14 DEG (1.70 MVA) X/R: 0.21					
VOLTAGE:	480.	EQUIV. IMPEDANCE=	0.1324 + J 0.0309	OHMS	
LOW VOLTAGE POWER CIRCUIT BREAKER			2.039 KA		
MOLDED CASE CIRCUIT BREAKER < 10KA			2.039 KA		
MOLDED CASE CIRCUIT BREAKER < 20KA			2.039 KA		
MOLDED CASE CIRCUIT BREAKER > 20KA			2.039 KA		
CBL-06-CLA-1510		BUS-06-CLA-1510	0.007 KA	ANG:	101.55
CBL-DSC 06-CLA-1510		MCC-10A	2.036 KA	ANG:	-12.96
DSC 06-CLA-1520 3P Duty: 1.234 KA AT -10.97 DEG (1.03 MVA) X/R: 0.19					
VOLTAGE:	480.	EQUIV. IMPEDANCE=	0.2205 + J 0.0427	OHMS	
LOW VOLTAGE POWER CIRCUIT BREAKER			1.234 KA		
MOLDED CASE CIRCUIT BREAKER < 10KA			1.234 KA		
MOLDED CASE CIRCUIT BREAKER < 20KA			1.234 KA		
MOLDED CASE CIRCUIT BREAKER > 20KA			1.234 KA		
CBL-06-CLA-1520		BUS-06-CLA-1520	0.007 KA	ANG:	101.55
CBL-DSC 06-CLA-1520		MCC-10B	1.231 KA	ANG:	-10.67
DSC 06-CLA-1530 3P Duty: 0.889 KA AT -8.39 DEG (0.74 MVA) X/R: 0.13					
VOLTAGE:	480.	EQUIV. IMPEDANCE=	0.3085 + J 0.0455	OHMS	
LOW VOLTAGE POWER CIRCUIT BREAKER			0.889 KA		
MOLDED CASE CIRCUIT BREAKER < 10KA			0.889 KA		
MOLDED CASE CIRCUIT BREAKER < 20KA			0.889 KA		
MOLDED CASE CIRCUIT BREAKER > 20KA			0.889 KA		
CBL-06-CLA-1530		BUS-06-CLA-1530	0.007 KA	ANG:	101.55
CBL-DSC 06-CLA-1530		MCC-10A	0.886 KA	ANG:	-7.97

THREE PHASE FAULT REPORT
(FOR APPLICATION OF LOW VOLTAGE BREAKERS)
PRE FAULT VOLTAGE: 1.0000
MODEL TRANSFORMER TAPS: NO

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DSC 06-PMP-1515      3P Duty: 2.796 KA AT -18.43 DEG ( 2.32 MVA) X/R: 0.31
VOLTAGE: 480. EQUIV. IMPEDANCE= 0.0940 + J 0.0313 OHMS
LOW VOLTAGE POWER CIRCUIT BREAKER 2.796 KA
MOLDED CASE CIRCUIT BREAKER < 10KA 2.796 KA
MOLDED CASE CIRCUIT BREAKER < 20KA 2.796 KA
MOLDED CASE CIRCUIT BREAKER > 20KA 2.796 KA
CBL-06-PMP-1515      BUS-06-PMP-1515      0.071 KA      ANG: 101.64
CBL-DSC 06-PMP-1515      MCC-10A      2.762 KA      ANG: -17.16

DSC 06-PMP-1525      3P Duty: 1.745 KA AT -16.42 DEG ( 1.45 MVA) X/R: 0.30
VOLTAGE: 480. EQUIV. IMPEDANCE= 0.1523 + J 0.0449 OHMS
LOW VOLTAGE POWER CIRCUIT BREAKER 1.745 KA
MOLDED CASE CIRCUIT BREAKER < 10KA 1.745 KA
MOLDED CASE CIRCUIT BREAKER < 20KA 1.745 KA
MOLDED CASE CIRCUIT BREAKER > 20KA 1.745 KA
CBL-06-PMP-1525      BUS-06-PMP-1525      0.071 KA      ANG: 101.64
CBL-DSC 06-PMP-1525      MCC-10B      1.713 KA      ANG: -14.34

DSC 06-PMP-1535      3P Duty: 1.280 KA AT -13.58 DEG ( 1.06 MVA) X/R: 0.22
VOLTAGE: 480. EQUIV. IMPEDANCE= 0.2104 + J 0.0508 OHMS
LOW VOLTAGE POWER CIRCUIT BREAKER 1.280 KA
MOLDED CASE CIRCUIT BREAKER < 10KA 1.280 KA
MOLDED CASE CIRCUIT BREAKER < 20KA 1.280 KA
MOLDED CASE CIRCUIT BREAKER > 20KA 1.280 KA
CBL-06-PMP-1535      BUS-06-PMP-1535      0.071 KA      ANG: 101.64
CBL-DSC 06-PMP-1535      MCC-10A      1.252 KA      ANG: -10.66

DSC 08-EF-1230      3P Duty: 3.506 KA AT -19.81 DEG ( 2.92 MVA) X/R: 0.35
VOLTAGE: 480. EQUIV. IMPEDANCE= 0.0744 + J 0.0268 OHMS
LOW VOLTAGE POWER CIRCUIT BREAKER 3.506 KA
MOLDED CASE CIRCUIT BREAKER < 10KA 3.506 KA
MOLDED CASE CIRCUIT BREAKER < 20KA 3.506 KA
MOLDED CASE CIRCUIT BREAKER > 20KA 3.506 KA
CBL-08-EF-1230      BUS-08-EF-1230      0.002 KA      ANG: 101.54
CBL-DSC 08-EF-1230      MCC-11A      3.505 KA      ANG: -19.79

DSC 09-MIX-1110      3P Duty: 0.799 KA AT -7.08 DEG ( 0.66 MVA) X/R: 0.12
VOLTAGE: 480. EQUIV. IMPEDANCE= 0.3442 + J 0.0427 OHMS
LOW VOLTAGE POWER CIRCUIT BREAKER 0.799 KA
MOLDED CASE CIRCUIT BREAKER < 10KA 0.799 KA
MOLDED CASE CIRCUIT BREAKER < 20KA 0.799 KA
MOLDED CASE CIRCUIT BREAKER > 20KA 0.799 KA
CBL-09-MIX-1110      BUS-09-MIX-1110      0.007 KA      ANG: 101.56
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THREE PHASE FAULT REPORT
(FOR APPLICATION OF LOW VOLTAGE BREAKERS)
PRE FAULT VOLTAGE: 1.0000
MODEL TRANSFORMER TAPS: NO

CONTRIBUTIONS TO DSC 09-MIX-1110			(CONTINUED)	
CBL-DSC 09-MIX-1110	MCC-11A	0.797 KA	ANG:	-6.60
DSC 09-PMP-903	3P Duty: 0.862 KA AT -12.26 DEG (0.72 MVA) X/R: 0.21			
VOLTAGE: 480. EQUIV. IMPEDANCE= 0.3142 + J 0.0683 OHMS				
LOW VOLTAGE POWER CIRCUIT BREAKER 0.862 KA				
MOLDED CASE CIRCUIT BREAKER < 10KA 0.862 KA				
MOLDED CASE CIRCUIT BREAKER < 20KA 0.862 KA				
MOLDED CASE CIRCUIT BREAKER > 20KA 0.862 KA				
CBL-09-PMP-903	BUS-09-PMP-903	0.007 KA	ANG:	101.56
CBL-DSC 09-PMP-903	MCC-11B	0.859 KA	ANG:	-11.83
DSC 11-EF-1530	3P Duty: 1.761 KA AT -11.90 DEG (1.46 MVA) X/R: 0.21			
VOLTAGE: 480. EQUIV. IMPEDANCE= 0.1539 + J 0.0325 OHMS				
LOW VOLTAGE POWER CIRCUIT BREAKER 1.761 KA				
MOLDED CASE CIRCUIT BREAKER < 10KA 1.761 KA				
MOLDED CASE CIRCUIT BREAKER < 20KA 1.761 KA				
MOLDED CASE CIRCUIT BREAKER > 20KA 1.761 KA				
CBL-11-EF-1530	BUS-11-EF-1530	0.011 KA	ANG:	101.57
CBL-DSC 11-EF-1530	MCC-6A	1.757 KA	ANG:	-11.59
DSC 11-EF-1540	3P Duty: 2.130 KA AT -14.17 DEG (1.77 MVA) X/R: 0.25			
VOLTAGE: 480. EQUIV. IMPEDANCE= 0.1262 + J 0.0319 OHMS				
LOW VOLTAGE POWER CIRCUIT BREAKER 2.130 KA				
MOLDED CASE CIRCUIT BREAKER < 10KA 2.130 KA				
MOLDED CASE CIRCUIT BREAKER < 20KA 2.130 KA				
MOLDED CASE CIRCUIT BREAKER > 20KA 2.130 KA				
CBL-11-EF-1540	BUS-11-EF-1540	0.011 KA	ANG:	101.57
CBL-DSC 11-EF-1540	MCC-6B	2.125 KA	ANG:	-13.92
DSC 11-EF-1560	3P Duty: 1.755 KA AT -12.05 DEG (1.46 MVA) X/R: 0.21			
VOLTAGE: 480. EQUIV. IMPEDANCE= 0.1544 + J 0.0330 OHMS				
LOW VOLTAGE POWER CIRCUIT BREAKER 1.755 KA				
MOLDED CASE CIRCUIT BREAKER < 10KA 1.755 KA				
MOLDED CASE CIRCUIT BREAKER < 20KA 1.755 KA				
MOLDED CASE CIRCUIT BREAKER > 20KA 1.755 KA				
CBL-11-EF-1560	BUS-11-EF-1560	0.004 KA	ANG:	101.55
CBL-DSC 11-EF-1560	MCC-6B	1.754 KA	ANG:	-11.95
DSC 11-PMP-2311	3P Duty: 1.371 KA AT -10.96 DEG (1.14 MVA) X/R: 0.19			
VOLTAGE: 480. EQUIV. IMPEDANCE= 0.1985 + J 0.0384 OHMS				

THREE PHASE FAULT REPORT
(FOR APPLICATION OF LOW VOLTAGE BREAKERS)
PRE FAULT VOLTAGE: 1.0000
MODEL TRANSFORMER TAPS: NO

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LOW VOLTAGE POWER CIRCUIT BREAKER	1.371 KA				
MOLDED CASE CIRCUIT BREAKER < 10KA	1.371 KA				
MOLDED CASE CIRCUIT BREAKER < 20KA	1.371 KA				
MOLDED CASE CIRCUIT BREAKER > 20KA	1.371 KA				
CBL-11-PMP-2311	BUS-11-PMP-2311	0.035 KA	ANG:	101.66	
CBL-DSC 11-PMP-2311	MCC-6A	1.357 KA	ANG:	-9.59	
DSC 11-PMP-2314 3P Duty: 1.760 KA AT -11.80 DEG (1.46 MVA) X/R: 0.21					
VOLTAGE: 480. EQUIV. IMPEDANCE= 0.1541 + J 0.0322 OHMS					
LOW VOLTAGE POWER CIRCUIT BREAKER	1.760 KA				
MOLDED CASE CIRCUIT BREAKER < 10KA	1.760 KA				
MOLDED CASE CIRCUIT BREAKER < 20KA	1.760 KA				
MOLDED CASE CIRCUIT BREAKER > 20KA	1.760 KA				
CBL-11-PMP-2314	BUS-11-PMP-2314	0.007 KA	ANG:	101.56	
CBL-DSC 11-PMP-2314	MCC-6A	1.757 KA	ANG:	-11.59	
DSC 11-PMP-2322 3P Duty: 1.369 KA AT -11.25 DEG (1.14 MVA) X/R: 0.19					
VOLTAGE: 480. EQUIV. IMPEDANCE= 0.1986 + J 0.0395 OHMS					
LOW VOLTAGE POWER CIRCUIT BREAKER	1.369 KA				
MOLDED CASE CIRCUIT BREAKER < 10KA	1.369 KA				
MOLDED CASE CIRCUIT BREAKER < 20KA	1.369 KA				
MOLDED CASE CIRCUIT BREAKER > 20KA	1.369 KA				
CBL-11-PMP-2322	BUS-11-PMP-2322	0.035 KA	ANG:	101.66	
CBL-DSC 11-PMP-2322	MCC-6B	1.355 KA	ANG:	-9.87	
DSC 11-PMP-2324 3P Duty: 2.128 KA AT -14.09 DEG (1.77 MVA) X/R: 0.25					
VOLTAGE: 480. EQUIV. IMPEDANCE= 0.1263 + J 0.0317 OHMS					
LOW VOLTAGE POWER CIRCUIT BREAKER	2.128 KA				
MOLDED CASE CIRCUIT BREAKER < 10KA	2.128 KA				
MOLDED CASE CIRCUIT BREAKER < 20KA	2.128 KA				
MOLDED CASE CIRCUIT BREAKER > 20KA	2.128 KA				
CBL-11-PMP-2324	BUS-11-PMP-2324	0.007 KA	ANG:	101.56	
CBL-DSC 11-PMP-2324	MCC-6B	2.125 KA	ANG:	-13.91	
DSC 11-PMP-2610 3P Duty: 1.361 KA AT -9.85 DEG (1.13 MVA) X/R: 0.17					
VOLTAGE: 480. EQUIV. IMPEDANCE= 0.2007 + J 0.0349 OHMS					
LOW VOLTAGE POWER CIRCUIT BREAKER	1.361 KA				
MOLDED CASE CIRCUIT BREAKER < 10KA	1.361 KA				
MOLDED CASE CIRCUIT BREAKER < 20KA	1.361 KA				
MOLDED CASE CIRCUIT BREAKER > 20KA	1.361 KA				
CBL-11-PMP-2610	BUS-11-PMP-2610	0.007 KA	ANG:	101.56	
CBL-DSC 11-PMP-2610	MCC-6A	1.358 KA	ANG:	-9.58	
DSC 11-PMP-2620 3P Duty: 1.358 KA AT -10.14 DEG (1.13 MVA) X/R: 0.18					

THREE PHASE FAULT REPORT
(FOR APPLICATION OF LOW VOLTAGE BREAKERS)
PRE FAULT VOLTAGE: 1.0000
MODEL TRANSFORMER TAPS: NO

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VOLTAGE:	480.	EQUIV. IMPEDANCE=	0.2008 + J 0.0359	OHMS	
LOW VOLTAGE POWER CIRCUIT BREAKER	1.358 KA				
MOLDED CASE CIRCUIT BREAKER < 10KA	1.358 KA				
MOLDED CASE CIRCUIT BREAKER < 20KA	1.358 KA				
MOLDED CASE CIRCUIT BREAKER > 20KA	1.358 KA				
CBL-11-PMP-2620	BUS-11-PMP-2620	0.007 KA	ANG:	101.56	
CBL-DSC 11-PMP-2620	MCC-6B	1.356 KA	ANG:	-9.86	
DSC 11-PMP-2711 3P Duty: 1.362 KA AT -9.99 DEG (1.13 MVA) X/R: 0.17					
VOLTAGE:	480.	EQUIV. IMPEDANCE=	0.2004 + J 0.0353	OHMS	
LOW VOLTAGE POWER CIRCUIT BREAKER	1.362 KA				
MOLDED CASE CIRCUIT BREAKER < 10KA	1.362 KA				
MOLDED CASE CIRCUIT BREAKER < 20KA	1.362 KA				
MOLDED CASE CIRCUIT BREAKER > 20KA	1.362 KA				
CBL-11-PMP-2711	BUS-11-PMP-2711	0.011 KA	ANG:	101.57	
CBL-DSC 11-PMP-2711	MCC-6A	1.358 KA	ANG:	-9.58	
DSC 11-PMP-2721 3P Duty: 1.661 KA AT -11.78 DEG (1.38 MVA) X/R: 0.21					
VOLTAGE:	480.	EQUIV. IMPEDANCE=	0.1634 + J 0.0341	OHMS	
LOW VOLTAGE POWER CIRCUIT BREAKER	1.661 KA				
MOLDED CASE CIRCUIT BREAKER < 10KA	1.661 KA				
MOLDED CASE CIRCUIT BREAKER < 20KA	1.661 KA				
MOLDED CASE CIRCUIT BREAKER > 20KA	1.661 KA				
CBL-11-PMP-2721	BUS-11-PMP-2721	0.011 KA	ANG:	101.57	
CBL-DSC 11-PMP-2721	MCC-6B	1.656 KA	ANG:	-11.44	
DSC 11-SF-1510 3P Duty: 1.761 KA AT -11.90 DEG (1.46 MVA) X/R: 0.21					
VOLTAGE:	480.	EQUIV. IMPEDANCE=	0.1539 + J 0.0325	OHMS	
LOW VOLTAGE POWER CIRCUIT BREAKER	1.761 KA				
MOLDED CASE CIRCUIT BREAKER < 10KA	1.761 KA				
MOLDED CASE CIRCUIT BREAKER < 20KA	1.761 KA				
MOLDED CASE CIRCUIT BREAKER > 20KA	1.761 KA				
CBL-11-SF-1510	BUS-11-SF-1510	0.011 KA	ANG:	101.57	
CBL-DSC 11-SF-1510	MCC-6A	1.757 KA	ANG:	-11.59	
DSC 11-SF-1520 3P Duty: 2.130 KA AT -14.17 DEG (1.77 MVA) X/R: 0.25					
VOLTAGE:	480.	EQUIV. IMPEDANCE=	0.1262 + J 0.0319	OHMS	
LOW VOLTAGE POWER CIRCUIT BREAKER	2.130 KA				
MOLDED CASE CIRCUIT BREAKER < 10KA	2.130 KA				
MOLDED CASE CIRCUIT BREAKER < 20KA	2.130 KA				
MOLDED CASE CIRCUIT BREAKER > 20KA	2.130 KA				
CBL-11-SF-1520	BUS-11-SF-1520	0.011 KA	ANG:	101.57	
CBL-DSC 11-SF-1520	MCC-6B	2.125 KA	ANG:	-13.92	

THREE PHASE FAULT REPORT
(FOR APPLICATION OF LOW VOLTAGE BREAKERS)
PRE FAULT VOLTAGE: 1.0000
MODEL TRANSFORMER TAPS: NO

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DSC 11-SF-1550      3P Duty: 1.756 KA AT -12.16 DEG ( 1.46 MVA) X/R: 0.22
VOLTAGE: 480. EQUIV. IMPEDANCE= 0.1543 + J 0.0332 OHMS
LOW VOLTAGE POWER CIRCUIT BREAKER 1.756 KA
MOLDED CASE CIRCUIT BREAKER < 10KA 1.756 KA
MOLDED CASE CIRCUIT BREAKER < 20KA 1.756 KA
MOLDED CASE CIRCUIT BREAKER > 20KA 1.756 KA
CBL-11-SF-1550      BUS-11-SF-1550      0.007 KA      ANG: 101.56
CBL-DSC 11-SF-1550      MCC-6B      1.753 KA      ANG: -11.95

DSC 11-SF-1575      3P Duty: 1.759 KA AT -11.69 DEG ( 1.46 MVA) X/R: 0.20
VOLTAGE: 480. EQUIV. IMPEDANCE= 0.1543 + J 0.0319 OHMS
LOW VOLTAGE POWER CIRCUIT BREAKER 1.759 KA
MOLDED CASE CIRCUIT BREAKER < 10KA 1.759 KA
MOLDED CASE CIRCUIT BREAKER < 20KA 1.759 KA
MOLDED CASE CIRCUIT BREAKER > 20KA 1.759 KA
CBL-11-SF-1575      BUS-11-SF-1575      0.004 KA      ANG: 101.55
CBL-DSC 11-SF-1575      MCC-6A      1.757 KA      ANG: -11.58

DSC 12-AUG-4615      3P Duty: 2.637 KA AT -15.10 DEG ( 2.19 MVA) X/R: 0.26
VOLTAGE: 480. EQUIV. IMPEDANCE= 0.1014 + J 0.0274 OHMS
LOW VOLTAGE POWER CIRCUIT BREAKER 2.637 KA
MOLDED CASE CIRCUIT BREAKER < 10KA 2.637 KA
MOLDED CASE CIRCUIT BREAKER < 20KA 2.637 KA
MOLDED CASE CIRCUIT BREAKER > 20KA 2.637 KA
CBL-12-AUG-4615      BUS-12-AUG-4615      0.176 KA      ANG: 101.79
CBL-DSC 12-AUG-4615      MCC-8A      2.563 KA      ANG: -11.59

DSC 12-AUG-4617      3P Duty: 3.213 KA AT -16.58 DEG ( 2.67 MVA) X/R: 0.31
VOLTAGE: 480. EQUIV. IMPEDANCE= 0.0827 + J 0.0246 OHMS
LOW VOLTAGE POWER CIRCUIT BREAKER 3.213 KA
MOLDED CASE CIRCUIT BREAKER < 10KA 3.213 KA
MOLDED CASE CIRCUIT BREAKER < 20KA 3.213 KA
MOLDED CASE CIRCUIT BREAKER > 20KA 3.213 KA
CBL-12-AUG-4617      BUS-12-AUG-4617      0.176 KA      ANG: 101.79
CBL-DSC 12-AUG-4617      MCC-8B      3.133 KA      ANG: -13.75

DSC 12-CON-4610      3P Duty: 1.455 KA AT -8.33 DEG ( 1.21 MVA) X/R: 0.13
VOLTAGE: 480. EQUIV. IMPEDANCE= 0.1884 + J 0.0276 OHMS
LOW VOLTAGE POWER CIRCUIT BREAKER 1.455 KA
MOLDED CASE CIRCUIT BREAKER < 10KA 1.455 KA
MOLDED CASE CIRCUIT BREAKER < 20KA 1.455 KA
MOLDED CASE CIRCUIT BREAKER > 20KA 1.455 KA
CBL-12-CON-4610      BUS-12-CON-4610      0.053 KA      ANG: 101.72

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THREE PHASE FAULT REPORT
(FOR APPLICATION OF LOW VOLTAGE BREAKERS)
PRE FAULT VOLTAGE: 1.0000
MODEL TRANSFORMER TAPS: NO

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CONTRIBUTIONS TO DSC 12-CON-4610 (CONTINUED)
CBL-DSC 12-CON-4610 MCC-8A 1.438 KA ANG: -6.35

DSC 12-CON-4620 3P Duty: 1.797 KA AT -9.22 DEG ( 1.49 MVA) X/R: 0.15
VOLTAGE: 480. EQUIV. IMPEDANCE= 0.1522 + J 0.0247 OHMS
LOW VOLTAGE POWER CIRCUIT BREAKER 1.797 KA
MOLDED CASE CIRCUIT BREAKER < 10KA 1.797 KA
MOLDED CASE CIRCUIT BREAKER < 20KA 1.797 KA
MOLDED CASE CIRCUIT BREAKER > 20KA 1.797 KA
CBL-12-CON-4620 BUS-12-CON-4620 0.053 KA ANG: 101.72
CBL-DSC 12-CON-4620 MCC-8B 1.779 KA ANG: -7.63

DSC 12-EF-2530 3P Duty: 4.154 KA AT -17.07 DEG ( 3.45 MVA) X/R: 0.31
VOLTAGE: 480. EQUIV. IMPEDANCE= 0.0638 + J 0.0196 OHMS
LOW VOLTAGE POWER CIRCUIT BREAKER 4.154 KA
MOLDED CASE CIRCUIT BREAKER < 10KA 4.154 KA
MOLDED CASE CIRCUIT BREAKER < 20KA 4.154 KA
MOLDED CASE CIRCUIT BREAKER > 20KA 4.154 KA
CBL-12-EF-2530 BUS-12-EF-2530 0.141 KA ANG: 101.74
CBL-DSC 12-EF-2530 MCC-8A 4.088 KA ANG: -15.34

DSC 12-EF-2540 3P Duty: 4.122 KA AT -17.97 DEG ( 3.43 MVA) X/R: 0.34
VOLTAGE: 480. EQUIV. IMPEDANCE= 0.0639 + J 0.0207 OHMS
LOW VOLTAGE POWER CIRCUIT BREAKER 4.122 KA
MOLDED CASE CIRCUIT BREAKER < 10KA 4.122 KA
MOLDED CASE CIRCUIT BREAKER < 20KA 4.122 KA
MOLDED CASE CIRCUIT BREAKER > 20KA 4.122 KA
CBL-12-EF-2540 BUS-12-EF-2540 0.141 KA ANG: 101.74
CBL-DSC 12-EF-2540 MCC-8B 4.054 KA ANG: -16.24

DSC 12-HPU-4750 3P Duty: 3.230 KA AT -15.88 DEG ( 2.68 MVA) X/R: 0.28
VOLTAGE: 480. EQUIV. IMPEDANCE= 0.0825 + J 0.0235 OHMS
LOW VOLTAGE POWER CIRCUIT BREAKER 3.230 KA
MOLDED CASE CIRCUIT BREAKER < 10KA 3.230 KA
MOLDED CASE CIRCUIT BREAKER < 20KA 3.230 KA
MOLDED CASE CIRCUIT BREAKER > 20KA 3.230 KA
CBL-12-HPU-4750 BUS-12-HPU-4750 0.176 KA ANG: 101.79
CBL-DSC 12-HPU-4750 MCC-8A 3.152 KA ANG: -13.04

DSC 12-SF-2510 3P Duty: 4.154 KA AT -17.07 DEG ( 3.45 MVA) X/R: 0.31
VOLTAGE: 480. EQUIV. IMPEDANCE= 0.0638 + J 0.0196 OHMS

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THREE PHASE FAULT REPORT
(FOR APPLICATION OF LOW VOLTAGE BREAKERS)
PRE FAULT VOLTAGE: 1.0000
MODEL TRANSFORMER TAPS: NO

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	LOW VOLTAGE POWER CIRCUIT BREAKER	4.154 KA			
	MOLDED CASE CIRCUIT BREAKER < 10KA	4.154 KA			
	MOLDED CASE CIRCUIT BREAKER < 20KA	4.154 KA			
	MOLDED CASE CIRCUIT BREAKER > 20KA	4.154 KA			
	CBL-12-SF-2510	BUS-12-SF-2510	0.141 KA	ANG:	101.74
	CBL-DSC 12-SF-2510	MCC-8A	4.088 KA	ANG:	-15.34
DSC 12-SF-2520	3P Duty: 4.122 KA AT -17.97 DEG (3.43 MVA) X/R: 0.34				
	VOLTAGE: 480. EQUIV. IMPEDANCE=	0.0639 + J 0.0207 OHMS			
	LOW VOLTAGE POWER CIRCUIT BREAKER	4.122 KA			
	MOLDED CASE CIRCUIT BREAKER < 10KA	4.122 KA			
	MOLDED CASE CIRCUIT BREAKER < 20KA	4.122 KA			
	MOLDED CASE CIRCUIT BREAKER > 20KA	4.122 KA			
	CBL-12-SF-2520	BUS-12-SF-2520	0.141 KA	ANG:	101.74
	CBL-DSC 12-SF-2520	MCC-8B	4.054 KA	ANG:	-16.24
DSC 12-SF-2550	3P Duty: 2.348 KA AT -8.64 DEG (1.95 MVA) X/R: 0.15				
	VOLTAGE: 480. EQUIV. IMPEDANCE=	0.1167 + J 0.0177 OHMS			
	LOW VOLTAGE POWER CIRCUIT BREAKER	2.348 KA			
	MOLDED CASE CIRCUIT BREAKER < 10KA	2.348 KA			
	MOLDED CASE CIRCUIT BREAKER < 20KA	2.348 KA			
	MOLDED CASE CIRCUIT BREAKER > 20KA	2.348 KA			
	CBL-12-SF-2550	BUS-12-SF-2550	0.002 KA	ANG:	101.54
	CBL-DSC 12-SF-2550	MCC-8A	2.348 KA	ANG:	-8.60
DSC 12-SF-2560	3P Duty: 1.156 KA AT -6.01 DEG (0.96 MVA) X/R: 0.10				
	VOLTAGE: 480. EQUIV. IMPEDANCE=	0.2384 + J 0.0251 OHMS			
	LOW VOLTAGE POWER CIRCUIT BREAKER	1.156 KA			
	MOLDED CASE CIRCUIT BREAKER < 10KA	1.156 KA			
	MOLDED CASE CIRCUIT BREAKER < 20KA	1.156 KA			
	MOLDED CASE CIRCUIT BREAKER > 20KA	1.156 KA			
	CBL-12-SF-2560	BUS-12-SF-2560	0.002 KA	ANG:	101.54
	CBL-DSC 12-SF-2560	MCC-8B	1.156 KA	ANG:	-5.92
DSC 12-SF-2570	3P Duty: 1.445 KA AT -7.15 DEG (1.20 MVA) X/R: 0.12				
	VOLTAGE: 480. EQUIV. IMPEDANCE=	0.1903 + J 0.0239 OHMS			
	LOW VOLTAGE POWER CIRCUIT BREAKER	1.445 KA			
	MOLDED CASE CIRCUIT BREAKER < 10KA	1.445 KA			
	MOLDED CASE CIRCUIT BREAKER < 20KA	1.445 KA			
	MOLDED CASE CIRCUIT BREAKER > 20KA	1.445 KA			
	CBL-12-SF-2570	BUS-12-SF-2570	0.021 KA	ANG:	101.61
	CBL-DSC 12-SF-2570	MCC-8A	1.438 KA	ANG:	-6.35
DSC 12-SF-2580	3P Duty: 1.162 KA AT -6.92 DEG (0.97 MVA) X/R: 0.11				

THREE PHASE FAULT REPORT
(FOR APPLICATION OF LOW VOLTAGE BREAKERS)
PRE FAULT VOLTAGE: 1.0000
MODEL TRANSFORMER TAPS: NO

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VOLTAGE:	480.	EQUIV. IMPEDANCE=	0.2367 + J 0.0287	OHMS	
LOW VOLTAGE POWER CIRCUIT BREAKER	1.162 KA				
MOLDED CASE CIRCUIT BREAKER < 10KA	1.162 KA				
MOLDED CASE CIRCUIT BREAKER < 20KA	1.162 KA				
MOLDED CASE CIRCUIT BREAKER > 20KA	1.162 KA				
CBL-12-SF-2580	BUS-12-SF-2580	0.021 KA	ANG:	101.61	
CBL-DSC 12-SF-2580	MCC-8B	1.156 KA	ANG:	-5.93	
DSC 12-SRW-4612 3P Duty: 1.445 KA AT -7.15 DEG (1.20 MVA) X/R: 0.12					
VOLTAGE:	480.	EQUIV. IMPEDANCE=	0.1903 + J 0.0239	OHMS	
LOW VOLTAGE POWER CIRCUIT BREAKER	1.445 KA				
MOLDED CASE CIRCUIT BREAKER < 10KA	1.445 KA				
MOLDED CASE CIRCUIT BREAKER < 20KA	1.445 KA				
MOLDED CASE CIRCUIT BREAKER > 20KA	1.445 KA				
CBL-12-SRW-4612	BUS-12-SRW-4612	0.021 KA	ANG:	101.61	
CBL-DSC 12-SRW-4612	MCC-8A	1.438 KA	ANG:	-6.35	
DSC 12-SRW-4613 3P Duty: 1.442 KA AT -7.48 DEG (1.20 MVA) X/R: 0.13					
VOLTAGE:	480.	EQUIV. IMPEDANCE=	0.1905 + J 0.0250	OHMS	
LOW VOLTAGE POWER CIRCUIT BREAKER	1.442 KA				
MOLDED CASE CIRCUIT BREAKER < 10KA	1.442 KA				
MOLDED CASE CIRCUIT BREAKER < 20KA	1.442 KA				
MOLDED CASE CIRCUIT BREAKER > 20KA	1.442 KA				
CBL-12-SRW-4613	BUS-12-SRW-4613	0.021 KA	ANG:	101.61	
CBL-DSC 12-SRW-4613	MCC-8B	1.435 KA	ANG:	-6.69	
DSC 19-PMP-1110 3P Duty: 1.920 KA AT -13.78 DEG (1.60 MVA) X/R: 0.23					
VOLTAGE:	480.	EQUIV. IMPEDANCE=	0.1402 + J 0.0344	OHMS	
LOW VOLTAGE POWER CIRCUIT BREAKER	1.920 KA				
MOLDED CASE CIRCUIT BREAKER < 10KA	1.920 KA				
MOLDED CASE CIRCUIT BREAKER < 20KA	1.920 KA				
MOLDED CASE CIRCUIT BREAKER > 20KA	1.920 KA				
CBL-19-PMP-1110	BUS-19-PMP-1110	0.141 KA	ANG:	101.74	
CBL-DSC 19-PMP-1110	MCC-8A	1.864 KA	ANG:	-9.86	
DSC 19-PMP-1120 3P Duty: 2.209 KA AT -14.45 DEG (1.84 MVA) X/R: 0.25					
VOLTAGE:	480.	EQUIV. IMPEDANCE=	0.1215 + J 0.0313	OHMS	
LOW VOLTAGE POWER CIRCUIT BREAKER	2.209 KA				
MOLDED CASE CIRCUIT BREAKER < 10KA	2.209 KA				
MOLDED CASE CIRCUIT BREAKER < 20KA	2.209 KA				
MOLDED CASE CIRCUIT BREAKER > 20KA	2.209 KA				
CBL-19-PMP-1120	BUS-19-PMP-1120	0.141 KA	ANG:	101.74	
CBL-DSC 19-PMP-1120	MCC-8B	2.151 KA	ANG:	-11.08	

THREE PHASE FAULT REPORT
(FOR APPLICATION OF LOW VOLTAGE BREAKERS)
PRE FAULT VOLTAGE: 1.0000
MODEL TRANSFORMER TAPS: NO

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DSC 21-EF-1250      3P Duty: 2.346 KA AT -8.20 DEG ( 1.95 MVA) X/R: 0.14
VOLTAGE: 480. EQUIV. IMPEDANCE= 0.1169 + J 0.0169 OHMS
LOW VOLTAGE POWER CIRCUIT BREAKER 2.346 KA
MOLDED CASE CIRCUIT BREAKER < 10KA 2.346 KA
MOLDED CASE CIRCUIT BREAKER < 20KA 2.346 KA
MOLDED CASE CIRCUIT BREAKER > 20KA 2.346 KA
CBL-21-EF-1250      BUS-21-EF-1250      0.004 KA      ANG: 101.55
CBL-DSC 21-EF-1250      MCC-9A      2.345 KA      ANG: -8.12

DSC 21-EF-1310      3P Duty: 1.158 KA AT -5.67 DEG ( 0.96 MVA) X/R: 0.09
VOLTAGE: 480. EQUIV. IMPEDANCE= 0.2381 + J 0.0236 OHMS
LOW VOLTAGE POWER CIRCUIT BREAKER 1.158 KA
MOLDED CASE CIRCUIT BREAKER < 10KA 1.158 KA
MOLDED CASE CIRCUIT BREAKER < 20KA 1.158 KA
MOLDED CASE CIRCUIT BREAKER > 20KA 1.158 KA
CBL-21-EF-1310      BUS-21-EF-1310      0.005 KA      ANG: 101.55
CBL-DSC 21-EF-1310      MCC-9A      1.157 KA      ANG: -5.42

DSC 2P-07      3P Duty: 0.949 KA AT -10.04 DEG ( 0.79 MVA) X/R: 0.14
VOLTAGE: 480. EQUIV. IMPEDANCE= 0.2875 + J 0.0509 OHMS
LOW VOLTAGE POWER CIRCUIT BREAKER 0.949 KA
MOLDED CASE CIRCUIT BREAKER < 10KA 0.949 KA
MOLDED CASE CIRCUIT BREAKER < 20KA 0.949 KA
MOLDED CASE CIRCUIT BREAKER > 20KA 0.949 KA
CBL-2P-07      BUS-2P-07      0.070 KA      ANG: -78.23
CBL-DSC 2P-07      MCC-5B      0.925 KA      ANG: -5.99

DSC 3G-01      3P Duty: 0.628 KA AT -6.06 DEG ( 0.52 MVA) X/R: 0.08
VOLTAGE: 480. EQUIV. IMPEDANCE= 0.4387 + J 0.0466 OHMS
LOW VOLTAGE POWER CIRCUIT BREAKER 0.628 KA
MOLDED CASE CIRCUIT BREAKER < 10KA 0.628 KA
MOLDED CASE CIRCUIT BREAKER < 20KA 0.628 KA
MOLDED CASE CIRCUIT BREAKER > 20KA 0.628 KA
CBL-3G-01      BUS-3G-01      0.021 KA      ANG: 101.61
CBL-DSC 3G-01      MCC-9A      0.622 KA      ANG: -4.20

DSC 3G-02      3P Duty: 0.735 KA AT -6.20 DEG ( 0.61 MVA) X/R: 0.09
VOLTAGE: 480. EQUIV. IMPEDANCE= 0.3748 + J 0.0407 OHMS
LOW VOLTAGE POWER CIRCUIT BREAKER 0.735 KA
MOLDED CASE CIRCUIT BREAKER < 10KA 0.735 KA
MOLDED CASE CIRCUIT BREAKER < 20KA 0.735 KA
MOLDED CASE CIRCUIT BREAKER > 20KA 0.735 KA
CBL-3G-02      BUS-3G-02      0.021 KA      ANG: 101.61
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THREE PHASE FAULT REPORT
(FOR APPLICATION OF LOW VOLTAGE BREAKERS)
PRE FAULT VOLTAGE: 1.0000
MODEL TRANSFORMER TAPS: NO

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	CONTRIBUTIONS TO DSC 3G-02	(CONTINUED)			
	CBL-DSC 3G-02	MCC-9B	0.729 KA	ANG:	-4.61
DSC 3G-03	3P Duty: 0.769 KA AT -6.04 DEG (0.64 MVA) X/R: 0.09				
	VOLTAGE: 480. EQUIV. IMPEDANCE= 0.3582 + J 0.0379 OHMS				
	LOW VOLTAGE POWER CIRCUIT BREAKER 0.769 KA				
	MOLDED CASE CIRCUIT BREAKER < 10KA 0.769 KA				
	MOLDED CASE CIRCUIT BREAKER < 20KA 0.769 KA				
	MOLDED CASE CIRCUIT BREAKER > 20KA 0.769 KA				
	CBL-3G-03	BUS-3G-03	0.021 KA	ANG:	101.61
	CBL-DSC 3G-03	MCC-9A	0.763 KA	ANG:	-4.52
DSC 3P-05	3P Duty: 0.971 KA AT -5.59 DEG (0.81 MVA) X/R: 0.09				
	VOLTAGE: 480. EQUIV. IMPEDANCE= 0.2841 + J 0.0278 OHMS				
	LOW VOLTAGE POWER CIRCUIT BREAKER 0.971 KA				
	MOLDED CASE CIRCUIT BREAKER < 10KA 0.971 KA				
	MOLDED CASE CIRCUIT BREAKER < 20KA 0.971 KA				
	MOLDED CASE CIRCUIT BREAKER > 20KA 0.971 KA				
	CBL-3P-05	BUS-3P-05	0.011 KA	ANG:	101.57
	CBL-DSC 3P-05	MCC-9A	0.968 KA	ANG:	-4.99
DSC 3P-07	3P Duty: 1.075 KA AT -6.01 DEG (0.89 MVA) X/R: 0.10				
	VOLTAGE: 480. EQUIV. IMPEDANCE= 0.2565 + J 0.0270 OHMS				
	LOW VOLTAGE POWER CIRCUIT BREAKER 1.075 KA				
	MOLDED CASE CIRCUIT BREAKER < 10KA 1.075 KA				
	MOLDED CASE CIRCUIT BREAKER < 20KA 1.075 KA				
	MOLDED CASE CIRCUIT BREAKER > 20KA 1.075 KA				
	CBL-3P-07	BUS-3P-07	0.011 KA	ANG:	101.57
	CBL-DSC 3P-07	MCC-9B	1.071 KA	ANG:	-5.47
DSC 4P-01	3P Duty: 2.195 KA AT -22.74 DEG (1.82 MVA) X/R: 0.42				
	VOLTAGE: 480. EQUIV. IMPEDANCE= 0.1165 + J 0.0488 OHMS				
	LOW VOLTAGE POWER CIRCUIT BREAKER 2.195 KA				
	MOLDED CASE CIRCUIT BREAKER < 10KA 2.195 KA				
	MOLDED CASE CIRCUIT BREAKER < 20KA 2.195 KA				
	MOLDED CASE CIRCUIT BREAKER > 20KA 2.195 KA				
	CBL-4P-01	BUS-4P-01	0.282 KA	ANG:	101.79
	CBL-DSC 4P-01	MCC-10A	2.048 KA	ANG:	-16.24
DSC 4P-02	3P Duty: 2.218 KA AT -24.73 DEG (1.84 MVA) X/R: 0.52				
	VOLTAGE: 480. EQUIV. IMPEDANCE= 0.1135 + J 0.0523 OHMS				

THREE PHASE FAULT REPORT
(FOR APPLICATION OF LOW VOLTAGE BREAKERS)
PRE FAULT VOLTAGE: 1.0000
MODEL TRANSFORMER TAPS: NO

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	LOW VOLTAGE POWER CIRCUIT BREAKER	2.218 KA			
	MOLDED CASE CIRCUIT BREAKER < 10KA	2.218 KA			
	MOLDED CASE CIRCUIT BREAKER < 20KA	2.218 KA			
	MOLDED CASE CIRCUIT BREAKER > 20KA	2.218 KA			
	CBL-4P-02	BUS-4P-02	0.282 KA	ANG:	101.79
	CBL-DSC 4P-02	MCC-10B	2.063 KA	ANG:	-18.43
DSC 4P-03	3P Duty: 2.275 KA AT -22.83 DEG (1.89 MVA) X/R: 0.42				
	VOLTAGE: 480. EQUIV. IMPEDANCE=	0.1122 + J 0.0473 OHMS			
	LOW VOLTAGE POWER CIRCUIT BREAKER	2.275 KA			
	MOLDED CASE CIRCUIT BREAKER < 10KA	2.275 KA			
	MOLDED CASE CIRCUIT BREAKER < 20KA	2.275 KA			
	MOLDED CASE CIRCUIT BREAKER > 20KA	2.275 KA			
	CBL-4P-03	BUS-4P-03	0.282 KA	ANG:	101.79
	CBL-DSC 4P-03	MCC-10A	2.128 KA	ANG:	-16.57
DSC SHB CRANE	3P Duty: 3.136 KA AT -13.68 DEG (2.61 MVA) X/R: 0.24				
	VOLTAGE: 480. EQUIV. IMPEDANCE=	0.0859 + J 0.0209 OHMS			
	LOW VOLTAGE POWER CIRCUIT BREAKER	3.136 KA			
	MOLDED CASE CIRCUIT BREAKER < 10KA	3.136 KA			
	MOLDED CASE CIRCUIT BREAKER < 20KA	3.136 KA			
	MOLDED CASE CIRCUIT BREAKER > 20KA	3.136 KA			
	CBL-DSC SHB CRANE	MCC-8B	3.136 KA	ANG:	-13.68
GEN SWBD [Ex]	3P Duty: 15.680 KA AT -86.61 DEG (13.04 MVA) X/R: 17.23				
	VOLTAGE: 480. EQUIV. IMPEDANCE=	0.0010 + J 0.0176 OHMS			
	LOW VOLTAGE POWER CIRCUIT BREAKER	17.735 KA			
	MOLDED CASE CIRCUIT BREAKER > 20KA	18.830 KA			
	CBL-GEN SWBD-2	21-GEN-1200	9.728 KA	ANG:	-266.23
	CBL-GEN SWBD-1 [Ex]	21-GEN-1100 [Ex]	5.952 KA	ANG:	92.78
LP-10	3P Duty: 1.679 KA AT -42.63 DEG (0.60 MVA) X/R: 0.92				
	VOLTAGE: 208. EQUIV. IMPEDANCE=	0.0526 + J 0.0484 OHMS			
	LOW VOLTAGE POWER CIRCUIT BREAKER	1.679 KA			
	MOLDED CASE CIRCUIT BREAKER < 10KA	1.679 KA			
	MOLDED CASE CIRCUIT BREAKER < 20KA	1.679 KA			
	MOLDED CASE CIRCUIT BREAKER > 20KA	1.679 KA			
	CBL-LP-10	BUS-Sec-XFMR-10	1.679 KA	ANG:	-42.63
LP-11	3P Duty: 1.477 KA AT -41.60 DEG (0.53 MVA) X/R: 0.90				
	VOLTAGE: 208. EQUIV. IMPEDANCE=	0.0608 + J 0.0540 OHMS			

THREE PHASE FAULT REPORT
(FOR APPLICATION OF LOW VOLTAGE BREAKERS)
PRE FAULT VOLTAGE: 1.0000
MODEL TRANSFORMER TAPS: NO

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	LOW VOLTAGE POWER CIRCUIT BREAKER	1.477 KA			
	MOLDED CASE CIRCUIT BREAKER < 10KA	1.477 KA			
	MOLDED CASE CIRCUIT BREAKER < 20KA	1.477 KA			
	MOLDED CASE CIRCUIT BREAKER > 20KA	1.477 KA			
	CBL-LP-11	BUS-Sec-XFMR-11	1.477 KA	ANG:	138.40
LP-13A	3P Duty: 4.119 KA AT -56.88 DEG (1.48 MVA) X/R: 1.53				
	VOLTAGE: 208. EQUIV. IMPEDANCE=	0.0159 + J 0.0244 OHMS			
	LOW VOLTAGE POWER CIRCUIT BREAKER	4.119 KA			
	MOLDED CASE CIRCUIT BREAKER < 10KA	4.119 KA			
	MOLDED CASE CIRCUIT BREAKER < 20KA	4.119 KA			
	MOLDED CASE CIRCUIT BREAKER > 20KA	4.119 KA			
	CBL-LP-13A	BUS-Sec-XFMR-13	4.119 KA	ANG:	-56.88
LP-13B	3P Duty: 3.902 KA AT -54.33 DEG (1.41 MVA) X/R: 1.39				
	VOLTAGE: 208. EQUIV. IMPEDANCE=	0.0179 + J 0.0250 OHMS			
	LOW VOLTAGE POWER CIRCUIT BREAKER	3.902 KA			
	MOLDED CASE CIRCUIT BREAKER < 10KA	3.902 KA			
	MOLDED CASE CIRCUIT BREAKER < 20KA	3.902 KA			
	MOLDED CASE CIRCUIT BREAKER > 20KA	3.902 KA			
	CBL-LP-13B	LP-13A	3.902 KA	ANG:	-54.33
LP-4PA	3P Duty: 0.273 KA AT -49.25 DEG (0.11 MVA) X/R: 1.16				
	VOLTAGE: 240. EQUIV. IMPEDANCE=	0.3313 + J 0.3844 OHMS			
	LOW VOLTAGE POWER CIRCUIT BREAKER	0.273 KA			
	MOLDED CASE CIRCUIT BREAKER < 10KA	0.273 KA			
	MOLDED CASE CIRCUIT BREAKER < 20KA	0.273 KA			
	MOLDED CASE CIRCUIT BREAKER > 20KA	0.273 KA			
	CBL-LP-4PA	BUS-Sec-XFMR-4PA	0.273 KA	ANG:	-229.25
LP-5	3P Duty: 4.129 KA AT -56.52 DEG (1.49 MVA) X/R: 1.51				
	VOLTAGE: 208. EQUIV. IMPEDANCE=	0.0160 + J 0.0243 OHMS			
	LOW VOLTAGE POWER CIRCUIT BREAKER	4.129 KA			
	MOLDED CASE CIRCUIT BREAKER < 10KA	4.129 KA			
	MOLDED CASE CIRCUIT BREAKER < 20KA	4.129 KA			
	MOLDED CASE CIRCUIT BREAKER > 20KA	4.129 KA			
	CBL-LP-5	BUS-Sec-XFMR-5	4.129 KA	ANG:	-56.52
LP-6	3P Duty: 1.694 KA AT -42.85 DEG (0.61 MVA) X/R: 0.93				
	VOLTAGE: 208. EQUIV. IMPEDANCE=	0.0520 + J 0.0482 OHMS			
	LOW VOLTAGE POWER CIRCUIT BREAKER	1.694 KA			
	MOLDED CASE CIRCUIT BREAKER < 10KA	1.694 KA			
	MOLDED CASE CIRCUIT BREAKER < 20KA	1.694 KA			
	MOLDED CASE CIRCUIT BREAKER > 20KA	1.694 KA			

THREE PHASE FAULT REPORT
(FOR APPLICATION OF LOW VOLTAGE BREAKERS)
PRE FAULT VOLTAGE: 1.0000
MODEL TRANSFORMER TAPS: NO

CONTRIBUTIONS TO LP-6		(CONTINUED)		
CBL-LP-6	BUS-Sec-XFMR-6	1.694 KA	ANG:	-42.85
LP-8	3P Duty: 4.235 KA AT -56.68 DEG (1.53 MVA) X/R: 1.52			
	VOLTAGE: 208. EQUIV. IMPEDANCE= 0.0156 + J 0.0237 OHMS			
	LOW VOLTAGE POWER CIRCUIT BREAKER 4.235 KA			
	MOLDED CASE CIRCUIT BREAKER < 10KA 4.235 KA			
	MOLDED CASE CIRCUIT BREAKER < 20KA 4.235 KA			
	MOLDED CASE CIRCUIT BREAKER > 20KA 4.235 KA			
CBL-LP-8	BUS-Sec-XFMR-8	4.235 KA	ANG:	-56.68
LP-9	3P Duty: 3.212 KA AT -40.69 DEG (1.16 MVA) X/R: 0.86			
	VOLTAGE: 208. EQUIV. IMPEDANCE= 0.0283 + J 0.0244 OHMS			
	LOW VOLTAGE POWER CIRCUIT BREAKER 3.212 KA			
	MOLDED CASE CIRCUIT BREAKER < 10KA 3.212 KA			
	MOLDED CASE CIRCUIT BREAKER < 20KA 3.212 KA			
	MOLDED CASE CIRCUIT BREAKER > 20KA 3.212 KA			
CBL-LP-9	BUS-Sec-XFMR-9	3.212 KA	ANG:	-40.69
MCC-10A	3P Duty: 10.498 KA AT -56.33 DEG (8.73 MVA) X/R: 1.47			
	VOLTAGE: 480. EQUIV. IMPEDANCE= 0.0146 + J 0.0220 OHMS			
	LOW VOLTAGE POWER CIRCUIT BREAKER 10.498 KA			
	MOLDED CASE CIRCUIT BREAKER < 20KA 10.498 KA			
	MOLDED CASE CIRCUIT BREAKER > 20KA 10.498 KA			
CBL-DSC 06-PMP-1515	DSC 06-PMP-1515	0.070 KA	ANG:	102.77
CBL-DSC 06-PMP-1535	DSC 06-PMP-1535	0.069 KA	ANG:	104.44
CBL-DSC 4P-01	DSC 4P-01	0.270 KA	ANG:	107.89
CBL-DSC 4P-03	DSC 4P-03	0.270 KA	ANG:	107.63
PI-0019	22-VFD-2203	0.140 KA	ANG:	103.34
CBL-DSC 06-CLA-1530	DSC 06-CLA-1530	0.007 KA	ANG:	101.97
PI-0014	06-VFD-1610	0.104 KA	ANG:	104.00
PI-0015	05-VFD-1630	0.102 KA	ANG:	107.56
PI-0016	06-VFD-1630	0.102 KA	ANG:	107.56
PI-0017	05-VFD-1510	1.331 KA	ANG:	-197.82
PI-0018	22-VFD-2201	0.140 KA	ANG:	102.89
CBL-DSC 06-CLA-1510	DSC 06-CLA-1510	0.007 KA	ANG:	101.72
CBL-MCC-10A	SWBD-MAIN-A	8.248 KA	ANG:	-59.38
MCC-10B	3P Duty: 9.098 KA AT -61.21 DEG (7.56 MVA) X/R: 1.90			
	VOLTAGE: 480. EQUIV. IMPEDANCE= 0.0147 + J 0.0267 OHMS			
	LOW VOLTAGE POWER CIRCUIT BREAKER 9.098 KA			
	MOLDED CASE CIRCUIT BREAKER < 10KA 9.320 KA			
	MOLDED CASE CIRCUIT BREAKER < 20KA 9.098 KA			
	MOLDED CASE CIRCUIT BREAKER > 20KA 9.098 KA			

THREE PHASE FAULT REPORT
(FOR APPLICATION OF LOW VOLTAGE BREAKERS)
PRE FAULT VOLTAGE: 1.0000
MODEL TRANSFORMER TAPS: NO

CONTRIBUTIONS TO MCC-10B		(CONTINUED)	
CBL-DSC 06-CLA-1520	DSC 06-CLA-1520	0.007 KA	ANG: 101.84
CBL-DSC 06-PMP-1525	DSC 06-PMP-1525	0.070 KA	ANG: 103.61
CBL-DSC 4P-02	DSC 4P-02	0.270 KA	ANG: 107.76
PI-0025	22-VFD-2022	0.140 KA	ANG: 103.23
PI-0021	06-VFD-1615	0.104 KA	ANG: 104.00
PI-0022	06-VFD-1625	0.103 KA	ANG: 105.80
PI-0023	06-VFD-1635	0.103 KA	ANG: 106.39
PI-0024	05-VFD-1520	0.052 KA	ANG: 103.50
CBL-DSC 04-PMP-1140	DSC 04-PMP-1140	0.271 KA	ANG: 107.37
CBL-MCC-10B	SWBD-MAIN-B	8.010 KA	ANG: -59.43
MCC-11A			
3P Duty: 9.178 KA AT -52.73 DEG (7.63 MVA) X/R: 1.36			
VOLTAGE: 480. EQUIV. IMPEDANCE= 0.0183 + J 0.0240 OHMS			
LOW VOLTAGE POWER CIRCUIT BREAKER 9.178 KA			
MOLDED CASE CIRCUIT BREAKER < 10KA 9.178 KA			
MOLDED CASE CIRCUIT BREAKER < 20KA 9.178 KA			
MOLDED CASE CIRCUIT BREAKER > 20KA 9.178 KA			
PI-0026	09-VFD-1100	0.514 KA	ANG: 104.24
CBL-DSC 09-MIX-1110	DSC 09-MIX-1110	0.007 KA	ANG: 102.02
CBL-DSC 08-EF-1230	DSC 08-EF-1230	0.002 KA	ANG: 101.56
CBL-MCC-11A	SWBD-MAIN-A	8.700 KA	ANG: -51.38
MCC-11B			
3P Duty: 4.012 KA AT -49.95 DEG (3.34 MVA) X/R: 1.33			
VOLTAGE: 480. EQUIV. IMPEDANCE= 0.0445 + J 0.0529 OHMS			
LOW VOLTAGE POWER CIRCUIT BREAKER 4.012 KA			
MOLDED CASE CIRCUIT BREAKER < 10KA 4.012 KA			
MOLDED CASE CIRCUIT BREAKER < 20KA 4.012 KA			
MOLDED CASE CIRCUIT BREAKER > 20KA 4.012 KA			
PI-0028	09-VFD-1200	0.514 KA	ANG: 104.24
CBL-DSC 09-PMP-903	DSC 09-PMP-903	0.007 KA	ANG: 101.94
CBL-MCC-11B	SWBD-MAIN-B	3.550 KA	ANG: -46.28
MCC-5A			
3P Duty: 15.588 KA AT -63.97 DEG (12.96 MVA) X/R: 2.05			
VOLTAGE: 480. EQUIV. IMPEDANCE= 0.0078 + J 0.0160 OHMS			
LOW VOLTAGE POWER CIRCUIT BREAKER 15.588 KA			
MOLDED CASE CIRCUIT BREAKER < 20KA 15.588 KA			
MOLDED CASE CIRCUIT BREAKER > 20KA 15.588 KA			
CBL-02-MIX-1210	BUS-02-MIX-1210	0.014 KA	ANG: 102.06
CBL-02-PMP-1211	BUS-02-PMP-1211	0.105 KA	ANG: 103.23
CBL-02-CON-1400	BUS-02-CON-1400	0.052 KA	ANG: 103.39
CBL-DSC 02-EF-1330	DSC 02-EF-1330	0.035 KA	ANG: 103.30
CBL-DSC 02-EF-1370	DSC 02-EF-1370	0.014 KA	ANG: 102.11
CBL-DSC 02-EF-1390	DSC 02-EF-1390	0.052 KA	ANG: 103.19

THREE PHASE FAULT REPORT
(FOR APPLICATION OF LOW VOLTAGE BREAKERS)
PRE FAULT VOLTAGE: 1.0000
MODEL TRANSFORMER TAPS: NO

CONTRIBUTIONS TO MCC-5A		(CONTINUED)		
CBL-DSC 02-SF-1350	DSC 02-SF-1350	0.052 KA	ANG:	104.16
CBL-DSC 02-SF-1570	DSC 02-SF-1570	0.004 KA	ANG:	101.68
CBL-MCC-5A	SWBD-MAIN-A	15.268 KA	ANG:	-63.70
MCC-5B				
3P Duty: 14.721 KA AT -64.97 DEG (12.24 MVA) X/R: 2.16				
VOLTAGE: 480. EQUIV. IMPEDANCE= 0.0080 + J 0.0171 OHMS				
LOW VOLTAGE POWER CIRCUIT BREAKER 14.721 KA				
MOLDED CASE CIRCUIT BREAKER < 20KA 14.721 KA				
MOLDED CASE CIRCUIT BREAKER > 20KA 14.721 KA				
CBL-02-PMP-1221	BUS-02-PMP-1221	0.105 KA	ANG:	103.23
CBL-02-MIX-1220	BUS-02-MIX-1220	0.014 KA	ANG:	102.06
CBL-DSC 02-EF-1320	DSC 02-EF-1320	0.014 KA	ANG:	102.38
CBL-DSC 02-EF-1340	DSC 02-EF-1340	0.035 KA	ANG:	103.30
CBL-DSC 02-EF-1380	DSC 02-EF-1380	0.014 KA	ANG:	102.24
CBL-DSC 02-EF-1396	DSC 02-EF-1396	0.032 KA	ANG:	102.72
CBL-DSC 02-SF-1310	DSC 02-SF-1310	0.014 KA	ANG:	102.24
CBL-DSC 02-SF-1360	DSC 02-SF-1360	0.052 KA	ANG:	104.16
CBL-DSC 02-SF-1575	DSC 02-SF-1575	0.004 KA	ANG:	101.69
CBL-DSC 02-SF-1810	DSC 02-SF-1810	0.004 KA	ANG:	101.68
CBL-DSC 03-EF-1420	DSC 03-EF-1420	0.014 KA	ANG:	101.98
CBL-DSC 03-SF-1410	DSC 03-SF-1410	0.011 KA	ANG:	102.36
CBL-DSC 03-SF-1575	DSC 03-SF-1575	0.004 KA	ANG:	101.81
CBL-DSC 2P-07	DSC 2P-07	0.069 KA	ANG:	105.79
CBL-MCC-5B	SWBD-MAIN-B	14.345 KA	ANG:	-64.66
MCC-6A				
3P Duty: 9.684 KA AT -60.20 DEG (8.05 MVA) X/R: 1.79				
VOLTAGE: 480. EQUIV. IMPEDANCE= 0.0142 + J 0.0248 OHMS				
LOW VOLTAGE POWER CIRCUIT BREAKER 9.684 KA				
MOLDED CASE CIRCUIT BREAKER < 10KA 9.773 KA				
MOLDED CASE CIRCUIT BREAKER < 20KA 9.684 KA				
MOLDED CASE CIRCUIT BREAKER > 20KA 9.684 KA				
CBL-DSC 11-PMP-2314	DSC 11-PMP-2314	0.007 KA	ANG:	101.76
CBL-DSC 11-PMP-2610	DSC 11-PMP-2610	0.007 KA	ANG:	101.83
CBL-DSC 11-PMP-2711	DSC 11-PMP-2711	0.011 KA	ANG:	101.97
CBL-DSC 11-SF-1510	DSC 11-SF-1510	0.011 KA	ANG:	101.87
CBL-DSC 11-SF-1575	DSC 11-SF-1575	0.004 KA	ANG:	101.65
PI-0005	11-VFD-2410	0.140 KA	ANG:	103.23
CBL-DSC 11-PMP-2311	DSC 11-PMP-2311	0.035 KA	ANG:	102.97
PI-0004	11-VFD-2312	0.347 KA	ANG:	103.70
CBL-DSC 11-EF-1530	DSC 11-EF-1530	0.011 KA	ANG:	101.87
CBL-MCC-6A	SWBD-MAIN-A	9.138 KA	ANG:	-59.19
MCC-6B				
3P Duty: 9.363 KA AT -60.98 DEG (7.78 MVA) X/R: 1.87				

THREE PHASE FAULT REPORT
(FOR APPLICATION OF LOW VOLTAGE BREAKERS)
PRE FAULT VOLTAGE: 1.0000
MODEL TRANSFORMER TAPS: NO

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VOLTAGE:	480.	EQUIV. IMPEDANCE=	0.0144 + J 0.0259	OHMS	
LOW VOLTAGE POWER CIRCUIT BREAKER	9.363 KA				
MOLDED CASE CIRCUIT BREAKER < 10KA	9.554 KA				
MOLDED CASE CIRCUIT BREAKER < 20KA	9.363 KA				
MOLDED CASE CIRCUIT BREAKER > 20KA	9.363 KA				
CBL-DSC 11-PMP-2322	DSC 11-PMP-2322	0.035 KA	ANG:	102.97	
CBL-DSC 11-PMP-2324	DSC 11-PMP-2324	0.007 KA	ANG:	101.72	
CBL-DSC 11-PMP-2620	DSC 11-PMP-2620	0.007 KA	ANG:	101.83	
CBL-DSC 11-PMP-2721	DSC 11-PMP-2721	0.011 KA	ANG:	101.89	
CBL-DSC 11-SF-1520	DSC 11-SF-1520	0.011 KA	ANG:	101.81	
CBL-DSC 11-SF-1550	DSC 11-SF-1550	0.007 KA	ANG:	101.76	
PI-0008	11-VFD-2321	0.348 KA	ANG:	103.28	
CBL-DSC 11-EF-1560	DSC 11-EF-1560	0.004 KA	ANG:	101.65	
PI-0007	11-VFD-2420	0.140 KA	ANG:	103.00	
CBL-DSC 11-EF-1540	DSC 11-EF-1540	0.011 KA	ANG:	101.81	
CBL-MCC-6B	SWBD-MAIN-B	8.807 KA	ANG:	-59.95	

MCC-8A 3P Duty: 21.300 KA AT -69.67 DEG (17.71 MVA) X/R: 2.75

VOLTAGE:	480.	EQUIV. IMPEDANCE=	0.0045 + J 0.0122	OHMS	
LOW VOLTAGE POWER CIRCUIT BREAKER	21.300 KA				
MOLDED CASE CIRCUIT BREAKER > 20KA	21.300 KA				
CBL-DSC 12-AUG-4615	DSC 12-AUG-4615	0.173 KA	ANG:	105.27	
CBL-DSC 12-CON-4610	DSC 12-CON-4610	0.052 KA	ANG:	103.68	
CBL-DSC 12-EF-2530	DSC 12-EF-2530	0.140 KA	ANG:	103.43	
CBL-DSC 12-HPU-4750	DSC 12-HPU-4750	0.173 KA	ANG:	104.58	
CBL-DSC 12-SF-2510	DSC 12-SF-2510	0.140 KA	ANG:	103.43	
CBL-DSC 12-SF-2550	DSC 12-SF-2550	0.002 KA	ANG:	101.58	
CBL-DSC 12-SF-2570	DSC 12-SF-2570	0.021 KA	ANG:	102.40	
CBL-DSC 12-SRW-4612	DSC 12-SRW-4612	0.021 KA	ANG:	102.40	
CBL-DSC 19-PMP-1110	DSC 19-PMP-1110	0.138 KA	ANG:	105.62	
PI-0010	12-VFD-3520	0.139 KA	ANG:	103.78	
PI-0009	12-VFD-3320	0.140 KA	ANG:	103.45	
CBL-12-VCP1-4110	12-VCP1-4110	0.908 KA	ANG:	103.67	
CBL-MCC-8A	SWBD-MAIN-A	19.269 KA	ANG:	-69.00	

MCC-8B 3P Duty: 19.533 KA AT -70.40 DEG (16.24 MVA) X/R: 2.84

VOLTAGE:	480.	EQUIV. IMPEDANCE=	0.0048 + J 0.0134	OHMS	
LOW VOLTAGE POWER CIRCUIT BREAKER	19.533 KA				
MOLDED CASE CIRCUIT BREAKER < 20KA	19.533 KA				
MOLDED CASE CIRCUIT BREAKER > 20KA	19.533 KA				
CBL-DSC 12-AUG-4617	DSC 12-AUG-4617	0.173 KA	ANG:	104.58	
CBL-DSC 12-CON-4620	DSC 12-CON-4620	0.052 KA	ANG:	103.29	
CBL-DSC 12-EF-2540	DSC 12-EF-2540	0.140 KA	ANG:	103.43	
CBL-DSC 12-SF-2520	DSC 12-SF-2520	0.140 KA	ANG:	103.43	

THREE PHASE FAULT REPORT
(FOR APPLICATION OF LOW VOLTAGE BREAKERS)
PRE FAULT VOLTAGE: 1.0000
MODEL TRANSFORMER TAPS: NO

CONTRIBUTIONS TO MCC-8B		(CONTINUED)			
CBL-DSC 12-SF-2560	DSC 12-SF-2560	0.002 KA	ANG:	101.63	
CBL-DSC 12-SF-2580	DSC 12-SF-2580	0.021 KA	ANG:	102.60	
CBL-DSC 12-SRW-4613	DSC 12-SRW-4613	0.021 KA	ANG:	102.40	
CBL-DSC 19-PMP-1120	DSC 19-PMP-1120	0.138 KA	ANG:	105.08	
PI-0012	12-VFD-3420	0.140 KA	ANG:	103.34	
CBL-12-VCP1-4210	12-VCP1-4210	0.908 KA	ANG:	103.67	
CBL-MCC-8B	SWBD-MAIN-B	17.808 KA	ANG:	-69.83	

MCC-9A 3P Duty: 22.689 KA AT -66.92 DEG (18.86 MVA) X/R: 2.35
VOLTAGE: 480. EQUIV. IMPEDANCE= 0.0048 + J 0.0112 OHMS

LOW VOLTAGE POWER CIRCUIT BREAKER 22.689 KA					
MOLDED CASE CIRCUIT BREAKER > 20KA 22.689 KA					
CBL-DSC 03-HSS-1140	DSC 03-HSS-1140	0.004 KA	ANG:	101.78	
CBL-DSC 03-HSS-1240	DSC 03-HSS-1240	0.004 KA	ANG:	101.81	
CBL-DSC 03-PMP-1510	DSC 03-PMP-1510	0.052 KA	ANG:	103.68	
CBL-DSC 03-PMP-1530	DSC 03-PMP-1530	0.052 KA	ANG:	104.65	
CBL-DSC 03-PMP-1610	DSC 03-PMP-1610	0.052 KA	ANG:	105.60	
CBL-DSC 03-SCC-1150	DSC 03-SCC-1150	0.005 KA	ANG:	101.80	
CBL-DSC 03-SCC-1350	DSC 03-SCC-1350	0.004 KA	ANG:	101.78	
CBL-DSC 03-SLC-1120	DSC 03-SLC-1120	0.005 KA	ANG:	101.75	
CBL-DSC 03-SLC-1320	DSC 03-SLC-1320	0.005 KA	ANG:	101.90	
CBL-DSC 21-EF-1250	DSC 21-EF-1250	0.004 KA	ANG:	101.63	
CBL-DSC 21-EF-1310	DSC 21-EF-1310	0.005 KA	ANG:	101.80	
CBL-DSC 3G-01	DSC 3G-01	0.021 KA	ANG:	103.46	
CBL-DSC 3G-03	DSC 3G-03	0.021 KA	ANG:	103.11	
CBL-DSC 3P-05	DSC 3P-05	0.011 KA	ANG:	102.17	
CBL-MCC-9A	SWBD-MAIN-A	22.448 KA	ANG:	-66.82	

MCC-9B 3P Duty: 20.751 KA AT -67.96 DEG (17.25 MVA) X/R: 2.48
VOLTAGE: 480. EQUIV. IMPEDANCE= 0.0050 + J 0.0124 OHMS

LOW VOLTAGE POWER CIRCUIT BREAKER 20.751 KA					
MOLDED CASE CIRCUIT BREAKER > 20KA 20.751 KA					
CBL-DSC 03-HSS-1340	DSC 03-HSS-1340	0.004 KA	ANG:	101.83	
CBL-DSC 03-PMP-1520	DSC 03-PMP-1520	0.069 KA	ANG:	105.15	
CBL-DSC 03-PMP-1620	DSC 03-PMP-1620	0.052 KA	ANG:	105.13	
CBL-DSC 03-SCC-1250	DSC 03-SCC-1250	0.005 KA	ANG:	101.81	
CBL-DSC 03-SLC-1220	DSC 03-SLC-1220	0.005 KA	ANG:	101.81	
CBL-DSC 3G-02	DSC 3G-02	0.021 KA	ANG:	103.18	
CBL-DSC 3P-07	DSC 3P-07	0.011 KA	ANG:	102.11	
CBL-MCC-9B	SWBD-MAIN-B	20.586 KA	ANG:	-67.90	

MTS-10 3P Duty: 8.192 KA AT -56.83 DEG (6.81 MVA) X/R: 1.58
VOLTAGE: 480. EQUIV. IMPEDANCE= 0.0185 + J 0.0283 OHMS

THREE PHASE FAULT REPORT
(FOR APPLICATION OF LOW VOLTAGE BREAKERS)
PRE FAULT VOLTAGE: 1.0000
MODEL TRANSFORMER TAPS: NO

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	LOW VOLTAGE POWER CIRCUIT BREAKER	8.192 KA			
	MOLDED CASE CIRCUIT BREAKER < 10KA	8.192 KA			
	MOLDED CASE CIRCUIT BREAKER < 20KA	8.192 KA			
	MOLDED CASE CIRCUIT BREAKER > 20KA	8.192 KA			
	MTS-10_	BUS-MTS-10 N	8.192 KA	ANG:	-56.83
MTS-11	3P Duty: 3.724 KA AT -47.24 DEG (3.10 MVA) X/R: 1.19				
	VOLTAGE: 480. EQUIV. IMPEDANCE=	0.0505 + J 0.0546 OHMS			
	LOW VOLTAGE POWER CIRCUIT BREAKER	3.724 KA			
	MOLDED CASE CIRCUIT BREAKER < 10KA	3.724 KA			
	MOLDED CASE CIRCUIT BREAKER < 20KA	3.724 KA			
	MOLDED CASE CIRCUIT BREAKER > 20KA	3.724 KA			
	MTS-11_	BUS-MTS-11 N	3.724 KA	ANG:	-47.24
MTS-13	3P Duty: 13.996 KA AT -65.55 DEG (11.64 MVA) X/R: 2.21				
	VOLTAGE: 480. EQUIV. IMPEDANCE=	0.0082 + J 0.0180 OHMS			
	LOW VOLTAGE POWER CIRCUIT BREAKER	13.996 KA			
	MOLDED CASE CIRCUIT BREAKER < 20KA	13.996 KA			
	MOLDED CASE CIRCUIT BREAKER > 20KA	13.996 KA			
	MTS-13_	BUS-MTS-13 N	13.996 KA	ANG:	-65.55
MTS-5	3P Duty: 12.772 KA AT -60.84 DEG (10.62 MVA) X/R: 1.81				
	VOLTAGE: 480. EQUIV. IMPEDANCE=	0.0106 + J 0.0189 OHMS			
	LOW VOLTAGE POWER CIRCUIT BREAKER	12.772 KA			
	MOLDED CASE CIRCUIT BREAKER < 20KA	12.772 KA			
	MOLDED CASE CIRCUIT BREAKER > 20KA	12.772 KA			
	MTS-5_	BUS-MTS-5 N	12.772 KA	ANG:	-60.84
MTS-6	3P Duty: 8.838 KA AT -59.87 DEG (7.35 MVA) X/R: 1.78				
	VOLTAGE: 480. EQUIV. IMPEDANCE=	0.0157 + J 0.0271 OHMS			
	LOW VOLTAGE POWER CIRCUIT BREAKER	8.838 KA			
	MOLDED CASE CIRCUIT BREAKER < 10KA	8.904 KA			
	MOLDED CASE CIRCUIT BREAKER < 20KA	8.838 KA			
	MOLDED CASE CIRCUIT BREAKER > 20KA	8.838 KA			
	MTS-6_	BUS-MTS-6 N	8.838 KA	ANG:	-59.87
MTS-8	3P Duty: 17.252 KA AT -65.71 DEG (14.34 MVA) X/R: 2.24				
	VOLTAGE: 480. EQUIV. IMPEDANCE=	0.0066 + J 0.0146 OHMS			
	LOW VOLTAGE POWER CIRCUIT BREAKER	17.252 KA			
	MOLDED CASE CIRCUIT BREAKER < 20KA	17.252 KA			
	MOLDED CASE CIRCUIT BREAKER > 20KA	17.252 KA			
	MTS-8_	BUS-MTS-8 N	17.252 KA	ANG:	-65.71
MTS-9	3P Duty: 17.065 KA AT -62.21 DEG (14.19 MVA) X/R: 1.90				

THREE PHASE FAULT REPORT
(FOR APPLICATION OF LOW VOLTAGE BREAKERS)
PRE FAULT VOLTAGE: 1.0000
MODEL TRANSFORMER TAPS: NO

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	VOLTAGE:	480.	EQUIV. IMPEDANCE=	0.0076 + J 0.0144	OHMS
	LOW VOLTAGE POWER CIRCUIT BREAKER			17.065 KA	
	MOLDED CASE CIRCUIT BREAKER < 20KA			17.065 KA	
	MOLDED CASE CIRCUIT BREAKER > 20KA			17.065 KA	
	MTS-9_		BUS-MTS-9 N	17.065 KA	ANG: -62.21
=====					
SHB CRANE	3P Duty:	2.898 KA AT	-13.04 DEG (	2.41 MVA)	X/R: 0.23
	VOLTAGE:	480.	EQUIV. IMPEDANCE=	0.0932 + J 0.0216	OHMS
	LOW VOLTAGE POWER CIRCUIT BREAKER			2.898 KA	
	MOLDED CASE CIRCUIT BREAKER < 10KA			2.898 KA	
	MOLDED CASE CIRCUIT BREAKER < 20KA			2.898 KA	
	MOLDED CASE CIRCUIT BREAKER > 20KA			2.898 KA	
	CBL-SHB CRANE		DSC SHB CRANE	2.898 KA	ANG: -13.04
=====					
SWBD-MAIN-A	3P Duty:	27.813 KA AT	-73.09 DEG (	23.12 MVA)	X/R: 3.39
	VOLTAGE:	480.	EQUIV. IMPEDANCE=	0.0029 + J 0.0095	OHMS
	LOW VOLTAGE POWER CIRCUIT BREAKER			27.813 KA	
	MOLDED CASE CIRCUIT BREAKER > 20KA			27.813 KA	
	CBL-SWBD-MAIN-A		BUS-Sec-HECO PRIMARY TX	22.078 KA	ANG: -74.90
	CBL-07-VFD-1110		07-VFD-1110	0.408 KA	ANG: 105.29
	CBL-MCC-11A		MCC-11A	0.504 KA	ANG: 105.52
	CBL-MCC-5A		MCC-5A	0.326 KA	ANG: 103.52
	CBL-MCC-6A		MCC-6A	0.550 KA	ANG: 104.34
	CBL-MCC-10A		MCC-10A	1.929 KA	ANG: -226.53
	CBL-MCC-8A		MCC-8A	1.993 KA	ANG: 104.43
	CBL-MCC-9A		MCC-9A	0.244 KA	ANG: 103.96
=====					
SWBD-MAIN-B	3P Duty:	24.937 KA AT	-73.75 DEG (	20.73 MVA)	X/R: 3.44
	VOLTAGE:	480.	EQUIV. IMPEDANCE=	0.0031 + J 0.0107	OHMS
	LOW VOLTAGE POWER CIRCUIT BREAKER			24.937 KA	
	MOLDED CASE CIRCUIT BREAKER > 20KA			24.937 KA	
	CBL-SWBD-MAIN-B		BUS-Sec-HILO SECONDARY TX	20.233 KA	ANG: -73.55
	CBL-07-VFD-1120		07-VFD-1120	0.408 KA	ANG: 105.29
	CBL-MCC-11B		MCC-11B	0.470 KA	ANG: -252.09
	CBL-MCC-9B		MCC-9B	0.167 KA	ANG: 104.47
	CBL-MCC-10B		MCC-10B	1.029 KA	ANG: -252.20
	CBL-MCC-6B		MCC-6B	0.558 KA	ANG: 104.04
	CBL-MCC-8B		MCC-8B	1.697 KA	ANG: 104.18
	CBL-MCC-5B		MCC-5B	0.380 KA	ANG: 103.79

UNBALANCED FAULT REPORT
(FOR APPLICATION OF LOW VOLTAGE BREAKERS)
PRE FAULT VOLTAGE: 1.0000
MODEL TRANSFORMER TAPS: NO

LOCATION VOLTAGE	FAULT DUTIES	KA (RMS)	X/R	EQUIVALENT (PU) FAULT IMPEDANCE	ASYM. KA AT 0.5 CYCLES * MAX. RMS	AVG. RMS *
05-VFD-1510	3P Duty:	10.489	1.46	Z1= 11.4677	10.631	10.560
	SLG DUTY:	6.713	1.25	Z2= 11.4677	6.756	
	480. VOLTS LN/LN:	9.083		Z0= 30.9366		
	LN/LN/GND:	9.680 (	4.926	GND RETURN KA)		
05-VFD-1520	3P Duty:	9.089	1.90	Z1= 13.2338	9.415	9.253
	SLG DUTY:	6.333	1.42	Z2= 13.2338	6.408	
	480. VOLTS LN/LN:	7.871		Z0= 30.8567		
	LN/LN/GND:	8.643 (	4.832	GND RETURN KA)		
05-VFD-1630	3P Duty:	10.486	1.46	Z1= 11.4707	10.629	10.557
	SLG DUTY:	6.713	1.25	Z2= 11.4707	6.756	
	480. VOLTS LN/LN:	9.081		Z0= 30.9366		
	LN/LN/GND:	9.678 (	4.926	GND RETURN KA)		
06-VFD-1610	3P Duty:	10.486	1.46	Z1= 11.4707	10.629	10.557
	SLG DUTY:	6.713	1.25	Z2= 11.4707	6.756	
	480. VOLTS LN/LN:	9.081		Z0= 30.9366		
	LN/LN/GND:	9.678 (	4.926	GND RETURN KA)		
06-VFD-1615	3P Duty:	9.089	1.90	Z1= 13.2336	9.415	9.253
	SLG DUTY:	6.333	1.42	Z2= 13.2336	6.408	
	480. VOLTS LN/LN:	7.871		Z0= 30.8567		
	LN/LN/GND:	8.643 (	4.832	GND RETURN KA)		
06-VFD-1625	3P Duty:	9.089	1.90	Z1= 13.2336	9.415	9.253
	SLG DUTY:	6.333	1.42	Z2= 13.2336	6.408	
	480. VOLTS LN/LN:	7.871		Z0= 30.8567		
	LN/LN/GND:	8.643 (	4.832	GND RETURN KA)		
06-VFD-1630	3P Duty:	10.486	1.46	Z1= 11.4707	10.629	10.557
	SLG DUTY:	6.713	1.25	Z2= 11.4707	6.756	
	480. VOLTS LN/LN:	9.081		Z0= 30.9366		
	LN/LN/GND:	9.678 (	4.926	GND RETURN KA)		
06-VFD-1635	3P Duty:	9.089	1.90	Z1= 13.2336	9.415	9.253
	SLG DUTY:	6.333	1.42	Z2= 13.2336	6.408	
	480. VOLTS LN/LN:	7.871		Z0= 30.8567		
	LN/LN/GND:	8.643 (	4.832	GND RETURN KA)		

UNBALANCED FAULT REPORT
(FOR APPLICATION OF LOW VOLTAGE BREAKERS)
PRE FAULT VOLTAGE: 1.0000
MODEL TRANSFORMER TAPS: NO

LOCATION VOLTAGE	FAULT DUTIES	KA (RMS)	X/R	EQUIVALENT (PU) FAULT IMPEDANCE	ASYM. KA AT 0.5 CYCLES * MAX. RMS	AVG. RMS *
07-VFD-1110	3P Duty:	17.030	1.16	Z1= 7.0627	17.106	17.069
	SLG DUTY:	13.368	0.80	Z2= 7.0627	13.374	
	480. VOLTS LN/LN:	14.749		Z0= 13.3458		
	LN/LN/GND:	17.129	(10.815	GND RETURN KA)		
07-VFD-1120	3P Duty:	16.058	1.28	Z1= 7.4903	16.177	16.118
	SLG DUTY:	13.045	0.86	Z2= 7.4903	13.054	
	480. VOLTS LN/LN:	13.907		Z0= 13.2722		
	LN/LN/GND:	16.419	(10.757	GND RETURN KA)		
09-VFD-1100	3P Duty:	9.169	1.36	Z1= 13.1184	9.258	9.214
	SLG DUTY:	6.361	1.02	Z2= 13.1184	6.374	
	480. VOLTS LN/LN:	7.940		Z0= 30.8953		
	LN/LN/GND:	8.742	(4.837	GND RETURN KA)		
09-VFD-1200	3P Duty:	4.010	1.33	Z1= 29.9946	4.046	4.028
	SLG DUTY:	2.466	0.93	Z2= 29.9946	2.469	
	480. VOLTS LN/LN:	3.473		Z0= 87.3025		
	LN/LN/GND:	3.751	(1.770	GND RETURN KA)		
11-VFD-2312	3P Duty:	9.674	1.79	Z1= 12.4333	9.961	9.818
	SLG DUTY:	6.932	1.40	Z2= 12.4333	7.009	
	480. VOLTS LN/LN:	8.378		Z0= 27.4436		
	LN/LN/GND:	9.199	(5.376	GND RETURN KA)		
11-VFD-2321	3P Duty:	9.354	1.87	Z1= 12.8594	9.674	9.514
	SLG DUTY:	6.836	1.44	Z2= 12.8594	6.921	
	480. VOLTS LN/LN:	8.100		Z0= 27.3636		
	LN/LN/GND:	8.953	(5.357	GND RETURN KA)		
11-VFD-2410	3P Duty:	9.674	1.79	Z1= 12.4338	9.961	9.818
	SLG DUTY:	6.931	1.40	Z2= 12.4338	7.009	
	480. VOLTS LN/LN:	8.378		Z0= 27.4436		
	LN/LN/GND:	9.198	(5.376	GND RETURN KA)		
11-VFD-2420	3P Duty:	9.353	1.87	Z1= 12.8599	9.673	9.514
	SLG DUTY:	6.835	1.44	Z2= 12.8599	6.921	
	480. VOLTS LN/LN:	8.100		Z0= 27.3636		
	LN/LN/GND:	8.953	(5.357	GND RETURN KA)		

UNBALANCED FAULT REPORT
(FOR APPLICATION OF LOW VOLTAGE BREAKERS)
PRE FAULT VOLTAGE: 1.0000
MODEL TRANSFORMER TAPS: NO

LOCATION VOLTAGE	FAULT DUTIES	KA (RMS)	X/R	EQUIVALENT (PU) FAULT IMPEDANCE	ASYM. KA AT 0.5 CYCLES * MAX. RMS	AVG. RMS *
12-VCP1-4110	3P Duty:	17.234	2.11	Z1= 6.9794	18.094	17.666
	SLG DUTY:	14.085	1.63	Z2= 6.9794	14.382	
	480. VOLTS LN/LN:	14.925		Z0= 11.8175		
	LN/LN/GND: 16.969 (11.838 GND RETURN KA)					
12-VCP1-4210	3P Duty:	16.102	2.23	Z1= 7.4698	17.035	16.572
	SLG DUTY:	13.622	1.70	Z2= 7.4698	13.956	
	480. VOLTS LN/LN:	13.945		Z0= 11.7375		
	LN/LN/GND: 16.073 (11.722 GND RETURN KA)					
12-VFD-3320	3P Duty:	21.253	2.74	Z1= 5.6596	23.298	22.287
	SLG DUTY:	19.190	2.30	Z2= 5.6596	20.401	
	480. VOLTS LN/LN:	18.405		Z0= 7.5284		
	LN/LN/GND: 21.155 (17.452 GND RETURN KA)					
12-VFD-3420	3P Duty:	19.493	2.84	Z1= 6.1706	21.513	20.516
	SLG DUTY:	18.282	2.38	Z2= 6.1706	19.546	
	480. VOLTS LN/LN:	16.881		Z0= 7.4497		
	LN/LN/GND: 19.736 (17.166 GND RETURN KA)					
12-VFD-3520	3P Duty:	21.253	2.74	Z1= 5.6596	23.298	22.287
	SLG DUTY:	19.190	2.30	Z2= 5.6596	20.401	
	480. VOLTS LN/LN:	18.405		Z0= 7.5284		
	LN/LN/GND: 21.155 (17.452 GND RETURN KA)					
21-GEN-1100 [Ex]	3P Duty:	15.586	17.20	Z1= 7.7175	24.084	20.087
	SLG DUTY:	15.381	16.00	Z2= 7.7175	23.581	
	480. VOLTS LN/LN:	13.497		Z0= 8.0283		
	LN/LN/GND: 15.612 (15.179 GND RETURN KA)					
21-GEN-1200	3P Duty:	15.862	18.96	Z1= 7.5830	24.757	20.580
	SLG DUTY:	15.786	18.19	Z2= 7.5830	24.536	
	480. VOLTS LN/LN:	13.737		Z0= 7.6924		
	LN/LN/GND: 15.875 (15.711 GND RETURN KA)					
21-PNL-1200	3P Duty:	2.375	0.20	Z1= 50.6499	2.375	2.375
	SLG DUTY:	1.415	0.14	Z2= 50.6499	1.415	
	480. VOLTS LN/LN:	2.057		Z0= 154.0120		
	LN/LN/GND: 2.155 (1.007 GND RETURN KA)					

UNBALANCED FAULT REPORT
(FOR APPLICATION OF LOW VOLTAGE BREAKERS)
PRE FAULT VOLTAGE: 1.0000
MODEL TRANSFORMER TAPS: NO

LOCATION VOLTAGE	FAULT DUTIES	KA (RMS)	X/R	EQUIVALENT (PU) FAULT IMPEDANCE	ASYM. KA AT 0.5 CYCLES * MAX. RMS	AVG. RMS *
21-VFD-1100	3P Duty:	13.493	0.99	Z1= 8.9144	13.517	13.505
	SLG DUTY:	9.981	0.72	Z2= 8.9144	9.982	
	480. VOLTS LN/LN:	11.685		Z0= 18.7659		
	LN/LN/GND:	13.214	(7.827	GND RETURN KA)		
21-VFD-1200	3P Duty:	12.878	1.08	Z1= 9.3401	12.917	12.897
	SLG DUTY:	9.798	0.77	Z2= 9.3401	9.801	
	480. VOLTS LN/LN:	11.153		Z0= 18.6925		
	LN/LN/GND:	12.772	(7.795	GND RETURN KA)		
21-VFD-1300	3P Duty:	13.493	0.99	Z1= 8.9144	13.517	13.505
	SLG DUTY:	9.981	0.72	Z2= 8.9144	9.982	
	480. VOLTS LN/LN:	11.685		Z0= 18.7659		
	LN/LN/GND:	13.214	(7.827	GND RETURN KA)		
22-VFD-2022	3P Duty:	9.089	1.90	Z1= 13.2335	9.416	9.253
	SLG DUTY:	6.333	1.42	Z2= 13.2335	6.408	
	480. VOLTS LN/LN:	7.871		Z0= 30.8567		
	LN/LN/GND:	8.643	(4.832	GND RETURN KA)		
22-VFD-2201	3P Duty:	10.486	1.46	Z1= 11.4706	10.629	10.557
	SLG DUTY:	6.713	1.25	Z2= 11.4706	6.756	
	480. VOLTS LN/LN:	9.081		Z0= 30.9366		
	LN/LN/GND:	9.678	(4.926	GND RETURN KA)		
22-VFD-2203	3P Duty:	10.486	1.46	Z1= 11.4706	10.629	10.557
	SLG DUTY:	6.713	1.25	Z2= 11.4706	6.756	
	480. VOLTS LN/LN:	9.081		Z0= 30.9366		
	LN/LN/GND:	9.678	(4.926	GND RETURN KA)		
23-VCP-2810	3P Duty:	2.496	0.28	Z1= 48.1941	2.496	2.496
	SLG DUTY:	1.482	0.22	Z2= 48.1941	1.482	
	480. VOLTS LN/LN:	2.161		Z0= 147.4045		
	LN/LN/GND:	2.267	(1.053	GND RETURN KA)		
ADMIN	3P Duty:	8.209	1.21	Z1= 14.6516	8.254	8.232
	SLG DUTY:	5.778	0.97	Z2= 14.6516	5.787	
	480. VOLTS LN/LN:	7.110		Z0= 33.4870		
	LN/LN/GND:	7.800	(4.435	GND RETURN KA)		

UNBALANCED FAULT REPORT
(FOR APPLICATION OF LOW VOLTAGE BREAKERS)
PRE FAULT VOLTAGE: 1.0000
MODEL TRANSFORMER TAPS: NO

LOCATION VOLTAGE	FAULT DUTIES	KA (RMS)	X/R	EQUIVALENT (PU) FAULT IMPEDANCE	ASYM. KA AT 0.5 CYCLES * MAX. RMS	AVG. RMS *
AHF-10A	3P Duty:	8.143	0.91	Z1= 14.7715	8.151	8.147
	SLG DUTY:	5.052	0.81	Z2= 14.7715	5.054	
480. VOLTS	LN/LN:	7.052		Z0= 42.0387		
	LN/LN/GND:	7.484	(3.655	GND RETURN KA)		
AHF-10B	3P Duty:	5.705	0.75	Z1= 21.0822	5.707	5.706
	SLG DUTY:	3.567	0.58	Z2= 21.0822	3.567	
480. VOLTS	LN/LN:	4.941		Z0= 59.4665		
	LN/LN/GND:	5.309	(2.583	GND RETURN KA)		
AHF-8A	3P Duty:	17.246	1.58	Z1= 6.9746	17.569	17.407
	SLG DUTY:	13.954	1.20	Z2= 6.9746	14.029	
480. VOLTS	LN/LN:	14.935		Z0= 12.1628		
	LN/LN/GND:	17.151	(11.607	GND RETURN KA)		
AHF-8B	3P Duty:	15.524	1.59	Z1= 7.7480	15.817	15.671
	SLG DUTY:	12.768	1.17	Z2= 7.7480	12.828	
480. VOLTS	LN/LN:	13.444		Z0= 13.1112		
	LN/LN/GND:	15.619	(10.714	GND RETURN KA)		
CL BLDG PORT GEN	3P Duty:	7.617	1.10	Z1= 15.7920	7.642	7.629
	SLG DUTY:	5.099	0.84	Z2= 15.7920	5.102	
480. VOLTS	LN/LN:	6.596		Z0= 39.6328		
	LN/LN/GND:	7.201	(3.810	GND RETURN KA)		
DCB 1 PORT GEN	3P Duty:	8.596	1.74	Z1= 13.9926	8.828	8.712
	SLG DUTY:	6.181	1.35	Z2= 13.9926	6.239	
480. VOLTS	LN/LN:	7.444		Z0= 30.7164		
	LN/LN/GND:	8.201	(4.800	GND RETURN KA)		
DP-10	3P Duty:	7.921	1.50	Z1= 15.1860	8.041	7.981
	SLG DUTY:	5.357	1.14	Z2= 15.1860	5.379	
480. VOLTS	LN/LN:	6.859		Z0= 37.3958		
	LN/LN/GND:	7.495	(4.024	GND RETURN KA)		
DP-13	3P Duty:	13.620	2.18	Z1= 8.8310	14.367	13.996
	SLG DUTY:	10.728	1.79	Z2= 8.8310	11.045	
480. VOLTS	LN/LN:	11.796		Z0= 16.0845		
	LN/LN/GND:	13.111	(8.821	GND RETURN KA)		

UNBALANCED FAULT REPORT
(FOR APPLICATION OF LOW VOLTAGE BREAKERS)
PRE FAULT VOLTAGE: 1.0000
MODEL TRANSFORMER TAPS: NO

LOCATION VOLTAGE	FAULT DUTIES	KA (RMS)	X/R	EQUIVALENT (PU) FAULT IMPEDANCE	ASYM. KA AT 0.5 CYCLES * MAX. RMS	AVG. RMS *
DP-5A	3P Duty:	15.064	1.95	Z1= 7.9846	15.652	15.359
	SLG DUTY:	12.194	1.56	Z2= 7.9846	12.408	
	480. VOLTS LN/LN:	13.046		Z0= 13.7754		
	LN/LN/GND:	14.734	(10.191	GND RETURN KA)		
DP-5B	3P Duty:	13.962	1.99	Z1= 8.6152	14.545	14.255
	SLG DUTY:	11.533	1.56	Z2= 8.6152	11.737	
	480. VOLTS LN/LN:	12.091		Z0= 14.2591		
	LN/LN/GND:	13.803	(9.762	GND RETURN KA)		
DP-6	3P Duty:	8.643	1.75	Z1= 13.9159	8.880	8.762
	SLG DUTY:	6.221	1.35	Z2= 13.9159	6.281	
	480. VOLTS LN/LN:	7.485		Z0= 30.4913		
	LN/LN/GND:	8.248	(4.834	GND RETURN KA)		
DP-8A-1	3P Duty:	19.814	2.34	Z1= 6.0707	21.126	20.475
	SLG DUTY:	17.288	1.89	Z2= 6.0707	17.899	
	480. VOLTS LN/LN:	17.159		Z0= 8.8232		
	LN/LN/GND:	19.735	(15.267	GND RETURN KA)		
DP-8A-2	3P Duty:	18.451	1.96	Z1= 6.5189	19.190	18.822
	SLG DUTY:	15.538	1.54	Z2= 6.5189	15.796	
	480. VOLTS LN/LN:	15.979		Z0= 10.3442		
	LN/LN/GND:	18.372	(13.329	GND RETURN KA)		
DP-8B-1	3P Duty:	18.052	2.39	Z1= 6.6630	19.316	18.690
	SLG DUTY:	16.240	1.91	Z2= 6.6630	16.833	
	480. VOLTS LN/LN:	15.634		Z0= 9.0138		
	LN/LN/GND:	18.267	(14.680	GND RETURN KA)		
DP-8B-2	3P Duty:	16.924	2.04	Z1= 7.1073	17.687	17.307
	SLG DUTY:	14.691	1.57	Z2= 7.1073	14.958	
	480. VOLTS LN/LN:	14.656		Z0= 10.5488		
	LN/LN/GND:	17.100	(12.874	GND RETURN KA)		
DP-9	3P Duty:	16.320	1.81	Z1= 7.3703	16.823	16.572
	SLG DUTY:	14.193	1.38	Z2= 7.3703	14.341	
	480. VOLTS LN/LN:	14.133		Z0= 10.9462		
	LN/LN/GND:	16.618	(12.429	GND RETURN KA)		

UNBALANCED FAULT REPORT
(FOR APPLICATION OF LOW VOLTAGE BREAKERS)
PRE FAULT VOLTAGE: 1.0000
MODEL TRANSFORMER TAPS: NO

LOCATION VOLTAGE	FAULT DUTIES	KA (RMS)	X/R	EQUIVALENT (PU) FAULT IMPEDANCE	ASYM. KA AT 0.5 CYCLES * MAX. RMS	AVG. RMS *
DSC 02-EF-1320	3P Duty:	0.960	0.11	Z1= 125.3408	0.960	0.960
	SLG DUTY:	0.561	0.08	Z2= 125.3408	0.561	
	480. VOLTS LN/LN:	0.831		Z0= 392.2390		
	LN/LN/GND:	0.864	(	0.397 GND RETURN KA)		
DSC 02-EF-1330	3P Duty:	1.151	0.13	Z1= 104.4718	1.151	1.151
	SLG DUTY:	0.672	0.09	Z2= 104.4718	0.672	
	480. VOLTS LN/LN:	0.997		Z0= 328.3242		
	LN/LN/GND:	1.039	(	0.474 GND RETURN KA)		
DSC 02-EF-1340	3P Duty:	1.150	0.14	Z1= 104.5823	1.150	1.150
	SLG DUTY:	0.672	0.09	Z2= 104.5823	0.672	
	480. VOLTS LN/LN:	0.996		Z0= 328.2699		
	LN/LN/GND:	1.039	(	0.474 GND RETURN KA)		
DSC 02-EF-1370	3P Duty:	1.416	0.13	Z1= 84.9723	1.416	1.416
	SLG DUTY:	0.831	0.10	Z2= 84.9723	0.831	
	480. VOLTS LN/LN:	1.226		Z0= 264.3675		
	LN/LN/GND:	1.276	(	0.588 GND RETURN KA)		
DSC 02-EF-1380	3P Duty:	1.143	0.12	Z1= 105.2235	1.143	1.143
	SLG DUTY:	0.670	0.09	Z2= 105.2235	0.670	
	480. VOLTS LN/LN:	0.990		Z0= 328.2699		
	LN/LN/GND:	1.030	(	0.474 GND RETURN KA)		
DSC 02-EF-1390	3P Duty:	1.778	0.19	Z1= 67.6408	1.778	1.778
	SLG DUTY:	1.044	0.14	Z2= 67.6408	1.044	
	480. VOLTS LN/LN:	1.540		Z0= 210.6975		
	LN/LN/GND:	1.613	(	0.738 GND RETURN KA)		
DSC 02-EF-1396	3P Duty:	1.490	0.18	Z1= 80.7020	1.490	1.490
	SLG DUTY:	0.876	0.13	Z2= 80.7020	0.876	
	480. VOLTS LN/LN:	1.291		Z0= 251.0205		
	LN/LN/GND:	1.349	(	0.619 GND RETURN KA)		
DSC 02-SF-1310	3P Duty:	1.143	0.12	Z1= 105.2235	1.143	1.143
	SLG DUTY:	0.670	0.09	Z2= 105.2235	0.670	
	480. VOLTS LN/LN:	0.990		Z0= 328.2699		
	LN/LN/GND:	1.030	(	0.474 GND RETURN KA)		

UNBALANCED FAULT REPORT
(FOR APPLICATION OF LOW VOLTAGE BREAKERS)
PRE FAULT VOLTAGE: 1.0000
MODEL TRANSFORMER TAPS: NO

LOCATION VOLTAGE	FAULT DUTIES	KA (RMS)	X/R	EQUIVALENT (PU) FAULT IMPEDANCE	ASYM. KA AT 0.5 CYCLES * MAX. RMS	AVG. RMS *
DSC 02-SF-1350	3P Duty:	1.157	0.14	Z1= 103.9234	1.157	1.157
	SLG DUTY:	0.674	0.09	Z2= 103.9234	0.674	
	480. VOLTS LN/LN:	1.002		Z0= 328.3242		
	LN/LN/GND:	1.047	(	0.475 GND RETURN KA)		
DSC 02-SF-1360	3P Duty:	1.156	0.14	Z1= 104.0267	1.156	1.156
	SLG DUTY:	0.673	0.09	Z2= 104.0267	0.673	
	480. VOLTS LN/LN:	1.001		Z0= 328.2699		
	LN/LN/GND:	1.047	(	0.475 GND RETURN KA)		
DSC 02-SF-1570	3P Duty:	1.412	0.13	Z1= 85.1786	1.412	1.412
	SLG DUTY:	0.830	0.10	Z2= 85.1786	0.830	
	480. VOLTS LN/LN:	1.223		Z0= 264.3675		
	LN/LN/GND:	1.271	(	0.588 GND RETURN KA)		
DSC 02-SF-1575	3P Duty:	1.279	0.15	Z1= 94.0269	1.279	1.279
	SLG DUTY:	0.753	0.11	Z2= 94.0269	0.753	
	480. VOLTS LN/LN:	1.108		Z0= 291.4100		
	LN/LN/GND:	1.153	(	0.533 GND RETURN KA)		
DSC 02-SF-1810	3P Duty:	1.410	0.14	Z1= 85.3097	1.410	1.410
	SLG DUTY:	0.830	0.10	Z2= 85.3097	0.830	
	480. VOLTS LN/LN:	1.221		Z0= 264.3127		
	LN/LN/GND:	1.271	(	0.588 GND RETURN KA)		
DSC 03-EF-1420	3P Duty:	1.851	0.17	Z1= 64.9877	1.851	1.851
	SLG DUTY:	1.093	0.12	Z2= 64.9877	1.093	
	480. VOLTS LN/LN:	1.603		Z0= 200.3788		
	LN/LN/GND:	1.675	(	0.775 GND RETURN KA)		
DSC 03-HSS-1140	3P Duty:	0.833	0.08	Z1= 144.4235	0.833	0.833
	SLG DUTY:	0.487	0.06	Z2= 144.4235	0.487	
	480. VOLTS LN/LN:	0.721		Z0= 451.6537		
	LN/LN/GND:	0.747	(	0.344 GND RETURN KA)		
DSC 03-HSS-1240	3P Duty:	0.730	0.08	Z1= 164.6797	0.730	0.730
	SLG DUTY:	0.427	0.06	Z2= 164.6797	0.427	
	480. VOLTS LN/LN:	0.633		Z0= 515.6439		
	LN/LN/GND:	0.655	(	0.302 GND RETURN KA)		

UNBALANCED FAULT REPORT
(FOR APPLICATION OF LOW VOLTAGE BREAKERS)
PRE FAULT VOLTAGE: 1.0000
MODEL TRANSFORMER TAPS: NO

LOCATION VOLTAGE	FAULT DUTIES	KA (RMS)	X/R	EQUIVALENT (PU) FAULT IMPEDANCE	ASYM. KA AT 0.5 CYCLES * MAX. RMS	AVG. RMS *
DSC 03-HSS-1340	3P Duty:	0.686	0.08	Z1= 175.3434	0.686	0.686
	SLG DUTY:	0.401	0.06	Z2= 175.3434	0.401	
	480. VOLTS LN/LN:	0.594		Z0= 548.8661		
	LN/LN/GND:	0.615	(0.283	GND RETURN KA)		
DSC 03-PMP-1510	3P Duty:	1.454	0.12	Z1= 82.7359	1.454	1.454
	SLG DUTY:	0.849	0.08	Z2= 82.7359	0.849	
	480. VOLTS LN/LN:	1.259		Z0= 259.6927		
	LN/LN/GND:	1.315	(0.600	GND RETURN KA)		
DSC 03-PMP-1520	3P Duty:	1.135	0.13	Z1= 106.0136	1.135	1.135
	SLG DUTY:	0.659	0.08	Z2= 106.0136	0.659	
	480. VOLTS LN/LN:	0.983		Z0= 336.4209		
	LN/LN/GND:	1.030	(0.464	GND RETURN KA)		
DSC 03-PMP-1530	3P Duty:	0.984	0.11	Z1= 122.2159	0.984	0.984
	SLG DUTY:	0.571	0.07	Z2= 122.2159	0.571	
	480. VOLTS LN/LN:	0.852		Z0= 387.6645		
	LN/LN/GND:	0.890	(0.402	GND RETURN KA)		
DSC 03-PMP-1610	3P Duty:	0.746	0.10	Z1= 161.2697	0.746	0.746
	SLG DUTY:	0.431	0.07	Z2= 161.2697	0.431	
	480. VOLTS LN/LN:	0.646		Z0= 515.6439		
	LN/LN/GND:	0.676	(0.303	GND RETURN KA)		
DSC 03-PMP-1620	3P Duty:	0.848	0.11	Z1= 141.9038	0.848	0.848
	SLG DUTY:	0.491	0.07	Z2= 141.9038	0.491	
	480. VOLTS LN/LN:	0.734		Z0= 451.6004		
	LN/LN/GND:	0.768	(0.345	GND RETURN KA)		
DSC 03-SCC-1150	3P Duty:	1.158	0.09	Z1= 103.8575	1.158	1.158
	SLG DUTY:	0.679	0.07	Z2= 103.8575	0.679	
	480. VOLTS LN/LN:	1.003		Z0= 323.6770		
	LN/LN/GND:	1.040	(0.480	GND RETURN KA)		
DSC 03-SCC-1250	3P Duty:	1.113	0.10	Z1= 108.0469	1.113	1.113
	SLG DUTY:	0.653	0.07	Z2= 108.0469	0.653	
	480. VOLTS LN/LN:	0.964		Z0= 336.4209		
	LN/LN/GND:	1.001	(0.462	GND RETURN KA)		

UNBALANCED FAULT REPORT
(FOR APPLICATION OF LOW VOLTAGE BREAKERS)
PRE FAULT VOLTAGE: 1.0000
MODEL TRANSFORMER TAPS: NO

LOCATION VOLTAGE	FAULT DUTIES	KA (RMS)	X/R	EQUIVALENT (PU) FAULT IMPEDANCE	ASYM. KA AT 0.5 CYCLES * MAX. RMS	AVG. RMS *
DSC 03-SCC-1350	3P Duty:	0.833	0.08	Z1= 144.4235	0.833	0.833
	SLG DUTY:	0.487	0.06	Z2= 144.4235	0.487	
	480. VOLTS LN/LN:	0.721		Z0= 451.6537		
	LN/LN/GND:	0.747 (	0.344	GND RETURN KA)		
DSC 03-SF-1410	3P Duty:	0.726	0.10	Z1= 165.7725	0.726	0.726
	SLG DUTY:	0.424	0.07	Z2= 165.7725	0.424	
	480. VOLTS LN/LN:	0.628		Z0= 520.1952		
	LN/LN/GND:	0.652 (	0.299	GND RETURN KA)		
DSC 03-SF-1575	3P Duty:	0.723	0.09	Z1= 166.2531	0.723	0.723
	SLG DUTY:	0.423	0.07	Z2= 166.2531	0.423	
	480. VOLTS LN/LN:	0.627		Z0= 520.1952		
	LN/LN/GND:	0.649 (	0.299	GND RETURN KA)		
DSC 03-SLC-1120	3P Duty:	1.439	0.10	Z1= 83.6153	1.439	1.439
	SLG DUTY:	0.845	0.08	Z2= 83.6153	0.845	
	480. VOLTS LN/LN:	1.246		Z0= 259.6927		
	LN/LN/GND:	1.294 (	0.599	GND RETURN KA)		
DSC 03-SLC-1220	3P Duty:	1.113	0.10	Z1= 108.0469	1.113	1.113
	SLG DUTY:	0.653	0.07	Z2= 108.0469	0.653	
	480. VOLTS LN/LN:	0.964		Z0= 336.4209		
	LN/LN/GND:	1.001 (	0.462	GND RETURN KA)		
DSC 03-SLC-1320	3P Duty:	0.833	0.08	Z1= 144.3367	0.833	0.833
	SLG DUTY:	0.487	0.06	Z2= 144.3367	0.487	
	480. VOLTS LN/LN:	0.722		Z0= 451.6537		
	LN/LN/GND:	0.748 (	0.344	GND RETURN KA)		
DSC 04-PMP-1140	3P Duty:	2.343	0.53	Z1= 51.3397	2.343	2.343
	SLG DUTY:	1.341	0.32	Z2= 51.3397	1.341	
	480. VOLTS LN/LN:	2.029		Z0= 167.8190		
	LN/LN/GND:	2.162 (	0.935	GND RETURN KA)		
DSC 06-CLA-1510	3P Duty:	2.039	0.21	Z1= 58.9920	2.039	2.039
	SLG DUTY:	1.192	0.19	Z2= 58.9920	1.192	
	480. VOLTS LN/LN:	1.766		Z0= 184.7328		
	LN/LN/GND:	1.834 (	0.842	GND RETURN KA)		

UNBALANCED FAULT REPORT
(FOR APPLICATION OF LOW VOLTAGE BREAKERS)
PRE FAULT VOLTAGE: 1.0000
MODEL TRANSFORMER TAPS: NO

LOCATION VOLTAGE	FAULT DUTIES	KA (RMS)	X/R	EQUIVALENT (PU) FAULT IMPEDANCE	ASYM. KA AT 0.5 CYCLES * MAX. RMS	AVG. RMS *
DSC 06-CLA-1520	3P Duty:	1.234	0.19	Z1= 97.4755	1.234	1.234
	SLG DUTY:	0.722	0.15	Z2= 97.4755	0.722	
	480. VOLTS LN/LN:	1.069		Z0= 305.4030		
	LN/LN/GND:	1.112	(0.510	GND RETURN KA)		
DSC 06-CLA-1530	3P Duty:	0.889	0.13	Z1= 135.3414	0.889	0.889
	SLG DUTY:	0.518	0.12	Z2= 135.3414	0.518	
	480. VOLTS LN/LN:	0.770		Z0= 426.4839		
	LN/LN/GND:	0.797	(0.365	GND RETURN KA)		
DSC 06-PMP-1515	3P Duty:	2.796	0.31	Z1= 43.0134	2.796	2.796
	SLG DUTY:	1.634	0.26	Z2= 43.0134	1.634	
	480. VOLTS LN/LN:	2.422		Z0= 135.0731		
	LN/LN/GND:	2.532	(1.153	GND RETURN KA)		
DSC 06-PMP-1525	3P Duty:	1.745	0.30	Z1= 68.9336	1.745	1.745
	SLG DUTY:	1.016	0.22	Z2= 68.9336	1.016	
	480. VOLTS LN/LN:	1.511		Z0= 217.7659		
	LN/LN/GND:	1.587	(0.716	GND RETURN KA)		
DSC 06-PMP-1535	3P Duty:	1.280	0.22	Z1= 93.9354	1.280	1.280
	SLG DUTY:	0.739	0.17	Z2= 93.9354	0.739	
	480. VOLTS LN/LN:	1.109		Z0= 300.9468		
	LN/LN/GND:	1.159	(0.519	GND RETURN KA)		
DSC 08-EF-1230	3P Duty:	3.506	0.35	Z1= 34.3049	3.506	3.506
	SLG DUTY:	2.105	0.27	Z2= 34.3049	2.105	
	480. VOLTS LN/LN:	3.036		Z0= 103.1763		
	LN/LN/GND:	3.206	(1.501	GND RETURN KA)		
DSC 09-MIX-1110	3P Duty:	0.799	0.12	Z1= 150.5605	0.799	0.799
	SLG DUTY:	0.466	0.09	Z2= 150.5605	0.466	
	480. VOLTS LN/LN:	0.692		Z0= 473.0255		
	LN/LN/GND:	0.718	(0.329	GND RETURN KA)		
DSC 09-PMP-903	3P Duty:	0.862	0.21	Z1= 139.5440	0.862	0.862
	SLG DUTY:	0.498	0.17	Z2= 139.5440	0.498	
	480. VOLTS LN/LN:	0.746		Z0= 446.0246		
	LN/LN/GND:	0.776	(0.350	GND RETURN KA)		

UNBALANCED FAULT REPORT
(FOR APPLICATION OF LOW VOLTAGE BREAKERS)
PRE FAULT VOLTAGE: 1.0000
MODEL TRANSFORMER TAPS: NO

LOCATION VOLTAGE	FAULT DUTIES	KA (RMS)	X/R	EQUIVALENT (PU) FAULT IMPEDANCE	ASYM. KA AT 0.5 CYCLES * MAX. RMS	AVG. RMS *
DSC 11-EF-1530	3P Duty:	1.761	0.21	Z1= 68.2869	1.761	1.761
	SLG DUTY:	1.037	0.16	Z2= 68.2869	1.037	
	480. VOLTS LN/LN:	1.525		Z0= 211.7244		
	LN/LN/GND:	1.592 (	0.734	GND RETURN KA)		
DSC 11-EF-1540	3P Duty:	2.130	0.25	Z1= 56.4799	2.130	2.130
	SLG DUTY:	1.261	0.19	Z2= 56.4799	1.261	
	480. VOLTS LN/LN:	1.844		Z0= 173.4782		
	LN/LN/GND:	1.933 (	0.895	GND RETURN KA)		
DSC 11-EF-1560	3P Duty:	1.755	0.21	Z1= 68.5381	1.755	1.755
	SLG DUTY:	1.035	0.16	Z2= 68.5381	1.035	
	480. VOLTS LN/LN:	1.520		Z0= 211.6664		
	LN/LN/GND:	1.587 (	0.734	GND RETURN KA)		
DSC 11-PMP-2311	3P Duty:	1.371	0.19	Z1= 87.7544	1.371	1.371
	SLG DUTY:	0.801	0.14	Z2= 87.7544	0.801	
	480. VOLTS LN/LN:	1.187		Z0= 275.4989		
	LN/LN/GND:	1.239 (	0.565	GND RETURN KA)		
DSC 11-PMP-2314	3P Duty:	1.760	0.21	Z1= 68.3375	1.760	1.760
	SLG DUTY:	1.036	0.16	Z2= 68.3375	1.036	
	480. VOLTS LN/LN:	1.524		Z0= 211.7244		
	LN/LN/GND:	1.590 (	0.734	GND RETURN KA)		
DSC 11-PMP-2322	3P Duty:	1.369	0.19	Z1= 87.8788	1.369	1.369
	SLG DUTY:	0.800	0.14	Z2= 87.8788	0.800	
	480. VOLTS LN/LN:	1.185		Z0= 275.4421		
	LN/LN/GND:	1.239 (	0.565	GND RETURN KA)		
DSC 11-PMP-2324	3P Duty:	2.128	0.25	Z1= 56.5163	2.128	2.128
	SLG DUTY:	1.261	0.19	Z2= 56.5163	1.261	
	480. VOLTS LN/LN:	1.843		Z0= 173.4782		
	LN/LN/GND:	1.931 (	0.895	GND RETURN KA)		
DSC 11-PMP-2610	3P Duty:	1.361	0.17	Z1= 88.4093	1.361	1.361
	SLG DUTY:	0.798	0.13	Z2= 88.4093	0.798	
	480. VOLTS LN/LN:	1.178		Z0= 275.4989		
	LN/LN/GND:	1.226 (	0.564	GND RETURN KA)		

UNBALANCED FAULT REPORT
(FOR APPLICATION OF LOW VOLTAGE BREAKERS)
PRE FAULT VOLTAGE: 1.0000
MODEL TRANSFORMER TAPS: NO

LOCATION VOLTAGE	FAULT DUTIES	KA (RMS)	X/R	EQUIVALENT (PU) FAULT IMPEDANCE	ASYM. KA AT 0.5 CYCLES * MAX. RMS	AVG. RMS *
DSC 11-PMP-2620	3P Duty:	1.358	0.18	Z1= 88.5426	1.358	1.358
	SLG DUTY:	0.798	0.14	Z2= 88.5426	0.798	
	480. VOLTS LN/LN:	1.176		Z0= 275.4421		
	LN/LN/GND:	1.225	(	0.564 GND RETURN KA)		
DSC 11-PMP-2711	3P Duty:	1.362	0.17	Z1= 88.3291	1.362	1.362
	SLG DUTY:	0.798	0.13	Z2= 88.3291	0.798	
	480. VOLTS LN/LN:	1.179		Z0= 275.4989		
	LN/LN/GND:	1.228	(	0.565 GND RETURN KA)		
DSC 11-PMP-2721	3P Duty:	1.661	0.21	Z1= 72.4332	1.661	1.661
	SLG DUTY:	0.978	0.16	Z2= 72.4332	0.978	
	480. VOLTS LN/LN:	1.438		Z0= 224.4116		
	LN/LN/GND:	1.502	(	0.693 GND RETURN KA)		
DSC 11-SF-1510	3P Duty:	1.761	0.21	Z1= 68.2869	1.761	1.761
	SLG DUTY:	1.037	0.16	Z2= 68.2869	1.037	
	480. VOLTS LN/LN:	1.525		Z0= 211.7244		
	LN/LN/GND:	1.592	(	0.734 GND RETURN KA)		
DSC 11-SF-1520	3P Duty:	2.130	0.25	Z1= 56.4799	2.130	2.130
	SLG DUTY:	1.261	0.19	Z2= 56.4799	1.261	
	480. VOLTS LN/LN:	1.844		Z0= 173.4782		
	LN/LN/GND:	1.933	(	0.895 GND RETURN KA)		
DSC 11-SF-1550	3P Duty:	1.756	0.22	Z1= 68.4869	1.756	1.756
	SLG DUTY:	1.036	0.16	Z2= 68.4869	1.036	
	480. VOLTS LN/LN:	1.521		Z0= 211.6664		
	LN/LN/GND:	1.589	(	0.734 GND RETURN KA)		
DSC 11-SF-1575	3P Duty:	1.759	0.20	Z1= 68.3879	1.759	1.759
	SLG DUTY:	1.036	0.16	Z2= 68.3879	1.036	
	480. VOLTS LN/LN:	1.523		Z0= 211.7244		
	LN/LN/GND:	1.589	(	0.734 GND RETURN KA)		
DSC 12-AUG-4615	3P Duty:	2.637	0.26	Z1= 45.6049	2.637	2.637
	SLG DUTY:	1.543	0.16	Z2= 45.6049	1.543	
	480. VOLTS LN/LN:	2.284		Z0= 143.2130		
	LN/LN/GND:	2.414	(	1.088 GND RETURN KA)		

UNBALANCED FAULT REPORT
(FOR APPLICATION OF LOW VOLTAGE BREAKERS)
PRE FAULT VOLTAGE: 1.0000
MODEL TRANSFORMER TAPS: NO

LOCATION VOLTAGE	FAULT DUTIES	KA (RMS)	X/R	EQUIVALENT (PU) FAULT IMPEDANCE	ASYM. KA AT 0.5 CYCLES * MAX. RMS	AVG. RMS *
DSC 12-AUG-4617	3P Duty:	3.213	0.31	Z1= 37.4341	3.213	3.213
	SLG DUTY:	1.903	0.19	Z2= 37.4341	1.903	
	480. VOLTS LN/LN:	2.783		Z0= 115.3827		
	LN/LN/GND:	2.953 (	1.348	GND RETURN KA)		
DSC 12-CON-4610	3P Duty:	1.455	0.13	Z1= 82.6521	1.455	1.455
	SLG DUTY:	0.849	0.09	Z2= 82.6521	0.849	
	480. VOLTS LN/LN:	1.260		Z0= 259.8757		
	LN/LN/GND:	1.316 (	0.599	GND RETURN KA)		
DSC 12-CON-4620	3P Duty:	1.797	0.15	Z1= 66.9174	1.797	1.797
	SLG DUTY:	1.055	0.10	Z2= 66.9174	1.055	
	480. VOLTS LN/LN:	1.557		Z0= 208.6461		
	LN/LN/GND:	1.629 (	0.746	GND RETURN KA)		
DSC 12-EF-2530	3P Duty:	4.154	0.31	Z1= 28.9581	4.154	4.154
	SLG DUTY:	2.485	0.20	Z2= 28.9581	2.485	
	480. VOLTS LN/LN:	3.597		Z0= 87.6893		
	LN/LN/GND:	3.815 (	1.768	GND RETURN KA)		
DSC 12-EF-2540	3P Duty:	4.122	0.34	Z1= 29.1788	4.122	4.122
	SLG DUTY:	2.480	0.21	Z2= 29.1788	2.480	
	480. VOLTS LN/LN:	3.570		Z0= 87.6306		
	LN/LN/GND:	3.800 (	1.768	GND RETURN KA)		
DSC 12-HPU-4750	3P Duty:	3.230	0.28	Z1= 37.2442	3.230	3.230
	SLG DUTY:	1.905	0.18	Z2= 37.2442	1.905	
	480. VOLTS LN/LN:	2.797		Z0= 115.4404		
	LN/LN/GND:	2.960 (	1.348	GND RETURN KA)		
DSC 12-SF-2510	3P Duty:	4.154	0.31	Z1= 28.9581	4.154	4.154
	SLG DUTY:	2.485	0.20	Z2= 28.9581	2.485	
	480. VOLTS LN/LN:	3.597		Z0= 87.6893		
	LN/LN/GND:	3.815 (	1.768	GND RETURN KA)		
DSC 12-SF-2520	3P Duty:	4.122	0.34	Z1= 29.1788	4.122	4.122
	SLG DUTY:	2.480	0.21	Z2= 29.1788	2.480	
	480. VOLTS LN/LN:	3.570		Z0= 87.6306		
	LN/LN/GND:	3.800 (	1.768	GND RETURN KA)		

UNBALANCED FAULT REPORT
(FOR APPLICATION OF LOW VOLTAGE BREAKERS)
PRE FAULT VOLTAGE: 1.0000
MODEL TRANSFORMER TAPS: NO

LOCATION VOLTAGE	FAULT DUTIES	KA (RMS)	X/R	EQUIVALENT (PU) FAULT IMPEDANCE	ASYM. KA AT 0.5 CYCLES * MAX. RMS	AVG. RMS *
DSC 12-SF-2550	3P Duty:	2.348	0.15	Z1= 51.2230	2.348	2.348
	SLG DUTY:	1.389	0.11	Z2= 51.2230	1.389	
	480. VOLTS LN/LN:	2.034		Z0= 157.5382		
	LN/LN/GND:	2.122 (	0.985	GND RETURN KA)		
DSC 12-SF-2560	3P Duty:	1.156	0.10	Z1= 104.0280	1.156	1.156
	SLG DUTY:	0.679	0.08	Z2= 104.0280	0.679	
	480. VOLTS LN/LN:	1.001		Z0= 323.7997		
	LN/LN/GND:	1.039 (	0.480	GND RETURN KA)		
DSC 12-SF-2570	3P Duty:	1.445	0.12	Z1= 83.2545	1.445	1.445
	SLG DUTY:	0.847	0.08	Z2= 83.2545	0.847	
	480. VOLTS LN/LN:	1.251		Z0= 259.8757		
	LN/LN/GND:	1.302 (	0.599	GND RETURN KA)		
DSC 12-SF-2580	3P Duty:	1.162	0.11	Z1= 103.4924	1.162	1.162
	SLG DUTY:	0.680	0.08	Z2= 103.4924	0.680	
	480. VOLTS LN/LN:	1.007		Z0= 323.7997		
	LN/LN/GND:	1.047 (	0.481	GND RETURN KA)		
DSC 12-SRW-4612	3P Duty:	1.445	0.12	Z1= 83.2545	1.445	1.445
	SLG DUTY:	0.847	0.08	Z2= 83.2545	0.847	
	480. VOLTS LN/LN:	1.251		Z0= 259.8757		
	LN/LN/GND:	1.302 (	0.599	GND RETURN KA)		
DSC 12-SRW-4613	3P Duty:	1.442	0.13	Z1= 83.4079	1.442	1.442
	SLG DUTY:	0.846	0.09	Z2= 83.4079	0.846	
	480. VOLTS LN/LN:	1.249		Z0= 259.8216		
	LN/LN/GND:	1.301 (	0.599	GND RETURN KA)		
DSC 19-PMP-1110	3P Duty:	1.920	0.23	Z1= 62.6355	1.920	1.920
	SLG DUTY:	1.116	0.14	Z2= 62.6355	1.116	
	480. VOLTS LN/LN:	1.663		Z0= 198.7864		
	LN/LN/GND:	1.753 (	0.785	GND RETURN KA)		
DSC 19-PMP-1120	3P Duty:	2.209	0.25	Z1= 54.4451	2.209	2.209
	SLG DUTY:	1.293	0.16	Z2= 54.4451	1.293	
	480. VOLTS LN/LN:	1.913		Z0= 170.9397		
	LN/LN/GND:	2.021 (	0.912	GND RETURN KA)		

UNBALANCED FAULT REPORT
(FOR APPLICATION OF LOW VOLTAGE BREAKERS)
PRE FAULT VOLTAGE: 1.0000
MODEL TRANSFORMER TAPS: NO

LOCATION VOLTAGE	FAULT DUTIES	KA (RMS)	X/R	EQUIVALENT (PU) FAULT IMPEDANCE	ASYM. KA AT 0.5 CYCLES * MAX. RMS	AVG. RMS *
DSC 21-EF-1250	3P Duty:	2.346	0.14	Z1= 51.2653	2.346	2.346
	SLG DUTY:	1.389	0.10	Z2= 51.2653	1.389	
	480. VOLTS LN/LN:	2.032		Z0= 157.3334		
	LN/LN/GND:	2.121	(0.986	GND RETURN KA)		
DSC 21-EF-1310	3P Duty:	1.158	0.09	Z1= 103.8575	1.158	1.158
	SLG DUTY:	0.679	0.07	Z2= 103.8575	0.679	
	480. VOLTS LN/LN:	1.003		Z0= 323.6770		
	LN/LN/GND:	1.040	(0.480	GND RETURN KA)		
DSC 2P-07	3P Duty:	0.949	0.14	Z1= 126.7204	0.949	0.949
	SLG DUTY:	0.549	0.09	Z2= 126.7204	0.549	
	480. VOLTS LN/LN:	0.822		Z0= 405.0337		
	LN/LN/GND:	0.862	(0.385	GND RETURN KA)		
DSC 3G-01	3P Duty:	0.628	0.08	Z1= 191.4580	0.628	0.628
	SLG DUTY:	0.365	0.06	Z2= 191.4580	0.365	
	480. VOLTS LN/LN:	0.544		Z0= 605.2314		
	LN/LN/GND:	0.565	(0.257	GND RETURN KA)		
DSC 3G-02	3P Duty:	0.735	0.09	Z1= 163.6487	0.735	0.735
	SLG DUTY:	0.428	0.06	Z2= 163.6487	0.428	
	480. VOLTS LN/LN:	0.637		Z0= 515.5908		
	LN/LN/GND:	0.662	(0.302	GND RETURN KA)		
DSC 3G-03	3P Duty:	0.769	0.09	Z1= 156.3339	0.769	0.769
	SLG DUTY:	0.448	0.06	Z2= 156.3339	0.448	
	480. VOLTS LN/LN:	0.666		Z0= 492.6073		
	LN/LN/GND:	0.692	(0.316	GND RETURN KA)		
DSC 3P-05	3P Duty:	0.971	0.09	Z1= 123.9019	0.971	0.971
	SLG DUTY:	0.568	0.07	Z2= 123.9019	0.568	
	480. VOLTS LN/LN:	0.841		Z0= 387.6645		
	LN/LN/GND:	0.872	(0.401	GND RETURN KA)		
DSC 3P-07	3P Duty:	1.075	0.10	Z1= 111.9297	1.075	1.075
	SLG DUTY:	0.630	0.07	Z2= 111.9297	0.630	
	480. VOLTS LN/LN:	0.931		Z0= 349.2183		
	LN/LN/GND:	0.967	(0.445	GND RETURN KA)		

UNBALANCED FAULT REPORT
(FOR APPLICATION OF LOW VOLTAGE BREAKERS)
PRE FAULT VOLTAGE: 1.0000
MODEL TRANSFORMER TAPS: NO

LOCATION VOLTAGE	FAULT DUTIES	KA (RMS)	X/R	EQUIVALENT (PU) FAULT IMPEDANCE	ASYM. KA AT 0.5 CYCLES * MAX. RMS	AVG. RMS *
DSC 4P-01	3P Duty:	2.195	0.42	Z1= 54.8062	2.195	2.195
	SLG DUTY:	1.243	0.28	Z2= 54.8062	1.243	
	480. VOLTS LN/LN:	1.901		Z0= 181.8093		
	LN/LN/GND:	2.013	(0.864	GND RETURN KA)		
DSC 4P-02	3P Duty:	2.218	0.52	Z1= 54.2329	2.218	2.218
	SLG DUTY:	1.265	0.31	Z2= 54.2329	1.265	
	480. VOLTS LN/LN:	1.921		Z0= 178.2633		
	LN/LN/GND:	2.047	(0.881	GND RETURN KA)		
DSC 4P-03	3P Duty:	2.275	0.42	Z1= 52.8597	2.275	2.275
	SLG DUTY:	1.291	0.28	Z2= 52.8597	1.291	
	480. VOLTS LN/LN:	1.971		Z0= 174.8436		
	LN/LN/GND:	2.086	(0.898	GND RETURN KA)		
DSC SHB CRANE	3P Duty:	3.136	0.24	Z1= 38.3544	3.136	3.136
	SLG DUTY:	1.881	0.17	Z2= 38.3544	1.881	
	480. VOLTS LN/LN:	2.716		Z0= 115.3827		
	LN/LN/GND:	2.859	(1.342	GND RETURN KA)		
GEN SWBD [Ex]	3P Duty:	15.680	17.23	Z1= 7.6711	24.234	20.211
	SLG DUTY:	15.516	15.32	Z2= 7.6711	23.670	
	480. VOLTS LN/LN:	13.579		Z0= 7.9152		
	LN/LN/GND:	15.709	(15.355	GND RETURN KA)		
LP-10	3P Duty:	1.679	0.92	Z1= 165.3013	1.681	1.680
	SLG DUTY:	1.703	0.89	Z2= 165.3013	1.704	
	208. VOLTS LN/LN:	1.454		Z0= 158.6323		
	LN/LN/GND:	1.717	(1.726	GND RETURN KA)		
LP-11	3P Duty:	1.477	0.90	Z1= 187.8894	1.479	1.478
	SLG DUTY:	1.520	0.85	Z2= 187.8894	1.521	
	208. VOLTS LN/LN:	1.279		Z0= 172.2249		
	LN/LN/GND:	1.532	(1.565	GND RETURN KA)		
LP-13A	3P Duty:	4.119	1.53	Z1= 67.3844	4.187	4.153
	SLG DUTY:	4.248	1.49	Z2= 67.3844	4.310	
	208. VOLTS LN/LN:	3.567		Z0= 61.2716		
	LN/LN/GND:	4.238	(4.385	GND RETURN KA)		

UNBALANCED FAULT REPORT
(FOR APPLICATION OF LOW VOLTAGE BREAKERS)
PRE FAULT VOLTAGE: 1.0000
MODEL TRANSFORMER TAPS: NO

LOCATION VOLTAGE	FAULT DUTIES	KA (RMS)	X/R	EQUIVALENT (PU) FAULT IMPEDANCE	ASYM. KA AT 0.5 CYCLES * MAX. RMS	AVG. RMS *
LP-13B	3P Duty:	3.902	1.39	Z1= 71.1386	3.944	3.923
	SLG DUTY:	3.872	1.26	Z2= 71.1386	3.899	
	208. VOLTS LN/LN:	3.379		Z0= 73.2214		
	LN/LN/GND:	4.034	(3.835	GND RETURN KA)		
LP-4PA	3P Duty:	0.273	1.16	Z1= 880.9971	0.273	0.274
	SLG DUTY:	0.264	1.09	Z2= 880.9971	0.264	
	240. VOLTS LN/LN:	0.236		Z0= 976.3798		
	LN/LN/GND:	0.275	(0.255	GND RETURN KA)		
LP-5	3P Duty:	4.129	1.51	Z1= 67.2184	4.194	4.162
	SLG DUTY:	4.306	1.49	Z2= 67.2184	4.369	
	208. VOLTS LN/LN:	3.576		Z0= 58.9630		
	LN/LN/GND:	4.251	(4.498	GND RETURN KA)		
LP-6	3P Duty:	1.694	0.93	Z1= 163.8210	1.696	1.695
	SLG DUTY:	1.713	0.89	Z2= 163.8210	1.715	
	208. VOLTS LN/LN:	1.467		Z0= 158.6323		
	LN/LN/GND:	1.731	(1.732	GND RETURN KA)		
LP-8	3P Duty:	4.235	1.52	Z1= 65.5392	4.303	4.269
	SLG DUTY:	4.330	1.48	Z2= 65.5392	4.392	
	208. VOLTS LN/LN:	3.668		Z0= 61.2716		
	LN/LN/GND:	4.331	(4.428	GND RETURN KA)		
LP-9	3P Duty:	3.212	0.86	Z1= 86.4197	3.214	3.213
	SLG DUTY:	3.231	0.83	Z2= 86.4197	3.233	
	208. VOLTS LN/LN:	2.782		Z0= 84.9903		
	LN/LN/GND:	3.276	(3.249	GND RETURN KA)		
MCC-10A	3P Duty:	10.498	1.47	Z1= 11.4571	10.642	10.570
	SLG DUTY:	6.718	1.25	Z2= 11.4571	6.761	
	480. VOLTS LN/LN:	9.092		Z0= 30.9225		
	LN/LN/GND:	9.689	(4.929	GND RETURN KA)		
MCC-10B	3P Duty:	9.098	1.90	Z1= 13.2203	9.426	9.263
	SLG DUTY:	6.338	1.42	Z2= 13.2203	6.413	
	480. VOLTS LN/LN:	7.879		Z0= 30.8426		
	LN/LN/GND:	8.651	(4.835	GND RETURN KA)		

UNBALANCED FAULT REPORT
(FOR APPLICATION OF LOW VOLTAGE BREAKERS)
PRE FAULT VOLTAGE: 1.0000
MODEL TRANSFORMER TAPS: NO

LOCATION VOLTAGE	FAULT DUTIES	KA (RMS)	X/R	EQUIVALENT (PU) FAULT IMPEDANCE	ASYM. KA AT 0.5 CYCLES * MAX. RMS	AVG. RMS *
MCC-11A	3P Duty:	9.178	1.36	Z1= 13.1058	9.267	9.223
	SLG DUTY:	6.365	1.02	Z2= 13.1058	6.379	
	480. VOLTS LN/LN:	7.948		Z0= 30.8813		
	LN/LN/GND:	8.750	(4.840	GND RETURN KA)		
MCC-11B	3P Duty:	4.012	1.33	Z1= 29.9836	4.047	4.029
	SLG DUTY:	2.467	0.93	Z2= 29.9836	2.470	
	480. VOLTS LN/LN:	3.474		Z0= 87.2885		
	LN/LN/GND:	3.753	(1.770	GND RETURN KA)		
MCC-5A	3P Duty:	15.588	2.05	Z1= 7.7162	16.304	15.948
	SLG DUTY:	12.779	1.65	Z2= 7.7162	13.060	
	480. VOLTS LN/LN:	13.500		Z0= 12.9389		
	LN/LN/GND:	15.265	(10.777	GND RETURN KA)		
MCC-5B	3P Duty:	14.721	2.16	Z1= 8.1708	15.508	15.117
	SLG DUTY:	12.426	1.71	Z2= 8.1708	12.740	
	480. VOLTS LN/LN:	12.749		Z0= 12.8590		
	LN/LN/GND:	14.592	(10.692	GND RETURN KA)		
MCC-6A	3P Duty:	9.684	1.79	Z1= 12.4205	9.972	9.829
	SLG DUTY:	6.937	1.40	Z2= 12.4205	7.015	
	480. VOLTS LN/LN:	8.387		Z0= 27.4295		
	LN/LN/GND:	9.208	(5.379	GND RETURN KA)		
MCC-6B	3P Duty:	9.363	1.87	Z1= 12.8467	9.684	9.524
	SLG DUTY:	6.841	1.44	Z2= 12.8467	6.926	
	480. VOLTS LN/LN:	8.108		Z0= 27.3495		
	LN/LN/GND:	8.962	(5.361	GND RETURN KA)		
MCC-8A	3P Duty:	21.300	2.75	Z1= 5.6469	23.365	22.345
	SLG DUTY:	19.231	2.30	Z2= 5.6469	20.451	
	480. VOLTS LN/LN:	18.447		Z0= 7.5148		
	LN/LN/GND:	21.204	(17.486	GND RETURN KA)		
MCC-8B	3P Duty:	19.533	2.84	Z1= 6.1580	21.570	20.564
	SLG DUTY:	18.318	2.39	Z2= 6.1580	19.592	
	480. VOLTS LN/LN:	16.916		Z0= 7.4362		
	LN/LN/GND:	19.778	(17.199	GND RETURN KA)		

UNBALANCED FAULT REPORT
(FOR APPLICATION OF LOW VOLTAGE BREAKERS)
PRE FAULT VOLTAGE: 1.0000
MODEL TRANSFORMER TAPS: NO

LOCATION VOLTAGE	FAULT DUTIES	KA (RMS)	X/R	EQUIVALENT (PU) FAULT IMPEDANCE	ASYM. KA AT 0.5 CYCLES * MAX. RMS	AVG. RMS *
MCC-9A	3P Duty:	22.689	2.35	Z1= 5.3013	24.208	23.454
	SLG DUTY:	21.630	1.93	Z2= 5.3013	22.451	
	480. VOLTS LN/LN:	19.649		Z0= 6.1565		
	LN/LN/GND:	23.403	(20.571	GND RETURN KA)		
MCC-9B	3P Duty:	20.751	2.48	Z1= 5.7963	22.336	21.551
	SLG DUTY:	20.529	2.03	Z2= 5.7963	21.438	
	480. VOLTS LN/LN:	17.971		Z0= 6.0768		
	LN/LN/GND:	21.858	(20.205	GND RETURN KA)		
MTS-10	3P Duty:	8.192	1.58	Z1= 14.6821	8.345	8.269
	SLG DUTY:	5.578	1.20	Z2= 14.6821	5.607	
	480. VOLTS LN/LN:	7.095		Z0= 35.7275		
	LN/LN/GND:	7.762	(4.204	GND RETURN KA)		
MTS-11	3P Duty:	3.724	1.19	Z1= 32.2952	3.743	3.734
	SLG DUTY:	2.273	0.85	Z2= 32.2952	2.274	
	480. VOLTS LN/LN:	3.225		Z0= 95.1323		
	LN/LN/GND:	3.474	(1.626	GND RETURN KA)		
MTS-13	3P Duty:	13.996	2.21	Z1= 8.5940	14.785	14.393
	SLG DUTY:	11.094	1.81	Z2= 8.5940	11.433	
	480. VOLTS LN/LN:	12.121		Z0= 15.4471		
	LN/LN/GND:	13.493	(9.159	GND RETURN KA)		
MTS-5	3P Duty:	12.772	1.81	Z1= 9.4177	13.160	12.967
	SLG DUTY:	10.225	1.41	Z2= 9.4177	10.341	
	480. VOLTS LN/LN:	11.061		Z0= 16.7021		
	LN/LN/GND:	12.549	(8.467	GND RETURN KA)		
MTS-6	3P Duty:	8.838	1.78	Z1= 13.6094	9.095	8.967
	SLG DUTY:	6.387	1.37	Z2= 13.6094	6.453	
	480. VOLTS LN/LN:	7.654		Z0= 29.5918		
	LN/LN/GND:	8.441	(4.974	GND RETURN KA)		
MTS-8	3P Duty:	17.252	2.24	Z1= 6.9718	18.261	17.760
	SLG DUTY:	15.192	1.76	Z2= 6.9718	15.612	
	480. VOLTS LN/LN:	14.941		Z0= 9.9598		
	LN/LN/GND:	17.399	(13.486	GND RETURN KA)		

UNBALANCED FAULT REPORT
(FOR APPLICATION OF LOW VOLTAGE BREAKERS)
PRE FAULT VOLTAGE: 1.0000
MODEL TRANSFORMER TAPS: NO

LOCATION VOLTAGE	FAULT DUTIES	KA (RMS)	X/R	EQUIVALENT (PU) FAULT IMPEDANCE	ASYM. KA AT 0.5 CYCLES * MAX. RMS	AVG. RMS *
MTS-9	3P Duty:	17.065	1.90	Z1= 7.0485	17.681	17.374
	SLG DUTY:	15.154	1.45	Z2= 7.0485	15.354	
480. VOLTS	LN/LN:	14.778		Z0= 9.9522		
	LN/LN/GND:	17.485	(13.494	GND RETURN KA)		
SHB CRANE	3P Duty:	2.898	0.23	Z1= 41.5047	2.898	2.898
	SLG DUTY:	1.734	0.17	Z2= 41.5047	1.734	
480. VOLTS	LN/LN:	2.510		Z0= 125.3793		
	LN/LN/GND:	2.638	(1.235	GND RETURN KA)		
SWBD-MAIN-A	3P Duty:	27.813	3.39	Z1= 4.3247	31.873	29.879
	SLG DUTY:	30.118	3.51	Z2= 4.3247	34.790	
480. VOLTS	LN/LN:	24.087		Z0= 3.3338		
	LN/LN/GND:	29.483	(32.834	GND RETURN KA)		
SWBD-MAIN-B	3P Duty:	24.937	3.44	Z1= 4.8234	28.677	26.841
	SLG DUTY:	27.959	3.57	Z2= 4.8234	32.420	
480. VOLTS	LN/LN:	21.596		Z0= 3.2611		
	LN/LN/GND:	27.110	(31.811	GND RETURN KA)		

F A U L T S T U D Y S U M M A R Y
(FOR APPLICATION OF LOW VOLTAGE BREAKERS)
PRE FAULT VOLTAGE: 1.0000
MODEL TRANSFORMER TAPS: NO

BUS RECORD NO NAME	VOLTAGE		A V A I L A B L E		F A U L T D U T I E S (KA)	
	L-L	3 PHASE	X/R	LINE/GRND	X/R	
=====						
05-VFD-1510	480.	10.489	1.46	6.713	1.25	
05-VFD-1520	480.	9.089	1.90	6.333	1.42	
05-VFD-1630	480.	10.486	1.46	6.713	1.25	
06-VFD-1610	480.	10.486	1.46	6.713	1.25	
06-VFD-1615	480.	9.089	1.90	6.333	1.42	
06-VFD-1625	480.	9.089	1.90	6.333	1.42	
06-VFD-1630	480.	10.486	1.46	6.713	1.25	
06-VFD-1635	480.	9.089	1.90	6.333	1.42	
07-VFD-1110	480.	17.030	1.16	13.368	0.80	
07-VFD-1120	480.	16.058	1.28	13.045	0.86	
09-VFD-1100	480.	9.169	1.36	6.361	1.02	
09-VFD-1200	480.	4.010	1.33	2.466	0.93	
11-VFD-2312	480.	9.674	1.79	6.932	1.40	
11-VFD-2321	480.	9.354	1.87	6.836	1.44	
11-VFD-2410	480.	9.674	1.79	6.931	1.40	
11-VFD-2420	480.	9.353	1.87	6.835	1.44	
12-VCP1-4110	480.	17.234	2.11	14.085	1.63	
12-VCP1-4210	480.	16.102	2.23	13.622	1.70	
12-VFD-3320	480.	21.253	2.74	19.190	2.30	
12-VFD-3420	480.	19.493	2.84	18.282	2.38	
12-VFD-3520	480.	21.253	2.74	19.190	2.30	
21-GEN-1100 [Ex]	480.	15.586	17.20	15.381	16.00	
21-GEN-1200	480.	15.862	18.96	15.786	18.19	
21-PNL-1200	480.	2.375	0.20	1.415	0.14	
21-VFD-1100	480.	13.493	0.99	9.981	0.72	
21-VFD-1200	480.	12.878	1.08	9.798	0.77	
21-VFD-1300	480.	13.493	0.99	9.981	0.72	
22-VFD-2022	480.	9.089	1.90	6.333	1.42	
22-VFD-2201	480.	10.486	1.46	6.713	1.25	
22-VFD-2203	480.	10.486	1.46	6.713	1.25	
23-VCP-2810	480.	2.496	0.28	1.482	0.22	
ADMIN	480.	8.209	1.21	5.778	0.97	
AHF-10A	480.	8.143	0.91	5.052	0.81	
AHF-10B	480.	5.705	0.75	3.567	0.58	
AHF-8A	480.	17.246	1.58	13.954	1.20	
AHF-8B	480.	15.524	1.59	12.768	1.17	
CL BLDG PORT GEN	480.	7.617	1.10	5.099	0.84	
DCB 1 PORT GEN	480.	8.596	1.74	6.181	1.35	
DP-10	480.	7.921	1.50	5.357	1.14	
DP-13	480.	13.620	2.18	10.728	1.79	
DP-5A	480.	15.064	1.95	12.194	1.56	
DP-5B	480.	13.962	1.99	11.533	1.56	
DP-6	480.	8.643	1.75	6.221	1.35	
DP-8A-1	480.	19.814	2.34	17.288	1.89	

FAULT STUDY SUMMARY
(FOR APPLICATION OF LOW VOLTAGE BREAKERS)

PRE FAULT VOLTAGE: 1.0000

MODEL TRANSFORMER TAPS: NO

BUS RECORD NO NAME	VOLTAGE	A V A I L A B L E		F A U L T D U T I E S (KA)	
	L-L	3 PHASE	X/R	LINE/GRND	X/R
=====					
DP-8A-2	480.	18.451	1.96	15.538	1.54
DP-8B-1	480.	18.052	2.39	16.240	1.91
DP-8B-2	480.	16.924	2.04	14.691	1.57
DP-9	480.	16.320	1.81	14.193	1.38
DSC 02-EF-1320	480.	0.960	0.11	0.561	0.08
DSC 02-EF-1330	480.	1.151	0.13	0.672	0.09
DSC 02-EF-1340	480.	1.150	0.14	0.672	0.09
DSC 02-EF-1370	480.	1.416	0.13	0.831	0.10
DSC 02-EF-1380	480.	1.143	0.12	0.670	0.09
DSC 02-EF-1390	480.	1.778	0.19	1.044	0.14
DSC 02-EF-1396	480.	1.490	0.18	0.876	0.13
DSC 02-SF-1310	480.	1.143	0.12	0.670	0.09
DSC 02-SF-1350	480.	1.157	0.14	0.674	0.09
DSC 02-SF-1360	480.	1.156	0.14	0.673	0.09
DSC 02-SF-1570	480.	1.412	0.13	0.830	0.10
DSC 02-SF-1575	480.	1.279	0.15	0.753	0.11
DSC 02-SF-1810	480.	1.410	0.14	0.830	0.10
DSC 03-EF-1420	480.	1.851	0.17	1.093	0.12
DSC 03-HSS-1140	480.	0.833	0.08	0.487	0.06
DSC 03-HSS-1240	480.	0.730	0.08	0.427	0.06
DSC 03-HSS-1340	480.	0.686	0.08	0.401	0.06
DSC 03-PMP-1510	480.	1.454	0.12	0.849	0.08
DSC 03-PMP-1520	480.	1.135	0.13	0.659	0.08
DSC 03-PMP-1530	480.	0.984	0.11	0.571	0.07
DSC 03-PMP-1610	480.	0.746	0.10	0.431	0.07
DSC 03-PMP-1620	480.	0.848	0.11	0.491	0.07
DSC 03-SCC-1150	480.	1.158	0.09	0.679	0.07
DSC 03-SCC-1250	480.	1.113	0.10	0.653	0.07
DSC 03-SCC-1350	480.	0.833	0.08	0.487	0.06
DSC 03-SF-1410	480.	0.726	0.10	0.424	0.07
DSC 03-SF-1575	480.	0.723	0.09	0.423	0.07
DSC 03-SLC-1120	480.	1.439	0.10	0.845	0.08
DSC 03-SLC-1220	480.	1.113	0.10	0.653	0.07
DSC 03-SLC-1320	480.	0.833	0.08	0.487	0.06
DSC 04-PMP-1140	480.	2.343	0.53	1.341	0.32
DSC 06-CLA-1510	480.	2.039	0.21	1.192	0.19
DSC 06-CLA-1520	480.	1.234	0.19	0.722	0.15
DSC 06-CLA-1530	480.	0.889	0.13	0.518	0.12
DSC 06-PMP-1515	480.	2.796	0.31	1.634	0.26
DSC 06-PMP-1525	480.	1.745	0.30	1.016	0.22
DSC 06-PMP-1535	480.	1.280	0.22	0.739	0.17

FAULT STUDY SUMMARY
(FOR APPLICATION OF LOW VOLTAGE BREAKERS)
PRE FAULT VOLTAGE: 1.0000
MODEL TRANSFORMER TAPS: NO

BUS RECORD NO NAME	VOLTAGE		A V A I L A B L E		F A U L T D U T I E S (KA)	
	L-L	3 PHASE	X/R	LINE/GRND	X/R	
=====						
DSC 08-EF-1230	480.	3.506	0.35	2.105	0.27	
DSC 09-MIX-1110	480.	0.799	0.12	0.466	0.09	
DSC 09-PMP-903	480.	0.862	0.21	0.498	0.17	
DSC 11-EF-1530	480.	1.761	0.21	1.037	0.16	
DSC 11-EF-1540	480.	2.130	0.25	1.261	0.19	
DSC 11-EF-1560	480.	1.755	0.21	1.035	0.16	
DSC 11-PMP-2311	480.	1.371	0.19	0.801	0.14	
DSC 11-PMP-2314	480.	1.760	0.21	1.036	0.16	
DSC 11-PMP-2322	480.	1.369	0.19	0.800	0.14	
DSC 11-PMP-2324	480.	2.128	0.25	1.261	0.19	
DSC 11-PMP-2610	480.	1.361	0.17	0.798	0.13	
DSC 11-PMP-2620	480.	1.358	0.18	0.798	0.14	
DSC 11-PMP-2711	480.	1.362	0.17	0.798	0.13	
DSC 11-PMP-2721	480.	1.661	0.21	0.978	0.16	
DSC 11-SF-1510	480.	1.761	0.21	1.037	0.16	
DSC 11-SF-1520	480.	2.130	0.25	1.261	0.19	
DSC 11-SF-1550	480.	1.756	0.22	1.036	0.16	
DSC 11-SF-1575	480.	1.759	0.20	1.036	0.16	
DSC 12-AUG-4615	480.	2.637	0.26	1.543	0.16	
DSC 12-AUG-4617	480.	3.213	0.31	1.903	0.19	
DSC 12-CON-4610	480.	1.455	0.13	0.849	0.09	
DSC 12-CON-4620	480.	1.797	0.15	1.055	0.10	
DSC 12-EF-2530	480.	4.154	0.31	2.485	0.20	
DSC 12-EF-2540	480.	4.122	0.34	2.480	0.21	
DSC 12-HPU-4750	480.	3.230	0.28	1.905	0.18	
DSC 12-SF-2510	480.	4.154	0.31	2.485	0.20	
DSC 12-SF-2520	480.	4.122	0.34	2.480	0.21	
DSC 12-SF-2550	480.	2.348	0.15	1.389	0.11	
DSC 12-SF-2560	480.	1.156	0.10	0.679	0.08	
DSC 12-SF-2570	480.	1.445	0.12	0.847	0.08	
DSC 12-SF-2580	480.	1.162	0.11	0.680	0.08	
DSC 12-SRW-4612	480.	1.445	0.12	0.847	0.08	
DSC 12-SRW-4613	480.	1.442	0.13	0.846	0.09	
DSC 19-PMP-1110	480.	1.920	0.23	1.116	0.14	
DSC 19-PMP-1120	480.	2.209	0.25	1.293	0.16	
DSC 21-EF-1250	480.	2.346	0.14	1.389	0.10	
DSC 21-EF-1310	480.	1.158	0.09	0.679	0.07	
DSC 2P-07	480.	0.949	0.14	0.549	0.09	
DSC 3G-01	480.	0.628	0.08	0.365	0.06	
DSC 3G-02	480.	0.735	0.09	0.428	0.06	
DSC 3G-03	480.	0.769	0.09	0.448	0.06	

F A U L T S T U D Y S U M M A R Y
(FOR APPLICATION OF LOW VOLTAGE BREAKERS)
PRE FAULT VOLTAGE: 1.0000
MODEL TRANSFORMER TAPS: NO

BUS RECORD NO NAME	VOLTAGE L-L	A V A I L A B L E 3 PHASE	F A U L T X/R	D U T I E S (KA) LINE/GRND	X/R
DSC 3P-05	480.	0.971	0.09	0.568	0.07
DSC 3P-07	480.	1.075	0.10	0.630	0.07
DSC 4P-01	480.	2.195	0.42	1.243	0.28
DSC 4P-02	480.	2.218	0.52	1.265	0.31
DSC 4P-03	480.	2.275	0.42	1.291	0.28
DSC SHB CRANE	480.	3.136	0.24	1.881	0.17
GEN SWBD [Ex]	480.	15.680	17.23	15.516	15.32
LP-10	208.	1.679	0.92	1.703	0.89
LP-11	208.	1.477	0.90	1.520	0.85
LP-13A	208.	4.119	1.53	4.248	1.49
LP-13B	208.	3.902	1.39	3.872	1.26
LP-4PA	240.	0.273	1.16	0.264	1.09
LP-5	208.	4.129	1.51	4.306	1.49
LP-6	208.	1.694	0.93	1.713	0.89
LP-8	208.	4.235	1.52	4.330	1.48
LP-9	208.	3.212	0.86	3.231	0.83
MCC-10A	480.	10.498	1.47	6.718	1.25
MCC-10B	480.	9.098	1.90	6.338	1.42
MCC-11A	480.	9.178	1.36	6.365	1.02
MCC-11B	480.	4.012	1.33	2.467	0.93
MCC-5A	480.	15.588	2.05	12.779	1.65
MCC-5B	480.	14.721	2.16	12.426	1.71
MCC-6A	480.	9.684	1.79	6.937	1.40
MCC-6B	480.	9.363	1.87	6.841	1.44
MCC-8A	480.	21.300	2.75	19.231	2.30
MCC-8B	480.	19.533	2.84	18.318	2.39
MCC-9A	480.	22.689	2.35	21.630	1.93
MCC-9B	480.	20.751	2.48	20.529	2.03
MTS-10	480.	8.192	1.58	5.578	1.20
MTS-11	480.	3.724	1.19	2.273	0.85
MTS-13	480.	13.996	2.21	11.094	1.81
MTS-5	480.	12.772	1.81	10.225	1.41
MTS-6	480.	8.838	1.78	6.387	1.37
MTS-8	480.	17.252	2.24	15.192	1.76
MTS-9	480.	17.065	1.90	15.154	1.45
SHB CRANE	480.	2.898	0.23	1.734	0.17
SWBD-MAIN-A	480.	27.813	3.39	30.118	3.51
SWBD-MAIN-B	480.	24.937	3.44	27.959	3.57

319 FAULTED BUSES, 438 BRANCHES, 122 CONTRIBUTIONS
UNBALANCED FAULTS REQUESTED

*** SHORT CIRCUIT STUDY COMPLETE ***

THREE PHASE MOMENTARY DUTY REPORT
PRE FAULT VOLTAGE: 1.0000
MODEL TRANSFORMER TAPS: NO

=====			
05-VFD-1510	VOLTAGE:	480.	(SEE LOW VOLTAGE REPORT)
05-VFD-1520	VOLTAGE:	480.	(SEE LOW VOLTAGE REPORT)
05-VFD-1630	VOLTAGE:	480.	(SEE LOW VOLTAGE REPORT)
06-VFD-1610	VOLTAGE:	480.	(SEE LOW VOLTAGE REPORT)
06-VFD-1615	VOLTAGE:	480.	(SEE LOW VOLTAGE REPORT)
06-VFD-1625	VOLTAGE:	480.	(SEE LOW VOLTAGE REPORT)
06-VFD-1630	VOLTAGE:	480.	(SEE LOW VOLTAGE REPORT)
06-VFD-1635	VOLTAGE:	480.	(SEE LOW VOLTAGE REPORT)
07-VFD-1110	VOLTAGE:	480.	(SEE LOW VOLTAGE REPORT)
07-VFD-1120	VOLTAGE:	480.	(SEE LOW VOLTAGE REPORT)
09-VFD-1100	VOLTAGE:	480.	(SEE LOW VOLTAGE REPORT)
09-VFD-1200	VOLTAGE:	480.	(SEE LOW VOLTAGE REPORT)
11-VFD-2312	VOLTAGE:	480.	(SEE LOW VOLTAGE REPORT)
11-VFD-2321	VOLTAGE:	480.	(SEE LOW VOLTAGE REPORT)
11-VFD-2410	VOLTAGE:	480.	(SEE LOW VOLTAGE REPORT)
11-VFD-2420	VOLTAGE:	480.	(SEE LOW VOLTAGE REPORT)
12-VCP1-4110	VOLTAGE:	480.	(SEE LOW VOLTAGE REPORT)
12-VCP1-4210	VOLTAGE:	480.	(SEE LOW VOLTAGE REPORT)
12-VFD-3320	VOLTAGE:	480.	(SEE LOW VOLTAGE REPORT)
12-VFD-3420	VOLTAGE:	480.	(SEE LOW VOLTAGE REPORT)
12-VFD-3520	VOLTAGE:	480.	(SEE LOW VOLTAGE REPORT)
21-GEN-1100 [Ex]	VOLTAGE:	480.	(SEE LOW VOLTAGE REPORT)

THREE PHASE MOMENTARY DUTY REPORT
PRE FAULT VOLTAGE: 1.0000
MODEL TRANSFORMER TAPS: NO

21-GEN-1200	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
21-PNL-1200	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
21-VFD-1100	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
21-VFD-1200	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
21-VFD-1300	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
22-VFD-2022	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
22-VFD-2201	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
22-VFD-2203	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
23-VCP-2810	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
ADMIN	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
AHF-10A	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
AHF-10B	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
AHF-8A	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
AHF-8B	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
CL BLDG PORT GEN	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
DCB 1 PORT GEN	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
DP-10	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
DP-13	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
DP-5A	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
DP-5B	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
DP-6	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)

THREE PHASE MOMENTARY DUTY REPORT
PRE FAULT VOLTAGE: 1.0000
MODEL TRANSFORMER TAPS: NO

=====		
DP-8A-1	VOLTAGE:	480. (SEE LOW VOLTAGE REPORT)
DP-8A-2	VOLTAGE:	480. (SEE LOW VOLTAGE REPORT)
DP-8B-1	VOLTAGE:	480. (SEE LOW VOLTAGE REPORT)
DP-8B-2	VOLTAGE:	480. (SEE LOW VOLTAGE REPORT)
DP-9	VOLTAGE:	480. (SEE LOW VOLTAGE REPORT)
DSC 02-EF-1320	VOLTAGE:	480. (SEE LOW VOLTAGE REPORT)
DSC 02-EF-1330	VOLTAGE:	480. (SEE LOW VOLTAGE REPORT)
DSC 02-EF-1340	VOLTAGE:	480. (SEE LOW VOLTAGE REPORT)
DSC 02-EF-1370	VOLTAGE:	480. (SEE LOW VOLTAGE REPORT)
DSC 02-EF-1380	VOLTAGE:	480. (SEE LOW VOLTAGE REPORT)
DSC 02-EF-1390	VOLTAGE:	480. (SEE LOW VOLTAGE REPORT)
DSC 02-EF-1396	VOLTAGE:	480. (SEE LOW VOLTAGE REPORT)
DSC 02-SF-1310	VOLTAGE:	480. (SEE LOW VOLTAGE REPORT)
DSC 02-SF-1350	VOLTAGE:	480. (SEE LOW VOLTAGE REPORT)
DSC 02-SF-1360	VOLTAGE:	480. (SEE LOW VOLTAGE REPORT)
DSC 02-SF-1570	VOLTAGE:	480. (SEE LOW VOLTAGE REPORT)
DSC 02-SF-1575	VOLTAGE:	480. (SEE LOW VOLTAGE REPORT)
DSC 02-SF-1810	VOLTAGE:	480. (SEE LOW VOLTAGE REPORT)
DSC 03-EF-1420	VOLTAGE:	480. (SEE LOW VOLTAGE REPORT)
DSC 03-HSS-1140	VOLTAGE:	480. (SEE LOW VOLTAGE REPORT)
DSC 03-HSS-1240	VOLTAGE:	480. (SEE LOW VOLTAGE REPORT)
DSC 03-HSS-1340	VOLTAGE:	480. (SEE LOW VOLTAGE REPORT)

THREE PHASE MOMENTARY DUTY REPORT
PRE FAULT VOLTAGE: 1.0000
MODEL TRANSFORMER TAPS: NO

DSC 03-PMP-1510	VOLTAGE:	480.	(SEE LOW VOLTAGE REPORT)
DSC 03-PMP-1520	VOLTAGE:	480.	(SEE LOW VOLTAGE REPORT)
DSC 03-PMP-1530	VOLTAGE:	480.	(SEE LOW VOLTAGE REPORT)
DSC 03-PMP-1610	VOLTAGE:	480.	(SEE LOW VOLTAGE REPORT)
DSC 03-PMP-1620	VOLTAGE:	480.	(SEE LOW VOLTAGE REPORT)
DSC 03-SCC-1150	VOLTAGE:	480.	(SEE LOW VOLTAGE REPORT)
DSC 03-SCC-1250	VOLTAGE:	480.	(SEE LOW VOLTAGE REPORT)
DSC 03-SCC-1350	VOLTAGE:	480.	(SEE LOW VOLTAGE REPORT)
DSC 03-SF-1410	VOLTAGE:	480.	(SEE LOW VOLTAGE REPORT)
DSC 03-SF-1575	VOLTAGE:	480.	(SEE LOW VOLTAGE REPORT)
DSC 03-SLC-1120	VOLTAGE:	480.	(SEE LOW VOLTAGE REPORT)
DSC 03-SLC-1220	VOLTAGE:	480.	(SEE LOW VOLTAGE REPORT)
DSC 03-SLC-1320	VOLTAGE:	480.	(SEE LOW VOLTAGE REPORT)
DSC 04-PMP-1140	VOLTAGE:	480.	(SEE LOW VOLTAGE REPORT)
DSC 06-CLA-1510	VOLTAGE:	480.	(SEE LOW VOLTAGE REPORT)
DSC 06-CLA-1520	VOLTAGE:	480.	(SEE LOW VOLTAGE REPORT)
DSC 06-CLA-1530	VOLTAGE:	480.	(SEE LOW VOLTAGE REPORT)
DSC 06-PMP-1515	VOLTAGE:	480.	(SEE LOW VOLTAGE REPORT)
DSC 06-PMP-1525	VOLTAGE:	480.	(SEE LOW VOLTAGE REPORT)
DSC 06-PMP-1535	VOLTAGE:	480.	(SEE LOW VOLTAGE REPORT)
DSC 08-EF-1230	VOLTAGE:	480.	(SEE LOW VOLTAGE REPORT)
DSC 09-MIX-1110	VOLTAGE:	480.	(SEE LOW VOLTAGE REPORT)

THREE PHASE MOMENTARY DUTY REPORT
PRE FAULT VOLTAGE: 1.0000
MODEL TRANSFORMER TAPS: NO

DSC 09-PMP-903	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
DSC 11-EF-1530	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
DSC 11-EF-1540	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
DSC 11-EF-1560	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
DSC 11-PMP-2311	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
DSC 11-PMP-2314	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
DSC 11-PMP-2322	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
DSC 11-PMP-2324	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
DSC 11-PMP-2610	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
DSC 11-PMP-2620	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
DSC 11-PMP-2711	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
DSC 11-PMP-2721	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
DSC 11-SF-1510	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
DSC 11-SF-1520	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
DSC 11-SF-1550	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
DSC 11-SF-1575	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
DSC 12-AUG-4615	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
DSC 12-AUG-4617	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
DSC 12-CON-4610	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
DSC 12-CON-4620	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
DSC 12-EF-2530	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
DSC 12-EF-2540	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)

THREE PHASE MOMENTARY DUTY REPORT
PRE FAULT VOLTAGE: 1.0000
MODEL TRANSFORMER TAPS: NO

DSC 12-HPU-4750	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
DSC 12-SF-2510	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
DSC 12-SF-2520	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
DSC 12-SF-2550	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
DSC 12-SF-2560	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
DSC 12-SF-2570	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
DSC 12-SF-2580	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
DSC 12-SRW-4612	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
DSC 12-SRW-4613	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
DSC 19-PMP-1110	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
DSC 19-PMP-1120	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
DSC 21-EF-1250	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
DSC 21-EF-1310	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
DSC 2P-07	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
DSC 3G-01	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
DSC 3G-02	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
DSC 3G-03	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
DSC 3P-05	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
DSC 3P-07	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
DSC 4P-01	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
DSC 4P-02	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
DSC 4P-03	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)

THREE PHASE MOMENTARY DUTY REPORT
PRE FAULT VOLTAGE: 1.0000
MODEL TRANSFORMER TAPS: NO

DSC SHB CRANE	VOLTAGE:	480.	(SEE LOW VOLTAGE REPORT)
GEN SWBD [Ex]	VOLTAGE:	480.	(SEE LOW VOLTAGE REPORT)
LP-10	VOLTAGE:	208.	(SEE LOW VOLTAGE REPORT)
LP-11	VOLTAGE:	208.	(SEE LOW VOLTAGE REPORT)
LP-13A	VOLTAGE:	208.	(SEE LOW VOLTAGE REPORT)
LP-13B	VOLTAGE:	208.	(SEE LOW VOLTAGE REPORT)
LP-4PA	VOLTAGE:	240.	(SEE LOW VOLTAGE REPORT)
LP-5	VOLTAGE:	208.	(SEE LOW VOLTAGE REPORT)
LP-6	VOLTAGE:	208.	(SEE LOW VOLTAGE REPORT)
LP-8	VOLTAGE:	208.	(SEE LOW VOLTAGE REPORT)
LP-9	VOLTAGE:	208.	(SEE LOW VOLTAGE REPORT)
MCC-10A	VOLTAGE:	480.	(SEE LOW VOLTAGE REPORT)
MCC-10B	VOLTAGE:	480.	(SEE LOW VOLTAGE REPORT)
MCC-11A	VOLTAGE:	480.	(SEE LOW VOLTAGE REPORT)
MCC-11B	VOLTAGE:	480.	(SEE LOW VOLTAGE REPORT)
MCC-5A	VOLTAGE:	480.	(SEE LOW VOLTAGE REPORT)
MCC-5B	VOLTAGE:	480.	(SEE LOW VOLTAGE REPORT)
MCC-6A	VOLTAGE:	480.	(SEE LOW VOLTAGE REPORT)
MCC-6B	VOLTAGE:	480.	(SEE LOW VOLTAGE REPORT)
MCC-8A	VOLTAGE:	480.	(SEE LOW VOLTAGE REPORT)
MCC-8B	VOLTAGE:	480.	(SEE LOW VOLTAGE REPORT)
MCC-9A	VOLTAGE:	480.	(SEE LOW VOLTAGE REPORT)

THREE PHASE MOMENTARY DUTY REPORT
PRE FAULT VOLTAGE: 1.0000
MODEL TRANSFORMER TAPS: NO

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MCC-9B	VOLTAGE:	480.	(SEE LOW VOLTAGE REPORT)
MTS-10	VOLTAGE:	480.	(SEE LOW VOLTAGE REPORT)
MTS-11	VOLTAGE:	480.	(SEE LOW VOLTAGE REPORT)
MTS-13	VOLTAGE:	480.	(SEE LOW VOLTAGE REPORT)
MTS-5	VOLTAGE:	480.	(SEE LOW VOLTAGE REPORT)
MTS-6	VOLTAGE:	480.	(SEE LOW VOLTAGE REPORT)
MTS-8	VOLTAGE:	480.	(SEE LOW VOLTAGE REPORT)
MTS-9	VOLTAGE:	480.	(SEE LOW VOLTAGE REPORT)
SHB CRANE	VOLTAGE:	480.	(SEE LOW VOLTAGE REPORT)
SWBD-MAIN-A	VOLTAGE:	480.	(SEE LOW VOLTAGE REPORT)
SWBD-MAIN-B	VOLTAGE:	480.	(SEE LOW VOLTAGE REPORT)

MOMENTARY DUTY SUMMARY REPORT

PRE FAULT VOLTAGE: 1.0000

MODEL TRANSFORMER TAPS: NO

SOLUTION METHOD: E/Z

BUS RECORD	VOLTAGE	* 3	P H A S E *	* * *	SLG * * *
NO NAME	L-L		KA X/R	KA	X/R

2 FAULTED BUSES, 438 BRANCHES, 122 CONTRIBUTIONS

UNBALANCED FAULTS REQUESTED

*** SHORT CIRCUIT STUDY COMPLETE ***

THREE PHASE INTERRUPTING DUTY REPORT
PRE FAULT VOLTAGE: 1.0000
MODEL TRANSFORMER TAPS: NO

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05-VFD-1510          VOLTAGE:  480. ( SEE LOW VOLTAGE REPORT )
05-VFD-1520          VOLTAGE:  480. ( SEE LOW VOLTAGE REPORT )
05-VFD-1630          VOLTAGE:  480. ( SEE LOW VOLTAGE REPORT )
06-VFD-1610          VOLTAGE:  480. ( SEE LOW VOLTAGE REPORT )
06-VFD-1615          VOLTAGE:  480. ( SEE LOW VOLTAGE REPORT )
06-VFD-1625          VOLTAGE:  480. ( SEE LOW VOLTAGE REPORT )
06-VFD-1630          VOLTAGE:  480. ( SEE LOW VOLTAGE REPORT )
06-VFD-1635          VOLTAGE:  480. ( SEE LOW VOLTAGE REPORT )
07-VFD-1110          VOLTAGE:  480. ( SEE LOW VOLTAGE REPORT )
07-VFD-1120          VOLTAGE:  480. ( SEE LOW VOLTAGE REPORT )
09-VFD-1100          VOLTAGE:  480. ( SEE LOW VOLTAGE REPORT )
09-VFD-1200          VOLTAGE:  480. ( SEE LOW VOLTAGE REPORT )
11-VFD-2312          VOLTAGE:  480. ( SEE LOW VOLTAGE REPORT )
11-VFD-2321          VOLTAGE:  480. ( SEE LOW VOLTAGE REPORT )
11-VFD-2410          VOLTAGE:  480. ( SEE LOW VOLTAGE REPORT )
11-VFD-2420          VOLTAGE:  480. ( SEE LOW VOLTAGE REPORT )
12-VCP1-4110         VOLTAGE:  480. ( SEE LOW VOLTAGE REPORT )
12-VCP1-4210         VOLTAGE:  480. ( SEE LOW VOLTAGE REPORT )
12-VFD-3320          VOLTAGE:  480. ( SEE LOW VOLTAGE REPORT )
12-VFD-3420          VOLTAGE:  480. ( SEE LOW VOLTAGE REPORT )
12-VFD-3520          VOLTAGE:  480. ( SEE LOW VOLTAGE REPORT )
21-GEN-1100 [Ex]     VOLTAGE:  480. ( SEE LOW VOLTAGE REPORT )
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THREE PHASE INTERRUPTING DUTY REPORT
PRE FAULT VOLTAGE: 1.0000
MODEL TRANSFORMER TAPS: NO

21-GEN-1200	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
21-PNL-1200	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
21-VFD-1100	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
21-VFD-1200	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
21-VFD-1300	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
22-VFD-2022	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
22-VFD-2201	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
22-VFD-2203	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
23-VCP-2810	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
ADMIN	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
AHF-10A	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
AHF-10B	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
AHF-8A	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
AHF-8B	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
CL BLDG PORT GEN	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
DCB 1 PORT GEN	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
DP-10	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
DP-13	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
DP-5A	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
DP-5B	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
DP-6	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
DP-8A-1	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)

THREE PHASE INTERRUPTING DUTY REPORT
PRE FAULT VOLTAGE: 1.0000
MODEL TRANSFORMER TAPS: NO

DP-8A-2	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
DP-8B-1	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
DP-8B-2	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
DP-9	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
DSC 02-EF-1320	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
DSC 02-EF-1330	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
DSC 02-EF-1340	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
DSC 02-EF-1370	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
DSC 02-EF-1380	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
DSC 02-EF-1390	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
DSC 02-EF-1396	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
DSC 02-SF-1310	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
DSC 02-SF-1350	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
DSC 02-SF-1360	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
DSC 02-SF-1570	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
DSC 02-SF-1575	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
DSC 02-SF-1810	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
DSC 03-EF-1420	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
DSC 03-HSS-1140	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
DSC 03-HSS-1240	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
DSC 03-HSS-1340	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
DSC 03-PMP-1510	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)

THREE PHASE INTERRUPTING DUTY REPORT
PRE FAULT VOLTAGE: 1.0000
MODEL TRANSFORMER TAPS: NO

DSC 03-PMP-1520	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
DSC 03-PMP-1530	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
DSC 03-PMP-1610	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
DSC 03-PMP-1620	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
DSC 03-SCC-1150	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
DSC 03-SCC-1250	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
DSC 03-SCC-1350	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
DSC 03-SF-1410	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
DSC 03-SF-1575	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
DSC 03-SLC-1120	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
DSC 03-SLC-1220	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
DSC 03-SLC-1320	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
DSC 04-PMP-1140	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
DSC 06-CLA-1510	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
DSC 06-CLA-1520	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
DSC 06-CLA-1530	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
DSC 06-PMP-1515	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
DSC 06-PMP-1525	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
DSC 06-PMP-1535	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
DSC 08-EF-1230	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
DSC 09-MIX-1110	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
DSC 09-PMP-903	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)

THREE PHASE INTERRUPTING DUTY REPORT
PRE FAULT VOLTAGE: 1.0000
MODEL TRANSFORMER TAPS: NO

```
=====
DSC 11-EF-1530      VOLTAGE:  480. ( SEE LOW VOLTAGE REPORT )
DSC 11-EF-1540      VOLTAGE:  480. ( SEE LOW VOLTAGE REPORT )
DSC 11-EF-1560      VOLTAGE:  480. ( SEE LOW VOLTAGE REPORT )
DSC 11-PMP-2311      VOLTAGE:  480. ( SEE LOW VOLTAGE REPORT )
DSC 11-PMP-2314      VOLTAGE:  480. ( SEE LOW VOLTAGE REPORT )
DSC 11-PMP-2322      VOLTAGE:  480. ( SEE LOW VOLTAGE REPORT )
DSC 11-PMP-2324      VOLTAGE:  480. ( SEE LOW VOLTAGE REPORT )
DSC 11-PMP-2610      VOLTAGE:  480. ( SEE LOW VOLTAGE REPORT )
DSC 11-PMP-2620      VOLTAGE:  480. ( SEE LOW VOLTAGE REPORT )
DSC 11-PMP-2711      VOLTAGE:  480. ( SEE LOW VOLTAGE REPORT )
DSC 11-PMP-2721      VOLTAGE:  480. ( SEE LOW VOLTAGE REPORT )
DSC 11-SF-1510       VOLTAGE:  480. ( SEE LOW VOLTAGE REPORT )
DSC 11-SF-1520       VOLTAGE:  480. ( SEE LOW VOLTAGE REPORT )
DSC 11-SF-1550       VOLTAGE:  480. ( SEE LOW VOLTAGE REPORT )
DSC 11-SF-1575       VOLTAGE:  480. ( SEE LOW VOLTAGE REPORT )
DSC 12-AUG-4615      VOLTAGE:  480. ( SEE LOW VOLTAGE REPORT )
DSC 12-AUG-4617      VOLTAGE:  480. ( SEE LOW VOLTAGE REPORT )
DSC 12-CON-4610      VOLTAGE:  480. ( SEE LOW VOLTAGE REPORT )
DSC 12-CON-4620      VOLTAGE:  480. ( SEE LOW VOLTAGE REPORT )
DSC 12-EF-2530       VOLTAGE:  480. ( SEE LOW VOLTAGE REPORT )
DSC 12-EF-2540       VOLTAGE:  480. ( SEE LOW VOLTAGE REPORT )
DSC 12-HPU-4750      VOLTAGE:  480. ( SEE LOW VOLTAGE REPORT )
```

THREE PHASE INTERRUPTING DUTY REPORT
PRE FAULT VOLTAGE: 1.0000
MODEL TRANSFORMER TAPS: NO

DSC 12-SF-2510	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
DSC 12-SF-2520	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
DSC 12-SF-2550	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
DSC 12-SF-2560	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
DSC 12-SF-2570	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
DSC 12-SF-2580	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
DSC 12-SRW-4612	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
DSC 12-SRW-4613	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
DSC 19-PMP-1110	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
DSC 19-PMP-1120	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
DSC 21-EF-1250	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
DSC 21-EF-1310	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
DSC 2P-07	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
DSC 3G-01	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
DSC 3G-02	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
DSC 3G-03	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
DSC 3P-05	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
DSC 3P-07	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
DSC 4P-01	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
DSC 4P-02	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
DSC 4P-03	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)
DSC SHB CRANE	VOLTAGE: 480. (SEE LOW VOLTAGE REPORT)

THREE PHASE INTERRUPTING DUTY REPORT
PRE FAULT VOLTAGE: 1.0000
MODEL TRANSFORMER TAPS: NO

GEN SWBD [Ex]	VOLTAGE:	480.	(SEE LOW VOLTAGE REPORT)
LP-10	VOLTAGE:	208.	(SEE LOW VOLTAGE REPORT)
LP-11	VOLTAGE:	208.	(SEE LOW VOLTAGE REPORT)
LP-13A	VOLTAGE:	208.	(SEE LOW VOLTAGE REPORT)
LP-13B	VOLTAGE:	208.	(SEE LOW VOLTAGE REPORT)
LP-4PA	VOLTAGE:	240.	(SEE LOW VOLTAGE REPORT)
LP-5	VOLTAGE:	208.	(SEE LOW VOLTAGE REPORT)
LP-6	VOLTAGE:	208.	(SEE LOW VOLTAGE REPORT)
LP-8	VOLTAGE:	208.	(SEE LOW VOLTAGE REPORT)
LP-9	VOLTAGE:	208.	(SEE LOW VOLTAGE REPORT)
MCC-10A	VOLTAGE:	480.	(SEE LOW VOLTAGE REPORT)
MCC-10B	VOLTAGE:	480.	(SEE LOW VOLTAGE REPORT)
MCC-11A	VOLTAGE:	480.	(SEE LOW VOLTAGE REPORT)
MCC-11B	VOLTAGE:	480.	(SEE LOW VOLTAGE REPORT)
MCC-5A	VOLTAGE:	480.	(SEE LOW VOLTAGE REPORT)
MCC-5B	VOLTAGE:	480.	(SEE LOW VOLTAGE REPORT)
MCC-6A	VOLTAGE:	480.	(SEE LOW VOLTAGE REPORT)
MCC-6B	VOLTAGE:	480.	(SEE LOW VOLTAGE REPORT)
MCC-8A	VOLTAGE:	480.	(SEE LOW VOLTAGE REPORT)
MCC-8B	VOLTAGE:	480.	(SEE LOW VOLTAGE REPORT)
MCC-9A	VOLTAGE:	480.	(SEE LOW VOLTAGE REPORT)
MCC-9B	VOLTAGE:	480.	(SEE LOW VOLTAGE REPORT)

THREE PHASE INTERRUPTING DUTY REPORT

PRE FAULT VOLTAGE: 1.0000

MODEL TRANSFORMER TAPS: NO

```
=====
MTS-10          VOLTAGE:  480. ( SEE LOW VOLTAGE REPORT )
MTS-11          VOLTAGE:  480. ( SEE LOW VOLTAGE REPORT )
MTS-13          VOLTAGE:  480. ( SEE LOW VOLTAGE REPORT )
MTS-5           VOLTAGE:  480. ( SEE LOW VOLTAGE REPORT )
MTS-6           VOLTAGE:  480. ( SEE LOW VOLTAGE REPORT )
MTS-8           VOLTAGE:  480. ( SEE LOW VOLTAGE REPORT )
MTS-9           VOLTAGE:  480. ( SEE LOW VOLTAGE REPORT )
SHB CRANE       VOLTAGE:  480. ( SEE LOW VOLTAGE REPORT )
SWBD-MAIN-A     VOLTAGE:  480. ( SEE LOW VOLTAGE REPORT )
SWBD-MAIN-B     VOLTAGE:  480. ( SEE LOW VOLTAGE REPORT )
```

I N T E R R U P T I N G D U T Y S U M M A R Y R E P O R T

PRE FAULT VOLTAGE: 1.0000

MODEL TRANSFORMER TAPS: NO

NACD OPTION: INTERPOLATED

BUS RECORD		VOLTAGE	NACD	* 3 P H A S E *		* * * S L G * * *	
NO NAME		L-L	RATIO	E/Z KA	X/R	E/Z KA	X/R

2 FAULTED BUSES, 438 BRANCHES, 122 CONTRIBUTIONS
 UNBALANCED FAULTS REQUESTED

*** SHORT CIRCUIT STUDY COMPLETE ***

5.0 UTILITY DATA

Summary of fault being displayed:
 Prefault voltage: Flat Bus V=1 p.u.
 Generator impedance: Subtransient
 MOV iteration: [Off]
 Enforce generator current limit [Off]
 Voltage controlled current source simulated [No]
 Type-3 wind plant simulated [No]
 Converter-interfaced resource simulated [No]
 ANSI x/r ratio calculation [Off]

=====									
=====									
1. Bus Fault on:		0 1101 Load		13.8 kV 3LG					
				FAULT CURRENT (A @ DEG)					
		+ SEQ		- SEQ		0 SEQ		A PHASE	
B PHASE		C PHASE							
1354.7@ -67.4		0.0@ 0.0		0.0@ 0.0		1354.7@			
-67.4	1354.7@ 172.6	1354.7@ 52.6							
THEVENIN IMPEDANCE (OHM)									
2.25902+j5.43003		2.25686+j5.45092		4.50836+j21.4737					
SHORT CIRCUIT MVA= 32.4				X/R RATIO= 2.40371		R0/X1=			
0.83026	X0/X1= 3.95462								

BUS	0 1101 Load	13.8KV	AREA	1	ZONE	1	TIER	0	(PREFault
V=1.000@	0.0 PU)								
			+ SEQ		- SEQ		0 SEQ		
A PHASE		B PHASE		C PHASE					
VOLTAGE (KV, L-G)		>	0.000@	0.0	0.000@	0.0	0.000@	0.0	
0.000@	0.0	0.000@	0.0	0.000@	0.0				
BRANCH CURRENT (A) TO		>							
0 1101		13.8 1L	1354.7@	112.6	0.0@	0.0	0.0@	0.0	
1354.7@	112.6	1354.7@ -7.4	1354.7@	-127.4					
CURRENT TO FAULT (A)			>	1354.7@ -67.4	0.0@	0.0	0.0@	0.0	
1354.7@	-67.4	1354.7@ 172.6	1354.7@	52.6					
THEVENIN IMPEDANCE (OHM)		>	5.88119@	67.4	5.89965@	67.5	21.9419@	78.1	

Summary of fault being displayed:
 Prefault voltage: Flat Bus V=1 p.u.
 Generator impedance: Subtransient
 MOV iteration: [Off]
 Enforce generator current limit [Off]
 Voltage controlled current source simulated [No]
 Type-3 wind plant simulated [No]
 Converter-interfaced resource simulated [No]

ANSI x/r ratio calculation [Off]

```
=====
2. Bus Fault on:          0 1101 Load          13.8 kV 2LG Type=B-C
                                FAULT CURRENT (A @ DEG)
                                0 SEQ              A PHASE
      + SEQ              - SEQ
B PHASE          C PHASE
    755.8@ -68.4    597.3@ 113.8    160.6@ 103.2    0.0@
0.0 1157.0@-169.3 1234.0@ 33.6
                                THEVENIN IMPEDANCE (OHM)
                                2.25902+j5.43003  2.25686+j5.45092  4.50836+j21.4737
                                SHORT CIRCUIT MVA= 29.5          X/R RATIO= 2.53196          R0/X1=
0.83026          X0/X1= 3.95462
-----
```

```
-----
BUS      0 1101 Load      13.8KV AREA 1 ZONE 1 TIER 0 (PREFault
V=1.000@ 0.0 PU)
      + SEQ              - SEQ              0 SEQ
A PHASE          B PHASE          C PHASE
VOLTAGE (KV, L-G) >          3.524@ 1.3    3.524@ 1.3    3.524@ 1.3
10.572@ 1.3    0.000@ 0.0    0.000@ 0.0
BRANCH CURRENT (A) TO >
    0 1101      13.8 1L    755.8@ 111.6    597.3@ -66.2    160.6@ -76.8
    0.0@ 0.0    1157.0@ 10.7    1234.0@-146.4
CURRENT TO FAULT (A) >          755.8@ -68.4    597.3@ 113.8    160.6@ 103.2
    0.0@ 0.0    1157.0@-169.3    1234.0@ 33.6
THEVENIN IMPEDANCE (OHM) >          5.88119@ 67.4    5.89965@ 67.5    21.9419@ 78.1
-----
```

Summary of fault being displayed:
Prefault voltage: Flat Bus V=1 p.u.
Generator impedance: Subtransient
MOV iteration: [Off]
Enforce generator current limit [Off]
Voltage controlled current source simulated [No]
Type-3 wind plant simulated [No]
Converter-interfaced resource simulated [No]
ANSI x/r ratio calculation [Off]

```
=====
3. Bus Fault on:          0 1101 Load          13.8 kV 1LG Type=A
                                FAULT CURRENT (A @ DEG)
                                0 SEQ              A PHASE
      + SEQ              - SEQ
B PHASE          C PHASE
    237.2@ -74.4    237.2@ -74.4    237.2@ -74.4    711.6@
-----
```

0.83026 X0/X1= 3.95462 SHORT CIRCUIT MVA= 17.0 X/R RATIO= 3.58531 R0/X1=

BUS	0 1101 Load	13.8KV	AREA	1	ZONE	1	TIER	0	(PREFault
V=1.000@	0.0 PU)								
			+ SEQ				- SEQ		0 SEQ
A PHASE	B PHASE		C PHASE						
VOLTAGE (KV, L-G)	>	6.585@	1.5			1.399@	173.1		5.205@-176.3
0.000@ 0.0	10.752@-136.4	10.081@	140.6						
BRANCH CURRENT (A) TO	>								
0 1101	13.8 1L	237.2@	105.6			237.2@	105.6		237.2@ 105.6
711.6@ 105.6	0.0@ 0.0	0.0@	0.0						
CURRENT TO FAULT (A)	>	237.2@	-74.4			237.2@	-74.4		237.2@ -74.4
711.6@ -74.4	0.0@ 0.0	0.0@	0.0						
THEVENIN IMPEDANCE (OHM)	>	5.88119@	67.4			5.89965@	67.5		21.9419@ 78.1

```
Summary of fault being displayed:
Prefault voltage: Flat Bus V=1 p.u.
Generator impedance: Subtransient
MOV iteration: [Off]
Enforce generator current limit [Off]
Voltage controlled current source simulated [No]
Type-3 wind plant simulated [No]
Converter-interfaced resource simulated [No]
ANSI x/r ratio calculation [Off]
```

4. Bus Fault on:		0 1101 Load	13.8 kV LL Type=B-C
		FAULT CURRENT (A @ DEG)	
+ SEQ		- SEQ	0 SEQ
B PHASE	C PHASE		A PHASE
676.3@ -67.5	676.3@ 112.5	0.0@ 0.0	0.0@ 0.0
1171.4@ -157.5	1171.4@ 22.5		
		THEVENIN IMPEDANCE (OHM)	
2.25902+j5.43003	2.25686+j5.45092	4.50836+j21.4737	
SHORT CIRCUIT MVA= 28.0		X/R RATIO= 2.40948	R0/X1=
0.83026	X0/X1= 3.95462		

BUS	0	1101	Load	13.8KV	AREA	1	ZONE	1	TIER	0	(PREFault
-----	---	------	------	--------	------	---	------	---	------	---	-----------

V=1.000@ 0.0 PU)

		+ SEQ	- SEQ	0 SEQ
A PHASE	B PHASE	C PHASE		
VOLTAGE (KV, L-G)	>	3.990@ 0.0	3.990@ 0.0	0.000@ 0.0
7.980@ 0.0	3.990@-180.0	3.990@-180.0		
BRANCH CURRENT (A) TO	>			
0 1101	13.8 1L	676.3@ 112.5	676.3@ -67.5	0.0@ 0.0
0.0@ 0.0	1171.4@ 22.5	1171.4@-157.5		
CURRENT TO FAULT (A)	>	676.3@ -67.5	676.3@ 112.5	0.0@ 0.0
0.0@ 0.0	1171.4@-157.5	1171.4@ 22.5		
THEVENIN IMPEDANCE (OHM)	>	5.88119@ 67.4	5.89965@ 67.5	21.9419@ 78.1

Summary of fault being displayed:
 Prefault voltage: Flat Bus V=1 p.u.
 Generator impedance: Subtransient
 MOV iteration: [Off]
 Enforce generator current limit [Off]
 Voltage controlled current source simulated [No]
 Type-3 wind plant simulated [No]
 Converter-interfaced resource simulated [No]
 ANSI x/r ratio calculation [Off]

=====									
=====									
1. Bus Fault on: 0 Kano 13 Load 13.8 kV 3LG									
FAULT CURRENT (A @ DEG)									
0 SEQ A PHASE									
+ SEQ - SEQ									
B PHASE C PHASE									
1640.2@ -69.2 0.0@ 0.0 0.0@ 0.0 1640.2@									
-69.2 1640.2@ 170.8 1640.2@ 50.8									
THEVENIN IMPEDANCE (OHM)									
1.72561+j4.54063 1.72345+j4.56151 3.28532+j17.8936									
SHORT CIRCUIT MVA= 39.2 X/R RATIO= 2.63132 R0/X1=									
0.72354 X0/X1= 3.94079									

BUS 0 Kano 13 Load 13.8KV AREA 1 ZONE 1 TIER 0 (PREFault									
V=1.000@ 0.0 PU)									
+ SEQ - SEQ 0 SEQ									
A PHASE B PHASE C PHASE									
VOLTAGE (KV, L-G) > 0.000@ 0.0 0.000@ 0.0 0.000@ 0.0									
0.000@ 0.0 0.000@ 0.0 0.000@ 0.0									
BRANCH CURRENT (A) TO >									
0 Kan 13 13.8 1L 1640.2@ 110.8 0.0@ 0.0 0.0@ 0.0									
1640.2@ 110.8 1640.2@ -9.2 1640.2@ -129.2									
CURRENT TO FAULT (A) > 1640.2@ -69.2 0.0@ 0.0 0.0@ 0.0									
1640.2@ -69.2 1640.2@ 170.8 1640.2@ 50.8									
THEVENIN IMPEDANCE (OHM) > 4.85747@ 69.2 4.87623@ 69.3 18.1927@ 79.6									

Summary of fault being displayed:
 Prefault voltage: Flat Bus V=1 p.u.
 Generator impedance: Subtransient
 MOV iteration: [Off]
 Enforce generator current limit [Off]
 Voltage controlled current source simulated [No]
 Type-3 wind plant simulated [No]
 Converter-interfaced resource simulated [No]

ANSI x/r ratio calculation [Off]

```
=====
2. Bus Fault on:          0 Kano 13 Load      13.8 kV 2LG Type=B-C
                                FAULT CURRENT (A @ DEG)
                                0 SEQ          A PHASE
      + SEQ          - SEQ
B PHASE          C PHASE
    914.6@ -70.2    723.2@ 112.0    193.8@ 101.7    0.0@
0.0    1401.9@-171.1    1491.9@ 31.9
                                THEVENIN IMPEDANCE (OHM)
                                1.72561+j4.54063    1.72345+j4.56151    3.28532+j17.8936
                                SHORT CIRCUIT MVA= 35.7          X/R RATIO= 2.77775          R0/X1=
0.72354    X0/X1= 3.94079
-----
```

```
-----
BUS      0 Kano 13 Load    13.8KV AREA 1 ZONE 1 TIER 0 (PREFault
V=1.000@ 0.0 PU)
      + SEQ          - SEQ          0 SEQ
A PHASE          B PHASE          C PHASE
VOLTAGE (KV, L-G) >    3.526@ 1.3    3.526@ 1.3    3.526@ 1.3
10.579@ 1.3    0.000@ 0.0    0.000@ 0.0
BRANCH CURRENT (A) TO >
    0 Kan 13    13.8 1L    914.6@ 109.8    723.2@ -68.0    193.8@ -78.3
    0.0@ 0.0    1401.9@ 8.9    1491.9@-148.1
CURRENT TO FAULT (A) >    914.6@ -70.2    723.2@ 112.0    193.8@ 101.7
    0.0@ 0.0    1401.9@-171.1    1491.9@ 31.9
THEVENIN IMPEDANCE (OHM) >    4.85747@ 69.2    4.87623@ 69.3    18.1927@ 79.6
-----
```

Summary of fault being displayed:
Prefault voltage: Flat Bus V=1 p.u.
Generator impedance: Subtransient
MOV iteration: [Off]
Enforce generator current limit [Off]
Voltage controlled current source simulated [No]
Type-3 wind plant simulated [No]
Converter-interfaced resource simulated [No]
ANSI x/r ratio calculation [Off]

```
=====
3. Bus Fault on:          0 Kano 13 Load      13.8 kV 1LG Type=A
                                FAULT CURRENT (A @ DEG)
                                0 SEQ          A PHASE
      + SEQ          - SEQ
B PHASE          C PHASE
    286.4@ -76.0    286.4@ -76.0    286.4@ -76.0    859.1@
-----
```

0.0@ 0.0	0.0@ 0.0		THEVENIN IMPEDANCE (OHM)
1.72561+j4.54063	1.72345+j4.56151	3.28532+j17.8936	

BUS	0	Kano	13	Load	13.8KV	AREA	1	ZONE	1	TIER	0	(PREFault
V=1.000@	0.0	0.0	PU)									
					+ SEQ						- SEQ	0 SEQ
A PHASE				B PHASE		C PHASE						
VOLTAGE (KV, L-G)					>	6.588@	1.4	1.396@			173.3	5.210@-176.4
0.000@	0.0	10.747@	-136.5			10.099@	140.6					
BRANCH CURRENT (A) TO					>							
0 Kan 13		13.8	1L			286.4@	104.0	286.4@			104.0	286.4@ 104.0
859.1@	104.0	0.0@	0.0			0.0@	0.0					
CURRENT TO FAULT (A)					>	286.4@	-76.0	286.4@			-76.0	286.4@ -76.0
859.1@	-76.0	0.0@	0.0			0.0@	0.0					
THEVENIN IMPEDANCE (OHM)					>	4.85747@	69.2	4.87623@			69.3	18.1927@ 79.6

```
Summary of fault being displayed:
Prefault voltage: Flat Bus V=1 p.u.
Generator impedance: Subtransient
MOV iteration: [Off]
Enforce generator current limit [Off]
Voltage controlled current source simulated [No]
Type-3 wind plant simulated [No]
Converter-interfaced resource simulated [No]
ANSI x/r ratio calculation [Off]
```

BUS	0	Kano	13	Load	13.8KV	AREA	1	ZONE	1	TIER	0	(PREFault
-----	---	------	----	------	--------	------	---	------	---	------	---	-----------

V=1.000@ 0.0 PU)

A PHASE	B PHASE	C PHASE	+ SEQ	- SEQ	0 SEQ
VOLTAGE (KV, L-G)	>	3.991@ 0.1	3.991@ 0.1	0.000@ 0.0	
7.983@ 0.1	3.991@-179.9	3.991@-179.9			
BRANCH CURRENT (A) TO	>	818.5@ 110.8	818.5@ -69.2	0.0@ 0.0	
0 Kan 13	13.8 1L	818.5@ 110.8	818.5@ -69.2	0.0@ 0.0	
0.0@ 0.0	1417.8@ 20.8	1417.8@-159.2			
CURRENT TO FAULT (A)	>	818.5@ -69.2	818.5@ 110.8	0.0@ 0.0	
0.0@ 0.0	1417.8@-159.2	1417.8@ 20.8			
THEVENIN IMPEDANCE (OHM)	>	4.85747@ 69.2	4.87623@ 69.3	18.1927@ 79.6	

6.0 APPLICABLE CODES AND STANDARDS

These standards and codes are referenced and the latest version followed where appropriate:

American National Standards Institute (ANSI)

- IEEE/ANSI C37.09-2021
- IEEE/ANSI C37.20.3-2023
- IEEE/ANSI C37.20.4-2013
- IEEE/ANSI C37.59-2018

Institute of Electrical and Electronics Engineers (IEEE)

- IEEE Std C37.010™-2016
- IEEE Std C37.13™-2015
- IEEE Std C37.41™-2017
- IEEE Std C57.12.00™-2021
- IEEE Std C57.12.01™-2020
- IEEE Std 3002™ Series
- IEEE Std 3002.3™-2018
- IEEE Std 3004™ Series
- IEEE Std 3004.5™-2014
- IEEE Std 946™-2020
- IEEE Std 1584™-2018

National Fire Protection Association (NFPA)

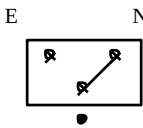
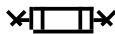
- NFPA-70®-2023: National Electrical Code (NEC)
- NFPA-70E®-2024: Standard for Electrical Safety in the Workplace

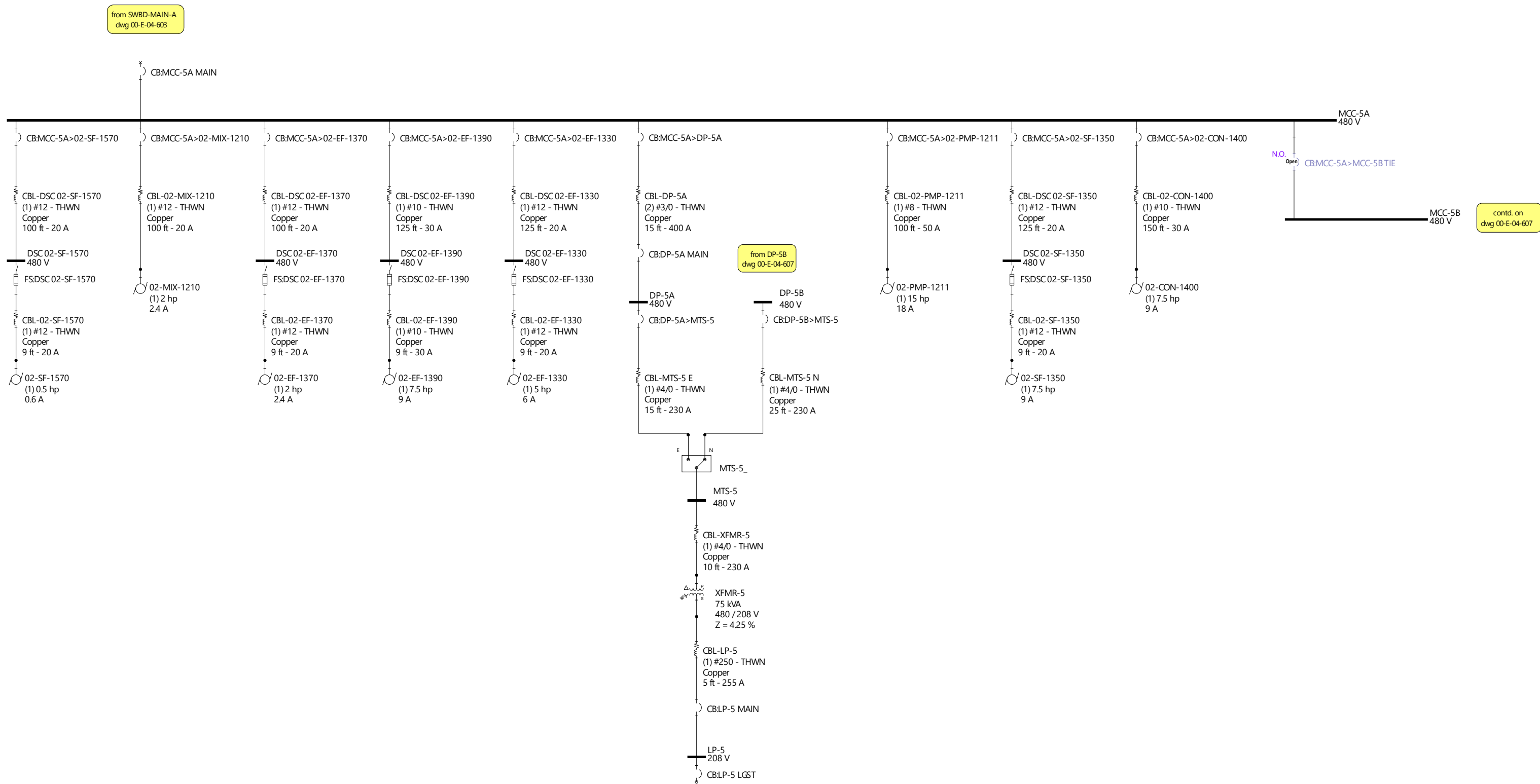
Occupational Safety and Health Administration (OSHA)

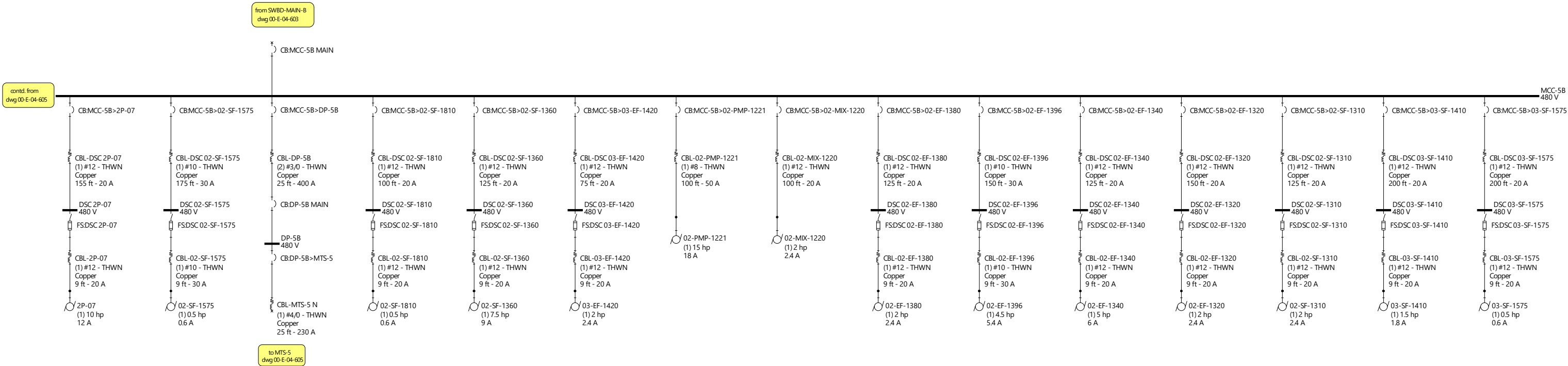
- Standards – 29 CFR
 - 1910.333 - Selection and use of work practices

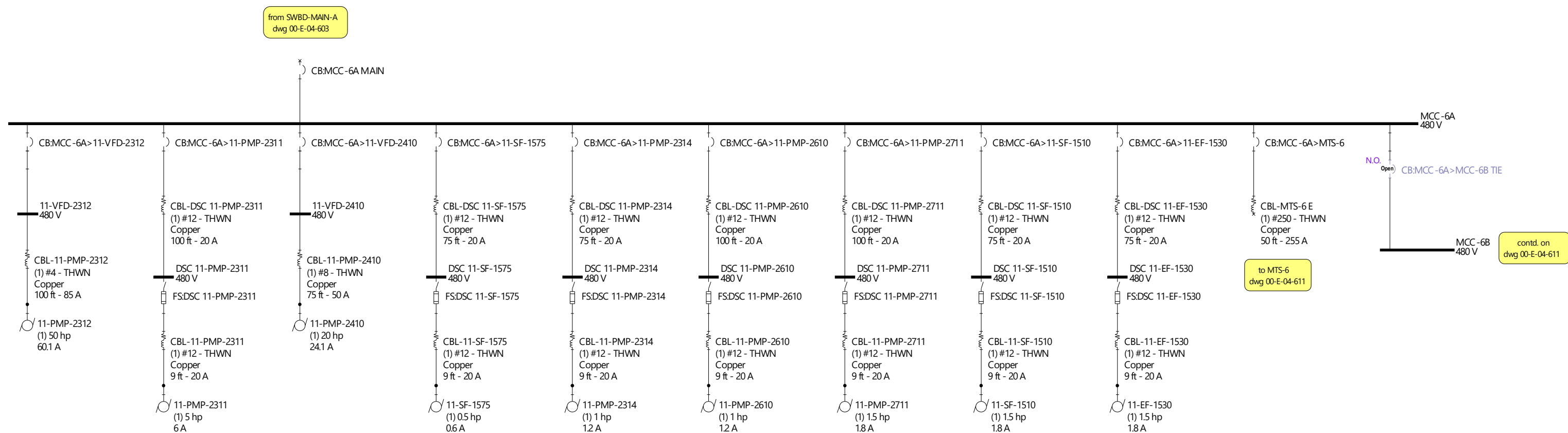
7.0 ONE-LINE DIAGRAMS

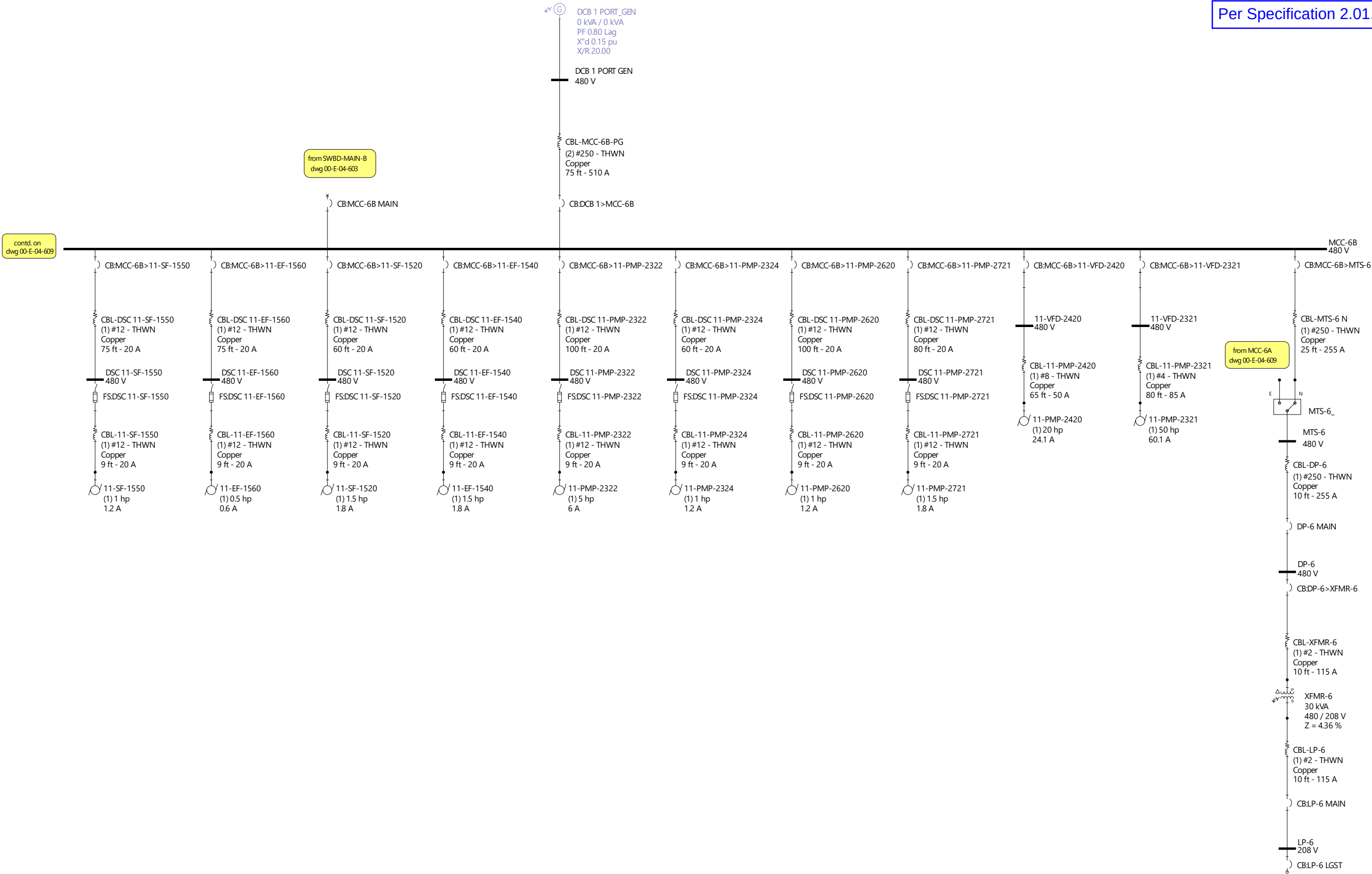
Attached one-line diagrams may include the following abbreviations and symbols.

Abbreviations	Description	Symbol
ATS MTS	Automatic Transfer Switch Manual Transfer Switch	
CBL	Cable / Feeder	
MCB / MN FCB CB / BKR	Main Circuit Breaker Feeder Circuit Breaker Circuit Breaker	
FU	Fuse	
B / BUS DSC / DISC PNL SWBD SWGR	Equipment Bus Disconnect Panel / Distribution Panel Switchboard Switchgear	
MTR	Motor	
R / RLY	Relay	
TX	Transformer	
UTIL	Utility Source	
SW BPS	Disconnect Switch Bolted Pressure Switch	
GEN	Generator	

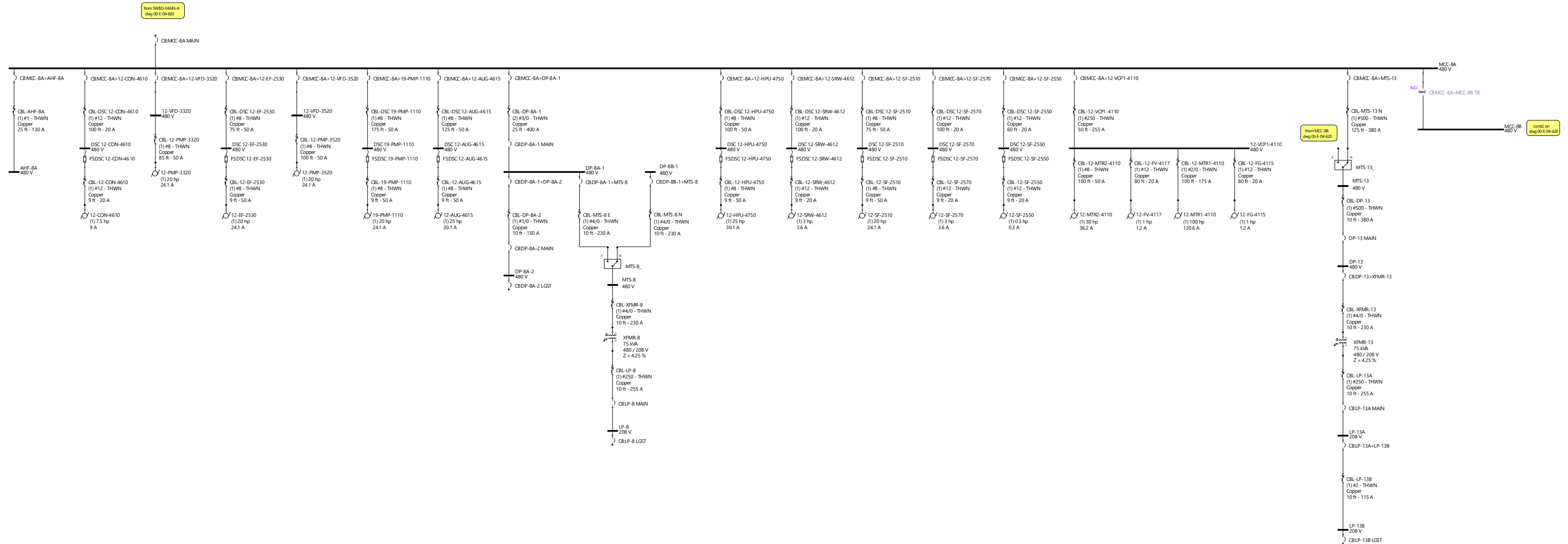




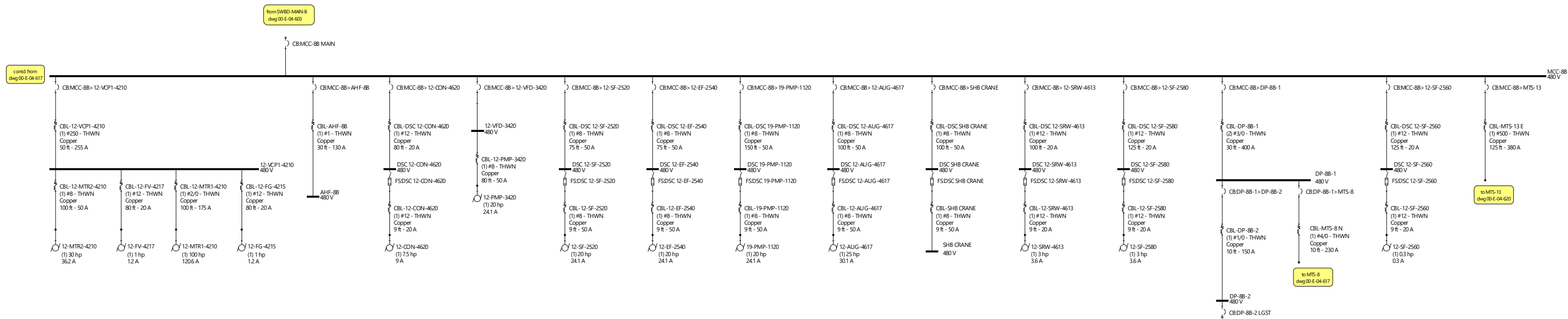


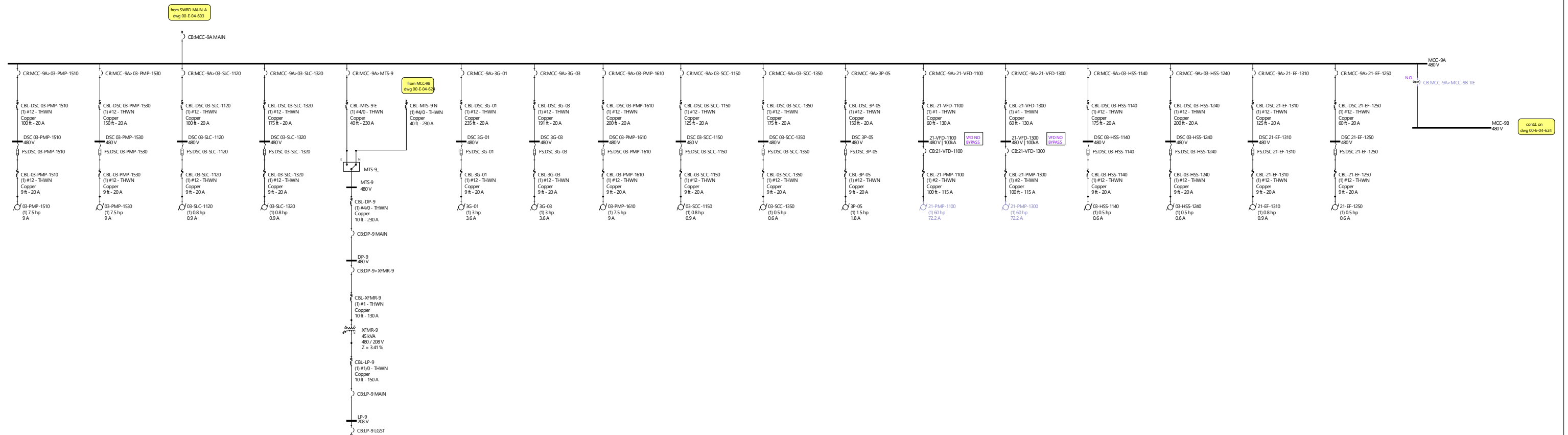


Per Specification 2.01.B.1.a.2



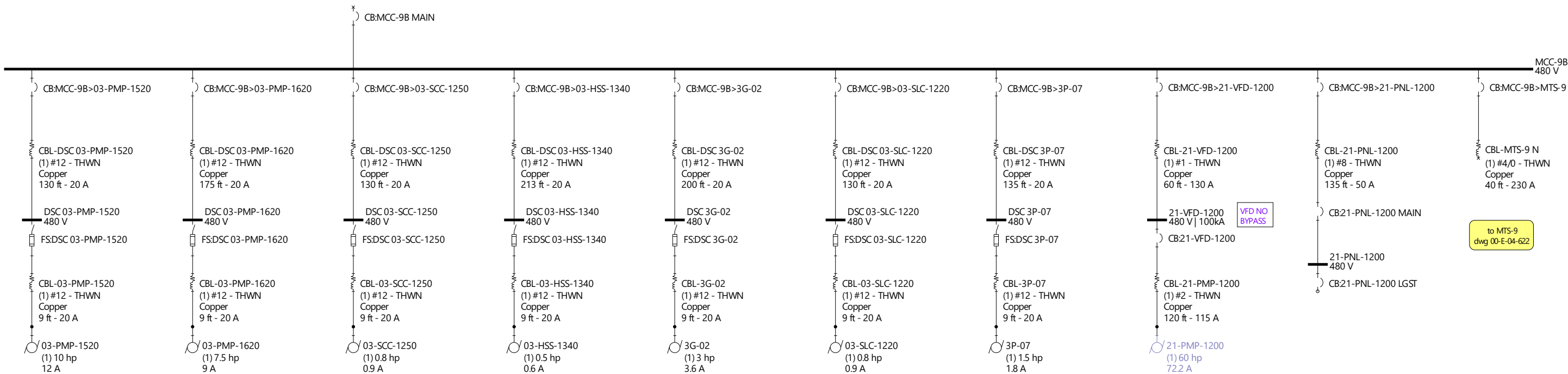
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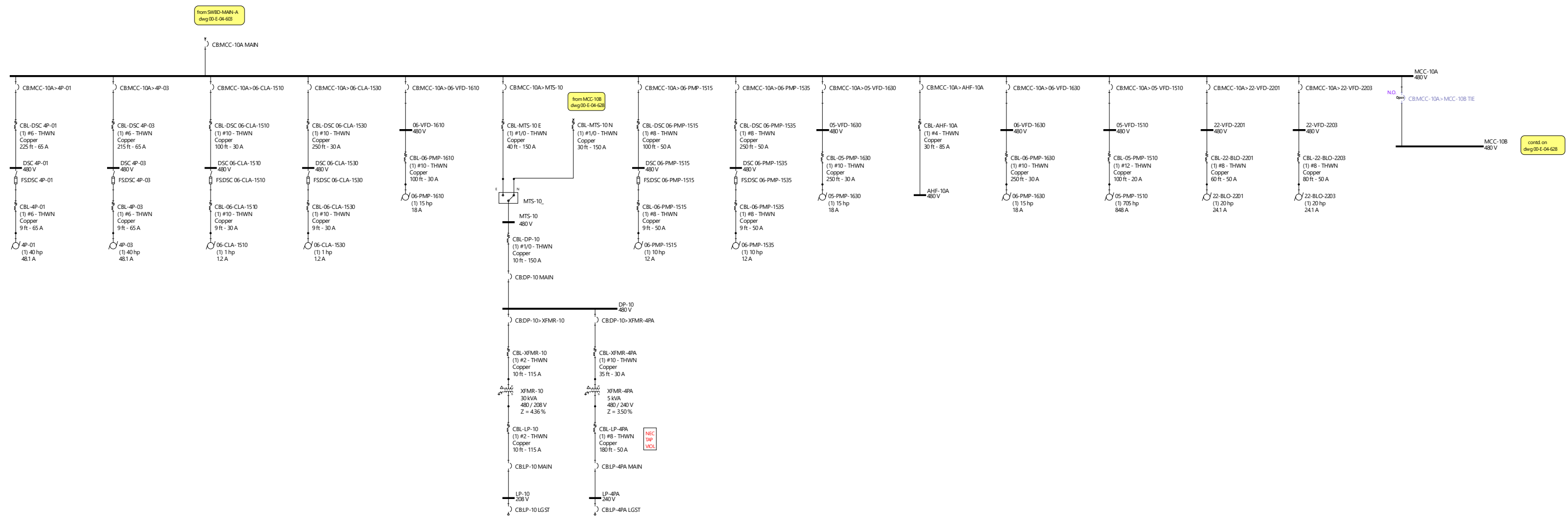


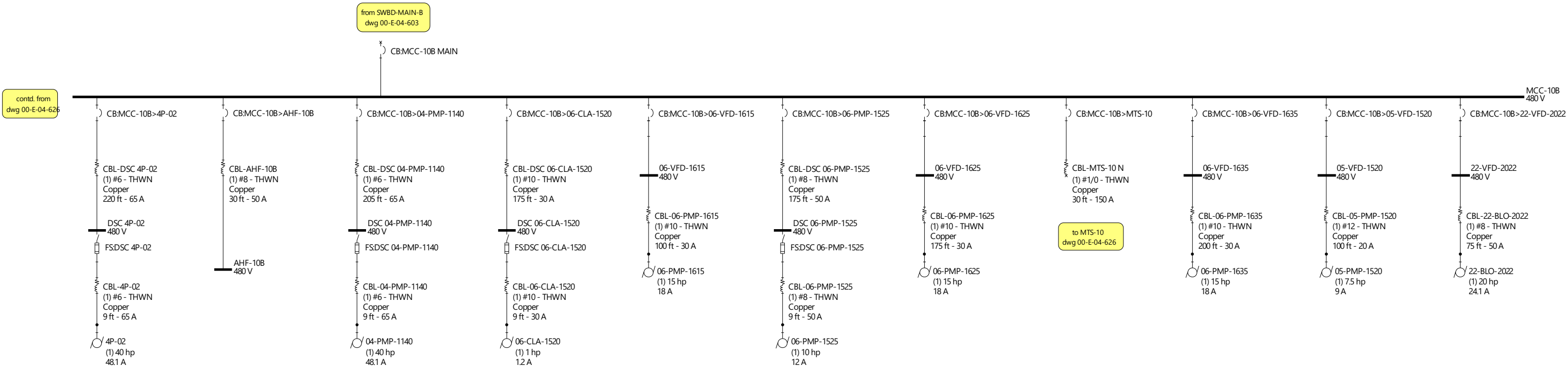


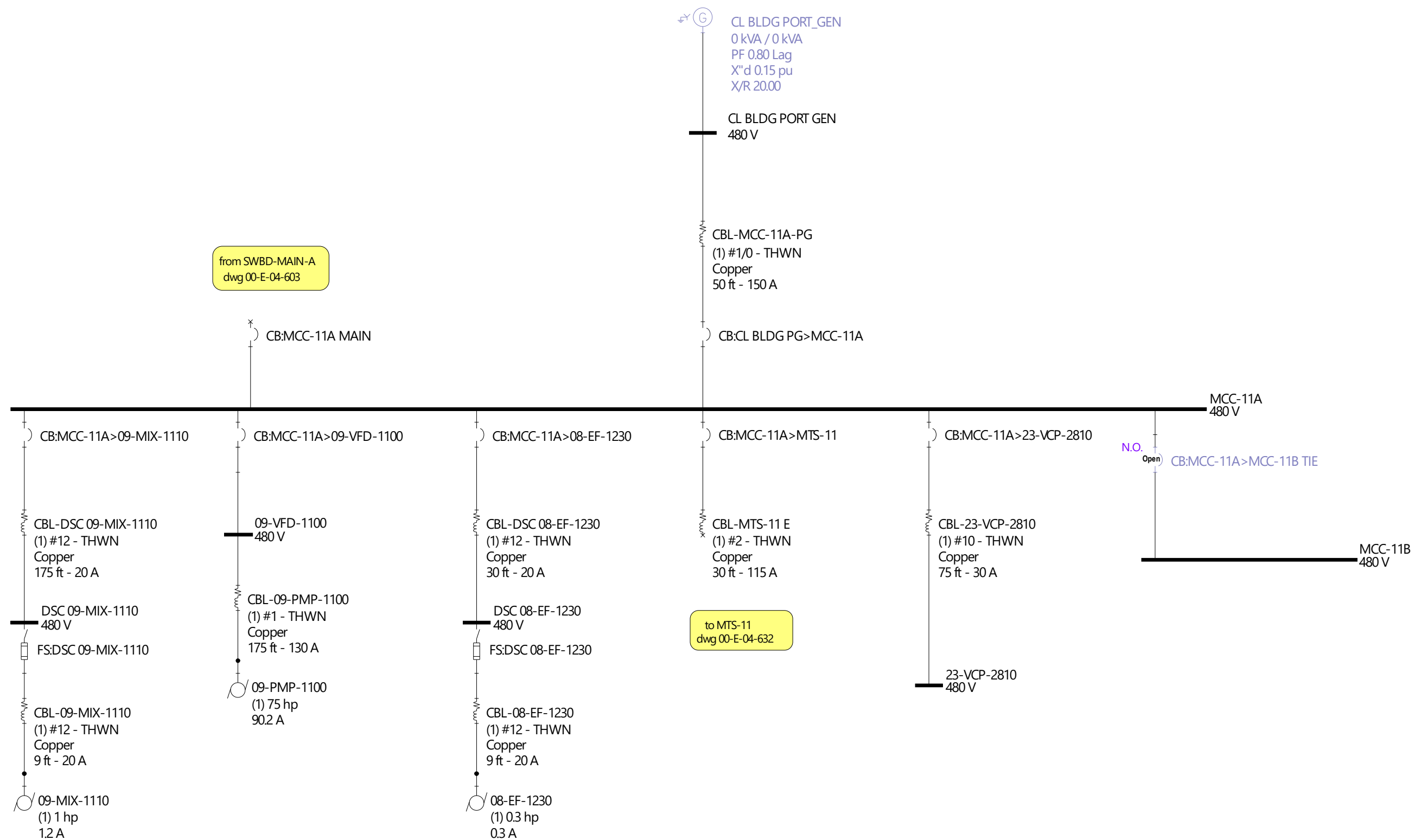
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from SWBD-MAIN-B
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from MCC-11A
dwg 00-E-04-630

